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Bowen et al.

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(54) **INSECT INHIBITORY PROTEINS**
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CPC **C12N 15/8286** (2013.01); **C07K 14/195** (2013.01); **C07K 14/32** (2013.01); **C07K 2319/00** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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(57) **ABSTRACT**

A pesticidal protein class exhibiting toxic activity against Coleopteran and Lepidopteran pest species is disclosed, and includes, but is not limited to, TIC7040, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389. DNA constructs are provided which contain a recombinant nucleic acid sequence encoding the TIC7040, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 pesticidal proteins. Transgenic plants, plant cells, seed, and plant parts resistant to Coleopteran and Lepidopteran infestation are provided which contain recombinant nucleic acid sequences encoding the TIC7040, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 pesticidal proteins of the present invention. Methods for detecting the presence of the recombinant nucleic acid sequences or the proteins of the present invention in a biological sample, and methods of controlling Coleopteran and Lepidopteran species pests using the TIC7040, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 pesticidal proteins are also provided.

31 Claims, No Drawings

Specification includes a Sequence Listing.

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INSECT INHIBITORY PROTEINS

REFERENCE TO RELATED APPLICATION

This application is a divisional of U.S. patent application Ser. No. 17/575,318, filed Jan. 13, 2022, which is a divisional of U.S. patent application Ser. No. 16/658,892, filed Oct. 21, 2019, now U.S. Pat. No. 11,225,672, which is a continuation of U.S. patent application Ser. No. 15/944,114, filed Apr. 3, 2018, now U.S. Pat. No. 10,465,205, which claims the benefit of U.S. Provisional Application No. 62/480,614, filed Apr. 3, 2017, each of which is herein incorporated by reference in its entirety.

INCORPORATION OF SEQUENCE LISTING

The file named "MONS443USD2_ST26.xml" containing a computer-readable form of the Sequence Listing was created on Mar. 13, 2024. This file is 398,784 bytes (measured in MS-Windows®) is contemporaneously filed by electronic submission (using the United States Patent Office Patent Center filing system), and is incorporated into this application by reference in its entirety.

FIELD OF THE INVENTION

The invention generally relates to the field of insect inhibitory proteins. A novel class of proteins exhibiting insect inhibitory activity against agriculturally-relevant pests of crop plants and seeds is disclosed. In particular, the disclosed protein are insecticidally active against agriculturally-relevant pests of crop plants and seeds, particularly Coleopteran and Lepidopteran species of insect pests. Plants, plant parts, and seeds containing a recombinant polynucleotide construct encoding one or more of the disclosed toxin proteins are provided.

BACKGROUND OF THE INVENTION

Improving crop yield from agriculturally significant plants including, among others, corn, soybean, sugarcane, rice, wheat, vegetables, and cotton, has become increasingly important. In addition to the growing need for agricultural products to feed, clothe and provide energy for a growing human population, climate-related effects and pressure from the growing population to use land other than for agricultural practices are predicted to reduce the amount of arable land available for farming. These factors have led to grim forecasts of food security, particularly in the absence of major improvements in plant biotechnology and agronomic practices. In light of these pressures, environmentally sustainable improvements in technology, agricultural techniques, and pest management are vital tools to expand crop production on the limited amount of arable land available for farming.

Insects, particularly insects within the Lepidoptera and Coleoptera orders, are considered a major cause of damage to field crops, thereby decreasing crop yields over infested areas. Historically, the intensive application of synthetic chemical insecticides was relied upon as the pest control agent in agriculture. Concerns for the environment and human health, in addition to emerging resistance issues, stimulated the research and development of biological pesticides. This research effort led to the progressive discovery and use of various entomopathogenic microbial species, including bacteria.

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The biological control paradigm shifted when the potential of entomopathogenic bacteria, especially bacteria belonging to the genus *Bacillus*, was discovered and developed as a biological pest control agent. Strains of the bacterium *Bacillus thuringiensis* (Bt) have been used as a source for pesticidal proteins since it was discovered that Bt strains show a high toxicity against specific insects. Bt strains are known to produce delta-endotoxins that are localized within parasporal crystalline inclusion bodies at the onset of sporulation and during the stationary growth phase (e.g., Cry proteins), and are also known to produce secreted insecticidal protein. Upon ingestion by a susceptible insect, delta-endotoxins as well as secreted toxins exert their effects at the surface of the midgut epithelium, disrupting the cell membrane, leading to cell disruption and death. Genes encoding insecticidal proteins have also been identified in bacterial species other than Bt, including other *Bacillus* and a diversity of additional bacterial species, such as *Brevibacillus laterosporus*, *Lysinibacillus sphaericus* ("Ls" formerly known as *Bacillus sphaericus*) and *Paenibacillus popilliae*.

Crystalline and secreted soluble insecticidal toxins are highly specific for their hosts and have gained worldwide acceptance as alternatives to chemical insecticides. For example, insecticidal toxin proteins have been employed in various agricultural applications to protect agriculturally important plants from insect infestations, decrease the need for chemical pesticide applications, and increase yields. Insecticidal toxin proteins are used to control agriculturally-relevant pests of crop plants by mechanical methods, such as spraying to disperse microbial formulations containing various bacteria strains onto plant surfaces, and by using genetic transformation techniques to produce transgenic plants and seeds expressing insecticidal toxin protein.

The use of transgenic plants expressing insecticidal toxin proteins has been globally adapted. For example, in 2012, 26.1 million hectares were planted with transgenic crops expressing Bt toxins (James, C., Global Status of Commercialized Biotech/GM Crops: 2012. ISAAA Brief No. 44). The global use of transgenic insect-protected crops and the limited number of insecticidal toxin proteins used in these crops has created a selection pressure for existing insect alleles that impart resistance to the currently-utilized insecticidal proteins.

The development of resistance in target pests to insecticidal toxin proteins creates the continuing need for discovery and development of new forms of insecticidal toxin proteins that are useful for managing the increase in insect resistance to transgenic crops expressing insecticidal toxin proteins. New protein toxins with improved efficacy and which exhibit control over a broader spectrum of susceptible insect species will reduce the number of surviving insects which can develop resistance alleles. In addition, the use in one plant of two or more transgenic insecticidal toxin proteins toxic to the same insect pest and displaying different modes of action reduces the probability of resistance in any single target insect species.

Thus, the inventors herein disclose a novel protein toxin family from *Bacillus laterosporus* along with similar toxin proteins, variant proteins, and exemplary recombinant proteins that exhibit insecticidal activity against target Lepidopteran and Coleopteran, particularly against Western Corn Rootworm and Northern Corn Rootworm.

SUMMARY OF THE INVENTION

Disclosed herein is a novel group of pesticidal proteins with insect inhibitory activity (toxin proteins), referred to

herein as TIC7040-related protein toxins, which are shown to exhibit inhibitory activity against one or more pests of crop plants. The TIC7040 protein and proteins in the TIC7040 protein toxin class can be used alone or in combination with other insecticidal proteins and toxic agents in formulations and in planta, thus providing alternatives to insecticidal proteins and insecticide chemistries currently in use in agricultural systems.

In one embodiment, disclosed in this application is a recombinant nucleic acid molecule comprising a heterologous promoter operably linked to a polynucleotide segment encoding a pesticidal protein or fragment thereof, wherein: (a) said pesticidal protein comprises the amino acid sequence of SEQ ID NO:2, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:12, SEQ ID NO:14, SEQ ID NO:16, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:30, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:43, SEQ ID NO:46, SEQ ID NO:48, SEQ ID NO:50, SEQ ID NO:53, SEQ ID NO:55, SEQ ID NO:57, SEQ ID NO:59, SEQ ID NO:61, SEQ ID NO:63, SEQ ID NO:65, SEQ ID NO:67, SEQ ID NO:69, SEQ ID NO:71, SEQ ID NO:73, SEQ ID NO:75, SEQ ID NO:77, SEQ ID NO:79, SEQ ID NO:81, SEQ ID NO:83, SEQ ID NO:85, SEQ ID NO:87, SEQ ID NO:89, SEQ ID NO:91, SEQ ID NO:93, SEQ ID NO:95, SEQ ID NO:97, SEQ ID NO:99, SEQ ID NO:101, SEQ ID NO:103, SEQ ID NO:105, SEQ ID NO:107, SEQ ID NO:109, SEQ ID NO:111, SEQ ID NO:113, SEQ ID NO:115, SEQ ID NO:117, SEQ ID NO:119, SEQ ID NO:121, SEQ ID NO:123, SEQ ID NO:125, or SEQ ID NO:127; or (b) said pesticidal protein comprises an amino acid sequence having: (i) at least 75% identity to SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:30, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:36, SEQ ID NO:38, SEQ ID NO:40, SEQ ID NO:42, SEQ ID NO:44, SEQ ID NO:46, SEQ ID NO:48, SEQ ID NO:50, SEQ ID NO:52, SEQ ID NO:54, SEQ ID NO:56, SEQ ID NO:58, SEQ ID NO:60, SEQ ID NO:62, SEQ ID NO:64, SEQ ID NO:66, SEQ ID NO:68, SEQ ID NO:70, SEQ ID NO:72, SEQ ID NO:74, SEQ ID NO:76, SEQ ID NO:78, SEQ ID NO:80, SEQ ID NO:82,

SEQ ID NO:84, SEQ ID NO:86, SEQ ID NO:88, SEQ ID NO:90, SEQ ID NO:92, SEQ ID NO:94, SEQ ID NO:96, SEQ ID NO:98, SEQ ID NO:100, SEQ ID NO:102, SEQ ID NO:104, SEQ ID NO:106, SEQ ID NO:108, SEQ ID NO:110, SEQ ID NO:112, SEQ ID NO:114, SEQ ID NO:116, SEQ ID NO:118, SEQ ID NO:120, SEQ ID NO:122, SEQ ID NO:124, OR SEQ ID NO:126; or (d) said recombinant nucleic acid molecule is in operable linkage with a vector, and said vector is selected from the group consisting of a plasmid, phagemid, bacmid, cosmid, and a bacterial or yeast artificial chromosome. The recombinant nucleic acid molecule can comprise a sequence that functions to express the pesticidal protein in a plant; or is expressed in a plant cell to produce a pesticidally effective amount of pesticidal protein.

In another embodiment of this application are host cells comprising a recombinant nucleic acid molecule of the application, wherein the host cell is selected from the group consisting of a bacterial and a plant cell. Contemplated host cells include *Agrobacterium*, *Rhizobium*, *Bacillus*, *Brevibacillus*, *Escherichia*, *Pseudomonas*, *Klebsiella*, *Pantoea*, and *Erwinia*. In certain embodiments said *Bacillus* species is *Bacillus cereus* or *Bacillus thuringiensis*, said *Brevibacillus* is *Brevibacillus laterosperus*, or said *Escherichia* is *Escherichia coli*. Contemplated plant host cells include a dicotyledonous cell and a monocotyledonous cell. Further contemplated plant host cells include an alfalfa, banana, barley, bean, broccoli, cabbage, Brassica, carrot, cassava, castor, cauliflower, celery, chickpea, Chinese cabbage, citrus, coconut, coffee, corn, clover, cotton (*Gossypium* sp.), a cucurbit, cucumber, Douglas fir, eggplant, eucalyptus, flax, garlic, grape, hops, leek, lettuce, Loblolly pine, millets, melons, nut, oat, olive, onion, ornamental, palm, pasture grass, pea, peanut, pepper, pigeonpea, pine, potato, poplar, pumpkin, Radiata pine, radish, rapeseed, rice, rootstocks, rye, safflower, shrub, sorghum, Southern pine, soybean, spinach, squash, strawberry, sugar beet, sugarcane, sunflower, sweet corn, sweet gum, sweet potato, switchgrass, tea, tobacco, tomato, triticale, turf grass, watermelon, and wheat plant cell.

In yet another embodiment, the pesticidal protein exhibits activity against Coleopteran insect, including Western Corn Rootworm, Southern Corn Rootworm, Northern Corn Rootworm, Mexican Corn Rootworm, Brazilian Corn Rootworm, Colorado Potato Beetle, Brazilian Corn Rootworm complex consisting of *Diabrotica viridula* and *Diabrotica speciosa*, Crucifer Flea Beetle, Striped Flea Beetle, and Western Black Flea Beetle.

In another embodiment, the pesticidal protein exhibits activity against a Lepidopteran insect, including Black Cutworm, Corn Earworm, Diamondback Moth, European Corn Borer, Fall Armyworm, Southern Armyworm, Soybean Looper, Southwestern Corn Borer, Tobacco Budworm, Velvetbean Caterpillar, Sugarcane Borer, Lesser Cornstalk Borer, Black Armyworm, Beet Armyworm, Old World Bollworm, Oriental leaf Worm, or Pink Bollworm.

Also contemplated in this application are plants comprising a recombinant nucleic acid molecule comprising a heterologous promoter operably linked to a polynucleotide segment encoding a pesticidal protein or fragment thereof, wherein: (a) said pesticidal protein comprises the amino acid sequence of SEQ ID NO:2, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:12, SEQ ID NO:14, SEQ ID NO:16, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:30, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:43, SEQ ID NO:46, SEQ ID NO:48, SEQ ID

Commodity products comprising a detectable amount of the recombinant nucleic acid molecules disclosed in this application are contemplated. Such commodity products include commodity corn bagged by a grain handler, corn flakes, corn cakes, corn flour, corn meal, corn syrup, corn oil, corn silage, corn starch, corn cereal, and the like, and corresponding cotton commodity products such as whole or processed cotton seed, cotton oil, lint, seeds and plant parts processed for feed or food, fiber, paper, biomasses, and fuel

products such as fuel derived from cotton oil or pellets derived from cotton gin waste, and corresponding soybean commodity products such as whole or processed soybean seed, soybean oil, soybean protein, soybean meal, soybean flour, soybean flakes, soybean bran, soybean milk, soybean cheese, soybean wine, animal feed comprising soybean, paper comprising soybean, cream comprising soybean, soybean biomass, and fuel products produced using soybean plants and soybean plant parts, and corresponding rice, wheat, sorghum, pigeon pea, peanut, fruit, melon, and vegetable commodity products including where applicable, juices, concentrates, jams, jellies, marmalades, and other edible forms of such commodity products containing a detectable amount of such polynucleotides and or polypeptides of this application.

Also contemplated in this application is a method of producing seed comprising the recombinant nucleic acid molecules disclosed in this application. The method comprises planting at least one of the seed comprising the recombinant nucleic acid molecules disclosed in this application; growing plant from the seed; and harvesting seed from the plants, wherein the harvested seed comprises the recombinant nucleic acid molecules in this application.

In another illustrative embodiment, a plant resistant to insect infestation is provided, wherein the cells of said plant comprise: (a) a recombinant nucleic acid molecule encoding an insecticidally effective amount of a pesticidal protein as set forth in SEQ ID NO:2, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:12, SEQ ID NO:14, SEQ ID NO:16, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:30, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:43, SEQ ID NO:46, SEQ ID NO:48, SEQ ID NO:50, SEQ ID NO:53, SEQ ID NO:55, SEQ ID NO:57, SEQ ID NO:59, SEQ ID NO:61, SEQ ID NO:63, SEQ ID NO:65, SEQ ID NO:67, SEQ ID NO:69, SEQ ID NO:71, SEQ ID NO:73, SEQ ID NO:75, SEQ ID NO:77, SEQ ID NO:79, SEQ ID NO:81, SEQ ID NO:83, SEQ ID NO:85, SEQ ID NO:87, SEQ ID NO:89, SEQ ID NO:91, SEQ ID NO:93, SEQ ID NO:95, SEQ ID NO:97, SEQ ID NO:99, SEQ ID NO:101, SEQ ID NO:103, SEQ ID NO:105, SEQ ID NO:107, SEQ ID NO:109, SEQ ID NO:111, SEQ ID NO:113, SEQ ID NO:115, SEQ ID NO:117, SEQ ID NO:119, SEQ ID NO:121, SEQ ID NO:123, SEQ ID NO:125, or SEQ ID NO:127; or (b) an insecticidally effective amount of a protein comprising an amino acid sequence having: (i) at least 75% identity to SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:63, SEQ ID NO:65, SEQ ID NO:67, SEQ ID NO:69, SEQ ID NO:75, SEQ ID NO:85, SEQ ID NO:87, SEQ ID NO:89, SEQ ID NO:91, SEQ ID NO:93, SEQ ID NO:95, SEQ ID NO:97, SEQ ID NO:99, SEQ ID NO:101, SEQ ID NO:103, SEQ ID NO:105, SEQ ID NO:107, or SEQ ID NO:109; or (ii) at least 80% to SEQ ID NO:111 SEQ ID NO: 113, SEQ ID NO:115, SEQ ID NO:119, SEQ ID NO:125, or SEQ ID NO:127; or (iii) at least 85% identity to SEQ ID NO:121 or SEQ ID NO:123; or (iv) at least 90% identity to SEQ ID NO:2, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:14, SEQ ID NO:34, SEQ ID NO:43, SEQ ID NO:46, SEQ ID NO:48, SEQ ID NO:50, SEQ ID NO:57, SEQ ID NO:71, SEQ ID NO:73, SEQ ID NO:77, SEQ ID NO:79, SEQ ID NO:81, or SEQ ID NO:117; or (v) at least 93% identity to SEQ ID NO:12, SEQ ID NO:16, SEQ ID NO:30, and SEQ ID NO:59; or (vi) at least 94% identity to SEQ ID NO:53 and SEQ ID NO:55; or (vii) at least 99% identity to SEQ ID NO:32, SEQ ID NO:61, or SEQ ID NO:83.

Also disclosed in this application are methods for controlling a Coleopteran or Lepidopteran species pest, and controlling a Coleopteran or Lepidopteran species pest infestation of a plant, particularly a crop plant. The method comprises, in one embodiment, (a) contacting the pest with an insecticidally effective amount of one or more pesticidal proteins as set forth in SEQ ID NO:2, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:12, SEQ ID NO:14, SEQ ID NO:16, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:30, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:43, SEQ ID NO:46, SEQ ID NO:48, SEQ ID NO:50, SEQ ID NO:53, SEQ ID NO:55, SEQ ID NO:57, SEQ ID NO:59, SEQ ID NO:61, SEQ ID NO:63, SEQ ID NO:65, SEQ ID NO:67, SEQ ID NO:69, SEQ ID NO:71, SEQ ID NO:73, SEQ ID NO:75, SEQ ID NO:77, SEQ ID NO:79, SEQ ID NO:81, SEQ ID NO:83, SEQ ID NO:85, SEQ ID NO:87, SEQ ID NO:89, SEQ ID NO:91, SEQ ID NO:93, SEQ ID NO:95, SEQ ID NO:97, SEQ ID NO:99, SEQ ID NO:101, SEQ ID NO:103, SEQ ID NO:105, SEQ ID NO:107, SEQ ID NO:109, SEQ ID NO:111, SEQ ID NO:113, SEQ ID NO:115, SEQ ID NO:117, SEQ ID NO:119, SEQ ID NO:121, SEQ ID NO:123, SEQ ID NO:125, or SEQ ID NO:127; or (b) contacting the pest with an insecticidally effective amount of one or more pesticidal proteins comprising an amino acid sequence having: (i) at least 75% identity to SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:63, SEQ ID NO:65, SEQ ID NO:67, SEQ ID NO:69, SEQ ID NO:75, SEQ ID NO:85, SEQ ID NO:87, SEQ ID NO:89, SEQ ID NO:91, SEQ ID NO:93, SEQ ID NO:95, SEQ ID NO:97, SEQ ID NO:99, SEQ ID NO:101, SEQ ID NO:103, SEQ ID NO:105, SEQ ID NO:107, or SEQ ID NO:109; or (ii) at least 80% to SEQ ID NO:111 SEQ ID NO: 113, SEQ ID NO:115, SEQ ID NO:119, SEQ ID NO:125, or SEQ ID NO:127; or (iii) at least 85% identity to SEQ ID NO:121 or SEQ ID NO:123; or (iv) at least 90% identity to SEQ ID NO:2, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:14, SEQ ID NO:34, SEQ ID NO:43, SEQ ID NO:46, SEQ ID NO:48, SEQ ID NO:50, SEQ ID NO:57, SEQ ID NO:71, SEQ ID NO:73, SEQ ID NO:77, SEQ ID NO:79, SEQ ID NO:81, or SEQ ID NO:117; or (v) at least 93% identity to SEQ ID NO:12, SEQ ID NO:16, SEQ ID NO:30, and SEQ ID NO:59; or (vi) at least 94% identity to SEQ ID NO:53 and SEQ ID NO:55; or (vii) at least 99% identity to SEQ ID NO:32, SEQ ID NO:61, or SEQ ID NO:83.

Further provided herein is a method of detecting the presence of a recombinant nucleic acid molecule comprising a polynucleotide segment encoding a pesticidal protein or fragment thereof, wherein: (a) said pesticidal protein comprises the amino acid sequence of SEQ ID NO:2, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:12, SEQ ID NO:14, SEQ ID NO:16, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:30, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:43, SEQ ID NO:46, SEQ ID NO:48, SEQ ID NO:50, SEQ ID NO:53, SEQ ID NO:55, SEQ ID NO:57, SEQ ID NO:59, SEQ ID NO:61, SEQ ID NO:63, SEQ ID NO:65, SEQ ID NO:67, SEQ ID NO:69, SEQ ID NO:71, SEQ ID NO:73, SEQ ID NO:75, SEQ ID NO:77, SEQ ID NO:79, SEQ ID NO:81, SEQ ID NO:83, SEQ ID NO:85, SEQ ID NO:87, SEQ ID NO:89, SEQ ID NO:91, SEQ ID NO:93, SEQ ID NO:95, SEQ ID NO:97, SEQ ID NO:99, SEQ ID NO:101, SEQ ID NO:103, SEQ ID NO:105, SEQ ID NO:107, SEQ ID NO:109, SEQ ID NO:111, SEQ ID NO:113, SEQ ID NO:115, SEQ ID NO:117, SEQ ID NO:119, SEQ ID NO:121, SEQ ID NO:123, SEQ ID NO:125, or SEQ ID NO:127; or (b) an insecticidally effective amount of a protein comprising an amino acid sequence having: (i) at least 75% identity to SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:63, SEQ ID NO:65, SEQ ID NO:67, SEQ ID NO:69, SEQ ID NO:75, SEQ ID NO:85, SEQ ID NO:87, SEQ ID NO:89, SEQ ID NO:91, SEQ ID NO:93, SEQ ID NO:95, SEQ ID NO:97, SEQ ID NO:99, SEQ ID NO:101, SEQ ID NO:103, SEQ ID NO:105, SEQ ID NO:107, or SEQ ID NO:109; or (ii) at least 80% to SEQ ID NO:111 SEQ ID NO: 113, SEQ ID NO:115, SEQ ID NO:119, SEQ ID NO:125, or SEQ ID NO:127; or (iii) at least 85% identity to SEQ ID NO:121 or SEQ ID NO:123; or (iv) at least 90% identity to SEQ ID NO:2, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:14, SEQ ID NO:34, SEQ ID NO:43, SEQ ID NO:46, SEQ ID NO:48, SEQ ID NO:50, SEQ ID NO:57, SEQ ID NO:71, SEQ ID NO:73, SEQ ID NO:77, SEQ ID NO:79, SEQ ID NO:81, or SEQ ID NO:117; or (v) at least 93% identity to SEQ ID NO:12, SEQ ID NO:16, SEQ ID NO:30, and SEQ ID NO:59; or (vi) at least 94% identity to SEQ ID NO:53 and SEQ ID NO:55; or (vii) at least 99% identity to SEQ ID NO:32, SEQ ID NO:61, or SEQ ID NO:83.

NO:117, SEQ ID NO:119, SEQ ID NO:121, SEQ ID NO:123, SEQ ID NO:125, or SEQ ID NO:127; or (b) said pesticidal protein comprises an amino acid sequence having: (i) at least 75% identity to SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:63, SEQ ID NO:65, SEQ ID NO:67, SEQ ID NO:69, SEQ ID NO:75, SEQ ID NO:85, SEQ ID NO:87, SEQ ID NO:89, SEQ ID NO:91, SEQ ID NO:93, SEQ ID NO:95, SEQ ID NO:97, SEQ ID NO:99, SEQ ID NO:101, SEQ ID NO:103, SEQ ID NO:105, SEQ ID NO:107, or SEQ ID NO:109; or (ii) at least 80% to SEQ ID NO:111 SEQ ID NO: 113, SEQ ID NO: 115, SEQ ID NO:119, SEQ ID NO:125, or SEQ ID NO:127; or (iii) at least 85% identity to SEQ ID NO:121 or SEQ ID NO:123; or (iv) at least 90% identity to SEQ ID NO:2, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:14, SEQ ID NO:34, SEQ ID NO:43, SEQ ID NO:46, SEQ ID NO:48, SEQ ID NO:50, SEQ ID NO:57, SEQ ID NO:71, SEQ ID NO:73, SEQ ID NO:77, SEQ ID NO:79, SEQ ID NO:81, or SEQ ID NO:117; or (v) at least 93% identity to SEQ ID NO:12, SEQ ID NO:16, SEQ ID NO:30, and SEQ ID NO:59; or (vi) at least 94% identity to SEQ ID NO:53 and SEQ ID NO:55; or (vii) at least 99% identity to SEQ ID NO:32, SEQ ID NO:61, or SEQ ID NO:83; or (c) said polynucleotide segment hybridizes to a polynucleotide having the nucleotide sequence of SEQ ID NO:1, SEQ ID NO:3, SEQ ID NO:5, SEQ ID NO:7, SEQ ID NO:9, SEQ ID NO:11, SEQ ID NO:13, SEQ ID NO:15, SEQ ID NO:17, SEQ ID NO:19, SEQ ID NO:21, SEQ ID NO:23, SEQ ID NO:25, SEQ ID NO:27, SEQ ID NO:29, SEQ ID NO:31, SEQ ID NO:33, SEQ ID NO:35, SEQ ID NO:36, SEQ ID NO:37, SEQ ID NO:38, SEQ ID NO:39, SEQ ID NO:40, SEQ ID NO:41, SEQ ID NO:42, SEQ ID NO:44, SEQ ID NO:45, SEQ ID NO:47, SEQ ID NO:49, SEQ ID NO:51, SEQ ID NO:52, SEQ ID NO:54, SEQ ID NO:56, SEQ ID NO:58, SEQ ID NO:60, SEQ ID NO:62, SEQ ID NO:64, SEQ ID NO:66, SEQ ID NO:68, SEQ ID NO:70, SEQ ID NO:72, SEQ ID NO:74, SEQ ID NO:76, SEQ ID NO:78, SEQ ID NO:80, SEQ ID NO:82, SEQ ID NO:84, SEQ ID NO:86, SEQ ID NO:88, SEQ ID NO:90, SEQ ID NO:92, SEQ ID NO:94, SEQ ID NO:96, SEQ ID NO:98, SEQ ID NO:100, SEQ ID NO:102, SEQ ID NO:104, SEQ ID NO:106, SEQ ID NO:108, SEQ ID NO:110, SEQ ID NO:112, SEQ ID NO:114, SEQ ID NO:116, SEQ ID NO:118, SEQ ID NO:120, SEQ ID NO:122, SEQ ID NO:124, or SEQ ID NO:126. In one embodiment of the invention, the method comprises contacting a sample of nucleic acids with a nucleic acid probe that hybridizes under stringent hybridization conditions with genomic DNA from a plant comprising a polynucleotide segment encoding a pesticidal protein or fragment thereof provided herein, and does not hybridize under such hybridization conditions with genomic DNA from an otherwise isogenic plant that does not comprise the segment, wherein the probe is homologous or complementary to SEQ ID NO:35, SEQ ID NO:36, SEQ ID NO:37, SEQ ID NO:38, SEQ ID NO:39, SEQ ID NO:40, SEQ ID NO:41, SEQ ID NO:42, SEQ ID NO:44, SEQ ID NO:45, SEQ ID NO:47, SEQ ID NO:49, SEQ ID NO:51, SEQ ID NO:52, SEQ ID NO:54, SEQ ID NO:56, SEQ ID NO:58, SEQ ID NO:60, SEQ ID NO:62, SEQ ID NO:64, SEQ ID NO:66, SEQ ID NO:68, SEQ ID NO:76, SEQ ID NO:78, SEQ ID NO:80, SEQ ID NO:82, SEQ ID NO:84, SEQ ID NO:86, SEQ ID NO:88, SEQ ID NO:90, SEQ ID NO:92, SEQ ID NO:94, SEQ ID NO:96, SEQ ID NO:98, SEQ ID NO:100, SEQ ID NO:102, SEQ ID NO:104, SEQ ID NO:106, SEQ ID NO:108, SEQ ID NO:110, SEQ ID NO:112, SEQ ID NO:114, SEQ ID NO:116, SEQ ID

NO:118, SEQ ID NO:120, SEQ ID NO:122, SEQ ID NO:124, OR SEQ ID NO:126, or a sequence that encodes a pesticidal protein comprising an amino acid sequence having: (i) at least 75% identity to SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:63, SEQ ID NO:65, SEQ ID NO:67, SEQ ID NO:69, SEQ ID NO:75, SEQ ID NO:85, SEQ ID NO:87, SEQ ID NO:89, SEQ ID NO:91, SEQ ID NO:93, SEQ ID NO:95, SEQ ID NO:97, SEQ ID NO:99, SEQ ID NO:101, SEQ ID NO:103, SEQ ID NO:105, SEQ ID NO:107, or SEQ ID NO:109; or (ii) at least 80% to SEQ ID NO:111 SEQ ID NO: 113, SEQ ID NO:115, SEQ ID NO:119, SEQ ID NO:125, or SEQ ID NO:127; or (iii) at least 85% identity to SEQ ID NO:121 or SEQ ID NO:123; or (iv) at least 90% identity to SEQ ID NO:2, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:14, SEQ ID NO:34, SEQ ID NO:43, SEQ ID NO:46, SEQ ID NO:48, SEQ ID NO:50, SEQ ID NO:57, SEQ ID NO:71, SEQ ID NO:73, SEQ ID NO:77, SEQ ID NO:79, SEQ ID NO:81, or SEQ ID NO:117; or (v) at least 93% identity to SEQ ID NO:12, SEQ ID NO:16, SEQ ID NO:30, and SEQ ID NO:59; or (vi) at least 94% identity to SEQ ID NO:53 and SEQ ID NO:55; or (vii) at least 99% identity to SEQ ID NO:32, SEQ ID NO:61, or SEQ ID NO:83. The method may further comprise (a) subjecting the sample and probe to stringent hybridization conditions; and (b) detecting hybridization of the probe with DNA of the sample.

Also provided by the invention are methods of detecting the presence of a pesticidal protein or fragment thereof in a sample comprising protein, wherein said pesticidal protein comprises the amino acid sequence of SEQ ID NO:2, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:12, SEQ ID NO:14, SEQ ID NO:16, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:30, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:43, SEQ ID NO:46, SEQ ID NO:48, SEQ ID NO:50, SEQ ID NO:53, SEQ ID NO:55, SEQ ID NO:57, SEQ ID NO:59, SEQ ID NO:61, SEQ ID NO:63, SEQ ID NO:65, SEQ ID NO:67, SEQ ID NO:69, SEQ ID NO:71, SEQ ID NO:73, SEQ ID NO:75, SEQ ID NO:77, SEQ ID NO:79, SEQ ID NO:81, SEQ ID NO:83, SEQ ID NO:85, SEQ ID NO:87, SEQ ID NO:89, SEQ ID NO:91, SEQ ID NO:93, SEQ ID NO:95, SEQ ID NO:97, SEQ ID NO:99, SEQ ID NO:101, SEQ ID NO:103, SEQ ID NO:105, SEQ ID NO:107, SEQ ID NO:109, SEQ ID NO:111, SEQ ID NO:113, SEQ ID NO:115, SEQ ID NO:117, SEQ ID NO:119, SEQ ID NO:121, SEQ ID NO:123, SEQ ID NO:125, or SEQ ID NO:127; or said pesticidal protein comprises an amino acid sequence having: (i) at least 75% identity to SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:63, SEQ ID NO:65, SEQ ID NO:67, SEQ ID NO:69, SEQ ID NO:75, SEQ ID NO:85, SEQ ID NO:87, SEQ ID NO:89, SEQ ID NO:91, SEQ ID NO:93, SEQ ID NO:95, SEQ ID NO:97, SEQ ID NO:99, SEQ ID NO:101, SEQ ID NO:103, SEQ ID NO:105, SEQ ID NO:107, or SEQ ID NO:109; or (ii) at least 80% to SEQ ID NO:111 SEQ ID NO: 113, SEQ ID NO:115, SEQ ID NO:119, SEQ ID NO:125, or SEQ ID NO:127; or (iii) at least 85% identity to SEQ ID NO:121 or SEQ ID NO:123; or (iv) at least 90% identity to SEQ ID NO:2, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:14, SEQ ID NO:34, SEQ ID NO:43, SEQ ID NO:46, SEQ ID NO:48, SEQ ID NO:50, SEQ ID NO:57, SEQ ID NO:71, SEQ ID NO:73, SEQ ID NO:77, SEQ ID NO:79, SEQ ID NO:81, or SEQ ID NO:117; or (v) at least 93% identity to SEQ ID NO:12, SEQ ID NO:16, SEQ ID NO:30, and

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SEQ ID NO:59; or (vi) at least 94% identity to SEQ ID NO:53 and SEQ ID NO:55; or (vii) at least 99% identity to SEQ ID NO:32, SEQ ID NO:61, or SEQ ID NO:83. In one embodiment, the method comprises: (a) contacting a sample with an immunoreactive antibody; and (b) detecting the presence of the protein. In some embodiments the step of detecting comprises an ELISA, or a Western blot.

BRIEF DESCRIPTION OF THE SEQUENCES

SEQ ID NO:1 is a nucleic acid sequence obtained from *Brevibacillus laterosporus* strain DSC005019 encoding a TIC7040 pesticidal protein sequence.

SEQ ID NO:2 is the amino acid sequence of the TIC7040 protein.

SEQ ID NO:3 is a nucleic acid sequence obtained from *Brevibacillus laterosporus* strain DSC005019 encoding a TIC7040HT pesticidal protein sequence.

SEQ ID NO:4 is the amino acid sequence of the TIC7040HT protein.

SEQ ID NO:5 is a nucleic acid sequence encoding a TIC7040_4 pesticidal protein sequence which comprises a C-terminal truncation relative to the TIC7040HT protein.

SEQ ID NO:6 is the amino acid sequence of the TIC7040_4 protein, consisting of amino acids 1 through 671 of TIC7040HT.

SEQ ID NO:7 is a nucleic acid sequence encoding a TIC7040_5 pesticidal protein sequence which comprises an N-terminal and C-terminal truncation relative to the TIC7040HT protein.

SEQ ID NO:8 is the amino acid sequence of the TIC7040_5 protein, comprising amino acids 13 through 611 of TIC7040HT.

SEQ ID NO:9 is a nucleic acid sequence encoding a TIC7040_6 pesticidal protein sequence which comprises an N-terminal and C-terminal truncation relative to the TIC7040HT protein.

SEQ ID NO:10 is the amino acid sequence of the TIC7040_6 protein, comprising amino acids 13 through 671 of TIC7040HT.

SEQ ID NO:11 is a nucleic acid sequence obtained from *Brevibacillus laterosporus* strain DSC005973 encoding a TIC7042 pesticidal protein sequence.

SEQ ID NO:12 is the amino acid sequence of the TIC7042 protein.

SEQ ID NO:13 is a nucleic acid sequence obtained from *Brevibacillus laterosporus* strain DSC006713 encoding a TIC7381 pesticidal protein sequence.

SEQ ID NO:14 is the amino acid sequence of the TIC7381 protein.

SEQ ID NO:15 is a nucleic acid sequence obtained from *Brevibacillus laterosporus* strain DSC007657 encoding a TIC7382 pesticidal protein sequence.

SEQ ID NO:16 is the amino acid sequence of the TIC7382 protein.

SEQ ID NO:17 is a nucleic acid sequence obtained from *Brevibacillus laterosporus* strain DSC008106 encoding a TIC7383 pesticidal protein sequence.

SEQ ID NO:18 is the amino acid sequence of the TIC7383 protein.

SEQ ID NO:19 is a nucleic acid sequence encoding a TIC7383_2 protein which comprises an N-terminal truncation relative to the TIC7383 protein.

SEQ ID NO:20 is the amino acid sequence of the TIC7383_2 protein, comprising amino acids 15 through 1256 of TIC7383.

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SEQ ID NO:21 is a nucleic acid sequence encoding a TIC7383_3 protein which comprises a C-terminal truncation relative to the TIC7383 protein.

SEQ ID NO:22 is the amino acid sequence of the TIC7383_3 protein and consists of amino acids 1 through 659 of TIC7383.

SEQ ID NO:23 is a nucleic acid sequence encoding a TIC7383_4 protein which comprises a C-terminal truncation relative to the TIC7383 protein.

SEQ ID NO:24 is the amino acid sequence of the TIC7383_4 protein and consists of amino acids 1 through 679 of TIC7383.

SEQ ID NO:25 is a nucleic acid sequence encoding a TIC7383_5 protein which comprises an N-terminal and C-terminal truncation relative to the TIC7383 protein.

SEQ ID NO:26 is the amino acid sequence of the TIC7383_5 protein and comprises amino acids 15 through 659 of TIC7383.

SEQ ID NO:27 is a nucleic acid sequence encoding a TIC7383_6 protein which comprises an N-terminal and C-terminal truncation relative to the TIC7383 protein.

SEQ ID NO:28 is the amino acid sequence of the TIC7383_6 protein, comprising amino acids 15 through 679 of TIC7383.

SEQ ID NO:29 is a nucleic acid sequence obtained from *Brevibacillus laterosporus* strain DSC007651 encoding a TIC7386 pesticidal protein sequence.

SEQ ID NO:30 is the amino acid sequence of the TIC7386 protein.

SEQ ID NO:31 is a nucleic acid sequence obtained from *Brevibacillus laterosporus* strain DSC007962 encoding a TIC7388 pesticidal protein sequence.

SEQ ID NO:32 is the amino acid sequence of the TIC7388 protein.

SEQ ID NO:33 is a nucleic acid sequence obtained from *Brevibacillus laterosporus* strain DSC006878 encoding a TIC7389 pesticidal protein sequence.

SEQ ID NO:34 is the amino acid sequence of the TIC7389 protein.

SEQ ID NO:35 is a synthetic coding sequence, CR-BREla.TIC7040.nno_Mc:1, encoding a TIC7040 pesticidal protein used for expression in plant cells.

SEQ ID NO:36 is a synthetic coding sequence, CR-BREla.TIC7040_10.nno_Mc:1, encoding a TIC7040HT pesticidal protein used for expression in plant cells.

SEQ ID NO:37 is a synthetic coding sequence, CR-BREla.TIC7040_10.nno_Mc:3, encoding a TIC7040HT pesticidal protein used for expression in plant cells.

SEQ ID NO:38 is a synthetic coding sequence, CR-BREla.TIC7040_10.nno_Mc:4, encoding a TIC7040HT pesticidal protein used for expression in plant cells.

SEQ ID NO:39 is a synthetic coding sequence, CR-BREla.TIC7040_10.nno_Mc:5, encoding a TIC7040HT pesticidal protein used for expression in plant cells.

SEQ ID NO:40 is a synthetic coding sequence, CR-BREla.TIC7040_10.nno_Mc:6, encoding a TIC7040HT pesticidal protein used for expression in plant cells.

SEQ ID NO:41 is a synthetic coding sequence, CR-BREla.TIC7040_10.nno_Mc:7, encoding a TIC7040HT pesticidal protein used for expression in plant cells.

SEQ ID NO:42 is a synthetic coding sequence, CR-BREla.TIC7040_1.nno_Mc:1, encoding a protein having an N-terminal and C-terminal truncation relative to the TIC7040HT protein used for expression in plant cells.

SEQ ID NO:43 is the amino acid sequence of the CR-BREla.TIC7040_1.nno_Mc:1 protein, comprising amino acids 15 through 651 of TIC7040HT.

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SEQ ID NO: 44 is a synthetic coding sequence, CR-BREla.TIC7040_2.nno_Mc:1, encoding a TIC7040_6 (SEQ ID NO:10) pesticidal protein sequence which comprises an N-terminal and C-terminal truncation relative to the TIC7040HT protein used for expression in plant cells.

SEQ ID NO:45 is a synthetic coding sequence, CR-BREla.TIC7040_11.nno_Mc:1, encoding a protein having an N-terminal and C-terminal truncation relative to the TIC7040HT protein used for expression in plant cells.

SEQ ID NO:46 is the amino acid sequence of the CR-BREla.TIC7040_11.nno_Mc:1 protein, comprising amino acids 14 through 671 of TIC7040HT.

SEQ ID NO:47 is a synthetic coding sequence, CR-BREla.TIC7040_12.nno_Mc:2, encoding a protein having a C-terminal truncation relative to the TIC7040HT protein used for expression in plant cells.

SEQ ID NO:48 is the amino acid sequence of the CR-BREla.TIC7040_12.nno_Mc:2 protein, consisting of amino acids 1 through 660 of TIC7040HT.

SEQ ID NO:49 is a synthetic coding sequence, CR-BREla.TIC7040_13.nno_Mc:1, which encodes a protein having a C-terminal truncation relative to the TIC7040HT protein used for expression in plant cells.

SEQ ID NO:50 is the amino acid sequence of the CR-BREla.TIC7040_13.nno_Mc:1 protein, consisting of amino acids 1 through 627 of TIC7040HT.

SEQ ID NO:51 is a synthetic coding sequence, CR-BREla.TIC7042_1.nno_Mc:1, which encodes a TIC7042 protein (SEQ ID NO:12) used for expression in plant cells.

SEQ ID NO:52 is a synthetic coding sequence, CR-BREla.TIC7042_1.nno_Mc:1, which encodes a protein having an N-terminal and C-terminal truncation relative to the TIC7042 protein used for expression in plant cells.

SEQ ID NO:53 is the amino acid sequence of the CR-BREla.TIC7042_1.nno_Mc:1 protein, comprising amino acids 11 through 646 of TIC7042.

SEQ ID NO:54 is a synthetic coding sequence, CR-BREla.TIC7042_2.nno_Mc:1, which encodes a protein having an N-terminal and C-terminal truncation relative to the TIC7042 protein used for expression in plant cells.

SEQ ID NO:55 is the amino acid sequence of the CR-BREla.TIC7042_2.nno_Mc:1 protein, comprising amino acids 11 through 665 of TIC7042.

SEQ ID NO:56 is a synthetic coding sequence, CR-BREla.TIC7381_1.nno_Mc:1, which encodes a TIC7381 protein wherein an additional alanine codon is inserted immediately following the initiating methionine codon used for expression in plant cells.

SEQ ID NO:57 is the amino acid sequence of CR-BREla.TIC7381_1.nno_Mc:1, wherein an additional alanine amino acid is inserted immediately following the initiating methionine relative to the TIC7381 protein sequence.

SEQ ID NO:58 is a synthetic coding sequence used for expression in plant cells, CR-BREla.TIC7382_1.nno_Mc:1, which encodes a TIC7382 protein wherein an additional alanine codon is inserted immediately following the initiating methionine codon.

SEQ ID NO:59 is the amino acid sequence of CR-BREla.TIC7382_1.nno_Mc:1, wherein an additional alanine amino acid is inserted immediately following the initiating methionine relative to the TIC7381 protein sequence.

SEQ ID NO:60 is a synthetic coding sequence used for expression in plant cells, CR-BREla.TIC7382_2.nno_Mc:1, which encodes a CR-BREla.TIC7382_2.nno_Mc:1 protein comprising a C-terminal truncation relative to the TIC7382

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protein and wherein an additional alanine codon is inserted immediately following the initiating methionine codon.

SEQ ID NO:61 is the amino acid sequence of CR-BREla.TIC7382_2.nno_Mc:1 comprising a C-terminal deletion and wherein an additional alanine amino acid is inserted immediately following the initiating methionine relative to the TIC7382 protein sequence, and comprises amino acids 1 through 659 of TIC7382.

SEQ ID NO:62 is a synthetic coding sequence, CR-BREla.TIC7383_1.nno_Mc:1, which encodes a TIC7383 protein wherein an additional alanine codon is inserted immediately following the initiating methionine codon used for expression in plant cells.

SEQ ID NO:63 is the amino acid sequence of CR-BREla.TIC7383_1.nno_Mc:1, wherein an additional alanine amino acid is inserted immediately following the initiating methionine relative to the TIC7383 protein sequence.

SEQ ID NO:64 is a synthetic coding sequence, CR-BREla.TIC7383_7.nno_Mc:1, which encodes a CR-BREla.TIC7383_7.nno_Mc:1 protein comprising an N-terminal and C-terminal truncation relative to the TIC7383 protein and wherein an additional alanine codon is inserted immediately following the initiating methionine codon and which is used for expression in plant cells.

SEQ ID NO:65 is the amino acid sequence of CR-BREla.TIC7383_7.nno_Mc:1 comprising an N-terminal and C-terminal deletion and wherein an additional alanine amino acid is inserted immediately following the initiating methionine relative to the TIC7383 protein sequence, and comprises amino acids 54 through 668 of TIC7383.

SEQ ID NO:66 is a synthetic coding sequence, CR-BREla.TIC7383_8.nno_Mc:1, which encodes a CR-BREla.TIC7383_8.nno_Mc:1 protein comprising a C-terminal truncation relative to the TIC7383 protein, wherein an additional alanine codon is inserted immediately following the initiating methionine codon and which is used for expression in plant cells.

SEQ ID NO:67 is the amino acid sequence of CR-BREla.TIC7383_8.nno_Mc:1 comprising a C-terminal deletion, and wherein an additional alanine amino acid is inserted immediately following the initiating methionine relative to the TIC7383 protein sequence, and comprises amino acids 1 through 661 of TIC7383.

SEQ ID NO:68 is a synthetic coding sequence, CR-BREla.TIC7383_9.nno_Mc:1, which encodes a CR-BREla.TIC7383_9.nno_Mc:1 protein comprising a C-terminal truncation relative to the TIC7383 protein, wherein an additional alanine codon is inserted immediately following the initiating methionine codon, and which is used for expression in plant cells.

SEQ ID NO:69 is the amino acid sequence of CR-BREla.TIC7383_9.nno_Mc:1, comprising a C-terminal deletion wherein an additional alanine amino acid is inserted immediately following the initiating methionine relative to the TIC7383 protein sequence, and comprises amino acids 1 through 668 of TIC7383.

SEQ ID NO:70 is a coding sequence encoding the tryptic core (TIC7040HT_Tryp) of the TIC7040HT protein as determined by mass spectrometry.

SEQ ID NO:71 is the amino acid sequence of the tryptic core (TIC7040HT_Tryp) of the TIC7040HT protein as determined by mass spectrometry, and comprises amino acids 43 through 624 of TIC7040HT.

SEQ ID NO:72 is a coding sequence encoding the chymotryptic core (TIC7040HT_Chymo) of the TIC7040HT protein as determined by mass spectrometry.

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SEQ ID NO:73 is the amino acid sequence of the chymotryptic core (TIC7040HT_Chymo) of the TIC7040HT protein as determined by mass spectrometry, and comprises amino acids 45 through 641 of TIC7040HT.

SEQ ID NO:74 is a coding sequence encoding the tryptic core (TIC7383_Tryp) of the TIC7383 protein as determined by mass spectrometry.

SEQ ID NO:75 is the amino acid sequence of the tryptic core (TIC7383_Tryp) of the TIC7383 protein as determined by mass spectrometry, and comprises amino acids 55 through 668 of TIC7383.

SEQ ID NO:76 is a synthetic coding sequence, CR-BREla.TIC7040_14.nno_Mc:1, encoding a CR-BREla.TIC7040_14.nno_Mc:1 protein having an N-terminal and C-terminal truncation relative to the TIC7040HT protein, wherein an additional alanine codon is inserted immediately following the initiating methionine codon, and which is used for expression in plant cells.

SEQ ID NO:77 is the amino acid sequence of the CR-BREla.TIC7040_14.nno_Mc:1 protein, wherein an additional alanine amino acid is inserted immediately following the initiating methionine, and comprises amino acids 52 through 660 of TIC7040HT.

SEQ ID NO:78 is a synthetic coding sequence, CR-BREla.TIC7381_2.nno_Mc:1, encoding a CR-BREla.TIC7381_2.nno_Mc:1 protein having a C-terminal truncation relative to the TIC7381 protein, wherein an additional alanine codon is inserted immediately following the initiating methionine codon, and which is used for expression in plant cells.

SEQ ID NO:79 is the amino acid sequence of the CR-BREla.TIC7381_2.nno_Mc:1 protein, wherein an additional alanine amino acid is inserted immediately following the initiating methionine, and comprises amino acids 1 through 658 of TIC7381.

SEQ ID NO:80 is a synthetic coding sequence, CR-BREla.TIC7381_3.nno_Mc:1, encoding a CR-BREla.TIC7381_3.nno_Mc:1 protein having an N-terminal and C-terminal truncation relative to the TIC7381 protein, wherein an additional alanine codon is inserted immediately following the initiating methionine codon, and which is used for expression in plant cells.

SEQ ID NO:81 is the amino acid sequence of the CR-BREla.TIC7381_3.nno_Mc:1 protein, wherein an additional alanine amino acid is inserted immediately following the initiating methionine, and comprises amino acids 50 through 658 of TIC7381.

SEQ ID NO:82 is a synthetic coding sequence, CR-BREla.TIC7382_3.nno_Mc:1, encoding a CR-BREla.TIC7382_3.nno_Mc:1 protein having an N-terminal and C-terminal truncation relative to the TIC7382 protein, wherein an additional alanine codon is inserted immediately following the initiating methionine codon, and which is used for expression in plant cells.

SEQ ID NO:83 is the amino acid sequence of the CR-BREla.TIC7382_3.nno_Mc:1 protein, wherein an additional alanine amino acid is inserted immediately following the initiating methionine, and comprises amino acids 52 through 659 of TIC7382.

SEQ ID NO:84 is a synthetic coding sequence, CR-BREla.TIC7383_19.nno_Mc:1, encoding a CR-BREla.TIC7383_19.nno_Mc:1 protein having an N-terminal and C-terminal truncation relative to the TIC7383 protein, and which is used for expression in plant cells.

SEQ ID NO:85 is the amino acid sequence of the CR-BREla.TIC7383_19.nno_Mc:1 protein, and comprises amino acids 15 through 668 of TIC7383.

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SEQ ID NO:86 is a synthetic coding sequence, CR-BREla.TIC7383_20.nno_Mc:1, encoding a CR-BREla.TIC7383_20.nno_Mc:1 protein having an N-terminal and C-terminal truncation relative to the TIC7383 protein, and which is used for expression in plant cells.

SEQ ID NO:87 is the amino acid sequence of the CR-BREla.TIC7383_20.nno_Mc:1 protein, and comprises amino acids 15 through 661 of TIC7383.

SEQ ID NO:88 is a synthetic coding sequence, CR-BREla.TIC7383_21.nno_Mc:1, encoding a CR-BREla.TIC7383_21.nno_Mc:1 protein having an N-terminal and C-terminal truncation relative to the TIC7383 protein, wherein an additional alanine codon is inserted immediately following the initiating methionine codon, and which is used for expression in plant cells.

SEQ ID NO:89 is the amino acid sequence of the CR-BREla.TIC7383_21.nno_Mc:1 protein, wherein an additional alanine amino acid is inserted immediately following the initiating methionine, and comprises amino acids 54 through 661 of TIC7383.

SEQ ID NO:90 is a synthetic coding sequence, CR-BREla.TIC7383_22.nno_Mc:1, encoding a CR-BREla.TIC7383_22.nno_Mc:1 protein having an N-terminal and C-terminal truncation relative to the TIC7383 protein, and which is used for expression in plant cells.

SEQ ID NO:91 is the amino acid sequence of the CR-BREla.TIC7383_22.nno_Mc:1 protein, and comprises amino acids 54 through 668 of TIC7383.

SEQ ID NO:92 is a synthetic coding sequence, CR-BREla.TIC7383_23.nno_Mc:1, encoding a CR-BREla.TIC7383_23.nno_Mc:1 protein having an N-terminal and C-terminal truncation relative to the TIC7383 protein, and which is used for expression in plant cells.

SEQ ID NO:93 is the amino acid sequence of the CR-BREla.TIC7383_23.nno_Mc:1 protein, and comprises amino acids 54 through 661 of TIC7383.

SEQ ID NO:94 is a synthetic coding sequence, CR-BREla.TIC7383_24.nno_Mc:2, encoding a CR-BREla.TIC7383_24.nno_Mc:2 protein having an N-terminal and C-terminal truncation relative to the TIC7383 protein, and which is used for expression in plant cells.

SEQ ID NO:95 is the amino acid sequence of the CR-BREla.TIC7383_24.nno_Mc:2 protein, and comprises amino acids 73 through 661 of TIC7383.

SEQ ID NO:96 is a synthetic coding sequence, CR-BREla.TIC7383_25.nno_Mc:3, encoding a CR-BREla.TIC7383_25.nno_Mc:3 protein having an N-terminal and C-terminal truncation relative to the TIC7383 protein, and which is used for expression in plant cells.

SEQ ID NO:97 is the amino acid sequence of the CR-BREla.TIC7383_25.nno_Mc:3 protein, and comprises amino acids 94 through 661 of TIC7383.

SEQ ID NO:98 is a synthetic coding sequence, CR-BREla.TIC7383_26.nno_Mc:1, encoding a CR-BREla.TIC7383_26.nno_Mc:1 protein having an N-terminal and C-terminal truncation relative to the TIC7383 protein, and which is used for expression in plant cells.

SEQ ID NO:99 is the amino acid sequence of the CR-BREla.TIC7383_26.nno_Mc:1 protein, and comprises amino acids 114 through 661 of TIC7383.

SEQ ID NO:100 is a synthetic coding sequence, CR-BREla.TIC7383_27.nno_Mc:1, encoding a CR-BREla.TIC7383_27.nno_Mc:1 protein having an N-terminal and C-terminal truncation relative to the TIC7383 protein, wherein an additional alanine codon is inserted immediately following the initiating methionine codon, and which is used for expression in plant cells.

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SEQ ID NO:101 is the amino acid sequence of the CR-BREla.TIC7383_27.nno_Mc:1 protein, wherein an additional alanine amino acid is inserted immediately following the initiating methionine, and comprises amino acids 54 through 658 of TIC7383.

SEQ ID NO:102 is a synthetic coding sequence, CR-BREla.TIC7383_28.nno_Mc:1, encoding a CR-BREla.TIC7383_28.nno_Mc:1 protein having an N-terminal and C-terminal truncation relative to the TIC7383 protein, and which is used for expression in plant cells.

SEQ ID NO:103 is the amino acid sequence of the CR-BREla.TIC7383_28.nno_Mc:1 protein, and comprises amino acids 15 through 658 of TIC7383.

SEQ ID NO:104 is a synthetic coding sequence, CR-BREla.TIC7383_29.nno_Mc:1, encoding a CR-BREla.TIC7383_29.nno_Mc:1 protein having a C-terminal truncation relative to the TIC7383 protein, wherein an additional alanine codon is inserted immediately following the initiating methionine codon, and which is used for expression in plant cells.

SEQ ID NO:105 is the amino acid sequence of the CR-BREla.TIC7383_29.nno_Mc:1 protein, wherein an additional alanine amino acid is inserted immediately following the initiating methionine, and comprises amino acids 1 through 963 of TIC7383.

SEQ ID NO:106 is a synthetic coding sequence, CR-BREla.TIC7383_30.nno_Mc:1, encoding a CR-BREla.TIC7383_30.nno_Mc:1 protein, wherein an additional alanine codon is inserted immediately following the initiating methionine codon, and comprising mutations to the codons corresponding amino acid positions 964, 966, and 968 relative to TIC7383, and which is used for expression in plant cells.

SEQ ID NO:107 is the amino acid sequence of the CR-BREla.TIC7383_30.nno_Mc:1 protein, wherein an additional alanine amino acid is inserted immediately following the initiating methionine, and comprises the mutations, K964A; R966A; K968A, relative to TIC7383.

SEQ ID NO:108 is a synthetic coding sequence, CR-BREla.TIC7383_31.nno_Mc:1, encoding a CR-BREla.TIC7383_31.nno_Mc:1 protein having a C-terminal truncation relative to the TIC7383 protein, wherein an additional alanine codon is inserted immediately following the initiating methionine codon, and comprising mutations to the codons corresponding amino acid positions 964, 966, and 968 relative to TIC7383, and which is used for expression in plant cells.

SEQ ID NO:109 is the amino acid sequence of the CR-BREla.TIC7383_31.nno_Mc:1 protein, wherein an additional alanine amino acid is inserted immediately following the initiating methionine, and comprising amino acids 1 through 1065, and also comprises the mutations, K964A; R966A; K968A, relative to TIC7383.

SEQ ID NO:110 is a synthetic coding sequence, CR-BREla.TIC7383_32.nno_Mc:1, encoding a CR-BREla.TIC7383_32.nno_Mc:1 protein, wherein an additional alanine codon is inserted immediately following the initiating methionine codon, and comprising a deletion of the codons corresponding amino acid positions 964 through 969 relative to TIC7383, and which is used for expression in plant cells.

SEQ ID NO:111 is the amino acid sequence of the CR-BREla.TIC7383_32.nno_Mc:1 protein, wherein an additional alanine amino acid is inserted immediately following the initiating methionine, and comprising a deletion of amino acids 964 through 969 relative to TIC7383.

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SEQ ID NO:112 is a synthetic coding sequence, GOI-TIC10743.nno_Mc:1, encoding a GOI-TIC10743.nno_Mc:1 chimeric protein comprised of domains one and two of TIC7383 and domain three of TIC7042.

SEQ ID NO:113 is the amino acid sequence of the GOI-TIC10743.nno_Mc:1 chimeric protein.

SEQ ID NO:114 is a synthetic coding sequence, GOI-TIC10744.nno_Mc:1, encoding a GOI-TIC10744.nno_Mc:1 chimeric protein comprised of domains one and two of TIC7383 and domain three of TIC7381.

SEQ ID NO:115 is the amino acid sequence of the GOI-TIC10744.nno_Mc:1 chimeric protein.

SEQ ID NO:116 is a synthetic coding sequence, GOI-TIC10745.nno_Mc:1, encoding a GOI-TIC10745.nno_Mc:1 chimeric protein comprised of domains one and two of TIC7383 and domain three of TIC7382.

SEQ ID NO:117 is the amino acid sequence of the GOI-TIC10745.nno_Mc:1 chimeric protein.

SEQ ID NO:118 is a synthetic coding sequence, GOI-TIC10746.nno_Mc:1, encoding a GOI-TIC10746.nno_Mc:1 chimeric protein comprised of domains one and two of TIC7382 and domain three of TIC7383.

SEQ ID NO:119 is the amino acid sequence of the GOI-TIC10746.nno_Mc:1 chimeric protein.

SEQ ID NO:120 is a synthetic coding sequence, GOI-TIC10747.nno_Mc:1, encoding a GOI-TIC10747.nno_Mc:1 chimeric protein comprised of domains one and two of TIC7381 and domain three of TIC7383.

SEQ ID NO:121 is the amino acid sequence of the GOI-TIC10747.nno_Mc:1 chimeric protein.

SEQ ID NO:122 is a synthetic coding sequence, GOI-TIC10748.nno_Mc:1, encoding a GOI-TIC10748.nno_Mc:1 chimeric protein comprised of domains one and two of TIC7042 and domain three of TIC7383.

SEQ ID NO:123 is the amino acid sequence of the GOI-TIC10748.nno_Mc:1 chimeric protein.

SEQ ID NO:124 is a synthetic coding sequence, TIC10764NTermExt1, encoding a TIC10764NTermExt1 chimeric protein comprised of domains one and two of TIC7382 and domain three of TIC7383 which also includes the N-terminal extension peptide derived from TIC7382. The N-terminal extension of TIC7382 comprises amino acids 1-51 of the TIC7382 toxin protein and is encoded by the first 153 nucleotides of the TIC7382 coding sequence.

SEQ ID NO:125 is the amino acid sequence of the TIC10764NTermExt1 chimeric protein.

SEQ ID NO:126 is a synthetic coding sequence, TIC10764NTermExt2, encoding a TIC10764NTermExt2 chimeric protein comprised of domains one and two of TIC7382 and domain three of TIC7383 which also includes the N-terminal extension peptide derived from TIC7383. The N-terminal extension of TIC7383 comprises amino acids 1-53 of the TIC7383 toxin protein and is encoded by the first 159 nucleotides of the TIC7383 coding sequence.

SEQ ID NO:127 is the amino acid sequence of the TIC10764NTermExt2 chimeric protein.

DETAILED DESCRIPTION OF THE INVENTION

The problem in the art of agricultural pest control can be characterized as a need for new toxin proteins that are

efficacious against target pests, exhibit broad spectrum toxicity against target pest species, are capable of being expressed in plants without causing undesirable agronomic issues, and provide an alternative mode of action compared to current toxins that are used commercially in plants.

Novel insecticidal protein classes, exemplified by TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 and related family members that provide resistance against Coleopteran and Lepidopteran insect pests, and more particularly against corn rootworm pest species, are disclosed. Also disclosed are synthetic coding sequences designed for expression in a plant cell that encode TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, and TIC7383 related family members, along with N-terminal and C-terminal truncation variants and chimeras of proteins in TIC7040 toxin class. Further disclosed are recombinant nucleic acid molecules comprising a promoter in operable linkage to a coding sequence encoding a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, or TIC7389 toxin protein, or related family members, or fragments thereof.

Reference in this application to TIC7040, "TIC7040 protein", "TIC7040 protein toxin", "TIC7040 toxin protein", "TIC7040 pesticidal protein", "TIC7040-related toxins", or "TIC7040-related toxin protein", and the like, refer to any novel pesticidal protein or insect inhibitory protein, that comprises, that consists of, that is substantially homologous to, that is similar to, or that is derived from any pesticidal protein or insect inhibitory protein sequence of TIC7040 (SEQ ID NO:2) and pesticidal or insect inhibitory segments thereof, or combinations thereof, that confer activity against Coleopteran pests or Lepidopteran pests, including any protein exhibiting pesticidal or insect inhibitory activity if alignment of such protein with TIC7040 results in amino acid sequence identity of any fraction percentage from about 90 to about 100 percent.

Reference in this application to TIC7040HT, "TIC7040HT protein", "TIC7040HT protein toxin", "TIC7040HT toxin protein", "TIC7040HT pesticidal protein", "TIC7040HT-related toxins", or "TIC7040HT-related toxin protein", and the like, refer to any novel pesticidal protein or insect inhibitory protein, that comprises, that consists of, that is substantially homologous to, that is similar to, or that is derived from any pesticidal protein or insect inhibitory protein sequence of TIC7040HT (SEQ ID NO:4) and pesticidal or insect inhibitory segments thereof, or combinations thereof, that confer activity against Coleopteran pests or Lepidopteran pests, including any protein exhibiting pesticidal or insect inhibitory activity if alignment of such protein with TIC7040HT results in amino acid sequence identity of any fraction percentage from about 90 to about 100 percent.

Reference in this application to TIC7042, "TIC7042 protein", "TIC7042 protein toxin", "TIC7042 toxin protein", "TIC7042 pesticidal protein", "TIC7042-related toxins", or "TIC7042-related toxin protein", and the like, refer to any novel pesticidal protein or insect inhibitory protein, that comprises, that consists of, that is substantially homologous to, that is similar to, or that is derived from any pesticidal protein or insect inhibitory protein sequence of TIC7042 (SEQ ID NO:12) and pesticidal or insect inhibitory segments thereof, or combinations thereof, that confer activity against Coleopteran pests or Lepidopteran pests, including any protein exhibiting pesticidal or insect inhibitory activity if alignment of such protein with TIC7042 results in amino acid sequence identity of any fraction percentage from about 93 to about 100 percent.

Reference in this application to TIC7381, "TIC7381 protein", "TIC7381 protein toxin", "TIC7381 toxin protein", "TIC7381 pesticidal protein", "TIC7381-related toxins", or "TIC7381-related toxin protein", and the like, refer to any novel pesticidal protein or insect inhibitory protein, that comprises, that consists of, that is substantially homologous to, that is similar to, or that is derived from any pesticidal protein or insect inhibitory protein sequence of TIC7381 (SEQ ID NO:14) and pesticidal or insect inhibitory segments thereof, or combinations thereof, that confer activity against Coleopteran pests or Lepidopteran pests, including any protein exhibiting pesticidal or insect inhibitory activity if alignment of such protein with TIC7381 results in amino acid sequence identity of any fraction percentage from about 90 to about 100 percent.

Reference in this application to TIC7382, "TIC7382 protein", "TIC7382 protein toxin", "TIC7382 toxin protein", "TIC7382 pesticidal protein", "TIC7382-related toxins", or "TIC7382-related toxin protein", and the like, refer to any novel pesticidal protein or insect inhibitory protein, that comprises, that consists of, that is substantially homologous to, that is similar to, or that is derived from any pesticidal protein or insect inhibitory protein sequence of TIC7382 (SEQ ID NO:16) and pesticidal or insect inhibitory segments thereof, or combinations thereof, that confer activity against Coleopteran pests or Lepidopteran pests, including any protein exhibiting pesticidal or insect inhibitory activity if alignment of such protein with TIC7382 results in amino acid sequence identity of any fraction percentage from about 93 to about 100 percent.

Reference in this application to TIC7383, "TIC7383 protein", "TIC7383 protein toxin", "TIC7383 toxin protein", "TIC7383 pesticidal protein", "TIC7383-related toxins", or "TIC7383-related toxin protein", and the like, refer to any novel pesticidal protein or insect inhibitory protein, that comprises, that consists of, that is substantially homologous to, that is similar to, or that is derived from any pesticidal protein or insect inhibitory protein sequence of TIC7383 (SEQ ID NO:18) and pesticidal or insect inhibitory segments thereof, or combinations thereof, that confer activity against Coleopteran pests or Lepidopteran pests, including any protein exhibiting pesticidal or insect inhibitory activity if alignment of such protein with TIC7383 results in amino acid sequence identity of any fraction percentage from about 75 to about 100 percent.

Reference in this application to TIC7386, "TIC7386 protein", "TIC7386 protein toxin", "TIC7386 toxin protein", "TIC7386 pesticidal protein", "TIC7386-related toxins", or "TIC7386-related toxin protein", and the like, refer to any novel pesticidal protein or insect inhibitory protein, that comprises, that consists of, that is substantially homologous to, that is similar to, or that is derived from any pesticidal protein or insect inhibitory protein sequence of TIC7386 (SEQ ID NO:30) and pesticidal or insect inhibitory segments thereof, or combinations thereof, that confer activity against Coleopteran pests or Lepidopteran pests, including any protein exhibiting pesticidal or insect inhibitory activity if alignment of such protein with TIC7386 results in amino acid sequence identity of any fraction percentage from about 75 to about 100 percent.

Reference in this application to TIC7388, "TIC7388 protein", "TIC7388 protein toxin", "TIC7388 toxin protein", "TIC7388 pesticidal protein", "TIC7388-related toxins", or "TIC7388-related toxin protein", and the like, refer to any novel pesticidal protein or insect inhibitory protein, that comprises, that consists of, that is substantially homologous to, that is similar to, or that is derived from any pesticidal

protein or insect inhibitory protein sequence of TIC7388 (SEQ ID NO:32) and pesticidal or insect inhibitory segments thereof, or combinations thereof, that confer activity against Coleopteran pests or Lepidopteran pests, including any protein exhibiting pesticidal or insect inhibitory activity if alignment of such protein with TIC7388 results in amino acid sequence identity of any fraction percentage from about 99 to about 100 percent.

Reference in this application to TIC7389, "TIC7389 protein", "TIC7389 protein toxin", "TIC7389 toxin protein", "TIC7389 pesticidal protein", "TIC7389-related toxins", or "TIC7389-related toxin protein", and the like, refer to any novel pesticidal protein or insect inhibitory protein, that comprises, that consists of, that is substantially homologous to, that is similar to, or that is derived from any pesticidal protein or insect inhibitory protein sequence of TIC7389 (SEQ ID NO:34) and pesticidal or insect inhibitory segments thereof, or combinations thereof, that confer activity against Coleopteran pests or Lepidopteran pests, including any protein exhibiting pesticidal or insect inhibitory activity if alignment of such protein with TIC7389 results in amino acid sequence identity of any fraction percentage from about 90 to about 100 percent.

The term "segment" or "fragment" is used in this application to describe consecutive amino acid or nucleic acid sequences that are shorter than the complete amino acid or nucleic acid sequence describing a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, or TIC7389 protein or related family member insecticidal protein. In specific embodiments, fragments of a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, or TIC7389 protein are provided comprising at least about 50, at least about 75, at least about 95, at least about 100, at least about 125, at least about 150, at least about 175, at least about 200, at least about 225, at least about 250, at least about 275, at least about 300, at least about 500, at least about 600, at least about 700, at least about 750, at least about 800, at least about 850, at least about 900, at least about 950, at least about 1000, at least about 1050, at least about 1100, at least about 1150, at least about 1200, at least about 1250, at least about 1300, or at least about 1350 contiguous amino acids, or longer, of a protein having insecticidal activity as disclosed herein. In certain embodiments, the invention provides fragments of a protein provided herein, having the activity of the full length sequence. Methods for producing such fragments from a starting molecule are well known in the art.

A segment or fragment exhibiting insect inhibitory activity is also disclosed in this application if alignment of such segment or fragment, with the corresponding section of the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, or TIC7389 protein set forth in SEQ ID NOs:2, 4, 12, 14, 16, 18, 20, 30, 32, and 34, results in amino acid sequence identity of any fraction percentage from about 75 to about 100 percent between the segment or fragment and the corresponding section of the TIC7383 and TIC7386 proteins; or respectively results in amino acid sequence identity of any fraction percentage from about 90 to about 100 percent between the segment or fragment and the corresponding section of the TIC7040, TIC7040HT, TIC7381, and TIC7389 proteins; or respectively results in amino acid sequence identity of any fraction percentage from about 93 to about 100 percent between the segment or fragment and the corresponding section of the TIC7042 and TIC7382 proteins; or respectively results in amino acid sequence identity of any fraction percentage

from about 99 to about 100 percent between the segment or fragment and the corresponding section of the TIC7388 protein.

Reference in this application to the terms "active" or "activity", "pesticidal activity" or "pesticidal" or "insecticidal activity", "insect inhibitory" or "insecticidal" refer to efficacy of a toxic agent, such as a protein toxin, in inhibiting (inhibiting growth, feeding, fecundity, or viability), suppressing (suppressing growth, feeding, fecundity, or viability), controlling (controlling the pest infestation, controlling the pest feeding activities on a particular crop containing an effective amount of the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, or TIC7389 protein or related family member insecticidal protein) or killing (causing the morbidity, mortality, or reduced fecundity of) a pest. These terms are intended to include the result of providing a pesticidally effective amount of a toxic protein to a pest where the exposure of the pest to the toxic protein results in morbidity, mortality, reduced fecundity, or stunting. These terms also include repulsion of the pest from the plant, a tissue of the plant, a plant part, seed, plant cells, or from the particular geographic location where the plant may be growing, as a result of providing a pesticidally effective amount of the toxic protein in or on the plant. In general, pesticidal activity refers to the ability of a toxic protein to be effective in inhibiting the growth, development, viability, feeding behavior, mating behavior, fecundity, or any measurable decrease in the adverse effects caused by an insect feeding on this protein, protein fragment, protein segment or polynucleotide of a particular target pest, including but not limited to insects of the order Lepidoptera, Coleoptera or Hemiptera. The toxic protein can be produced by the plant or can be applied to the plant or to the environment within the location where the plant is located. The terms "bioactivity", "effective", "efficacious" or variations thereof are also terms interchangeably utilized in this application to describe the effects of proteins of the present invention on target insect pests.

A pesticidally effective amount of a toxic agent, when provided in the diet of a target pest, exhibits pesticidal activity when the toxic agent contacts the pest. A toxic agent can be a pesticidal protein or one or more chemical agents known in the art. Pesticidal or insecticidal chemical agents and pesticidal or insecticidal protein agents can be used alone or in combinations with each other.

Chemical agents include, but are not limited to, dsRNA molecules targeting specific genes for suppression in a target pest, organochlorides, organophosphates, carbamates, pyrethroids, neonicotinoids, and ryanoids. Pesticidal or insecticidal protein agents include the protein toxins set forth in this application, as well as other proteinaceous toxic agents including those that target Lepidopteran and Coleopteran pest species, as well as protein toxins that are used to control other plant pests such as Cry proteins available in the art for use in controlling Homopteran or Hemipteran species.

It is intended that reference to a pest, particularly a pest of a crop plant, means insect pests of crop plants, particularly those that are controlled by a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, or TIC7389 protein or related family member insecticidal protein. Reference to a pest can also include Homopteran and Hemipteran insect pests of plants, as well as nematodes and fungi when toxic agents targeting these pests are co-localized or present together with a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7388, TIC7389 proteins or related family member insecticidal protein, or a protein that is about 75 to about 100

percent identical to TIC7383 and TIC7386; or a protein that is about 90 to about 100 percent identical to TIC7040, TIC7040HT, TIC7381, and TIC7389; or a protein that is about 93 to about 100 percent identical to TIC7042, TIC7386, and TIC7382; or a protein that is about 99 to about 100 percent identical to TIC7388.

The TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389 and related family member insecticidal proteins exhibit insecticidal activity towards insect pests from the Coleopteran and Lepidopteran insect species, including adults, pupae, larvae, and neonates.

The insects of the order Lepidoptera include, but are not limited to, armyworms, cutworms, loopers, and heliothines in the Family Noctuidae, e.g., fall armyworm (*Spodoptera frugiperda*), Beet armyworm (*Spodoptera exigua*), Black armyworm (*Spodoptera exempta*), Bertha armyworm (*Mamestra configurata*), Southern armyworm (*Spodoptera eridania*), Black cutworm (*Agrotis ipsilon*), Cabbage looper (*Trichoplusia ni*), Soybean looper (*Pseudoplusia includens*), Velvetbean caterpillar (*Anticarsia gemmatilis*), Green cloverworm (*Hypena scabra*), Tobacco budworm (*Heliothis virescens*), Granulate cutworm (*Agrotis subterranea*), Armyworm (*Pseudaletia unipuncta*), Western cutworm (*Agrotis orthogonia*); borers, casebearers, webworms, cone-worms, cabbageworms and skeletonizers from the Family Pyralidae, e.g., European corn borer (*Ostrinia nubilalis*), Navel orangeworm (*Amyelois transitella*), Corn root webworm (*Crambus caliginosellus*), Sod webworm (*Herpetogramma licarsisalis*), Sunflower moth (*Homoeosoma electellum*), Lesser cornstalk borer (*Elasmopalpus lignosellus*); leafrollers, budworms, seed worms, and fruit worms in the Family Tortricidae, e.g., Codling moth (*Cydia pomonella*), Grape berry moth (*Endopiza viteana*), Oriental fruit moth (*Grapholitha molesta*), Sunflower bud moth (*Suleima helianthana*); and many other economically important Lepidoptera, e.g., Diamondback moth (*Plutella xylostella*), Pink bollworm (*Pectinophora gossypiella*) and Gypsy moth (*Lymantria dispar*). Other insect pests of order Lepidoptera include, e.g., Cotton leaf worm (*Alabama argillacea*), Fruit tree leaf roller (*Archips argyrospila*), European leafroller (*Archips rosana*) and other *Archips* species, (*Chilo suppressalis*, Asiatic rice borer, or rice stem borer), Rice leaf roller (*Cnaphalocrocis medinalis*), Corn root webworm (*Crambus caliginosellus*), Bluegrass webworm (*Crambus teterrellus*), Southwestern corn borer (*Diatraea grandiosella*), Sugarcane borer (*Diatraea saccharalis*), Spiny bollworm (*Earias insulana*), Spotted bollworm (*Earias vittella*), Old World bollworm (*Helicoverpa armigera*), Corn earworm (*Helicoverpa zea*, also known as soybean podworm and cotton bollworm), Western bean cutworm (*Striacosta albicosta*), European grape vine moth (*Lobesia botrana*), Citrus leafminer (*Phyllocnistis citrella*), large white butterfly (*Pieris brassicae*), small white butterfly (*Pieris rapae*, also known as imported cabbageworm), Tobacco cutworm (*Spodoptera litura*, also known as cluster caterpillar), and Tomato leafminer (*Tuta absoluta*).

The insects of the order Coleoptera include, but are not limited to, *Agriotes* spp., *Anthonomus* spp., *Atomaria linearis*, *Chaetocnema tibialis*, *Cosmopolites* spp., *Curculio* spp., *Dermestes* spp., *Diabrotica* spp., *Epilachna* spp., *Eremnus* spp., *Leptinotarsa decemlineata*, *Lissorhoptrus* spp., *Melolontha* spp., *Oryzaephilus* spp., *Otiiorhynchus* spp., *Phlyctinus* spp., *Popillia* spp., *Psylliodes* spp., *Rhizopertha* spp., *Scarabeidae*, *Sitophilus* spp., *Sitotroga* spp., *Tenebrio* spp., *Tribolium* spp. and *Trogoderma* spp, particularly when the pest is Western Corn Rootworm (*Diabrotica*

virgifera, WCR), Northern Corn Rootworm (*Diabrotica barberi*, NCR), Mexican Corn Rootworm (*Diabrotica virgifera zaeae*, MCR), Brazilian Corn Rootworm (*Diabrotica balteata*, BZR), Southern Corn Rootworm (*Diabrotica undecimpunctata howardii*, SCR), Colorado potato beetle (*Leptinotarsa decemlineata*, CPB), a Brazilian Corn Rootworm complex (BCR, consisting of *Diabrotica viridula* and *Diabrotica speciosa*), Crucifer Flea Beetle (*Phyllotreta cruciferae*), Striped Flea Beetle (*Phyllotreta striolata*), and Western Black Flea Beetle (*Phyllotreta pusilla*).

The insects of the order Hemiptera include, but are not limited to, Stink Bugs of the family Pentatomidae: Green Stink Bugs from the genus *Chinavia* (*Chinavia hilaris*, *Chinavia marginata*, and *Chinavia pensylvanica*), Stink bugs of the genus *Chlorochroa* (*Chlorochroa granulosa*, *Chlorochroa kaneii*, *Chlorochroa ligata*, *Chlorochroa lineate*, *Chlorochroa opuntiae*, *Chlorochroa persimilis*, *Chlorochroa rossiana*, *Chlorochroa sayi*, *Chlorochroa uhleri*, *Chlorochroa belfragii*, *Chlorochroa faceta*, *Chlorochroa osborni*, *Chlorochroa saucia*, and *Chlorochroa senilis*), Southern Green Stink Bug (*Nezara viridula*), Stink Bugs from the genus *Edessa* (*Edessa mediatibunda*, *Edessa bifida*, and *Edessa florida*), the Neotropical Brown Stink Bug (*Euschistus heros*), stink bugs from the genus *Euschistus* (*Euschistus acuminatus*, *Euschistus bififormis*, *Euschistus conspersus*, *Euschistus crenator*, *Euschistus egglestoni*, *Euschistus ictericus*, *Euschistus inflatus*, *Euschistus latimarginatus*, *Euschistus obscurus*, *Euschistus politus*, *Euschistus quadrator*, *Euschistus sevus*, *Euschistus strenuous*, *Euschistus tristigmus*, and *Euschistus variolarius*), Brown Marmorated Stink Bug (*Halyomorpha halys*), Red-Shouldered Stink Bug (*Thyanta accerra*), stink bugs of the genus *Thyanta* (*Thyanta calceata*, *Thyanta custator*, *Thyanta pallidovirens*, *Thyanta perditior*, *Thyanta maculata*, and *Thyanta pseudocasta*), the Green Belly Stink Bug (*Dichelops melacanthus*) and other stink bugs of the genus *Dichelops* (*Dichelops avilapirei*, *Dichelops bicolor*, *Dichelops dimidatus*, *Dichelops furcatus*, *Dichelops furcifrons*, *Dichelops lobatus*, *Dichelops miriamae*, *Dichelops nigrum*, *Dichelops peruanus*, *Dichelops phoenix*, and *Dichelops saltensis*), the Red Banded Stink Bug (*Piezodorus guildinii*) as well as *Piezodorus lituratus*; and insects of the family of Plataspididae such as Kudzu Bug (*Megacopta cribraria*), Western tarnished plant bug (*Lygus hesperus*), and Tarnished plant bug (*Lygus lineolaris*).

Reference in this application to an "isolated DNA molecule", or an equivalent term or phrase, is intended to mean that the DNA molecule is one that is present alone or in combination with other compositions, but not within its natural environment. For example, nucleic acid elements such as a coding sequence, intron sequence, untranslated leader sequence, promoter sequence, transcriptional termination sequence, and the like, that are naturally found within the DNA of the genome of an organism are not considered to be "isolated" so long as the element is within the genome of the organism and at the location within the genome in which it is naturally found. However, each of these elements, and subparts of these elements, would be "isolated" within the scope of this disclosure so long as the element is not within the genome of the organism and at the location within the genome in which it is naturally found. Similarly, a nucleotide sequence encoding an insecticidal protein or any naturally occurring insecticidal variant of that protein would be an isolated nucleotide sequence so long as the nucleotide sequence was not within the DNA of the bacterium from which the sequence encoding the protein is naturally found. A synthetic nucleotide sequence encoding the amino acid

sequence of the naturally occurring insecticidal protein would be considered to be isolated for the purposes of this disclosure. For the purposes of this disclosure, any transgenic nucleotide sequence, i.e., the nucleotide sequence of the DNA inserted into the genome of the cells of a plant or bacterium, or present in an extrachromosomal vector, would be considered to be an isolated nucleotide sequence whether it is present within the plasmid or similar structure used to transform the cells, within the genome of the plant or bacterium, or present in detectable amounts in tissues, progeny, biological samples or commodity products derived from the plant or bacterium.

As described further herein, an open reading frame (ORF) encoding TIC7040 (SEQ ID NO:1), was discovered in DNA obtained from *Brevibacillus laterosporus* strain DSC005019, which encodes the protein toxin presented as SEQ ID NO:2. Upon amplification using primers designed from the TIC7040 ORF, a slightly longer coding sequence was amplified using DNA obtained from *Brevibacillus laterosporus* strain DSC005019, herein referred to as TIC7040HT (SEQ ID NO:3), which encodes the protein presented as SEQ ID NO:4. Bioassay using microbial host cell-derived TIC7040HT protein demonstrated activity against the Coleopteran pests Western Corn Rootworm (*Diabrotica virgifera virgifera*, WCR), Cry3Bb-resistant Western Corn Rootworm (*Diabrotica virgifera virgifera*, WCRHP), Northern Corn Rootworm (*Diabrotica barberi*, NCR), Southern Corn Rootworm (*Diabrotica undecimpunctata howardii*, SCR), and Colorado potato beetle (*Leptinotarsa decemlineata*, CPB), and the Lepidopteran insect pests Corn Earworm (*Helicoverpa zea*, (CEW) also known as Soybean Podworm and Cotton Bollworm), Diamondback Moth (*Plutella xylostella*, DBM), European Corn Borer (*Ostrinia nubilalis*, ECB), Fall Armyworm (*Spodoptera frugiperda*, FAW), Soybean Looper (*Pseudoplusia includes*, SBL), Southwestern Corn Borer (*Diatraea grandiosella*, SWCB), and Velvetbean Caterpillar (*Anticarsia gemmatilis*, VBC).

As described further herein, an ORF encoding TIC7042 (SEQ ID NO:11), was discovered in DNA obtained from *Brevibacillus laterosporus* strain DSC005973 which encodes the protein toxin presented as SEQ ID NO:12. Bioassay using microbial host cell-derived TIC7042 protein demonstrated activity against the Coleopteran pests WCR, WCRHP, NCR, SCR, and CPB, and the Lepidopteran insect pests BCW, CEW, DBM, FAW, SBL, and VBC.

As described further herein, an ORF encoding TIC7381 (SEQ ID NO:13), was discovered in DNA obtained from *Brevibacillus laterosporus* strain DSC006713 which encodes the protein toxin presented as SEQ ID NO:14. Bioassay using microbial host cell-derived TIC7381 protein demonstrated activity against the Coleopteran pests WCR, WCRHP, NCR, SCR, and CPB, and the Lepidopteran insect pests CEW, DBM, ECB, FAW, SBL, SWCB, and VBC.

As described further herein, an ORF encoding TIC7382 (SEQ ID NO:15), was discovered in DNA obtained from *Brevibacillus laterosporus* strain DSC007657 which encodes the protein toxin presented as SEQ ID NO:16. Bioassay using microbial host cell-derived TIC7382 protein demonstrated activity against the Coleopteran pests WCR, WCRHP, NCR, SCR, and CPB, and the Lepidopteran insect pests DBM, ECB, SBL, SWCB, and VBC.

As described further herein, an ORF encoding TIC7383 (SEQ ID NO:17), was discovered in DNA obtained from *Brevibacillus laterosporus* strain DSC008106 which encodes the protein toxin presented as SEQ ID NO:18. Bioassay using microbial host cell-derived TIC7383 protein

demonstrated activity against the Coleopteran pests WCR, WCRHP, NCR, SCR, and CPB, and the Lepidopteran insect pests CEW, DBM, ECB, and VBC.

As described further herein, ORF's encoding TIC7386 (SEQ ID NO:29), TIC7388 (SEQ ID NO:31), and TIC7389 (SEQ ID NO:33) were discovered in DNA obtained from *Brevibacillus laterosporus* strains DSC007651, DSC007962, and DSC006878, respectively, and encode the proteins presented herein as SEQ ID NO:30, SEQ ID NO:32, and SEQ ID NO:34. Bioassay using microbial host cell-derived TIC7389 protein demonstrated activity against the Lepidopteran insect pest SBL and Coleopteran pests WCR and CPB.

Further described herein, chymotryptic and tryptic digests of TIC7040HT, and tryptic digests of TIC7383 maintained activity against WCR, relative to the full length TIC7040HT and TIC7383. In addition, truncations of TIC7040HT and TIC7383 maintained activity, and in some cases increased activity, against WCR and CPB, relative to the full length TIC7040HT and TIC7383.

Further described herein are chimeric toxins comprised of domains derived from TIC7381, TIC7382, TIC7383, and TIC7042. The chimeric toxin, TIC10743 (SEQ ID NO:113, encoded by SEQ ID NO:112) is comprised of domains one and two of TIC7383 and domain three of TIC7042. The chimeric toxin, TIC10744 (SEQ ID NO:115, encoded by SEQ ID NO:114) is comprised of domains one and two of TIC7383 and domain three of TIC7381. The chimeric toxin, TIC10745 (SEQ ID NO:117, encoded by SEQ ID NO:116) is comprised of domains one and two of TIC7383 and domain three of TIC7382. The chimeric toxin, TIC10746 (SEQ ID NO:119, encoded by SEQ ID NO:118) is comprised of domains one and two of TIC7382 and domain three of TIC7383. The chimeric toxin, TIC10747 (SEQ ID NO:121, encoded by SEQ ID NO:120) is comprised of domains one and two of TIC7381 and domain three of TIC7383. The chimeric toxin, TIC10748 (SEQ ID NO:123, encoded by SEQ ID NO:122) is comprised of domains one and two of TIC7042 and domain three of TIC7383. The chimeric toxin, TIC10746NTermExt1 (SEQ ID NO:125, encoded by SEQ ID NO:124) is comprised of domains one and two of TIC7382, domain three of TIC7383, and the N-terminal extension of TIC7382. The N-terminal extension of TIC7382 comprises amino acids 1-51 of the TIC7382 toxin protein and is encoded by the first 153 nucleotides of the TIC7382 coding sequence. The chimeric toxin, TIC10746NTermExt2 (SEQ ID NO:127, encoded by SEQ ID NO:126) is comprised of domains one and two of TIC7382, domain three of TIC7383, and the N-terminal extension of TIC7383. The N-terminal extension of TIC7383 comprises amino acids 1-53 of the TIC7383 toxin protein and is encoded by the first 159 nucleotides of the TIC7383 coding sequence.

For expression in plant cells, TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, or TIC7383, or variants, truncation variants, or chimeras thereof can be expressed to reside in the cytosol or targeted to various organelles of the plant cell. For example, targeting a protein to the chloroplast may result in increased levels of expressed protein in a transgenic plant while preventing off-phenotypes from occurring. Targeting may also result in an increase in pest resistance efficacy in the transgenic event. A target peptide or transit peptide is a short (3-70 amino acids long) peptide chain that directs the transport of a protein to a specific region in the cell, including the nucleus, mitochondria, endoplasmic reticulum (ER), chloroplast, apoplast, peroxisome and plasma membrane. Some target peptides are

cleaved from the protein by signal peptidases after the proteins are transported. For targeting to the chloroplast, proteins contain transit peptides which are around 40-50 amino acids. For descriptions of the use of chloroplast transit peptides, see U.S. Pat. Nos. 5,188,642 and 5,728,925. Many chloroplast-localized proteins are expressed from nuclear genes as precursors and are targeted to the chloroplast by a chloroplast transit peptide (CTP). Examples of such isolated chloroplast proteins include, but are not limited to, those associated with the small subunit (SSU) of ribulose-1,5,-bisphosphate carboxylase, ferredoxin, ferredoxin oxidoreductase, the light-harvesting complex protein I and protein II, thioredoxin F, enolpyruvyl shikimate phosphate synthase (EPSPS), and transit peptides described in U.S. Pat. No. 7,193,133. It has been demonstrated in vivo and in vitro that non-chloroplast proteins may be targeted to the chloroplast by use of protein fusions with a heterologous CTP and that the CTP is sufficient to target a protein to the chloroplast. Incorporation of a suitable chloroplast transit peptide such as the *Arabidopsis thaliana* EPSPS CTP (CTP2) (See, Klee et al., *Mol. Gen. Genet.* 210:437-442, 1987) or the *Petunia hybrida* EPSPS CTP (CTP4) (See, della-Cioppa et al., *Proc. Natl. Acad. Sci. USA* 83:6873-6877, 1986) has been shown to target heterologous EPSPS protein sequences to chloroplasts in transgenic plants (See, U.S. Pat. Nos. 5,627,061; 5,633,435; and 5,312,910; and EP 0218571; EP 189707; EP 508909; and EP 924299). For targeting TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, and TIC7383 and variants, truncation variants and chimeras thereof to the chloroplast, a sequence encoding a chloroplast transit peptide is placed 5' in operable linkage and in frame to a synthetic coding sequence encoding the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, and TIC7383 protein, or variants, truncation variants, or chimeras thereof, that has been designed for optimal expression in plant cells.

It is contemplated that additional toxin protein sequences related to TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 can be created by using the naturally occurring amino acid sequence of TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 to create novel proteins and with novel properties. The TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 toxin proteins can be aligned with other proteins similar to TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 to combine differences at the amino acid sequence level into novel amino acid sequence variants and making appropriate changes to the recombinant nucleic acid sequence encoding the variants.

This disclosure further contemplates that improved variants of the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 protein toxin classes can be engineered in planta by using various gene editing methods known in the art. Such technologies used for genome editing include, but are not limited to, ZFN (zinc-finger nuclease), meganucleases, TALEN (Transcription activator-like effector nucleases), and CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats)/Cas (CRISPR-associated) systems. These genome editing methods can be used to alter the toxin protein coding sequence transformed within a plant cell to a different toxin coding sequence. Specifically, through these methods, one or more codons within the toxin coding sequence is altered to engineer a new protein amino acid sequence. Alternatively, a fragment within the coding

sequence is replaced or deleted, or additional DNA fragments are inserted into the coding sequence, to engineer a new toxin coding sequence. The new coding sequence can encode a toxin protein with new properties such as increased activity or spectrum against insect pests, as well as provide activity against an insect pest species wherein resistance has developed against the original insect toxin protein. The plant cell comprising the gene edited toxin coding sequence can be used by methods known in the art to generate whole plants expressing the new toxin protein.

It is also contemplated that fragments of the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 proteins or protein variants thereof can be truncated forms wherein one or more amino acids are deleted from the N-terminal end, C-terminal end, the middle of the protein, or combinations thereof with insect inhibitory activity. These fragments can be naturally occurring or synthetic variants of TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 or protein variants thereof, but should retain the insect inhibitory activity of TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389.

Proteins that resemble the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 proteins can be identified by comparison to each other using various computer-based algorithms known in the art. For example, amino acid sequence identities of proteins related to TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 can be analyzed using a Clustal W alignment using these default parameters: Weight matrix: blosum, Gap opening penalty: 10.0, Gap extension penalty: 0.05, Hydrophilic gaps: On, Hydrophilic residues: GPSNDQERK, Residue-specific gap penalties: On (Thompson, et al (1994) *Nucleic Acids Research*, 22:4673-4680). Percent amino acid identity is further calculated by the product of 100% multiplied by (amino acid identities/length of subject protein). Other alignment algorithms are also available in the art and provide results similar to those obtained using a Clustal W alignment.

It is intended that a protein exhibiting insect inhibitory activity against a Lepidopteran or Coleopteran insect species is related to TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 if alignment of such query protein with TIC7383 or TIC7386 exhibits at least 75% to about 100% amino acid identity along the length of the query protein that is about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 100% amino acid sequence identity (or any fraction of a percentage in this range) between query and subject protein; or if alignment of such query protein with TIC7040HT, TIC7381, or TIC7389 exhibits at least 90% to about 100% amino acid identity along the length of the query protein that is about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 100% amino acid sequence identity (or any fraction of a percentage in this range) between query and subject protein; or if alignment of such query protein with TIC7042, or TIC7382 exhibits at least 93% to about 100% amino acid identity along the length of the query protein that is about 93%, 94%, 95%, 96%, 97%, 98%, 99%, 100% amino acid sequence identity (or any fraction of a percentage in this range) between query and subject protein; or if alignment of such query protein with TIC7040HT, TIC7381, or TIC7388 exhibits at least 99% to about 100%

amino acid identity along the length of the query protein (or any fraction of a percentage in this range) between query and subject protein.

Exemplary proteins TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 were aligned with each other using a Clustal W

algorithm. A pair-wise matrix of percent amino acid sequence identities for each of the full-length proteins was created, as reported in Table 1. The number of identical amino acids between two sequences is indicated in parenthesis.

TABLE 1

Pair-wise matrix display of exemplary proteins TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389.									
Toxin	TIC7042 SEQ ID NO: 12	TIC7386 SEQ ID NO: 30	TIC7388 SEQ ID NO: 32	TIC7382 SEQ ID NO: 16	TIC7040 SEQ ID NO: 2	TIC7040HT SEQ ID NO: 4	TIC7381 SEQ ID NO: 14	TIC7389 SEQ ID NO: 34	TIC7383 SEQ ID NO: 18
TIC7042 SEQ ID NO: 12	—	99.1 (1254)	92.3 (1169)	87.8 (1111)	85.7 (1085)	86.3 (1093)	86.7 (1097)	86.7 (1098)	73.3 (928)
TIC7386 SEQ ID NO: 30	98 (1254)	—	91.5 (1170)	86.7 (1109)	85.1 (1089)	85.7 (1096)	86 (1100)	86.1 (1101)	72.6 (928)
TIC7388 SEQ ID NO: 32	92.1 (1169)	92.2 (1170)	—	93.7 (1189)	88.8 (1127)	89.4 (1135)	89.8 (1140)	89.9 (1141)	76.1 (966)
TIC7382 SEQ ID NO: 16	89.2 (1111)	89 (1109)	95.4 (1189)	—	90.6 (1129)	91.4 (1139)	91.2 (1136)	91.5 (1140)	76.2 (949)
TIC7040 SEQ ID NO: 2	86.2 (1085)	86.5 (1089)	89.5 (1127)	89.7 (1129)	—	99.9 (1258)	98.5 (1240)	99 (1247)	76.6 (965)
TIC7040HT SEQ ID NO: 4	84.4 (1093)	84.6 (1096)	87.6 (1135)	88 (1139)	97.1 (1258)	—	96.6 (1251)	97.2 (1259)	74.8 (969)
TIC7381 SEQ ID NO: 14	86.4 (1097)	86.7 (1100)	89.8 (1140)	89.5 (1136)	97.7 (1240)	98.6 (1251)	—	99.5 (1263)	76.5 (971)
TIC7389 SEQ ID NO: 34	83.2 (1098)	83.5 (1101)	86.5 (1141)	86.4 (1140)	94.5 (1247)	95.5 (1259)	95.8 (1263)	—	73.9 (975)
TIC7383 SEQ ID NO: 18	73.9 (928)	73.9 (928)	76.9 (966)	75.6 (949)	76.8 (965)	77.1 (969)	77.3 (971)	77.6 (975)	—

In addition to percent identity, the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 proteins can also be related by primary structure (conserved amino acid motifs), by length (about 1243 to about 1259 amino acids) and by other characteristics. Bioinformatic analysis suggests that TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 belong to the Cry43 family of proteins. Characteristics of the T5C7040, TIC7040HT, T8C7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 proteins are reported in Table 2.

TABLE 2

Selected characteristics of the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 proteins and related family member proteins.									
Protein	Molecular Weight (in Daltons)	Amino Acid Length	Isoelectric Point	Charge at PH 7.0	No. of Strongly Basic (–) Amino Acids	No. of Strongly Acidic Amino Acids	No. of Hydrophobic Amino Acids	No. of Polar Amino Acids	
TIC7040	142822.03	1259	5.8063	–13.0	154	151	618	641	
TIC7040HT	146821.27	1295	5.8643	–14.0	161	156	636	659	
TIC7040_4	75400.92	671	7.0053	2.5	68	61	333	338	
TIC7040_5	67268.98	600	7.5498	4.0	61	53	303	297	
TIC7040_6	74075.56	660	7.0051	2.5	67	60	332	338	
TIC7040HT_Tryp	65150.72	582	8.1625	5.5	59	50	294	288	
TIC7040HT_Chymo	66845.58	597	7.8790	4.5	59	51	302	295	
CR-BREla.TIC7040_1.nno_Mc:1	71805.04	640	6.7230	1.0	62	57	325	315	

TABLE 2-continued

Selected characteristics of the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 proteins and related family member proteins.								
Protein	Molecular Weight (in Daltons)	Amino Acid Length	Isoelectric Point	Charge at PH 7.0	No. of Strongly Basic (–) Amino Acids	No. of Strongly Acidic Amino Acids	No. of Hydrophobic Amino Acids	No. of Polar Amino Acids
CR-BREla.TIC7040_11.nno_Mc:1	65281.91	583	8.1625	5.5	59	50	295	288
CR-BREla.TIC7040_12.nno_Mc:2	74125.45	660	6.5260	0.0	64	60	329	331
CR-BREla.TIC7040_13.nno_Mc:1	69983.90	624	7.2157	3.0	62	55	310	314
CR-BREla.TIC7040_14.nno_Mc:1	68315.18	611	6.8814	1.5	59	54	311	300
TIC7042	143469.54	1266	6.1143	–9.0	158	148	617	649
CR-BREla.TIC7042_1.nno_Mc:1	71715.51	637	6.6825	1.0	62	56	312	325
CR-BREla.TIC7042_2.nno_Mc:1	73986.03	657	6.9127	2.5	67	59	319	338
TIC7381	143793.24	1269	6.2077	–7.0	162	150	626	643
CR-BREla.TIC7381_1.nno_Mc:1	143850.30	1270	6.2077	–7.0	162	150	627	643
CR-BREla.TIC7381_2.nno_Mc:1	73892.21	659	6.5260	0.0	64	60	331	328
CR-BREla.TIC7381_3.nno_Mc:1	68253.10	611	6.8814	1.5	59	54	312	299
TIC7382	140890.86	1246	5.7572	–13.0	149	147	614	632
CR-BREla.TIC7382_1.nno_Mc:1	140961.94	1247	5.7572	–13.0	149	147	615	632
CR-BREla.TIC7382_2.nno_Mc:1	73858.00	660	6.6112	0.5	63	58	329	331
CR-BREla.TIC7382_3.nno_Mc:1	67924.61	610	6.9577	2.0	58	52	310	300
TIC7383	142470.64	1256	5.5492	–17.5	151	152	627	629
TIC7383_2	140903.05	1243	5.5471	–17.5	150	151	626	617
TIC7383_3	74526.97	659	5.7950	–3.0	61	61	343	316
TIC7383_4	76846.56	679	5.9941	–2.5	66	65	350	329
TIC7383_5	72959.38	646	5.7903	–3.0	60	60	342	304
TIC7383_6	75278.97	666	5.9913	–2.5	65	64	349	317
TIC7383_Tryp	69242.24	614	5.8225	–2.5	57	57	326	288
CR-BREla.TIC7383_1.nno_Mc:1	142541.72	1257	5.5492	–17.5	151	152	628	629
CR-BREla.TIC7383_7.nno_Mc:1	69545.62	617	5.8225	–2.5	57	57	328	289
CR-BREla.TIC7383_8.nno_Mc:1	74772.20	662	5.7950	–3.0	61	61	344	318
CR-BREla.TIC7383_9.nno_Mc:1	75669.20	669	5.5878	–4.0	62	63	347	322
CR-BREla.TIC7383_19.nno_Mc:1	74030.53	655	5.5815	–4.0	61	62	345	310
CR-BREla.TIC7383_20.nno_Mc:1	73133.53	648	5.7903	–3.0	60	60	342	306
CR-BREla.TIC7383_21.nno_Mc:1	68648.63	610	6.0997	–1.5	56	55	325	285
CR-BREla.TIC7383_22.nno_Mc:1	69474.54	616	5.8225	–2.5	57	57	327	289
CR-BREla.TIC7383_23.nno_Mc:1	68577.55	609	6.0997	–1.5	56	55	324	285
CR-BREla.TIC7383_24.nno_Mc:2	66643.51	590	6.6807	0.5	55	52	315	275
CR-BREla.TIC7383_25.nno_Mc:3	64647.09	569	6.3851	–0.5	54	52	299	270
CR-BREla.TIC7383_26.nno_Mc:1	62376.52	549	6.6802	0.5	54	51	287	262
CR-BREla.TIC7383_27.nno_Mc:1	68417.42	607	6.0997	–1.5	56	55	324	283
CR-BREla.TIC7383_28.nno_Mc:1	72902.33	645	5.7903	–3.0	60	60	341	304
CR-BREla.TIC7383_29.nno_Mc:1	108909.43	964	5.0905	–17.0	104	112	489	475
CR-BREla.TIC7383_30.nno_Mc:1	142342.42	1257	5.3793	–20.5	148	152	631	626
CR-BREla.TIC7383_31.nno_Mc:1	120993.09	1066	5.4956	–13.0	122	124	538	528
CR-BREla.TIC7383_32.nno_Mc:1	141715.71	1251	5.3252	–21.5	146	151	628	623
TIC7386	144952.03	1279	6.1338	–9.0	161	150	621	658
TIC7388	143562.87	1269	6.0794	–9.5	157	148	629	640
TIC7389	149398.97	1319	5.6940	–17.5	162	160	652	667
GOI-TIC10743.nno_Mc:1	69704.53	617	6.0297	–2.0	57	56	314	303
GOI-TIC10744.nno_Mc:1	69405.21	616	6.0308	–2.0	57	56	317	299
GOI-TIC10745.nno_Mc:1	69732.59	617	6.2812	–1.0	58	56	314	303
GOI-TIC10746.nno_Mc:1	67737.64	610	6.6433	0.5	57	53	324	286
GOI-TIC10747.nno_Mc:1	68393.51	612	6.8012	1.0	59	58	323	289
GOI-TIC10748.nno_Mc:1	68118.77	609	6.7284	1.0	59	54	312	297
TIC10746NTermExt1	73599.95	659	6.3342	–1.0	62	59	342	317
TIC10746NTermExt2	73790.14	661	6.3342	–1.0	62	59	342	319

As described further in the Examples of this application, recombinant nucleic acid molecule sequences encoding TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, and TIC7383, and variants, truncation variants, and chimeras thereof were designed for use in plants. Exemplary plant-optimized recombinant nucleic acid molecule sequences that were designed for use in plants are presented in Table 8 of Example 5, along with the corresponding nucleotide and protein sequences, description and modifications.

Expression cassettes and vectors containing these recombinant nucleic acid molecule sequences can be constructed and introduced into corn, soybean, cotton or other plant cells in accordance with transformation methods and techniques known in the art. For example, *Agrobacterium*-mediated transformation is described in U.S. Patent Application Pub-

lications 2009/0138985A1 (soybean), 2008/0280361A1 (soybean), 2009/0142837A1 (corn), 2008/0282432 (cotton), 2008/0256667 (cotton), 2003/0110531 (wheat), 2001/0042257 A1 (sugar beet), U.S. Pat. No. 5,750,871 (canola), 7,026,528 (wheat), and 6,365,807 (rice), and in Arencibia et al. (1998) Transgenic Res. 7:213-222 (sugarcane), all of which are incorporated herein by reference in their entirety. Transformed cells can be regenerated into transformed plants that express TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, or TIC7383. To test pesticidal activity, bioassays are performed in the presence of Lepidoptera pest larvae using plant leaf disks obtained from transformed plants, as described in the Examples. To test pesticidal activity against Coleopteran pests, transformed plants of R₀ and F₁ generation are used in root worm assay, as described

in the Examples. To test pesticidal activity against Hemipteran pests, pods, corn ears or leaves of transformed plants are used in assay, either from tissue removed from the plant or remaining on the plant as described in the Examples.

As an alternative to traditional transformation methods, a DNA sequence, such as a transgene, expression cassette(s), etc., may be inserted or integrated into a specific site or locus within the genome of a plant or plant cell via site-directed integration. Recombinant DNA construct(s) and molecule(s) of this disclosure may thus include a donor template sequence comprising at least one transgene, expression cassette, or other DNA sequence for insertion into the genome of the plant or plant cell. Such donor template for site-directed integration may further include one or two homology arms flanking an insertion sequence (i.e., the sequence, transgene, cassette, etc., to be inserted into the plant genome). The recombinant DNA construct(s) of this disclosure may further comprise an expression cassette(s) encoding a site-specific nuclease and/or any associated protein(s) to carry out site-directed integration. These nuclease expressing cassette(s) may be present in the same molecule or vector as the donor template (in cis) or on a separate molecule or vector (in trans). Several methods for site-directed integration are known in the art involving different proteins (or complexes of proteins and/or guide RNA) that cut the genomic DNA to produce a double strand break (DSB) or nick at a desired genomic site or locus. As understood in the art, during the process of repairing the DSB or nick introduced by the nuclease enzyme, the donor template DNA may become integrated into the genome at the site of the DSB or nick. The presence of the homology arm(s) in the donor template may promote the adoption and targeting of the insertion sequence into the plant genome during the repair process through homologous recombination, although an insertion event may occur through non-homologous end joining (NHEJ). Examples of site-specific nucleases that may be used include zinc-finger nucleases, engineered or native meganucleases, TALE-endonucleases, and RNA-guided endonucleases (e.g., Cas9 or Cpf1). For methods using RNA-guided site-specific nucleases (e.g., Cas9 or Cpf1), the recombinant DNA construct(s) will also comprise a sequence encoding one or more guide RNAs to direct the nuclease to the desired site within the plant genome.

Recombinant nucleic acid molecule compositions that encode TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 or related family member insecticidal proteins are contemplated. For example, TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 or related family member insecticidal proteins can be expressed with recombinant DNA constructs in which a polynucleotide molecule with an ORF encoding the protein is operably linked to genetic expression elements such as a promoter and any other regulatory element necessary for expression in the system for which the construct is intended. Non-limiting examples include a plant-functional promoter operably linked to the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 or related family member insecticidal protein encoding sequences for expression of the protein in plants or a Bt-functional promoter operably linked to a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 or related family member insecticidal protein encoding sequences for expression of the protein in a Bt bacterium or other *Bacillus* species. Other elements can be operably linked to the TIC7040,

TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 or related family member insecticidal protein encoding sequences including, but not limited to, enhancers, introns, untranslated leaders, encoded protein immobilization tags (HIS-tag), translocation peptides (i.e., plastid transit peptides, signal peptides), polypeptide sequences for post-translational modifying enzymes, ribosomal binding sites, and RNAi target sites. Exemplary recombinant polynucleotide molecules provided herewith include, but are not limited to, a heterologous promoter operably linked to a polynucleotide such as SEQ ID NO:1, SEQ ID NO:3, SEQ ID NO:5, SEQ ID NO:7, SEQ ID NO:9, SEQ ID NO:11, SEQ ID NO:13, SEQ ID NO:15, SEQ ID NO:17, SEQ ID NO:19, SEQ ID NO:21, SEQ ID NO:23, SEQ ID NO:25, SEQ ID NO:27, SEQ ID NO:29, SEQ ID NO:31, SEQ ID NO:33, SEQ ID NO:35, SEQ ID NO:36, SEQ ID NO:37, SEQ ID NO:38, SEQ ID NO:39, SEQ ID NO:40, SEQ ID NO:41, SEQ ID NO:42, SEQ ID NO:44, SEQ ID NO:45, SEQ ID NO:47, SEQ ID NO:49, SEQ ID NO:51, SEQ ID NO:52, SEQ ID NO:54, SEQ ID NO:56, SEQ ID NO:58, SEQ ID NO:60, SEQ ID NO:62, SEQ ID NO:64, SEQ ID NO:66, SEQ ID NO:68, SEQ ID NO:70, SEQ ID NO:72, SEQ ID NO:74, SEQ ID NO:76, SEQ ID NO:78, SEQ ID NO:80, SEQ ID NO:82, SEQ ID NO:84, SEQ ID NO:86, SEQ ID NO:88, SEQ ID NO:90, SEQ ID NO:92, SEQ ID NO:94, SEQ ID NO:96, SEQ ID NO:98, SEQ ID NO:100, SEQ ID NO:102, SEQ ID NO:104, SEQ ID NO:106, SEQ ID NO:108, SEQ ID NO:110, SEQ ID NO:112, SEQ ID NO:114, SEQ ID NO:116, SEQ ID NO:118, SEQ ID NO:120, SEQ ID NO:122, SEQ ID NO:124, and SEQ ID NO:126 that encodes a polypeptide or protein having the amino acid sequence as set forth in SEQ ID NO:2, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:12, SEQ ID NO:14, SEQ ID NO:16, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:28, SEQ ID NO:30, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:43, SEQ ID NO:46, SEQ ID NO:48, SEQ ID NO:50, SEQ ID NO:53, SEQ ID NO:55, SEQ ID NO:57, SEQ ID NO:59, SEQ ID NO:61, SEQ ID NO:63, SEQ ID NO:65, SEQ ID NO:67, SEQ ID NO:69, SEQ ID NO:71, SEQ ID NO:73, SEQ ID NO:75, SEQ ID NO:77, SEQ ID NO:79, SEQ ID NO:81, SEQ ID NO:83, SEQ ID NO:85, SEQ ID NO:87, SEQ ID NO:89, SEQ ID NO:91, SEQ ID NO:93, SEQ ID NO:95, SEQ ID NO:97, SEQ ID NO:99, SEQ ID NO:101, SEQ ID NO:103, SEQ ID NO:105, SEQ ID NO:107, SEQ ID NO:109, SEQ ID NO:111, SEQ ID NO:113, SEQ ID NO:115, SEQ ID NO:117, SEQ ID NO:119, SEQ ID NO:121, SEQ ID NO:123, SEQ ID NO:125, and SEQ ID NO:127. A heterologous promoter can also be operably linked to synthetic DNA coding sequences encoding a plastid targeted or untargeted TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, variants, truncation variants, and chimeras thereof and untargeted TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, variants, truncation variants, and chimeras thereof or related family member insecticidal proteins. The codons of a recombinant nucleic acid molecule encoding for proteins disclosed herein can be substituted by synonymous codons (known in the art as a silent substitution).

A recombinant DNA construct comprising an encoding sequence for TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389 or related family member insecticidal protein can further comprise a region of DNA that encodes for one or more insect inhibitory agents which can be configured to concomitantly express or co-express with a DNA sequence encoding a TIC7040,

TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389 or related family member insecticidal protein, a protein different from a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389 or a related family member insecticidal protein, an insect inhibitory dsRNA molecule, or an ancillary protein. Ancillary proteins include, but are not limited to, co-factors, enzymes, binding-partners, or other agents that function to aid in the effectiveness of an insect inhibitory agent, for example, by aiding its expression, influencing its stability in plants, optimizing free energy for oligomerization, augmenting its toxicity, and increasing its spectrum of activity. An ancillary protein may facilitate the uptake of one or more insect inhibitory agents, for example, or potentiate the toxic effects of the toxic agent.

A recombinant DNA construct can be assembled so that all proteins or dsRNA molecules are expressed from one promoter or each protein or dsRNA molecules is under separate promoter control or some combination thereof. The TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389 and related family member proteins of this invention can be expressed from a multi-gene expression system in which one or more TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389 or related family member proteins is expressed from a common nucleotide segment which also contains other open reading frames and promoters, depending on the type of expression system selected. For example, a bacterial multi-gene expression system can utilize a single promoter to drive expression of multiply-linked/tandem open reading frames from within a single operon (i.e., polycistronic expression). In another example, a plant multi-gene expression system can utilize multiply-unlinked expression cassettes each expressing a different protein or other agent such as one or more dsRNA molecules.

Recombinant nucleic acid molecules or recombinant DNA constructs comprising a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389 or related family member protein encoding sequence can be delivered to host cells by vectors, e.g., a plasmid, baculovirus, synthetic chromosome, virion, cosmid, phagemid, phage, or viral vector. Such vectors can be used to achieve stable or transient expression of a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389 or related family member protein encoding sequence in a host cell, or subsequent expression of the encoded polypeptide. An exogenous recombinant polynucleotide or recombinant DNA construct that comprises a protein encoding sequence and that is introduced into a host cell is referred herein as a "transgene."

Transgenic bacteria, transgenic plant cells, transgenic plants, and transgenic plant parts that contain a recombinant polynucleotide that expresses any one or more of the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389 or related family member protein encoding sequences are provided herein. The term "bacterial cell" or "bacterium" can include, but is not limited to, an *Agrobacterium*, a *Bacillus*, an *Escherichia*, a *Salmonella*, a *Pseudomonas*, or a *Rhizobium* cell. The term "plant cell" or "plant" can include but is not limited to a dicotyledonous cell or a monocotyledonous cell. Contemplated plants and plant cells include, but are not limited to, alfalfa, banana, barley, bean, broccoli, cabbage, brassica, carrot, cassava, castor, cauliflower, celery, chickpea, Chinese cabbage, citrus, coconut, coffee, corn, clover, cotton, a cucurbit, cucumber, Douglas fir, eggplant, eucalyptus, flax,

garlic, grape, hops, leek, lettuce, Loblolly pine, millets, melons, nut, oat, olive, onion, ornamental, palm, pasture grass, pea, peanut, pepper, pigeonpea, pine, potato, poplar, pumpkin, Radiata pine, radish, rapeseed, rice, rootstocks, rye, safflower, shrub, sorghum, Southern pine, soybean, spinach, squash, strawberry, sugar beet, sugarcane, sunflower, sweet corn, sweet gum, sweet potato, switchgrass, tea, tobacco, tomato, triticale, turf grass, watermelon, and wheat plant cell or plant. In certain embodiments, transgenic plants and transgenic plant parts regenerated from a transgenic plant cell are provided. In certain embodiments, the transgenic plants can be obtained from a transgenic seed, by cutting, snapping, grinding or otherwise disassociating the part from the plant. In certain embodiments, the plant part can be a seed, a boll, a leaf, a flower, a stem, a root, or any portion thereof, or a non-regenerable portion of a transgenic plant part. As used in this context, a "non-regenerable" portion of a transgenic plant part is a portion that cannot be induced to form a whole plant or that cannot be induced to form a whole plant that is capable of sexual and/or asexual reproduction. In certain embodiments, a non-regenerable portion of a plant part is a portion of a transgenic seed, boll, leaf, flower, stem, or root.

Methods of making transgenic plants that comprise insect, Coleoptera- or Lepidoptera-inhibitory amounts of a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389, variant, truncation variant, chimera or related family member protein are provided. Such plants can be made by introducing a recombinant polynucleotide that encodes any TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389, variant, truncation variant, chimera or related family member protein provided in this application into a plant cell, and selecting a plant derived from said plant cell that expresses an insect, Coleoptera- or Lepidoptera-inhibitory amount of the proteins. Plants can be derived from the plant cells by regeneration, seed, pollen, or meristem transformation techniques. Methods for transforming plants are known in the art.

Processed plant products, wherein the processed product comprises a detectable amount of a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389, variant, truncation variant, chimera or related family member protein, an insect inhibitory segment or fragment thereof, or any distinguishing portion thereof, are also disclosed in this application. In certain embodiments, the processed product is selected from the group consisting of plant parts, plant biomass, oil, meal, sugar, animal feed, flour, flakes, bran, lint, hulls, processed seed, and seed. In certain embodiments, the processed product is non-regenerable. The plant product can comprise commodity or other products of commerce derived from a transgenic plant or transgenic plant part, where the commodity or other products can be tracked through commerce by detecting nucleotide segments or expressed RNA or proteins that encode or comprise distinguishing portions of a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389, variant, truncation variant, chimera or related family member protein.

Plants expressing a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389, variant, truncation variant, chimera or related family member protein can be crossed by breeding with transgenic events expressing other toxin proteins and/or expressing other transgenic traits such as herbicide tolerance

genes, genes conferring yield or stress tolerance traits, and the like, or such traits can be combined in a single vector so that the traits are all linked.

As described further in the Examples, sequences encoding TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, and TIC7383, variants of TIC7040HT, TIC7042, TIC7381, TIC7382, and TIC7383, truncation variants of TIC7040HT, TIC7042, TIC7381, TIC7382, and TIC7383, and chimeras of TIC7383, TIC7042, TIC7381 and TIC7382 were designed for use in plants and are presented in Table 8 in Example 5.

Expression cassettes and vectors containing these synthetic or artificial nucleotide sequences can be constructed and introduced into corn, cotton, and soybean plant cells in accordance with transformation methods and techniques which are known in the art. Transformed cells are regenerated into transformed plants that are observed to be expressing TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, or TIC7383; or variants, truncation variants or chimeras thereof. To test pesticidal activity, bioassays are performed in the presence of Lepidopteran, Coleopteran and Hemipteran pests.

As further described in the Examples, sequences encoding TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, variants, truncation variants, chimeras or related family member proteins and sequences having a substantial percentage identity to these proteins can be identified using methods known to those of ordinary skill in the art such as polymerase chain reaction (PCR), thermal amplification and hybridization. For example, a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, or TIC7383 protein and variants, truncation variants, chimeras or related family member proteins can be used to produce antibodies that bind specifically to related proteins, and can be used to screen for and to find other protein members that are closely related.

Furthermore, nucleotide sequences encoding TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, or TIC7383 and variants, truncation variants, chimeras or related family member proteins can be used as probes and primers for screening to identify other members of the class using thermal-cycle or isothermal amplification and hybridization methods. For example, oligonucleotides derived from sequences as set forth as SEQ ID NO:35, SEQ ID NO:36, SEQ ID NO:37, SEQ ID NO:38, SEQ ID NO:39, SEQ ID NO:40, SEQ ID NO:41, SEQ ID NO:42, SEQ ID NO:44, SEQ ID NO:45, SEQ ID NO:47, SEQ ID NO:49, SEQ ID NO:51, SEQ ID NO:52, SEQ ID NO:54, SEQ ID NO:56, SEQ ID NO:58, SEQ ID NO:60, SEQ ID NO:62, SEQ ID NO:64, SEQ ID NO:66, SEQ ID NO:68, SEQ ID NO:76, SEQ ID NO:78, SEQ ID NO:80, SEQ ID NO:82, SEQ ID NO:84, SEQ ID NO:86, SEQ ID NO:88, SEQ ID NO:90, SEQ ID NO:92, SEQ ID NO:94, SEQ ID NO:96, SEQ ID NO:98, SEQ ID NO:100, SEQ ID NO:102, SEQ ID NO:104, SEQ ID NO:106, SEQ ID NO:108, SEQ ID NO:110, SEQ ID NO:112, SEQ ID NO:114, SEQ ID NO:116, SEQ ID NO:118, SEQ ID NO:120, SEQ ID NO:122, SEQ ID NO:124, and SEQ ID NO:126 can be used to determine the presence or absence of a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, or TIC7383 protein, or variants, truncation variants, chimeras, or related family member protein transgene in a deoxyribonucleic acid sample derived from a commodity product. Given the sensitivity of certain nucleic acid detection methods that employ oligonucleotides, it is anticipated that oligonucleotides derived from the sequences as set forth as SEQ ID NO:35, SEQ ID NO:36, SEQ ID NO:37, SEQ ID NO:38, SEQ ID NO:39, SEQ ID NO:40, SEQ ID NO:41, SEQ ID

NO:42, SEQ ID NO:44, SEQ ID NO:45, SEQ ID NO:47, SEQ ID NO:49, SEQ ID NO:51, SEQ ID NO:52, SEQ ID NO:54, SEQ ID NO:56, SEQ ID NO:58, SEQ ID NO:60, SEQ ID NO:62, SEQ ID NO:64, SEQ ID NO:66, SEQ ID NO:68, SEQ ID NO:76, SEQ ID NO:78, SEQ ID NO:80, SEQ ID NO:82, SEQ ID NO:84, SEQ ID NO:86, SEQ ID NO:88, SEQ ID NO:90, SEQ ID NO:92, SEQ ID NO:94, SEQ ID NO:96, SEQ ID NO:98, SEQ ID NO:100, SEQ ID NO:102, SEQ ID NO:104, SEQ ID NO:106, SEQ ID NO:108, SEQ ID NO:110, SEQ ID NO:112, SEQ ID NO:114, SEQ ID NO:116, SEQ ID NO:118, SEQ ID NO:120, SEQ ID NO:122, SEQ ID NO:124, and SEQ ID NO:126 can be used to detect a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, or TIC7383, variants, truncation variants, or chimeric transgenes thereof in commodity products derived from pooled sources where only a fraction of the commodity product is derived from a transgenic plant containing any of SEQ ID NO:35, SEQ ID NO:36, SEQ ID NO:37, SEQ ID NO:38, SEQ ID NO:39, SEQ ID NO:40, SEQ ID NO:41, SEQ ID NO:42, SEQ ID NO:44, SEQ ID NO:45, SEQ ID NO:47, SEQ ID NO:49, SEQ ID NO:51, SEQ ID NO:52, SEQ ID NO:54, SEQ ID NO:56, SEQ ID NO:58, SEQ ID NO:60, SEQ ID NO:62, SEQ ID NO:64, SEQ ID NO:66, SEQ ID NO:68, SEQ ID NO:76, SEQ ID NO:78, SEQ ID NO:80, SEQ ID NO:82, SEQ ID NO:84, SEQ ID NO:86, SEQ ID NO:88, SEQ ID NO:90, SEQ ID NO:92, SEQ ID NO:94, SEQ ID NO:96, SEQ ID NO:98, SEQ ID NO:100, SEQ ID NO:102, SEQ ID NO:104, SEQ ID NO:106, SEQ ID NO:108, SEQ ID NO:110, SEQ ID NO:112, SEQ ID NO:114, SEQ ID NO:116, SEQ ID NO:118, SEQ ID NO:120, SEQ ID NO:122, SEQ ID NO:124, or SEQ ID NO:126. It is further recognized that such oligonucleotides can be used to introduce nucleotide sequence variation in SEQ ID NO:35, SEQ ID NO:36, SEQ ID NO:37, SEQ ID NO:38, SEQ ID NO:39, SEQ ID NO:40, SEQ ID NO:41, SEQ ID NO:42, SEQ ID NO:44, SEQ ID NO:45, SEQ ID NO:47, SEQ ID NO:49, SEQ ID NO:51, SEQ ID NO:52, SEQ ID NO:54, SEQ ID NO:56, SEQ ID NO:58, SEQ ID NO:60, SEQ ID NO:62, SEQ ID NO:64, SEQ ID NO:66, SEQ ID NO:68, SEQ ID NO:76, SEQ ID NO:78, SEQ ID NO:80, SEQ ID NO:82, SEQ ID NO:84, SEQ ID NO:86, SEQ ID NO:88, SEQ ID NO:90, SEQ ID NO:92, SEQ ID NO:94, SEQ ID NO:96, SEQ ID NO:98, SEQ ID NO:100, SEQ ID NO:102, SEQ ID NO:104, SEQ ID NO:106, SEQ ID NO:108, SEQ ID NO:110, SEQ ID NO:112, SEQ ID NO:114, SEQ ID NO:116, SEQ ID NO:118, SEQ ID NO:120, SEQ ID NO:122, SEQ ID NO:124, or SEQ ID NO:126. Such "mutagenesis" oligonucleotides are useful for identification of TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389 or related family member amino acid sequence variants exhibiting a range of insect inhibitory activity or varied expression in transgenic plant host cells.

Nucleotide sequence homologs, e.g., insecticidal proteins encoded by nucleotide sequences that hybridize to each or any of the sequences disclosed in this application under hybridization conditions, are also an embodiment of the present invention. The invention also provides a method for detecting a first nucleotide sequence that hybridizes to a second nucleotide sequence, wherein the first nucleotide sequence (or its reverse complement sequence) encodes a pesticidal protein or pesticidal fragment thereof and hybridizes under stringent hybridization conditions to the second nucleotide sequence. In such case, the second nucleotide sequence can be the nucleotide sequence selected from the group consisting of: SEQ ID NO:1, SEQ ID NO:3, SEQ ID

NO:5, SEQ ID NO:7, SEQ ID NO:9, SEQ ID NO:11, SEQ ID NO:13, SEQ ID NO:15, SEQ ID NO:17, SEQ ID NO:19, SEQ ID NO:21, SEQ ID NO:23, SEQ ID NO:25, SEQ ID NO:27, SEQ ID NO:29, SEQ ID NO:31, SEQ ID NO:33, SEQ ID NO:35, SEQ ID NO:36, SEQ ID NO:37, SEQ ID NO:38, SEQ ID NO:39, SEQ ID NO:40, SEQ ID NO:41, SEQ ID NO:42, SEQ ID NO:44, SEQ ID NO:45, SEQ ID NO:47, SEQ ID NO:49, SEQ ID NO:51, SEQ ID NO:52, SEQ ID NO:54, SEQ ID NO:56, SEQ ID NO:58, SEQ ID NO:60, SEQ ID NO:62, SEQ ID NO:64, SEQ ID NO:66, SEQ ID NO:68, SEQ ID NO:70, SEQ ID NO:72, SEQ ID NO:74, SEQ ID NO:76, SEQ ID NO:78, SEQ ID NO:80, SEQ ID NO:82, SEQ ID NO:84, SEQ ID NO:86, SEQ ID NO:88, SEQ ID NO:90, SEQ ID NO:92, SEQ ID NO:94, SEQ ID NO:96, SEQ ID NO:98, SEQ ID NO:100, SEQ ID NO:102, SEQ ID NO:104, SEQ ID NO:106, SEQ ID NO:108, SEQ ID NO:110, SEQ ID NO:112, SEQ ID NO:114, SEQ ID NO:116, SEQ ID NO:118, SEQ ID NO:120, SEQ ID NO:122, SEQ ID NO:124, and SEQ ID NO:126 under stringent hybridization conditions. Nucleotide coding sequences hybridize to one another under appropriate hybridization conditions and the proteins encoded by these nucleotide sequences cross react with antiserum raised against any one of the other proteins. Stringent hybridization conditions, as defined herein, comprise at least hybridization at 42° C. followed by two washes for five minutes each at room temperature with 2×SSC, 0.1% SDS, followed by two washes for thirty minutes each at 65° C. in 0.5×SSC, 0.1% SDS. Washes at even higher temperatures constitute even more stringent conditions, e.g., hybridization conditions of 68° C., followed by washing at 68° C., in 2×SSC containing 0.1% SDS.

One skilled in the art will recognize that, due to the redundancy of the genetic code, many other sequences are capable of encoding proteins related to TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, or TIC7389, and those sequences, to the extent that they function to express pesticidal proteins either in *Bacillus* strains or in plant cells, are embodiments of the present invention, recognizing of course that many such redundant coding sequences will not hybridize under these conditions to the native *Brevibacillus* sequences encoding TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, or TIC7389. This application contemplates the use of these and other identification methods known to those of ordinary skill in the art, to identify TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389 or related family member protein-encoding sequences and sequences having a substantial percentage identity thereto.

Methods of controlling insects, in particular Lepidoptera or Coleoptera infestations of crop plants, with the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, or TIC7389 or related family member proteins are also disclosed in this application. Such methods can comprise growing a plant comprising an insect-, Coleoptera-, or Lepidoptera-inhibitory amount of a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389, variant, truncation variant, chimera or related family member toxin protein. In certain embodiments, such methods can further comprise any one or more of: (i) applying any composition comprising or encoding a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389, variant, truncation variant, chimera or related family member toxin protein to a plant or a seed that gives rise to a plant; and (ii) transforming a plant or a plant cell that gives rise to a plant with

a polynucleotide encoding a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389, variant, truncation variant, chimera or related family member toxin protein. In general, it is contemplated that a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389, variant, truncation variant or related family member toxin protein can be provided in a composition, provided in a microorganism, or provided in a transgenic plant to confer insect inhibitory activity against Lepidopteran or Coleopteran insects.

In certain embodiments, a recombinant nucleic acid molecule of a TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389, variant, truncation variant, chimera or related family member toxin protein is the insecticidally active ingredient of an insect inhibitory composition prepared by culturing recombinant *Bacillus* or any other recombinant bacterial cell transformed to express the protein. Such a composition can be prepared by desiccation, lyophilization, homogenization, extraction, filtration, centrifugation, sedimentation, or concentration of a culture of such recombinant cells expressing/producing said recombinant polypeptide. Such a process can result in a *Bacillus* or other entomopathogenic bacterial cell extract, cell suspension, cell homogenate, cell lysate, cell supernatant, cell filtrate, or cell pellet. By obtaining the recombinant polypeptides so produced, a composition that includes the recombinant polypeptides can include bacterial cells, bacterial spores, and parasporal inclusion bodies and can be formulated for various uses, including as agricultural insect inhibitory spray products or as insect inhibitory formulations in diet bioassays.

In one embodiment, to reduce the likelihood of resistance development, an insect inhibitory composition comprising TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389, variant, truncation variant, chimera or related family member can further comprise at least one additional polypeptide known to those of ordinary skill in the art that exhibits insect inhibitory activity against the same Lepidopteran or Coleopteran insect species, but which is different from the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389, variant, truncation variant, chimera or related family member toxin protein. Possible additional polypeptides for such a composition include an insect inhibitory protein and an insect inhibitory dsRNA molecule. One example for the use of such ribonucleotide sequences to control insect pests is described in Baum, et al. (U.S. Patent Publication 2006/0021087 A1).

Such additional polypeptide for the control of Lepidopteran pests may be selected from the group consisting of an insect inhibitory protein, such as, but not limited to, Cry1A (U.S. Pat. No. 5,880,275), Cry1Ab, Cry1Ac, Cry1A.105, Cry1Ae, Cry1B (U.S. patent Ser. No. 10/525,318), Cry1C (U.S. Pat. No. 6,033,874), Cry1D, Cry1Da and variants thereof, Cry1E, Cry1F, and Cry1A/F chimeras (U.S. Pat. Nos. 7,070,982; 6,962,705; and 6,713,063), Cry1G, Cry1H, Cry1I, Cry1J, Cry1K, Cry1L, Cry1-type chimeras such as, but not limited to, TIC836, TIC860, TIC867, TIC869, and TIC1100 (International Application Publication WO2016/061391 (A2)), TIC2160 (International Application Publication WO2016/061392(A2)), Cry2A, Cry2Ab (U.S. Pat. No. 7,064,249), Cry2Ae, Cry4B, Cry6, Cry7, Cry8, Cry9, Cry15, Cry43A, Cry43B, Cry51Aa1, ET66, TIC400, TIC800, TIC834, TIC1415, Vip3A, VIP3Ab, VIP3B, AXMI-001, AXMI-002, AXMI-030, AXMI-035, AND AXMI-045 (U.S. Patent Publication 2013-0117884

A1), AXMI-52, AXMI-58, AXMI-88, AXMI-97, AXMI-102, AXMI-112, AXMI-117, AXMI-100 (U.S. Patent Publication 2013-0310543 A1), AXMI-115, AXMI-113, AXMI-005 (U.S. Patent Publication 2013-0104259 A1), AXMI-134 (U.S. Patent Publication 2013-0167264 A1), AXMI-150 (U.S. Patent Publication 2010-0160231 A1), AXMI-184 (U.S. Patent Publication 2010-0004176 A1), AXMI-196, AXMI-204, AXMI-207, AXMI-209 (U.S. Patent Publication 2011-0030096 A1), AXMI-218, AXMI-220 (U.S. Patent Publication 2014-0245491 A1), AXMI-221z, AXMI-222z, AXMI-223z, AXMI-224z, AXMI-225z (U.S. Patent Publication 2014-0196175 A1), AXMI-238 (U.S. Patent Publication 2014-0033363 A1), AXMI-270 (U.S. Patent Publication 2014-0223598 A1), AXMI-345 (U.S. Patent Publication 2014-0373195 A1), AXMI-335 (International Application Publication WO2013/134523(A2)), DIG-3 (U.S. Patent Publication 2013-0219570 A1), DIG-5 (U.S. Patent Publication 2010-0317569 A1), DIG-11 (U.S. Patent Publication 2010-0319093 A1), AfIP-1A and derivatives thereof (U.S. Patent Publication 2014-0033361 A1), AfIP-1B and derivatives thereof (U.S. Patent Publication 2014-0033361 A1), PIP-1A/PIB (U.S. Patent Publication 2014-0007292 A1), PSEEN3174 (U.S. Patent Publication 2014-0007292 A1), AECFG-592740 (U.S. Patent Publication 2014-0007292 A1), Pput_1063 (U.S. Patent Publication 2014-0007292 A1), DIG-657 (International Application Publication WO2015/195594(A2)), Pput_1064 (U.S. Patent Publication 2014-0007292 A1), GS-135 and derivatives thereof (U.S. Patent Publication 2012-0233726 A1), GS153 and derivatives thereof (U.S. Patent Publication 2012-0192310 A1), GS154 and derivatives thereof (U.S. Patent Publication 2012-0192310 A1), GS155 and derivatives thereof (U.S. Patent Publication 2012-0192310 A1), SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2012-0167259 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2012-0047606 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2011-0154536 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2011-0112013 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2010-0192256 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2010-0077507 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2010-0077508 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2009-0313721 A1, SEQ ID NO:2 or 4 and derivatives thereof as described in U.S. Patent Publication 2010-0269221 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Pat. No. 7,772,465 (B2), CF161_0085 and derivatives thereof as described in WO2014/008054 A2, Lepidopteran toxic proteins and their derivatives as described in US Patent Publications US2008-0172762 A1, US2011-0055968 A1, and US2012-0117690 A1; SEQ ID NO:2 and derivatives thereof as described in U.S. Pat. No. 7,510,878(B2), SEQ ID NO:2 and derivatives thereof as described in U.S. Pat. No. 7,812,129(B1), Cry71Aa1 and Cry72Aa1 (US Patent Publication US2016-0230187 A1), Axmi422 (US Patent Publication US2016-0201082 A1), Axmi440 (US Patent Publication US2016-0185830 A1), Axmi281 (US Patent Publication 2016-0177332 A1), BT-0044, BT-0051, BT-0068, BT-0128 and variants thereof (WO 2016-094159 A1), BT-009, BT-0012, BT-0013, BT-0023, BT0067 and variants thereof (WO 2016-094165 A1), Cry1JP578V, Cry1JP51, Cry1JP51P578V (WO 2016-061208 A1); and the like.

Such additional polypeptide for the control of Coleopteran pests may be selected from the group consisting of an

insect inhibitory protein, such as, but not limited to, Cry3Bb (U.S. Pat. No. 6,501,009), Cry1C variants, Cry3A variants, Cry3, Cry3B, Cry34/35, 5307, AXMI134 (U.S. Patent Publication 2013-0167264 A1) AXMI-184 (U.S. Patent Publication 2010-0004176 A1), AXMI-205 (U.S. Patent Publication 2014-0298538 A1), AXMI-207 (U.S. Patent Publication 2013-0303440 A1), AXMI-218, AXMI-220 (U.S. Patent Publication 20140245491A1), AXMI-221z, AXMI-223z (U.S. Patent Publication 2014-0196175 A1), AXMI-279 (U.S. Patent Publication 2014-0223599 A1), AXMI-R1 and variants thereof (U.S. Patent Publication 2010-0197592 A1, TIC407, TIC417, TIC431, TIC807, TIC853, TIC901, TIC1201, TIC3131, DIG-10 (U.S. Patent Publication 2010-0319092 A1), eHIPs (U.S. Patent Application Publication No. 2010/0017914), IP3 and variants thereof (U.S. Patent Publication 2012-0210462 A1) 7-Hexatoxin-Hv1a (U.S. Patent Application Publication 2014-0366227 A1), PHI-4 variants (U.S. Patent Application Publication 2016-0281105 A1), PIP-72 variants (WO 2016-144688 A1), PIP-45 variants, PIP-64 variants, PIP-74 variants, PIP-75 variants, and PIP-77 variants (WO 2016-144686 A1), DIG-305 (WO 2016109214 A1), PIP-47 variants (U.S. Patent Publication 2016-0186204 A1), DIG-17, DIG-90, DIG-79 (WO 2016-057123 A1), DIG-303 (WO 2016-070079 A1); and the like.

In other embodiments, such composition/formulation can further comprise at least one additional polypeptide that exhibits insect inhibitory activity to an insect that is not inhibited by an otherwise insect inhibitory protein of the present invention to expand the spectrum of insect inhibition obtained, e.g., an additional polypeptide that exhibits insect inhibitory activity to Hemipterans or Thysanopterans.

The possibility for insects to develop resistance to certain insecticides has been documented in the art. One insect resistance management strategy is to employ transgenic crops that express two distinct insect inhibitory agents that operate through different modes of action. Therefore, any insects with resistance to either one of the insect inhibitory agents can be controlled by the other insect inhibitory agent. Another insect resistance management strategy employs the use of plants that are not protected to the targeted Coleopteran or Lepidopteran pest species to provide a refuge for such unprotected plants. One particular example is described in U.S. Pat. No. 6,551,962, which is incorporated by reference in its entirety.

Other embodiments such as topically applied pesticidal chemistries that are designed for controlling pests that are also controlled by the proteins disclosed herein to be used with proteins in seed treatments, spray on, drip on, or wipe on formulations can be applied directly to the soil (a soil drench), applied to growing plants expressing the proteins disclosed herein, or formulated to be applied to seed containing one or more transgenes encoding one or more of the proteins disclosed. Such formulations for use in seed treatments can be applied with various stickers and tackifiers known in the art. Such formulations can contain pesticides that are synergistic in mode of action with the proteins disclosed, so that the formulation pesticides act through a different mode of action to control the same or similar pests that can be controlled by the proteins disclosed, or that such pesticides act to control pests within a broader host range or plant pest species that are not effectively controlled by the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389, variant, truncation variant, chimera, or related family member pesticidal proteins.

The aforementioned composition/formulation can further comprise an agriculturally-acceptable carrier, such as a bait,

a powder, dust, pellet, granule, spray, emulsion, a colloidal suspension, an aqueous solution, a *Bacillus* spore/crystal preparation, a seed treatment, a recombinant plant cell/plant tissue/seed/plant transformed to express one or more of the proteins, or bacterium transformed to express one or more of the proteins. Depending on the level of insect inhibitory or insecticidal inhibition inherent in the recombinant polypeptide and the level of formulation to be applied to a plant or diet assay, the composition/formulation can include various by weight amounts of the recombinant polypeptide, e.g. from 0.0001% to 0.001% to 0.01% to 1% to 99% by weight of the recombinant polypeptide.

EXAMPLES

In view of the foregoing, those of skill in the art should appreciate that changes can be made in the specific aspects which are disclosed and still obtain a like or similar result without departing from the spirit and scope of the invention. Thus, disclosed specific structural and functional details are not to be interpreted as limiting. It should be understood that the entire disclosure of each cited reference is incorporated within this disclosure.

Example 1

Discovery of TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, TIC7389

This Example describes the discovery of the pesticidal proteins TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389.

Sequences encoding novel *Brevibacillus laterosporus* (B1) pesticidal proteins were identified, cloned, sequence confirmed and tested in insect bioassay. The pesticidal proteins TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, TIC7386, TIC7388, and TIC7389 were isolated from the *Brevibacillus laterosporus* strains listed in Table 3, and represent novel pesticidal proteins belonging to the Cry43 family of toxins.

TABLE 3

Novel Cry43 pesticidal toxin proteins and corresponding <i>Brevibacillus laterosporus</i> strains.				
Toxin	DNA SEQ ID NO:	Protein SEQ ID NO:	<i>Brevibacillus laterosporus</i> strains	Length (a.a.)
TIC7040	1	2	DSC005019	1259
TIC7040HT	3	4	DSC005019	1295
TIC7042	11	12	DSC005973	1266
TIC7381	13	14	DSC006713	1269
TIC7382	15	16	DSC007657	1246
TIC7383	17	18	DSC008106	1256
TIC7386	29	30	DSC007651	1279
TIC7388	31	32	DSC007962	1269
TIC7389	33	34	DSC006878	1319

Polymerase chain reaction (PCR) primers were designed based upon contigs derived from sequencing of each *Brevibacillus laterosporus* strains listed in Table 3. Amplicons of the full length coding sequence for each protein toxin was produced using total DNA isolated from each strain listed in Table 3. With respect to TIC7040, a coding sequence of 3,888 bp was produced through amplification and differed from the predicted coding sequence of 3,780 bp. The ampli-

fied coding sequence and corresponding amino acid sequence were designated "TIC7040HT" to distinguish it from the original predicted contig, TIC7040. Each of the amplicons, with the exception of TIC7040, were cloned using methods known in the art into *Bacillus thuringiensis* (Bt) expression vectors in operable linkage with a Bt expressible promoter.

Example 2

TIC7040HT, TIC7042, TIC7381, TIC7382, and TIC7383 Demonstrate Coleopteran and Lepidopteran Activity in Insect Bioassay

This Example illustrates inhibitory activity exhibited by TIC7040HT, TIC7042, TIC7381, TIC7382, and TIC7383 proteins against various species of Coleoptera and Lepidoptera.

The pesticidal proteins TIC7040HT, TIC7042, TIC7381, TIC7382, and TIC7383 were expressed in Bt and assayed for toxicity against various species of Lepidoptera and Coleoptera. Preparations of each toxin were assayed against the Coleopteran species Western Corn Rootworm (*Diabrotica virgifera virgifera*, WCR), Cry3Bb-resistant Western Corn Rootworm (*Diabrotica virgifera virgifera*, WCRHP), Northern Corn Rootworm (*Diabrotica barberi*, NCR), Southern Corn Rootworm (*Diabrotica undecimpunctata howardii*, SCR), and Colorado potato beetle (*Leptinotarsa decemlineata*, CPB). Preparations of each toxin were also assayed against the Lepidopteran species Black Cutworm (*Agrotis ipsilon*, BCW), Corn Earworm (*Helicoverpa zea*, CEW), also known as Soybean Podworm and Cotton Bollworm), Diamondback Moth (*Plutella xylostella*, DBM), European Corn Borer (*Ostrinia nubilalis*, ECB), Fall Armyworm (*Spodoptera frugiperda*, FAW), Southern Armyworm (*Spodoptera eridania*, SAW), Soybean Looper (*Pseudoplusia includes*, SBL), Southwestern Corn Borer (*Diatraea grandiosella*, SWCB), tobacco budworm (*Heliothis virescens*, TBW), and Velvetbean Caterpillar (*Anticarsia gemmatilis*, VBC). The toxin preparations were also assayed against the Hemipteran species Tarnished plant bug (*Lygus lineolaris*), Western tarnished plant bug (*Lygus hesperus*), and Neotropical Brown Stink Bug (*Euschistus heros*).

Transformed Bt expressing TIC7040HT, TIC7042, TIC7381, TIC7382, or TIC7383 were grown for twenty four (24) hours, and spores as well as solubilized proteins were added to the insect diet for assay. Mortality and stunting were evaluated by comparing the growth and development of insects on a diet with a culture from a Bt strain expressing TIC7040HT, TIC7042, TIC7381, TIC7382, or TIC7383 to insects on a diet with an untreated control culture. Activity was not observed for Hemipteran insect pests for TIC7040HT, TIC7042, TIC7381, TIC7382, or TIC7383. Activity was observed for Coleopteran and Lepidopteran insect pests. The bioassay activity with respect to stunting (S) and mortality (M) observed for each protein is presented in Tables 4 (Coleoptera) and 5 (Lepidoptera), wherein "+" indicates activity, an empty cell indicates no activity observed, and "NT" indicates the toxin was not assayed against that specific insect pest.

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TABLE 4

Bioassay activity of TIC7040HT, TIC7042, TIC7381, TIC7382, and TIC7383 against Coleopteran insect pests.										
Toxin	WCR		WCRHP		NCR		SCR		CPB	
	S	M	S	M	S	M	S	M	S	M
TIC7040HT	+	+	+	+	+	+	+		+	+
TIC7042	+	+	+	+	+	+	+		+	+
TIC7381	+	+	+	+	+	+	+	+	+	+
TIC7382	+	+	+	+	+	+	+		+	+
TIC7383	+	+	+	+	+	+	+		+	+
TIC7389	+	+	NT	NT	NT	NT	NT	NT	+	+

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TABLE 5

Bioassay activity of TIC7040HT, TIC7042, TIC7381, TIC7382, and TIC7383 against Lepidopteran insect pests.																				
Toxin	BCW		CEW		DBM		ECB		FAW		SAW		SBL		SWCB		TBW		VBC	
	S	M	S	M	S	M	S	M	S	M	S	M	S	M	S	M	S	M	S	M
TIC7040HT			+		+	+	+		+				+		+				+	+
TIC7042	+		+		+	+			+				+						+	+
TIC7381			+		+	+	+		+	+			+	+	+				+	
TIC7382					+	+	+	+					+		+	+			+	
TIC7383			+		+	+	+												+	

As can be seen in Tables 4 and 5, the pesticidal proteins TIC7040HT, TIC7042, TIC7381, TIC7382, and TIC7383 demonstrated activity against many of the Coleopteran and Lepidopteran insect pest species. TIC7040HT demonstrated activity against the Coleopteran pests WCR, WCRHP, NCR, SCR, and CPB; and the Lepidopteran insect pests CEW, DBM, ECB, FAW, SBL, and VBC. TIC7042 demonstrated activity against the Coleopteran pests WCR, WCRHP, NCR, SCR, and CPB; and the Lepidopteran insect pests BCW, CEW, DBM, ECB, FAW, SBL, and VBC. Mortality caused by TIC7042 against WCR was very high. TIC7381 demonstrated activity against the Coleopteran pests WCR, WCRHP, NCR, SCR, and CPB; and the Lepidopteran insect pests CEW, DBM, ECB, FAW, SBL, SWCB, and VBC. TIC7382 demonstrated activity against the Coleopteran pests WCR, WCRHP, NCR, SCR, and CPB; and the Lepidopteran insect pests DBM, ECB, SBL, SWCB, and VBC. TIC7383 demonstrated activity against the Coleopteran pests WCR, WCRHP, NCR, SCR, and CPB and; the Lepidopteran insect pests CEW, DBM, ECB, and VBC.

The pesticidal protein TIC7389 demonstrated activity against the Coleopteran pests, WCR and CPB. TIC7389 was also assayed against the Lepidopteran insect pest species Corn Earworm, Fall Armyworm, and Soybean Looper, as well as the Hemipteran species Tarnished plant bug and Western tarnished plant bug. Stunting was observed for SBL, but not the other two Lepidopteran species and the two Hemipteran insect pests.

The insect toxins TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, and TIC7389 demonstrate activity against a variety of Coleopteran and Lepidopteran insect pest species. The insect toxin TIC7389 demonstrates toxicity to the Lepidopteran insect pest Soybean Looper.

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Example 3

Tryptic Digested TIC7040HT and TIC7383, and Chymotryptic Digested TIC7040HT Demonstrate Activity Against Western Corn Rootworm

This Example illustrates inhibitory activity exhibited by tryptic and chymotryptic digested TIC7040HT, and tryptic digested TIC7383 proteins against Western Corn Rootworm. Protein samples of TIC7040HT were subjected to tryptic and chymotryptic digest in separate reactions. Likewise, TIC7383 was subjected to a tryptic digest. The tryptic and chymotryptic digests were performed using methods known in the art. The digested proteins were analyzed by mass spectrometry to determine the resulting protein fragments.

The resulting tryptic and chymotryptic fragments of TIC7040HT, and the resulting tryptic fragment of TIC7383 are presented in Table 6.

The digested proteins were used in bioassay against Western Corn Rootworm. Table 6 shows the assay of activity for each digested protein preparation, and the full length TIC7040HT and TIC7383 proteins.

TABLE 6

Bioassay activity of TIC7040HT, tryptic and chymotryptic digested TIC7040HT, TIC7383, and tryptic digested TIC7383 against Western Corn Rootworm.						
Toxin	DNA SEQ ID NO:	Protein SEQ ID NO:	Amino Acid Position Relative to Full Length Protein	Stunting	Mortality	
TIC7040HT	3	4		+	+	
TIC7040HT_Tryp	70	71	43-624	+	+	
TIC7040HT_Chymo	72	73	45-641	+	+	
TIC7383	17	18		+	+	
TIC7383_Tryp	74	75	55-668	+	+	

As can be seen in Table 6, the tryptic and chymotryptic digested TIC7040HT and the tryptic digested TIC7383 proteins retained activity against Western Corn Rootworm.

Example 4

Truncations of TIC7040HT and TIC7383 Demonstrate Coleopteran Activity in Insect Bioassay

This Example illustrates inhibitory activity exhibited by truncations of the TIC7040HT and TIC7383 proteins against various species of Coleoptera.

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Coding sequences encoding truncations of TIC7040HT and TIC7383 were produced using methods known in the art and cloned into bacterial expression vectors to be expressed in Bt. The truncated proteins were provided in insect diets and assayed for activity against Coleopteran insect pests. The amino acid positions of the truncated TIC7040HT and TIC7383 toxins relative to the full length TIC7040HT and TIC7383 are shown in Table 7.

Transformed Bt expressing truncations of TIC7040HT and TIC7383 were grown for twenty four (24) hours and spores, as well as solubilized proteins, were added to insect diet for assay. Mortality and stunting were evaluated by comparing the growth and development of insects on a diet with a culture from the Bt strain expressing truncations of TIC7040HT or TIC7383 to insects on a diet with an untreated control culture.

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The truncations of TIC7040HT and TIC7383 were assayed for toxicity against various species of Coleoptera. Preparations of each toxin were assayed against the Coleopteran species Western Corn Rootworm, Cry3Bb-resistant Western Corn Rootworm, and Colorado potato beetle.

Activity was observed for Coleopteran insect pests. The bioassay activity with respect to stunting (S) and mortality (M) observed for each protein is presented in Table 7, wherein “+” indicates activity, an empty cell indicates no activity observed, and “NT” indicates the toxin was not assayed against that specific insect pest. The activity of the full length TIC7040HT and TIC7383 is also provided for comparison to the truncated protein activity.

TABLE 7

Bioassay activity of truncations of TIC7040HT and TIC7383 against Coleopteran insect pests.													
Toxin	DNA SEQ ID	Protein SEQ ID	Amino Acid Positions Relative to Full Length Protein	WCR		WCRHP		NCR		SCR		CPB	
				S	M	S	M	S	M	S	M	S	M
TIC7040HT	3	4		+	+	+	+	+	+	+	+	+	+
TIC7040HT_5	7	8	13-611	+	+			NT	NT	NT	NT	+	+
TIC7040HT_6	9	10	13-671	+	+			NT	NT	NT	NT	+	+
TIC7383	17	18		+	+	+	+	+	+	+	+	+	+
TIC7383_2	19	20	12-1256	+	+			NT	NT	NT	NT	+	+
TIC7383_3	21	22	1-659	+	+	+		NT	NT	NT	NT	+	+
TIC7383_4	23	24	1-679	+	+			NT	NT	NT	NT	+	+
TIC7383_5	25	26	15-659	+	+			NT	NT	NT	NT	+	+
TIC7383_6	27	28	15-679	+	+			NT	NT	NT	NT	+	+

As can be seen in Table 7, truncations of TIC7040HT and TIC7383 maintained activity against WCR and CPB.

Example 5

Design of Synthetic Coding Sequences Encoding TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, and Truncation Variants of TIC7040HT, TIC7042, TIC7382, and TIC7383 for Expression in Plant Cells

Synthetic or artificial coding sequences were constructed for use in expression of TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, and truncation variants of TIC7040HT, TIC7042, TIC7382, and TIC7383, in plants. These synthetic coding sequences were cloned into a binary plant transformation vectors, and used to transform plant cells. The synthetic nucleic acid sequences were synthesized according to methods generally described in U.S. Pat. No. 5,500,365, avoiding certain inimical problem sequences such as ATTTA and A/T rich plant polyadenylation sequences while preserving the amino acid sequence of the native Bt protein. The synthetic coding sequences for the TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, and truncation variants of TIC7040HT, TIC7042, TIC7382, and TIC7383 pesticidal proteins are presented Table 8.

TABLE 8

Synthetic coding sequences used for expression in plant cells encoding TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, and truncation variants of TIC7040HT, TIC7042, TIC7382, and TIC7383.						
Description	DNA SEQ ID NO:	Protein SEQ ID NO:	Additional Alanine Residue after Initiating Methionine	N-terminal Truncation	C-Terminal Truncation	Amino Acid Position Relative to Full Length Protein and Mutations
CR-BREla.TIC7040.nno_Mc:1	35	2	No	No	No	
CR-BREla.TIC7040_10.nno_Mc:1	36	4	No	No	No	
CR-BREla.TIC7040_10.nno_Mc:3	37	4	No	No	No	
CR-BREla.TIC7040_10.nno_Mc:4	38	4	No	No	No	
CR-BREla.TIC7040_10.nno_Mc:5	39	4	No	No	No	
CR-BREla.TIC7040_10.nno_Mc:6	40	4	No	No	No	
CR-BREla.TIC7040_10.nno_Mc:7	41	4	No	No	No	
CR-BREla.TIC7040_1.nno_Mc:1	42	43	No	Yes	Yes	15-651
CR-BREla.TIC7040_2.nno_Mc:1	44	10	No	Yes	Yes	13-671
CR-BREla.TIC7040_11.nno_Mc:1	45	46	No	Yes	Yes	14-671
CR-BREla.TIC7040_12.nno_Mc:2	47	48	No	No	Yes	1-660
CR-BREla.TIC7040_13.nno_Mc:1	49	50	No	No	Yes	1-627
CR-BREla.TIC7040_14.nno_Mc:1	76	77	Yes	Yes	Yes	52-660
CR-BREla.TIC7042.nno_Mc:1	51	12	No	No	No	
CR-BREla.TIC7042_1.nno_Mc:1	52	53	No	Yes	Yes	11-646
CR-BREla.TIC7042_2.nno_Mc:1	54	55	No	Yes	Yes	11-665
CR-BREla.TIC7381_1.nno_Mc:1	56	57	Yes	No	No	
CR-BREla.TIC7381_2.nno_Mc:1	78	79	Yes	No	Yes	1-658
CR-BREla.TIC7381_3.nno_Mc:1	80	81	Yes	Yes	Yes	50-658
CR-BREla.TIC7382_1.nno_Mc:1	58	59	Yes	No	No	
CR-BREla.TIC7382_2.nno_Mc:1	60	61	Yes	No	Yes	1-659
CR-BREla.TIC7382_3.nno_Mc:1	82	83	Yes	Yes	Yes	52-659
CR-BREla.TIC7383_1.nno_Mc:1	62	63	Yes	No	No	
CR-BREla.TIC7383_7.nno_Mc:1	64	65	Yes	Yes	Yes	54-668
CR-BREla.TIC7383_8.nno_Mc:1	66	67	Yes	No	Yes	1-661
CR-BREla.TIC7383_9.nno_Mc:1	68	69	Yes	No	Yes	1-668
CR-BREla.TIC7383_19.nno_Mc:1	84	85	No	Yes	Yes	15-668
CR-BREla.TIC7383_20.nno_Mc:1	86	87	No	Yes	Yes	15-661
CR-BREla.TIC7383_21.nno_Mc:1	88	89	Yes	Yes	Yes	54-661
CR-BREla.TIC7383_22.nno_Mc:1	90	91	No	Yes	Yes	54-668
CR-BREla.TIC7383_23.nno_Mc:1	92	93	No	Yes	Yes	54-661
CR-BREla.TIC7383_24.nno_Mc:2	94	95	No	Yes	Yes	73-661
CR-BREla.TIC7383_25.nno_Mc:3	96	97	No	Yes	Yes	94-661
CR-BREla.TIC7383_26.nno_Mc:1	98	99	No	Yes	Yes	114-661
CR-BREla.TIC7383_27.nno_Mc:1	100	101	Yes	Yes	Yes	54-658
CR-BREla.TIC7383_28.nno_Mc:1	102	103	No	Yes	Yes	15-658
CR-BREla.TIC7383_29.nno_Mc:1	104	105	Yes	No	Yes	1-963
CR-BREla.TIC7383_30.nno_Mc:1	106	107	Yes	No	No	K964A; R966A; K968A
CR-BREla.TIC7383_31.nno_Mc:1	108	109	Yes	No	Yes	1-1065; K964A; R966A; K968A
CR-BREla.TIC7383_32.nno_Mc:1	110	111	Yes	No	No	Deletion 964-969

Example 6

Expression Cassettes for Expression of TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, and Truncation Variants of TIC7040HT, TIC7042, TIC7382, and TIC7383 in Plant Cells

A variety of plant expression cassettes were designed with the sequences as set forth in Table 8. Such expression cassettes are useful for transient expression in plant protoplasts or transformation of plant cells. Typical expression cassettes are designed with respect to the eventual placement of the protein within the plant cell. For a plastid targeted protein, the synthetic TIC7040, TIC7040HT, TIC7042,

TIC7381, TIC7382, TIC7383 or truncation variant of TIC7040HT, TIC7042, TIC7382, or TIC7383 pesticidal protein coding sequences are operably linked in frame with a chloroplast targeting signal peptide coding sequence. The resulting plant transformation vectors comprise a first transgene cassette for expression of the pesticidal protein which comprises a constitutive promoter, operably linked 5' to a leader, operably linked 5' to an intron (or optionally no intron), operably linked 5' to a synthetic coding sequence encoding a plastid targeted or untargeted TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, or truncation variant of TIC7040HT, TIC7042, TIC7382, or TIC7383 protein, which is in turn operably linked 5' to a 3' UTR and, a second transgene cassette for the selection of

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transformed plant cells using glyphosate or antibiotic selection. All of the elements described above are arranged contiguously often with additional sequence provided for the construction of the expression cassette such as restriction endonuclease sites or ligation independent cloning sites.

Example 7

TIC7382 Provides Efficacious Resistance to Western Corn Rootworm when Expressed in Stably Transformed Corn Plants

This Example illustrates the inhibitory activity exhibited by truncations of TIC7382 against Coleoptera, such as Western Corn Rootworm, when expressed in plants and provided as a diet to the respective insect pest.

Binary plant transformation vectors comprising transgene cassettes designed to express untargeted TIC7382 (CR-BREla.TIC7382_1.nno_Mc:1) and the truncation variants CR-BREla.TIC7382_2.nno_Mc:1 and CR-BREla.TIC7382_3.nno_Mc:1 were cloned using methods known in the art. The plant transformation vectors comprised a first transgene cassette for expression of the TIC7382 pesticidal protein or one of the truncation variants which comprised a root preferred promoter, operably linked 5' to a leader, operably linked 5' to an intron, operably linked 5' to a synthetic coding sequence encoding TIC7382 or the truncation variants, each of which comprised an additional alanine residue immediately following the initiating methionine, which was in turn was operably linked 5' to a 3' UTR and, a second transgene cassette for the selection of transformed plant cells using glyphosate. The resulting vectors were used to stably transform corn plants using methods known in the art. Single T-DNA insertion events were selected and grown. Pesticidal activity was assayed against Western Corn Rootworm feeding on the roots of the stably transformed corn plants.

R_0 stably transformed plants were used to assay for Coleopteran resistance as well as generating F_1 progeny. Multiple single copy events were selected from each binary vector transformation. A portion of those events arising from each binary vector transformation were used in the Coleopteran assay, while another portion of events were used to generate F_1 progeny for further testing.

The R_0 assay plants were transplanted to eight inch pots. The plants were inoculated with eggs from Western Corn Rootworm. The eggs were incubated for approximately ten (10) days prior to inoculation to allow hatching to occur four (4) days after inoculation to ensure a sufficient number of larvae survive and are able to attack the corn roots. The transformed plants were inoculated at approximately V2 to V3 stage. The plants were grown after infestation for approximately twenty eight (28) days. The plants were removed from the pots with the roots being carefully washed to remove all soil. The damage to the roots was assessed using a damage rating scale of 1-5, as presented in Table 9. Comparison was also made to the negative control to assure the assay has been performed properly. Low root damage scores indicate resistance conferred by the TIC7382 protein

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or the truncation variants to WCR. An RDR score of 1.0 to 2.5 represents good efficacy, an RDR score of 2.6 to 3.5 represents medium efficacy, and an RDR score of 3.6 to 5.0 represents low efficacy.

TABLE 9

R_0 root damage rating scores.		
Root Damage Score	Description	
1	No visible feeding	
2	Some feeding; no pruning	
3	Pruning of at least one root	
4	Entire node pruned	
5	More than one node pruned	

For F_1 assay, eggs from Western Corn Rootworm were incubated for approximately ten (10) days to allow hatching within four (4) days after inoculation. The plants were inoculated at approximately V2 to V3 stage. Each pot was inoculated with about two thousand eggs. The plants were grown after infestation for approximately twenty eight (28) days. The plants were removed from the pots with the roots being carefully washed to remove all soil. The damage to the roots were assessed using a damage rating scale of 0-3, as presented in Table 10. Comparison was made to the negative control to assure the assay was performed properly. Low root damage scores indicate resistance conferred TIC7382, or the truncation to WCR. An RDR at F_1 of 0.0 to 0.75 represents good efficacy, an RDR of 0.76 to 1.5 represents medium efficacy, and an RDR of 1.6 to 3.0 represents low efficacy.

TABLE 10

F_1 root damage rating scores.		
Root Damage Score	Description	
0	No visible feeding	
0.01-0.09	Feeding scars and tracks	
0.1-0.9	Root pruning, but less than a full node	
1.0-1.9	At least a full node (or equivalent) destroyed to within 1.5 inches of plant	
2.0-2.9	Two or more nodes gone	
3	Three or more nodes gone	

Table 11 shows the average Root Damage Rating (RDR) assayed for the TIC7382 protein and truncation variants.

TABLE 11

Average Root Damage Rating (RDR) of transgenic corn plants expressing TIC7382 or truncation variants.

Description	Coding Sequence SEQ ID NO:	Protein SEQ ID NO:	Amino Acid Position Relative to Full Length Protein	Average R ₀ RDR	Average F ₁ RDR
CR-BREla.TIC7382_1.nno_Mc:1	58	59		3.7	2.4
CR-BREla.TIC7382_2.nno_Mc:1	60	61	1-658	3.3	1.1
CR-BREla.TIC7382_3.nno_Mc:1	82	83	50-658	1.6	1.0

As can be seen in Table 11, a C-terminal truncation of the TIC7382 protein giving rise to the truncation variant, CR-BREla.TIC7382_2.nno_Mc:1, improved efficacy relative to the Root Damage Ratings of CR-BREla.TIC7382_1.nno_Mc:1. Truncation of the TIC7382 protein at both the N-terminus and C-terminus giving rise to the truncation variant, CR-BREla.TIC7382_3.nno_Mc:1, resulted in less damage to the corn roots and further improved efficacy as demonstrated by a lower R₀ Root Damage Rating when compared to the Root Damage Ratings of CR-BREla.TIC7382_1.nno_Mc:1 and CR-BREla.TIC7382_2.nno_Mc:1.

Example 8

Assay of Activity of TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, and Truncation Variants of TIC7040HT, TIC7042, TIC7382, or TIC7383 Against Coleopteran Corn Rootworm Pests when Expressed in Stably Transformed Corn Plants

This Example illustrates the inhibitory activity of TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, or truncation variants of TIC7040HT, TIC7042, TIC7382, or TIC7383 against different Coleopteran species that feed on corn roots.

Binary plant transformation vectors comprising transgene cassettes designed to express both plastid targeted and untargeted TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, or truncation variants of TIC7040HT, TIC7042, TIC7382, or TIC7383 are cloned using methods known in the art and comprise the sequences as shown in Table 8. The resulting vectors are used to stably transform corn plants using methods known in the art. Single T-DNA insertion events are selected and grown. Pesticidal activity is assayed against the Coleopteran pests Western Corn Rootworm (*Diabrotica virgifera virgifera*, WCR), Northern Corn Rootworm (*Diabrotica barberi*, NCR), Mexican Corn Rootworm (*Diabrotica virgifera zea*, MCR), Brazilian Corn Rootworm (*Diabrotica balteata*, BZR), Southern Corn Rootworm (*Diabrotica undecimpunctata howardii*, SCR), Colorado potato beetle (*Leptinotarsa decemlineata*, CPB), or a Brazilian Corn Rootworm complex (BCR, consisting of *Diabrotica viridula* and *Diabrotica speciosa*) feeding on the roots of the stably transformed corn plants.

R₀ stably transformed plants are used to assay for Coleopteran resistance as well as generating F₁ progeny. Multiple single copy events are selected from each binary vector transformation. A portion of the events arising from each binary vector transformation are used in the R₀ Coleopteran

assay, while another portion of events are used to generate F₁ progeny for further testing.

The R₀ assay plants are transplanted to eight inch pots. The plants are inoculated with eggs from Western Corn Rootworm, Northern Corn Rootworm, or Southern Corn Rootworm. The eggs are incubated for approximately ten (10) days prior to inoculation to allow hatching to occur four (4) days after inoculation to ensure a sufficient number of larvae survive and are able to attack the corn roots. The transformed plants are inoculated at approximately V2 to V3 stage. The plants are grown after infestation for approximately twenty eight (28) days. The plants are removed from the pots with the roots being carefully washed to remove all soil. The damage to the roots is assessed using a damage rating scale of 1-5, as presented in Table 9 in Example 5. Comparison is also made to the negative controls to assure the assay has been performed properly. Multiple R₀ events for each binary vector transformation are used in the Coleopteran assay. Low root damage scores indicate resistance conferred by TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, or truncation variants of TIC7040HT, TIC7042, TIC7382, or TIC7383 to the tested Coleopteran pest.

A portion of the R₀ stably transformed events arising from each binary vector transformation are used to produce F₁ progeny. The R₀ stably transformed plants are allowed to self-fertilize, producing F₁ progeny. The F₁ seed is planted. Heterozygous plants are identified through molecular methods known in the art and used for assay against Coleopteran pests, as well as ELISA expression measurements of toxin protein. A portion of the heterozygous F₁ progeny from each event are used for insect assay, while another portion is used to measure toxin protein expression.

Eggs from Western Corn Rootworm, Northern Corn Rootworm, or Southern Corn Rootworm are incubated for approximately ten (10) days to allow hatching within four (4) days after inoculation. For WCR, each pot is inoculated with about two thousand eggs. For NCR, less eggs may be used due to the lower availability of eggs from this species. The plants are inoculated at approximately V2 to V3 stage. The plants are grown after infestation for approximately twenty eight (28) days. The plants are removed from the pots with the roots being carefully washed to remove all soil. The damage to the roots are assessed using a damage rating scale of 0-3, as presented in Table 10 in Example 5. Comparison is made to the negative control to assure the assay has been performed properly. Low root damage scores indicate resistance conferred by TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, or truncation variants of TIC7040HT, TIC7042, TIC7382, or TIC7383 protein to the Coleopteran pest.

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Example 9

Assay of Activity of TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, or Truncation Variants of TIC7040HT, TIC7042, TIC7382, or TIC7383 Against Lepidopteran Pests when Expressed in Stably Transformed Corn, Soybean, or Cotton Plants

This Example illustrates the assay of activity against various Lepidopteran pest species fed tissue from stably transformed corn, soybean or cotton plants expressing TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, or truncation variants of TIC7040HT, TIC7042, TIC7382, or TIC7383.

Binary plant transformation vectors comprising transgene cassettes designed to express both plastid targeted and untargeted TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, or truncation variants of TIC7040HT, TIC7042, TIC7382, or TIC7383 are cloned using methods known in the art and comprise the coding sequences as presented in Table 8.

Corn, soybean, or cotton is transformed with the binary transformation vectors described above using an *Agrobacterium*-mediated transformation method. The transformed cells are induced to form plants by methods known in the art. Bioassays using plant leaf disks are performed analogous to those described in U.S. Pat. No. 8,344,207. A non-transformed corn, soybean, or cotton plant is used to obtain tissue to be used as a negative control. Multiple transformation events from each binary vector are assessed against Lepidopteran pests such as, but not limited to, Black Cutworm, Corn Earworm, Diamondback Moth, European Corn Borer, Fall Armyworm, Southern Armyworm, Soybean Looper, Southwestern Corn Borer, Tobacco budworm, and Velvetbean Caterpillar. Those insects demonstrating stunting and/or mortality in the insect bioassay are determined to be susceptible to the effects of the tested TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, or truncation variant of TIC7040HT, TIC7042, TIC7382, or TIC7383 insect toxin pesticidal protein.

Example 10

Assay of Activity of TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, or Truncation Variants of TIC7040HT, TIC7042, TIC7382, or TIC7383 Against Flea Beetle Pests when Expressed in Stably Transformed Canola Plants

This Example illustrates the assay of activity against various species of Flea Beetle when allowed to feed on whole transgenic canola plants or tissues derived from transgenic canola plants expressing TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, or truncation variants of TIC7040HT, TIC7042, TIC7382, or TIC7383.

Binary plant transformation vectors comprising transgene cassettes designed to express both plastid targeted and untargeted TIC7040, TIC7040HT, TIC7042, TIC7381, TIC7382, TIC7383, or truncation variants of TIC7040HT, TIC7042, TIC7382, or TIC7383 are cloned using methods known in the art and comprise the coding sequences as presented in Table 8.

The resulting binary transformation vectors are used to stably transform canola plant cells using methods known in the art. The transformed cells are induced to form plants.

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Bioassays using plant leaf disks are performed analogous to those described in U.S. Pat. No. 8,344,207 using field collected Flea Beetles. A non-transformed canola plant is used to obtain tissue to be used as a negative control. Multiple transformation events from each binary vector are assessed against Coleopteran Flea Beetle pests such as, but not limited to, Crucifer Flea Beetle (*Phyllotreta cruciferae*), Striped Flea Beetle (*Phyllotreta striolata*), and Western Black Flea Beetle (*Phyllotreta pusilla*). Flea Beetle mortality is determined each day as the Beetles continue to feed. Leaf discs are changed every two (2) to three (3) days over a twelve (12) day period to assure fresh material is available to the Flea Beetles for feeding, and to reduce any impact of protein degradation in the sample.

Alternatively, transformed canola plants can be planted in a field where Flea Beetle infestations are present. The plants can be housed in a tent to prevent those Flea Beetles that emerge from the soil from escaping the experimental plots. Damage ratings of the canola leaves can be taken to determine which plants experienced lesser damage and demonstrated resistance to the Flea Beetles.

Example 11

Truncation of TIC7383 Improves Efficacy Against Corn Rootworm in Stably Transformed Corn Plants

This Example illustrates the improvement of efficacy of the TIC7383 through truncations at the N-terminus, the C-terminus, or both termini.

Binary plant transformation vectors comprising transgene cassettes designed to express untargeted TIC7383 and truncation variants were cloned using method known in the art. The plant transformation vectors comprised a first transgene cassette for expression of the TIC7383 pesticidal protein or the truncation variants which comprised a root preferred promoter, operably linked 5' to a leader, operably linked 5' to an intron, operably linked 5' to a synthetic coding sequence encoding TIC7383 or the truncation variants, some of which comprised an additional alanine residue immediately following the initiating methionine, which was in turn operably linked 5' to a 3' UTR and, a second transgene cassette for the selection of transformed plant cells using glyphosate. The resulting vectors were used to stably transform corn plants using methods known in the art. Single T-DNA insertion events were selected and grown. Pesticidal activity was assayed against Western Corn Rootworm feeding on the roots of the stably transformed corn plants.

R₀ stably transformed plants were used to assay for resistance to Western Corn Rootworm as well as generating F₁ progeny. Multiple single copy events were selected from each binary vector transformation. A portion of those events arising from each binary vector transformation were used in the Coleopteran assay, while another portion of events were used to generate F₁ progeny for further testing. R₀ and F₁ Root Damage Rating scores were determined using the Root Damage Rating scores as presented in Tables 9 and 10, respectively, presented in Example 7. Table 12 shows the average Root Damage Rating scores obtained from the R₀ and F₁ stably transformed corn plants expressing TIC7383 and truncation variants. Protein expression levels of TIC7383 and the corresponding truncation variants are also shown in Table 12 and are expressed as parts per million (ppm).

As can be seen in Table 12, truncations at the N-terminus, C-terminus, or both termini improved efficacy in some of the truncation variants as demonstrated by lower average Root

Damage Rating scores relative to the full length TIC7383 protein (CR-BREla.TIC7383_1.nno_Mc:1). Truncations of the TIC7383 toxin also improved expression within the plant in most instances as well.

TABLE 12

Average Root Damage Rating (RDR) scores of stably transformed corn plants expressing TIC7383 and truncation variants.					
Description	Additional Alanine Residue after Initiating Methionine	Amino Acid Position Relative to CR-BREla.TIC7383_1.nno_Mc:1	Protein Expression (ppin)	Average R ₀ RDR	Average F ₁ RDR
CR-BREla.TIC7383_1.nno_Mc:1	Yes		11.9	3.8	2.2
CR-BREla.TIC7383_7.nno_Mc:1	Yes	54-668	421.3	2.9	1.6
CR-BREla.TIC7383_8.nno_Mc:1	Yes	1-661	289.2	2.4	
CR-BREla.TIC7383_9.nno_Mc:1	Yes	1-668	317.2	3.3	
CR-BREla.TIC7383_19.nno_Mc:1	No	15-668	23.7	3.3	
CR-BREla.TIC7383_20.nno_Mc:1	No	15-661	29.1	3.9	
CR-BREla.TIC7383_21.nno_Mc:1	Yes	54-661	617.4	2.7	
CR-BREla.TIC7383_22.nno_Mc:1	No	54-668	451.3	2.6	
CR-BREla.TIC7383_23.nno_Mc:1	No	54-661	343.1	2.8	
CR-BREla.TIC7383_24.nno_Mc:2	No	73-661	356.5	2.9	
CR-BREla.TIC7383_26.nno_Mc:1	No	114-661	8.5	3.9	
CR-BREla.TIC7383_27.nno_Mc:1	Yes	54-658	356.2	2.9	
CR-BREla.TIC7383_28.nno_Mc:1	No	15-658	25.4	3.7	

Example 12

Assay of Activity of Chimeras of TIC7381, TIC7382, TIC7383 and TIC7042 Against Western Corn Rootworm in Stably Transformed Corn Plants

This Example illustrates the design of chimeras of TIC7381, TIC7382, TIC7383, and TIC7042 and the assay of activity against Western Corn Rootworm (WCR) in stably transformed corn plants expressing the chimeras.

Chimeras of TIC7381, TIC7382, TIC7383, and TIC7042 were designed wherein domains one and two (D1D2) of one toxin were combined with the third domain (D3) of another toxin. Table 13 below shows the composition of each chimera.

TABLE 13

Composition of the TIC7381, TIC7382, TIC7383, and TIC7042 chimeras.				
Description	Nucleotide SEQ ID NO:	Protein SEQ ID NO:	D1D2	D3
GOI-TIC10743.nno_Mc:1	112	113	TIC7383	TIC7042
GOI-TIC10744.nno_Mc:1	114	115	TIC7383	TIC7381
GOI-TIC10745.nno_Mc:1	116	117	TIC7383	TIC7382
GOI-TIC10746.nno_Mc:1	118	119	TIC7382	TIC7383
GOI-TIC10747.nno_Mc:1	120	121	TIC7381	TIC7383
GOI-TIC10748.nno_Mc:1	122	123	TIC7042	TIC7383

Binary plant transformation vectors comprising transgene cassettes designed to express untargeted chimeras of TIC7381, TIC7382, TIC7383, and TIC7042 were cloned using methods known in the art. The plant transformation vectors comprised a first transgene cassette for expression of the chimeric toxin pesticidal protein which comprised a root preferred promoter, operably linked 5' to a leader, operably linked 5' to an intron, operably linked 5' to a synthetic coding sequence encoding the chimeras of TIC7381, TIC7382, TIC7383, and TIC7042, which comprised an additional

alanine residue immediately following the initiating methionine, which was in turn was operably linked 5' to a 3' UTR and, a second transgene cassette for the selection of transformed plant cells using glyphosate. The resulting vectors

were used to stably transform corn plants using methods known in the art. Single T-DNA insertion events were selected and grown. Pesticidal activity was assayed against Western Corn Rootworm feeding on the roots of the stably transformed corn plants.

R₀ stably transformed plants were used to assay for resistance to Western Corn Rootworm as well as generating F₁ progeny. Multiple single copy events were selected from each binary vector transformation. A portion of those events arising from each binary vector transformation were used in the Coleopteran assay, while another portion of events were used to generate F₁ progeny for further testing. R₀ and F₁ Root Damage Rating scores were determined using the Root Damage Rating scores as presented in Tables 9 and 10, respectively, presented in Example 7. Table 14 shows the average Root Damage Rating scores obtained from the R₀ and F₁ stably transformed corn plants expressing the chimeras of TIC7381, TIC7382, TIC7383, and TIC7042 or truncation variants, wherein "NT" indicates not tested. Comparison is made to a TIC7382 variant truncated at the N-terminus and C-terminus protoxin domain (CR-BREla.TIC7382_3.nno_Mc:1) and a TIC7383 variant truncated at the N-terminus and C-terminus protoxin domain (CR-BREla.TIC7383_7.nno_Mc:1).

TABLE 14

Average Root Damage Rating (RDR) scores of stably transformed corn plants expressing chimeras of TIC7381, TIC7382, TIC7383, and TIC7042 against Western Corn Rootworm.		
Description	Average R ₀ RDR	Average F ₁ RDR
Negative Control		2.8
GOI-TIC10743.nno_Mc:1	3.4	2
GOI-TIC10744.nno_Mc:1	3.3	2.3
GOI-TIC10745.nno_Mc:1	3.8	2.6
GOI-TIC10746.nno_Mc:1	1.3	0.4
GOI-TIC10747.nno_Mc:1	NT	NT
GOI-TIC10748.nno_Mc:1	3.3	2.7

TABLE 14-continued

Average Root Damage Rating (RDR) scores of stably transformed corn plants expressing chimeras of TIC7381, TIC7382, TIC7383, and TIC7042 against Western Corn Rootworm.		
Description	Average R ₀ RDR	Average F ₁ RDR
CR-BREla.TIC7382_3.nno_Mc:1	1.6	1.0
CR-BREla.TIC7383_7.nno_Mc:1	3	1.5

As can be seen in Table 14, the chimeric toxin GOI-TIC10746.nno_Mc:1 comprised of domains one and two of TIC7382 and domain three of TIC7383 gave better efficacy at R₀ and F₁ relative to the negative control, CR-BREla.TIC7382_3.nno_Mc:1, and CR-BREla.TIC7383_7.nno_Mc:1. Four F₁ events expressing GOI-TIC10746.nno_Mc:1 were included in the assay. Root Damage Ratings scores ranged from 0.11 to 0.79 with (RDR: 0.11, 0.20, 0.30, and 0.79). The one GOI-TIC10746.nno_Mc:1 expressing event with an RDR of 0.11 was equal to a commercial control. In addition to the data presented above, protein isolated from bacteria expressing the chimeric toxin GOI-TIC10746.nno_Mc:1 also demonstrated activity against Northern Corn Rootworm when presented in a diet bioassay.

Example 13

Assay of Activity of Chimeras of TIC7382 and
TIC7383 Corn Rootworm in Stably Transformed
Corn Plants

This Example illustrates the design of chimeras of TIC7382 and TIC7383 and the assay of activity against Western Corn Rootworm (WCR) in stably transformed corn plants expressing the chimeras.

Synthetic coding sequences encoding chimeric toxins comprised of domains one and two of TIC7382 and domain three of TIC7383 designed for expression in plants are cloned using methods known in the art. The chimeric toxin,

TIC10746NTermExt1 (SEQ ID NO:125, encoded by SEQ ID NO:124) comprises the N-terminal extension of TIC7382. The chimeric toxin, TIC10746NTermExt2 (SEQ ID NO:127, encoded by SEQ ID NO:126) comprises the N-terminal extension of TIC7383. The synthetic coding sequences are cloned in binary plant transformation vectors similar to those described in Example 12 and used to transform corn plants. Root assays are performed as previously described in Example 8 for R₀ and F₁ plants after infestation of such corn rootworm species as Western Corn Rootworm (*Diabrotica virgifera virgifera*, WCR), Northern Corn Rootworm (*Diabrotica barberi*, NCR), Mexican Corn Rootworm (*Diabrotica virgifera zea*, MCR), Brazilian Corn Rootworm (*Diabrotica balteata*, BZR), Southern Corn Rootworm (*Diabrotica undecimpunctata howardii*, SCR), Colorado potato beetle (*Leptinotarsa decemlineata*, CPB), or a Brazilian Corn Rootworm complex (BCR, consisting of *Diabrotica viridula* and *Diabrotica speciosa*). Average low Root Damage Rating (RDR) scores of R₀ and F₁ plants relative to non-transformed corn plant controls are demonstrative of resistant to the corn rootworm species conferred by expression of the chimeric toxins.

All of the compositions disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the compositions of this invention have been described in terms of the foregoing illustrative embodiments, it will be apparent to those of skill in the art that variations, changes, modifications, and alterations may be applied to the compositions described herein, without departing from the true concept, spirit, and scope of the invention. More specifically, it will be apparent that certain agents that are both chemically and physiologically related may be substituted for the agents described herein while the same or similar results would be achieved. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope, and concept of the invention as defined by the appended claims.

All publications and published patent documents cited in the specification are incorporated herein by reference to the same extent as if each individual publication or patent application was specifically and individually indicated to be incorporated by reference.

SEQUENCE LISTING

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catcagctag	aaaaagcaag	aacgagcttg	aacgatttgt	ttaccaatga	atcgaaaaat	2040
gtgttaaaaa	aatacacgac	cgattatgat	atagatcaag	ctgcacaaat	ggtagaatgt	2100
gtatctgatg	aatgtgcaaa	tgctaaaaatg	atcctattag	atgaagtaaa	atatgcgaaa	2160
caactcagcg	aagcccgcga	tctactctg	aatggtaatt	ttgaatacca	agatagagat	2220
ggggagaaatc	catggaaaac	aagtccaaat	gttaccatcc	aggaaaaata	ccccattttt	2280
aaaggccggt	atctcagtat	gtcgggtgca	aactatatcg	aggcaacaaa	tgatactttc	2340
cccacttatg	cataccaaaa	aatagatgaa	gcaaaattaa	aacctatac	ccgtttataa	2400
gttcgagggt	ttgttggaat	tagtaaaagt	ttagagtgtg	tgattaaacg	gtatgatgaa	2460
gaagtggatg	cgatttttaa	tgtaacaaat	gatataccac	atgctccgac	acctttctgc	2520
gggtgatttg	atcgatgcga	gccacattct	tatatctcta	tgaatccaga	atgtcaccat	2580
gatgtaataa	ataacattga	aatatcctct	ccttgccaac	acaataagat	gttggaatac	2640
gctgatata	cttctcgcca	tagtgaatta	ggtaaaaaac	gtggaatttg	tcatgaatct	2700
catcattttg	aattccatat	tgatacagga	aaaatcgatt	tgaacgaaaa	tttgggaatt	2760
tgggttatat	ttaaaatgat	ttccacagat	gggtacgcaa	cattagataa	tttgggaagt	2820
attgaagagg	gtcctttagg	agccgaatca	ttagaacgtg	tgaagaaga	agaaaagaaa	2880
tggaacatc	acatggcaac	caagtgttca	gaaactaaac	ttgcatatca	tgctgcaaaa	2940
caagcgctgg	tgggggtatt	cacaaacact	aaatatgata	gattaaagtt	cgaacacacc	3000
atatccataa	tctcttttgc	tgattatctc	gtgcagtgca	ttccgtatgt	atataataaa	3060
tggttacca	atgttccagg	tatgaattac	gatattctata	cagaattaaa	aaactctgtt	3120
acgggagctt	tcaatttata	tgatcagcga	aatattataa	aaaatggaga	ctttaatcgc	3180
gggtcctatg	attggcatgc	gacacctcat	gcaagatgag	agcaaataga	taatagggtc	3240
gtgctggtgc	ttccaaatta	tgctgccaat	gtttcacaa	agggttggtt	agaacacaat	3300
cgtggttatg	tattacgtgt	aacggcgaaa	aaagaaggcc	ctggaattgg	atatgttaca	3360
ttcagtgatt	gtgcaaataa	tatagaaaaa	ctgacattta	cttcttgcca	ttatggtaca	3420
aacgaagtga	catatgagca	atctaattat	catcacagag	gagtagccga	cgaacaatct	3480
aattatccca	cagacggagt	accgtacgaa	caacatggtt	gtcatacaga	cggagtagccg	3540
tacgaacaat	ctaattatcc	tacagacgga	gtaccgtacg	aacaacatgg	ttgtcataca	3600
gacggagtgc	cgtacgaaca	acatggttgt	catcacagag	gagtagccga	cgaacaacat	3660
ggttgtcata	cagacggagt	accatacaaa	caacatggtt	gtcgtacaga	cggagtacta	3720
tacaaacaac	atggtttgtc	tacagacaga	tcaagagatg	aacaacttga	ttatgtgaca	3780
aaaacgattg	atgtattccc	tgatactgat	aaagtacgta	tcgacattgg	agaaaccgaa	3840
ggtaccttta	aagtagaaag	tgtggaactg	atttttatgg	aagagtaa		3888

SEQ ID NO: 4 moltype = AA length = 1295

-continued

FEATURE Location/Qualifiers
 REGION 1..1295
 note = The amino acid sequence of the TIC7040HT protein.
 source 1..1295
 mol_type = protein
 organism = Brevibacillus laterosporus

SEQUENCE: 4

MNQNQNQNK	ELQIIEPSSD	SLLYSHNNYP	YATDPNTVLE	GRNYKEWLNK	CTDNYTDALQ	60
GPEATAISKG	AVSAAISIST	KVLSLLGVPF	AAQIGQLWTF	ILNALWPSDN	TQWEEFMRHV	120
EELINQRIAD	YARSKALAE	TGLGNNLDLY	IEALEDWKRN	PTSQQAQDRV	KDRFRIADGL	180
FEAYMPSFRV	SGYEVPLLT	YAAAAHLHL	LLRDCSIYGI	QWGFSTQNVN	ENYNRQIRHT	240
AEYANHCTTW	YQTGLERLRG	TNASSWVPYN	RFRREMTLTV	LDICSLFSNY	DYRSYPAEVR	300
AELTREIYTD	PVVSTSLWMN	NAPSFGEIEN	LAIAPRTVT	WLNSTRISTG	TLQGWSGSNR	360
YWAAHMQNFS	ETNSGNIGFD	GPLYGSTVGT	IIRDDNYEMV	NRDIYTTITSE	AVAALWPTGQ	420
IVLGVASARF	TLRLNHNLT	QALVYENPIS	SSFNRSTLTR	ELPGENS DRP	TSSDYSHRLT	480
SITAFRAGSN	GTIPVFGWTS	ISVNRDNILE	RNKITQFPGV	KSHTLNNCQV	VRGTGFTGGD	540
WLRPNNGNSF	RLTITSFSSQ	SYRIRLRYAS	AANTSLRISS	SAAGISSTTV	PLTSTITSLP	600
QTAVPYEAFR	VIDLPITFTT	ATQSNYTFDF	VLQNPNSANV	FIDRFEPVPI	GGSLSEYETK	660
HQLEKARKAV	NDLPTNESKN	VLKKYTTDYD	IDQAANLVEC	VSEDCANAKM	ILLDEVKYAK	720
QLSEARNLLL	NGNFEYQDRD	GENPWKTSPN	VTIQENNPFI	KGRYLSMSG	NYIEATNDTF	780
PTYAYQKIDE	ETKLKPYTRYK	VRGFVGNISK	LELLIKRYDE	EVDAILNVPN	DIPHAPTPFC	840
GGFDRCKPHS	YIPMNECHH	DVINNIEISS	PCQHNKMLDN	ADISSRHSSEL	GKRGICHS	900
HHFEFPHIDTG	KIDLNLNGLI	WVIFKICSTD	GYATLDNLEV	IEEGPLGAES	LERVKRREKK	960
WKHHMEHKCS	ETKLAYHAAK	QALVGLFTNT	KYDRLKFFET	ISNLFADYL	VQSIPIVYNK	1020
WLPDVPGMNY	DIYTELKNLF	TGAFNLYDQR	NIKNGDFNR	GLMHWHTAPH	ARVEQIDNRS	1080
VLVLPNYAAN	VSQEVCLLEHN	RGYVLRVTAK	KEGPGIGYVT	FSDCANNIEK	LTFTSCDYGT	1140
NEVTYEQSNY	HTDGVPEYQS	NYPTDGVPEY	QHGCHTDGVP	YEQSNYPTDG	VPYEQHGCHT	1200
DGVPEYQHG	HTDGVPEYQH	GCHTDGVPEY	QHGCHTDGVL	YKQHGCHTDR	SRDEQLDYVT	1260
KTIDVFPDTD	KVRIDIGETE	GTFKVESVEL	IFMEE			1295

SEQ ID NO: 5 moltype = DNA length = 2016
 FEATURE Location/Qualifiers
 misc_feature 1..2016
 note = A nucleic acid encoding a TIC7040HT_4 pesticidal protein sequence which comprises a C-terminal truncation relative to the TIC7040HT protein.
 source 1..2016
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 5

atgaatcaga	atcagaatca	aaataaaaaat	gaactgcata	tcatagaacc	ttcaagtgtat	60
tctttgcttt	atagtcacaa	caattatccg	tatgcaactg	atccaaatag	agtattagaa	120
ggtaggaatt	ataaagagtg	gctaaataag	tgacacagata	attatacaga	cgctttacag	180
gggtccgaag	ctactgctat	atcaaaaagga	gctgtctctg	ctgcgatttc	tatcagcacc	240
aaagtctcta	gtttattagg	tgttccgttt	gcagctcaaa	tcggacaact	ttggaccttc	300
atattgaatg	gtttatggcg	ttcagacaat	actcaatggg	aagagttcat	gagacatgta	360
gaagaactca	ttaaccacag	aatagccgat	tatgcaagaa	gtaaggcact	tgacagaatta	420
acgggtttag	gtaataactt	agatttatat	atagaagctc	ttgaagattg	gaaacgaaat	480
cctactagtc	aacaagcgaa	agaccgtgta	aaagatagat	tccgtatagc	agatgggtta	540
tttgaagcgt	atatgccttc	atttagagta	tcagggttatg	aagtaccatt	attaacagtgt	600
tatgcagcgc	ctgcaaatct	ccatttactt	ttattaagag	attgctctat	ttacggaatc	660
caatggggat	ttagtcaaac	gaatgttaac	gagaattaca	atcgccaaat	aagacacacc	720
gcagagtatg	caaatcattg	tacaacttgg	tacaaaactg	gtttagaaag	attgcgaggt	780
accaatgctt	ccagttgggt	cccttataat	cgtttccgaa	gagaaatgac	gttaactgta	840
ttggatattt	gttctctatt	ttcaaatatt	gattatcgta	gttatccagc	agaggtaagg	900
gcagagctta	caagagaaat	ttatcaggac	ccagttgtaa	gcactagctt	gtggatgaat	960
aatgcaccat	cattcggaga	aatagaaaat	ctagcaatta	ggcgcccaag	aaccgttact	1020
tggttaaat	ctacaagaat	ttctacaggg	accttcaggg	gctggagtgg	ttctaacaga	1080
tattgggcag	ctcacatgca	aaacttttca	gaaaccaatt	caggaaatat	aggatttgac	1140
ggctcctct	atgggtcgac	ggtaggtact	attattcggt	atgataatta	cgaaatgggt	1200
aaccgagata	tttacaccat	tacttcagag	gctgttgccg	ccctttggcc	aactgggtcaa	1260
attgtgttgg	gagtcgcttc	ggctagattt	actttaagaa	atcttaacaa	taactcttaca	1320
caggcgctgg	tgtatgagaa	cccaataagt	tcaagtttta	ataggtcaac	tttaactcgt	1380
gaattacctg	gggaaaactc	agataggcca	acttcaagcg	actatagtc	tagactaacg	1440
tctattacag	cttttcgagc	tggaaagta	gggacgattc	cggttttttg	atggacatct	1500
ataagtgtta	atcgtagcaa	tatacttgag	cgaaacaaaa	taacacaaat	cccaggcggt	1560
aagtcacaca	ctctcaacaa	ttgtcaagta	gttagaggta	ctggattttac	aggaggagac	1620
tggttgagac	caataataaa	tggttcattt	agattaaacta	ttacttcatt	ctcagagcaa	1680
tcttaccgaa	tccgcttacg	ttatgcttcc	gcagcaataa	cttctttgcg	tatatcttct	1740
tctgcagcgc	gtattttctc	cacaaccgtt	cgcgttacct	caacaataac	atcactgcca	1800
caactcgtcg	taccatagta	agcttttaga	gttatagatt	tacctattac	ttttacaaca	1860
gctacccaaa	gtaattatag	ttttgatttt	gttctccaaa	atccatcaaa	cgcaaatgta	1920
ttcattgata	gatttgaatt	tgttccaatt	gggggttctt	tgtctgagta	tgaacccaaa	1980
catcagctag	aaaaagcaag	gaaagcggtg	aactaa			2016

SEQ ID NO: 6 moltype = AA length = 671
 FEATURE Location/Qualifiers
 REGION 1..671
 note = The amino acid sequence of the TIC7040HT_4 protein,

-continued

consisting of amino acids 1 through 671 of TIC7040HT.

source 1..671
mol_type = protein
organism = synthetic construct

SEQUENCE: 6

MNQNQNQKN	ELQIEPSSD	SLLYSHNNYP	YATDPNTVLE	GRNYKEWLNK	CTDNYTDALQ	60
GPEATAISKG	AVSAAISIST	KVLSLLGVFP	AAQIGQLWTF	ILNALWPSDN	TQWEEFMRHV	120
EELINQRIAD	YARSKALAE	TGLGNLDLY	IEALEDWKRN	PTSQQAQDRV	KDRFRIADGL	180
FEAYMPSFRV	SGYEVPLLT	YAAAAHLHL	LLRDCSIYGI	QWGFSTQNVN	ENYNRQIRHT	240
AEYANHCTTW	YQTGLERLRG	TNASSWVPYN	RFRREMTLTV	LDICSLFSNY	DYRSYPAEVR	300
AELTREIYTD	PVSTSLWMN	NAPSFGEIEN	LAIRAPRTVT	WLNSTRISTG	TLQGWSGSNR	360
YWAAHMQNFS	ETNSGNIGFD	GPLYGSTVGT	IIRDDNYEMV	NRDIYITISE	AVAALWPTGQ	420
IVLGVASARF	TLRNLNNLTL	QALVYENPIS	SSFNRSTLTR	ELPGENS DRP	TSSDYSHRLT	480
SITAFRAGSN	GTIPVFGWTS	ISVNRDNILE	RNKITQFPVG	KSHTLNNCQV	VRGTGFTGGD	540
WLRPNNGSF	RLTITSFSSQ	SYRIRLRYAS	AANTSLRISS	SAAGISSTTV	PLTSTITSLP	600
QTAVPYEAER	VIDLPITFTT	ATQSNYTFDF	VLQNPSNANV	FIDRFEFVPI	GGSLSEYETK	660
HQLEKARKAV	N					671

SEQ ID NO: 7
FEATURE
misc_feature

moltype = DNA length = 1923
Location/Qualifiers
1..1923
note = A nucleic acid encoding a TIC7040HT_5 pesticidal protein sequence which comprises an N-terminal and C-terminal truncation relative to the TIC7040HT protein.

source 1..1923
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 7

atgcaaatca	tagaaccttc	aagtgattct	ttgctttata	gtcacaca	ttatccgtat	60
gcaactgac	caaatacagt	attagaaggt	aggaattata	aagagtggtc	aaataagtc	120
acagataa	atcacagcgc	tttacagggt	cccgaaagct	ctgctatata	aaaaggagct	180
gtctctgctg	atgattctct	cagcaccaaa	gttcttagtt	tattagtggt	tccgtttgca	240
gctcaaatcg	gacaactttg	gaccttca	ttgaatcgct	tatggccttc	agacaatact	300
caatgggaag	agttcatgag	acatgtagaa	gaactcataa	accaacgaat	agccgattat	360
gcaagaagta	aggcacttgc	agaattaacg	ggtttaggta	ataacttaga	tttatatata	420
gaagctcttg	aagattggaa	cgaaatcct	actagtcaac	aagcgaaaga	ccgtgtaaaa	480
gatagattcc	gtatagcaga	tggtttattt	gaagcgtata	tcgcttcatt	tagagtagca	540
ggttatgaag	taccattatt	aacagtggt	gcagccgctg	caaatctcca	tttactttta	600
ttagagagatt	gctctattta	cggaaatcca	tggggattta	gtcaaacgaa	tgttaacgag	660
aattacaatc	gccaaataag	acacaccgca	gagtagtcaa	atcattgtac	aacttggtac	720
caaatcggtt	tagaaagatt	gcgaggtacc	aatgcttcca	gttgggtccc	ttataatcgt	780
ttccgaagag	aaatgacggt	aactgtattg	gatattttgt	ccttattttc	aaattatgat	840
tatcgtagtt	atccagcaga	ggtaagggca	gagcttacaa	gagaaattta	tacggaccca	900
ggtgtaagca	ctagcttggtg	gatgaataat	gcaccatcat	tcggagaaat	agaaaatcta	960
gcaattaggg	cgccaagaac	cgttacttgg	ttaaattcta	caagaatttc	tacaggggac	1020
ttgcagggct	ggagtgggtc	taacagatat	tgggcagctc	acatgcaaaa	cttttcagaa	1080
accaattcag	gaaatatagg	atttgacggt	cctctctatg	ggtcgacggt	aggtactatt	1140
attcgtagtg	ataattacga	aatggtgaac	cgagatat	acaccattac	ttcagaggct	1200
gttgcgccc	tttggccaac	tgggtcaaat	gtgttgggag	tcgcttcggc	tagatttact	1260
ttagaaatc	ttacaataa	tcttacacag	gcgctgggtg	atgagaaccc	aataagttca	1320
agtttttaata	gggcaacttt	aactcgtgaa	ttacctgggg	aaaactcaga	tagggccaact	1380
tcaagcgact	atagtcagat	attacagctt	ttcgagctgg	aagtaatggg		1440
acgattccgg	tttttggatg	gacatctata	agtgtaatac	gtgacaatat	acttgagcga	1500
aacaaataa	cacaatcccc	aggcgttaag	tcacacactc	tcaacaattg	tcaagtagtt	1560
agaggtactg	gatttacagg	aggagactgg	ttgagaccaa	ataataatgg	ttcatttaga	1620
ttactatata	cttcattctc	gagccaatct	taccgaatcc	gccttacgta	tgttcccgca	1680
gcaaataact	ctttgcgtat	atcttcttct	gcagccggtg	tttcttccac	aaccgttccg	1740
cttacctcaa	caataacatc	actgcccaca	actgctgtac	catatgaagc	ttttagagtt	1800
atagatttac	ctattacttt	tacaacagct	acccaaagta	attatacttt	tgattttggt	1860
ctccaaaatc	catcaaacgc	aaatgtattc	attgatagat	ttgaatttgt	tccaattggg	1920
taa						1923

SEQ ID NO: 8
FEATURE
REGION

moltype = AA length = 600
Location/Qualifiers
1..600
note = The amino acid sequence of the TIC7040HT_5 protein, comprising amino acids 13 through 611 of TIC7040HT.

source 1..600
mol_type = protein
organism = synthetic construct

SEQUENCE: 8

MQIEPSSDS	LLYSHNNYPY	ATDPNTVLEG	RNYKEWLNK	TDNYTDALQ	PEATAISKGA	60
VSAAISISTK	VLSLLGVFFA	AQIGQLWTFI	LNALWPSDNT	QWEEFMRHVE	ELINQRIADY	120
ARSKALAE	TLGNLDLYI	ELEDWKRN	TSQQAQDRV	DRFRIADGLF	EAYMPSFRVS	180
GYEVPLLT	VYAAAHLHL	LRDCSIYGIQ	WGFSTQNVNE	NYNRQIRHTA	EYANHCTTWY	240
QTGLERLRGT	NASSWVPYNR	RFRREMTLTVL	DICSLFSNYD	YRSYPAEVRA	ELTREIYTPD	300
VVSTSLWMN	APSFGEIENL	AIRAPRTVTW	LNSTRISTGT	LQGWSGSNRY	WAHMQNFSE	360
TNSGNIGFDG	PLYGSTVGTI	IRDDNYEMVN	RDIYITITSEA	VAALWPTGQI	VLGVASARFT	420
LRNLNNLTLQ	ALVYENPISS	SFNRSTLTRE	LPGENSDRPT	SSDYSHRLTS	ITAFRAGSNG	480

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TIPVFGWTSI SVNRDNILER NKITQFPQVK SHTLNNQVQV RGTGFTGGDW LRPNNNGSFR 540
LTITSFSSQS YRIRLRYASA ANTSRLRISS AAGISSTTVP LTSTITSLPQ TAVPYEAFRV 600

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SEQ ID NO: 9      moltype = DNA length = 1983
FEATURE          Location/Qualifiers
misc_feature     1..1983
                  note = A nucleic acid encoding a TIC7040HT_6 pesticidal
                  protein sequence which comprises an N-terminal and
                  C-terminal truncation relative to the TIC7040HT protein.
source           1..1983
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 9
atgcaaatca tagaaccttc aagtgtattct ttgctttata gtcacaacaa ttatccgtat 60
gcaactgata caaatcacgt attagaaggt aggaattata aagagtggct aaataagtgc 120
acagataatt atacagacgc ttacaggggt cccgaagcta ctgctatata aaaaggagct 180
gtctctgtcg cgattttctat cagcaccaaa gttcttagtt tattaggtgt tccgtttgca 240
gctcaaatcg gacaactttg gaccttcata ttgaatgcgt tatggccttc agacaatact 300
caatgggaag agttcatgag acatgtagaa gaactcataa accaacgaat agccgattat 360
gcaagaagta aggcacttgc agaattaacg ggtttaggtg ataacttaga tttatatata 420
gaagctcttg aagattggaa acgaaatcct actagtcaac aagcgaaaga ccgtgtaaaa 480
gatatattcc gtatagcaga tggtttattt gaagcgtata tgccttcatt tagagtatca 540
ggttatgaag taccattatt aacagtgtat gcagccgctg caaatctcca tttactttta 600
ttaagagatt gctctattta cgggaatccaa tggggattta gtcaaacgaa tgttaacgag 660
aattacaatc gccaaataag acacaccgca gagtatgcaa atcattgtac aacttggtac 720
caaatcggtt tagaaagatt gcgaggtacc aatgcttcca gttgggtccc ttataatcgt 780
ttccgaagag aaatgacgtt aactgtattg gatatttgtt ccttattttc aaattatgat 840
tatcgtagtt atccagcaga ggtaagggca gagcttacaa gagaatttta tacggacca 900
gttgaagca ctactctgtg gatgaataat gcaccatcat tcggagaaat agaaaatcta 960
gcaattaggg cgccaagaac cgttacttgg ttaaatctta caagaatttc tacaggggacc 1020
ttgcagggct ggagtgtgtt taacagatat tgggcagctc acatgcaaaa cttttcagaa 1080
accaattcag gaaatatagg atttgacggt cctctctatg ggtcgacggt aggtactatt 1140
attcgtgatg ataattacga aatggtgaac cgagatatat acaccattac ttcagaggct 1200
gttgccgccc tttggccaac tgggtcaaatt gtgttgggag tcgcttcggc tagatttact 1260
ttaagaaatc ttaacaataa tcttacacag gcgctgggtg atgagaaccc aataagttca 1320
agttttaata ggtcaacttt aactcgtgaa ttacctgggg aaaactcaga tagggccaact 1380
tcaagcgact atagtcatag actaacgtct attacagctt ttcgagctgg aagtaatggg 1440
acgattccgg tttttggatg gacatctata agtggttaac gtgacaatat acttgagcga 1500
aacaaaataa cacaattccc aggcgttaag tcacacactc tcaacaattg tcaagtagtt 1560
agaggtactg gatttacagg aggagactgg ttgagaccaa ataataatgg ttcatttaga 1620
ttaactatta cttcattctc gagccaatct taccgaatcc gcttacgtta tgcctccgca 1680
gcaaatactt ctttgcgatg atcttcttct gcagccggtg tttcttccac aaccgttccg 1740
cttacctcaa caataaacac actgcccaca actgctgtac catatgaagc ttttagagtt 1800
atagattttc ctattacttt tacaacagct acccaaaagta attatacttt tgattttgtt 1860
ctccaaaatc catcaaacgc aaatgtattc attgatagat ttgaatttgt tccaattggg 1920
ggttctttgt ctgagtatga aaccaaacat cagctagaaa aagcaaggaa agcgggtgac 1980
taa                                                1983

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SEQ ID NO: 10     moltype = AA length = 660
FEATURE          Location/Qualifiers
REGION           1..660
                  note = The amino acid sequence of the TIC7040HT_6 protein,
                  comprising amino acids 13 through 671 of TIC7040HT.
source           1..660
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 10
MQIIEPSSDS LLYSHNNYPY ATDPNTVLEG RNYKEWLKNC TDNYTDALQG PEATAISKGA 60
VSAAISISTK VLSLLGVPPA AQIGQLWTFI LNALWPSDNT QWEEFMRHVE ELINQRIADY 120
ARSKALAEIT GLGNLNDLYI EALDWRKRN TSQQAQDRVK DRFRIADGLF EAYMPSFRVS 180
GYEVPLLTIV AAAANLHLLL LRDCSIYGIQ WGFSTQTNVNE NYNRQIRHTA EYANHCTTWY 240
QTGLERLRGT NASSWVPYNR FPREMTLTVL DICSLFSNYD YRSYPAEVRA ELTREIYTDP 300
VVSTSLWMNN APSFGEIENL AIRAPRTVTW LNSTRISTGT LQWWSGSNRY WAAHMQNFSE 360
TNSGNIGFDG PLYGSTVGTI IRDDNYEMVN RDIYTITSEA VAALWPTGQI VLGVASARFT 420
LRNLNNNLTO ALVYENPISS SFNRSTLTRE LPGENS DRPT SSDYSHRLTS ITAFRAGSNG 480
TIPVFGWTSI SVNRDNILER NKITQFPQVK SHTLNNQVQV RGTGFTGGDW LRPNNNGSFR 540
LTITSFSSQS YRIRLRYASA ANTSRLRISS AAGISSTTVP LTSTITSLPQ TAVPYEAFRV 600
IDLPTITFTA TQSNYTFDFV LQNPNSNANVF IDRFEFVPIG GSLSEYETKH QLEKARKAVN 660

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SEQ ID NO: 11     moltype = DNA length = 3801
FEATURE          Location/Qualifiers
misc_feature     1..3801
                  note = A nucleic acid encoding a TIC7042 pesticidal protein
                  sequence obtained from Brevibacillus laterosporus species
                  DSC005973.
source           1..3801
                  mol_type = unassigned DNA
                  organism = Brevibacillus laterosporus

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SEQ ID NO: 12 moltype = AA length = 1266
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 REGION 1..1266
 note = The amino acid sequence of the TIC7042 protein.
 source 1..1266
 mol_type = protein
 organism = Brevibacillus laterosporus

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 MPSFAVRNFE QLLTVYASA ANIHLFLLRD SSIYGLDWGL SQTNVNNENYN RQIRHAATYA 240
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SEQ ID NO: 13          moltype = DNA length = 3810
FEATURE                Location/Qualifiers
misc_feature            1..3810
                        note = A nucleic acid encoding a TIC7381 pesticidal protein
                        sequence obtained from Brevibacillus laterosporus species
                        DSC006713.
source                  1..3810
                        mol_type = unassigned DNA
                        organism = Brevibacillus laterosporus

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SEQ ID NO: 14      moltype = AA  length = 1269
FEATURE           Location/Qualifiers
REGION            1..1269
                  note = The amino acid sequence of the TIC7381 protein.
source            1..1269
                  mol_type = protein
                  organism = Brevibacillus laterosporus

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AYMPSFRVSG YEVP LLTVYA AANLHLLLL RDCSYGIQW GFSQTNVNN YNRQIRHTAE 240
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VLPNYAANVS QVECLEHNRG YVLRVTAKKE GPGIGYVTF SDCANNIEKL FTS CDYGTNE 1140
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SEQ ID NO: 15      moltype = DNA  length = 3741
FEATURE           Location/Qualifiers
misc_feature       1..3741
                  note = A nucleic acid encoding a TIC7382 pesticidal protein
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source            1..3741
                  mol_type = unassigned DNA
                  organism = Brevibacillus laterosporus

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SEQ ID NO: 16      moltype = AA length = 1246
FEATURE           Location/Qualifiers
REGION            1..1246
                  note = The amino acid sequence of the TIC7382 protein.
source            1..1246
                  mol_type = protein
                  organism = Brevibacillus laterosporus

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SEQUENCE: 16
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EELINQRIAD YARNKALAEI TGLGNLDLY IEALDDWKRN PTSQEAQTRV IDRFRIVDGL 180
FEAYIPSPAV SGYVQLLTIV YAAANLHLL LLRDSTIYGI DWGLSQTNVN DYNRQIRLT 240
ATYANHCTTW YQTGLERLRG SNASSWVTYN RFRREMTLTV LDICSLFSNY DYRSYPAEVR 300
GEITREIYTD PVGVGVDSAS PSFGEIENLA IRAPRTVTWL NSTRISTGTL SGWSGSNRYW 360
AAHMQNFSET NSGNIGFDGP LYGSTVGTIH RTDDYDMGNR DIYTITSQAV LGLWPTGQRV 420
LGVASARFTL RNLFNMLTQV LVYENPISSN FGSSTLTHEL SGENSDRPTS SDYSHRLTSI 480
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RPNNGTFPRL TITSFSSQSY RIRLRATSV GNTSLVISS DAGISSTIP LTSTITSLPQ 600
TVPYQAFRVV DLPITFTPT TQRNYTFDFR LQNPNSANVF IDRFEFVPIG GSLSEYETKH 660
QLEKARKAVN DLFTNESKNV LKKETTDYDI DQANLVECI SDECANAKMI LLDEVKYAKQ 720
LSEARNLLLN GNFDNLDNRG ENPWKTSNPV TIQENNPIFK GRYLSMSGAN NIEATNEIFP 780
TYVYQKIDEA KLKPYTRYKV RGFVGSSKDL ELLVTRYDEE VDAILNLVND IPHAPPPFCG 840
GFDRCKPHSY PPMNPECHD VINNIEISS COHNKMLDNA DIFSRHSELG KKHGICHESH 900
HPEFHIDTCK IDLNENLGIW VIFKICSTDG YATLDNLEVI EEGPLGAESL ERVKRREKKW 960
KHHMEHKCSE TKLAYHAAKQ ALVGLFTNTK YDRLEFETTI SNILFADYLV QSIPIVYNKW 1020
LPDVPGMNYD IYTELKNLFT GAFNLYDQRN IIKNGDFNRG LMHWHPHFA RVEQIDNRSV 1080
LVLPNYAANV SQEVCLHNHR GYVLRVTAKK EGPIGIYVTF SDCANNIEKL TFTSCDYGTN 1140
EVPYEQSGYG TNEVPYEQSG YGTNEVPYEQ SGYRTDGVFY KQHGCRDGV PYEQSGCRTD 1200
RSRDEQLDYV TKTIDVFPDT DKVRIDIGET EGTFKVESVE LFCMEE 1246

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SEQ ID NO: 17      moltype = DNA length = 3771
FEATURE           Location/Qualifiers
misc_feature       1..3771
                  note = A nucleic acid encoding a TIC7383 pesticidal protein
                      sequence obtained from Brevibacillus laterosporus species
                      DSC008106.
source            1..3771
                  mol_type = unassigned DNA
                  organism = Brevibacillus laterosporus
SEQUENCE: 17

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SEQ ID NO: 18      moltype = AA length = 1256
FEATURE           Location/Qualifiers
REGION            1..1256
                  note = The amino acid sequence of the TIC7383 protein.
source            1..1256
                  mol_type = protein
                  organism = Brevibacillus laterosporus

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SEQUENCE: 18
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HVEELINQRI ADYARNKALA ELTGLGNLND LYIEALEDWK RNPTSQQAKD RVIDRFRRIAD 180
GLFEGYMPFS RVSGYEVPLL TVYAAAANLH LLLLRDCSTY GIQWGFSTQN VNENYNRQIR 240
HTAEYANHCT TWYQTGLERL RGTNAFSWIN YNRFREMTL TVLDVCSLFS NYDYRTYPE 300
VRAELTREYI DPTIGFQNSP LPGVVPNWYD YARSAEIEI IAIAPRPTVT WLNSTTIYT 360
RLNGYNNSNY YWAGFRQNS ETNSGSSPFG PDLGLTPNY RIETLDMVNR DIYSIYSRVV 420

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TGFTGGNWLRL	SSRVTSFRL	NVYSPSVQTY	RMRIYASPL	GNSTLGISST	DAGISFTSFP	600
LPSTIGSMPS	TVPYEAFRVL	DIPITVTVAS	QRNYNFIDFI	LNPSVGAVYI	DRIEFVPVGS	660
SVFEYETKHE	LEKAKKAVND	LFTNESKNML	KKDTTDYDID	QAADLVECVS	DECAHAKMIL	720
LDEVKYAKQL	SEARNLLQNG	NFKVLDVDNN	NPWTTSNVNT	IQENNPIFKG	HYLSMSGANA	780
IEATNEVFPT	VYVQKIESK	LKPTRYKVR	GFIGQSKDVE	LLVTRYDEEV	DAILNVPNDL	840
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KKHGICHESH	HFEFHIDTGH	IDLVENLGIW	VIFKICSTDG	YATLDNLEVI	EEGPLGAESL	960
ERVKRREKKW	KHNMEHKCSE	TKHAYHAAKQ	AVEALFTNFK	DERLKFETTI	SNILSAEYLV	1020
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NVQQIDGISV	LVLPNWGANV	SQEVCLKHNR	GYVLRVTAKE	EGHGKGYVTI	SDCANQVEKL	1140
SFTSRDYSYD	GVPEYQSNYP	TDGVSYGQHG	CNIDRVPEYQ	SGYPTDGVVPY	EQSGYRTDGV	1200
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SEQ ID NO: 19 moltype = DNA length = 3732
FEATURE Location/Qualifiers
misc_feature 1..3732
 note = A nucleic acid encoding a TIC7383.2 protein which
 comprises an N-terminal truncation relative to the TIC7383
 protein.
source 1..3732
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 19

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atggaagagt aa 3732

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SEQ ID NO: 20      moltype = AA  length = 1243
FEATURE           Location/Qualifiers
REGION            1..1243
                  note = The amino acid sequence of the TIC7383_2 protein,
                  comprising amino acids 15 through 1256 of TIC7383.
source            1..1243
                  mol_type = protein
                  organism = synthetic construct

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VSAAISISTK VLGLLGVPFA AQIGQLWTFI LNALWPSDNT QWEEFMRHVE ELINQRIADY 120
ARNKALABEL GLGNLMDLYI EALWDKRNK TSQQAQDRVI DRFRIADGLF EGYMPSFRVS 180
GYEVPLLTIV AAAANLHLLL LRDCSIYGIQ WGFSSQTNVNE NYNRQIRHTA EYANHCTTWY 240
QTGLEERLGT NAFSWINYNR FRREMTLTVL DVCSLFSNYD YRTYPEVTRA ELTREIYTD 300
IGFQNSPLPG VVPNWYDYAR SFAEINIAT RAPRTVTWLN STTIYTGRLN GYNNSNYWA 360
GFRQNFSETN SGSSFNPGDL GDLPNRYRIE TLDMVNRDIY SIYSRVVSQS WPIGNVKLFG 420
VSSSTLSFRD LNNSSSGTLV YENPTNFSSQ YLTTEFPGEN SERPTFTDYS HRLTCLTRIG 480
AGNYGLVLCA GTWSSSVERD NRLQDPKITO YPAVKGFNLD GFTVVKGTGF TGNWLRSSR 540
VTGSFRLNVY SPSVQTYRMR IRYASPLGNS TLGISSTDAG ISFTSFPLPS TIGSMPSTVP 600
YEAFRVLDIP ITVTVASQRN YNFIPIILNP SVGAVYIDRI EFVVPVGSVF EYETKHELEK 660
AKKAVNDLFT NESKNMLKDD TTDYDIDQAA DLVECVSDEC AHAKMILLDE VKYAKQLSEA 720
RNLLQGNFVK VLDVDNNNPW TTSFNVITQE NNPIFKGHYL SMSGANAIEA TNEVPPTYVY 780
QKIEESKLKP YTRYKVRGFI GQSKDVELLV TRYDEEVDAI LNVPNLDKYA VPTHLSGEPN 840
RCKPHTYPAT DPRCHDDVID KIDISSPCQN NIMLSDADIS SLHSGLGKKH GICHESHHE 900
FHIDTGKIDL VENLGIWVIF KICSTDGYAT LDNLEVIEEG PLGAESLERV KREKKWKHN 960
MEHKCSETHK AYHAAKQAVE ALPTNFKDER LKFETTISNI LSAEYLVQSI PYVYNKWLSD 1020
VPGMNYDIYT ELKNRIWQAF NLYDQRNIK NGHFNHGLMH WHATPHANVQ QIDGISVLVL 1080
PNWGANVSQE VCLKHNRGVY LRVTAKEEGH GKGYVTISDC ANQVEKLSFT SRDYSTDGVP 1140
YEQSNYPTDG VSYGQHGQNI DRVPYEQSGY PTDGVPYEQS GYRTDGVPEYK QHGCHSDGSR 1200
EEQHGVTYTK IDVPFDTDKV RIDIGETEGT FKVESVELIC MEE 1243

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SEQ ID NO: 21      moltype = DNA  length = 1980
FEATURE           Location/Qualifiers
misc_feature       1..1980
                  note = A nucleic acid encoding a TIC7383_3 protein which
                  comprises a C-terminal truncation relative to the TIC7383
                  protein.
source            1..1980
                  mol_type = other DNA
                  organism = Brevibacillus laterosporus

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SEQUENCE: 21
atgaatcaga atcagaatca gaatacaaat aaaaatgaac tgcaaatcat agaacttca 60
agtgtattct tgctttatag tcacaacaat tatccgtatg caactgatcc aaatacagta 120
tagaaggtaa ggaattataa agagtggcta aataagtcta cagataatta tacagacgct 180
ttacaagggtc cgaagctac tgctatatca aaaggagctg tctctgctgc gatttctatc 240
agcaccaaag ttcttggttt attaggtgtt ccatttgcag ctcaaatcgg gcaactttgg 300
accttcatat tgaatgcgtt atggccttca gacaatactc aatgggaaga gttcatgaga 360
catgtagaag aaactcataa ccaacgaata gccgattatg caagaaataa ggcacttgca 420
gaattaacgg ggtaggttaa taacttagat ttatatatag aggctcttga agattggaaa 480
cgaaatccta ctagtcaaca agcgaagac cgtgtaaatg atagattccg tatagcagat 540
gggttatttg aagggtatat gccttcattt agagtatcag gttatgaagt accattatta 600
acagtgtatg cagccgctgc aaatctccat ttacttttat taagagattg ctctatttac 660
ggaatccaat ggggatttag tcaaacgaat gttaacgaga attacaatcg ccaataaaga 720
cacaccgcaag agtatgctaa ctattgtaca acttggtacc aaactggttt agaaagattg 780
cgaggtacca atgctttcag ttggatcaat tataatcgat tccgtagaga aatgacgtta 840
accgtatttg atgtttgttc attattttca aattatgatt atcgtacgta tccaacagag 900
gtaagagcag agcttacgag agaaatttat acggacccaa taggttttca aaatagtcct 960
cttctcgccg ttgttctctaa ttggtacgat tacgcacgat cggtcgaga gatagaaaat 1020
atagccattc gagcgccacg aactgttact tgggtaaatt ctacaactat ttatacaggt 1080
agattaaatg gctataacaa tagtaattat tattggcgag gtttcaggca aaatttttca 1140
gaaaccaatt caggcagttc atttaacggt cctgacttag gggatttaac acctaattat 1200
cgtatagaaa cattggatat ggtgaatcgg gatatttact ccattttatc aagagttggt 1260
tcacaatctt ggccaatttg caacgttaaa ttggttgagg tctcttcac tactctttca 1320
ttcagagatc ttaacaataa ttcttcaggg acgctggtat atgaaaacc gacaaatttt 1380
agtagccaat atctaactac cgaattccct ggggaaaact cagaaaagac aacttttacc 1440
gattatagtc acagactaac ttgtctcaca cgtattgggg ctggaaatta tggattgggt 1500
ctatgcgccg gctggacatc tagtagtggt gagcgtgaca acagactcca gccggacaaa 1560
ataacacaat acccgctgtg taaggattc aacctcgatg gttttacagt agtaaaagg 1620
actgggttta cagggggaaa ttggttgaga tctagtcgtg ttacaggtag ctttagacta 1680

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aatgttttatt caccgtctgt ccaaacttat cgcattgcgta tacgtttatgc ttctccactg 1740
ggaaatttcta ctctagggtat atctttctact gatgctggta ttagtttcac aagttttcca 1800
cttcctctcaa caataggatc aatgccatca actgtaccat acgaagcttt tagagttcta 1860
gatataccta tcaactgttac agtagctagt caaagaaatt ataattttat ttctgatatt 1920
ctaaatccat cagtcggagc agtatacatt gacagaattg aattcgttcc agttgggtaa 1980

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SEQ ID NO: 22      moltype = AA length = 659
FEATURE           Location/Qualifiers
REGION           1..659
                  note = The amino acid sequence of the TIC7383_3 protein and
                  consists of amino acids 1 through 659 of TIC7383.
source           1..659
                  mol_type = protein
                  organism = Brevibacillus laterosporus

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```

SEQUENCE: 22
MNQNQNQNQN KNEIQIIEPS SDSLLYSHNN YPYATDPNTV LEGRNYKEWL NKCTDNYTDA 60
LQGPEATAIS KGAVSAAISI STKVLGLLGV PFAAQIGQLW TFILNALWPS DNTQWEEFMR 120
HVEELINQRI ADYARNKALA ELTGGLGNLD LYIEALEDWK RNPTSQQAKD RVIDRFRRIAD 180
GLFEGYMPSE RVSGYEVPLL TVYAAANLH LLLLRDCSIY GIQWGFSTQN VNENYNRQIR 240
HTAEYANHCT TWYQTGLERL RGTNAFSSWIN YNRFRRMTL TVLDVCSLFS NYDYRTYPTTE 300
VRAELTREIY TDPIGFQNSP LPGVVPNWYD YARSPAEIEN IAIRAPRTVT WLNSTTIYTG 360
RLNGYNNNSY YWAGFRQNFES ETNSGSSSFG PDLGLTPNY RIETLDMVNR DIYSIYSRVV 420
SQSWPIGNVK LFGVSSSTLS FRDLNNNSSG TLVYENPTNF SSQYLTTTEFP GENSERPTFT 480
DYSHRLTCLT RIGAGNYGLV LCAAGTSSSV ERDNRLQPKD ITQYPAVKGF NLDGFTVVKG 540
TGFTGGNWLRL SSRVTGSFRL NVYSPSVQTY RMRIRYASPL GNSTLGISST DAGISPTSFP 600
LPSTIGSMPS TVPYEAFRVL DIPITVTVAS QRNYNFIFDI LNPSVGAVYI DRIEFVPPV 659

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SEQ ID NO: 23      moltype = DNA length = 2040
FEATURE           Location/Qualifiers
misc_feature      1..2040
                  note = A nucleic acid encoding a TIC7383_4 protein which
                  comprises a C-terminal truncation relative to the TIC7383
                  protein.
source           1..2040
                  mol_type = other DNA
                  organism = Brevibacillus laterosporus

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SEQUENCE: 23
atgaatcaga atcagaatca gaatacaaat aaaaatgaac tgcaaatcat agaaccctta 60
agtgaattctt tgcttttatg tcacaacaat tatccgtatg caactgatcc aaatacagta 120
tagaaggta ggaattataa agagtggcta aataagtgtg cagataatta tacagacgct 180
ttacagggtc ccgaagctac tgctatatca aaaggagctg tctctgctgc gatttctatc 240
agcaccaaaag ttcttggttt attagtggtt ccatctgcag ctcaaatcgg gcaactttgg 300
accttcattt tgaatgcggt atggccttca gacaataact aatgggaaga gttcatgaga 360
catgtagaag aactcataaa ccaacgaata gccgattatg caagaaaataa ggcacttgca 420
gaattaacgg ggtaggtaa taacttagat ttatatatag aggcctctga agattggaaa 480
cgaaatcccta ctagtcaaca agcgaagac cgtgtaatat atagattccg tatagcagat 540
ggttttattg aagggatat gcttcattt agagtatcag gttatgaagt accattatta 600
acagtgtagt cagccgctgc aaatctccat ttacttttat taagagattg ctctatttac 660
ggaatccaat ggggattgat tcaaacgaat gttaacgaga attacaatcg ccaataaaga 720
cacaccgcag agtatgcaaa tcattgtaca acttggtacc aaactggttt agaaagattg 780
cgaggtagca atgctttcag ttggatcaat tataatcgat tccgtagaga aatgacgtta 840
accgtattgg atgtttgttc attattttca aattatgatt atcgtacgta tccaacagag 900
gtaagagcag agcttacgag agaaatttat acggacccaa taggttttca aaatagctct 960
cttcctggcg ttgttcctaa ttggtacgat tacgcacgat cgttcgcaga gatagaaaat 1020
atagccattc gagcgccacg aactgttact tgggttaaat ctacaactat ttatacaggt 1080
agattaaaat gctataacaa tagtaattat tattgggcag gtttcaggca aaatttttca 1140
gaaaccaatt caggcagttc atttaacggt cctgacttag gggatttaac acctaatat 1200
cgtagataaa cattggatat ggtgaatcgg gatatttact ccatatttcc aagagttggt 1260
tcacaatctt ggccaattgg caacgttaaa ttgtttggag tctcttcac tactctttca 1320
ttcagagatc ttaacaataa ttcttcaggg acgctgggat atgaaaacc gacaaatatt 1380
agtagccaat atctaactac cgaattccct ggggaaaact cagaaagacc aacttttacc 1440
gattatagtc acagactaac ttgtctcaca cgtattgggg ctggaaaatta tggattgggt 1500
ctatgcgccg gctggacatc tagtagtggt gagcgtgaca acagactcca gccggacaaa 1560
ataacacaaat acccgctgtg taaggattc aacctcgatg gttttacagt agtaaaaggt 1620
actgggttta cagggggaaa ttgggtgaga tctagtcgtg ttacaggtag ctttagacta 1680
aatgtttatt caccgtctgt ccaaacttat cgcattgcgta tacgtttatgc ttctccactg 1740
ggaaatttcta ctctagggtat atctttctact gatgctggta ttagtttcac aagttttcca 1800
cttcctctcaa caataggatc aatgccatca actgtaccat acgaagcttt tagagttcta 1860
gatataccta tcaactgttac agtagctagt caaagaaatt ataattttat ttctgatatt 1920
ctaaatccat cagtcggagc agtatacatt gacagaattg aattcgttcc agttgggtct 1980
tctgtatttg aatatgaaac caagcatgag ctagaaaaag ctaagaaagc ggtgaactaa 2040

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SEQ ID NO: 24      moltype = AA length = 679
FEATURE           Location/Qualifiers
REGION           1..679
                  note = The amino acid sequence of the TIC7383_4 protein and
                  consists of amino acids 1 through 679 of TIC7383.
source           1..679
                  mol_type = protein

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organism = Brevibacillus laterosporus

SEQUENCE: 24
MNQNQNQNQN KNEQLIIEPS SDSLlySHNN YPYATDPNTV LEGRNYKEWL NKCTDNYTDA 60
LQPEATAIS KGAVSAATSI STKVLGLLGV PPAAQIGQLW TFIlnALWPS DNTQWEEFMR 120
HVEELINQRI ADYARNKALA ELTGLGNNLD LYIEALEDWK RNPTSQQAKD RVIDRFRRIAD 180
GLFEGYMPSP RVSGYEVPLL TVYAAAANLH LLLLRDCSIY GIQWGFQSTN VNYNYNRQIR 240
HTAEYANHCT TWYQTGLERL RGTNAFSWIN YNRFRREMTL TVLDVCSLFS NYDYRTPTE 300
VRAELTFREIY TDIPIGFQNSP LPGVVPNWYD YARSFAEIEI IAIAPRTVTW WLNSTTIYTG 360
RLNGYNNNSNY YWAGFRQNSF ETNSGSSFNQ PDLGLTPNY RIETLDMVNR DIYSIYSRVV 420
SCSWPLIGNVK LFGVSSSTLV FRDLNNSSSG TLVYENPTNF SSQYLTFTEP GENSERPTPT 480
DYSHRLTCLT RIGAGNYGLV LCAGWTSSSV ERDNRLQDPK ITQYPAVKGF NLDGFTVVKG 540
TGFTGGNWLR SSRVTGSFRL NVYSPSVQTY RMRIRYASPL GNSTLGISST DAGISFTSFP 600
LPSTIGSMPS TVPYEAFRVL DIPITVTVAS QRNYNFIPI DI LNPVSGAVYI DRIEFVPVGS 660
SVFEYETKHE LEKAKKAVN
SVFEYETKHE LEKAKKAVN 679

SEQ ID NO: 25
FEATURE
misc_feature
note = A nucleic acid encoding a TIC7383_5 protein which
comprises an N-terminal and C-terminal truncation relative
to the TIC7383 protein.
source
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 25
atgcaaatca tagaactctt aagtgattct ttgctttata gtcacaacaa ttatccgtag 60
gcaactgata caaacactagt attagaaggat aggaattata aagagtggtt aaataagtg 120
acagataaatt atacagacgc tttacagggt cccgaagcta ctgctatatac aaaaggagct 180
gtctctgctg cgatttctat cagcaccaaa gttcttggtt tattaggtgt tccatttgca 240
gtccaaatcg ggcaactcttg gaccttcata ttgaatgcgt ttggccttc agacaatact 300
caatgggaag agttctatgag acatgtagaa gaactcataa accaacgaat agccgattat 360
gcaagaaata aggcactctg agaattaacg ggggttagta ataacttaga ttatatata 420
gaggctcttg aagattggaa acgaaatcct actagtcaac aagcgaaaga ccgtgtaata 480
gatagattcc gtatagcaga tgggtttatt gaagggtata tgccttcatt tagagtatca 540
ggttatgaag taccattatt aacagtgtat gcagcgctg caaatctcaa ttactttta 600
ttaagagatt gctctattta cggaaatccta tggggattta gtcaaacgaa tgtaaagag 660
aattacaatc gccaaaataag acacaccgca gagtatgcaa atcattgtac aacttggtag 720
caaatcggtt tagaagattg gcgaggtacc aatgctttca gttggatcaa ttataatcga 780
ttccgtagag aagtagacgtt aaccgtattg gatgtttgtt cattattttc aaattatgat 840
tatcgtagct atccaacaga ggtaagagca gagcttacga gagaatttta tacggacca 900
ataggttttc aatatagctc tcttctctggc gttgttccta attggtacga ttacgcacga 960
tcgttctcag agatagaaaa tatagccatt cgagcgccac gaactgttac ttggttaaat 1020
tctacaacta tttatacagg tagattaaat ggctataaca atagtaatta ttattgggca 1080
gggttcagcg aaaaatttttc agaaaaccaa tcaggcagtt catttaacgg tctgactta 1140
ggggatttaa accctaatta tcgtatagaa acattggata tgggtgaatcg ggatatttac 1200
tccatttatt caagagttgt ttcacaatct tggccaattg gcaacgttaa attggttgg 1260
gtctcttcat ctactctttc attcagagat cttaaacaata attcttcagg gacgtcggt 1320
tctgaaaaac cgacaaaatt tagtagccaa tatctaacta cgaattccc tggggaatac 1380
tcgaaaagac caactttac cgattatagt cacagactaa cttgtctcac acgtattggg 1440
gctggaaaat atggattggt tctatgcgcc ggctggacat ctagttagtg tgagcgtgac 1500
aacagactcc agccgggacaa aataaaccaa tacccgctg ttaaaggatt caacctcgat 1560
gggtttacag tagtaaaagg tactgggttt acagggggaa attggttgag atctagtcgt 1620
gttacaggtg gctttagact aaatgtttat tcaccgtctg tccaaaacta tcgcatgctg 1680
atacgtttat cttctccact gggaaattct actctaggta tatcttctac tgatgctggt 1740
attagtttca caagttttcc acttccctca acaataggat caatgccatc aactgtacca 1800
tacgaagctt ttagagttct agatatacct atcactgtta cagtagctag tcaaaagaaat 1860
tataatttta ttttcgatat tctaaatcca tcagtcggag cagtatacat tgacagaatt 1920
gaattcgttc cagttgggta a 1941

SEQ ID NO: 26
FEATURE
REGION
note = The amino acid sequence of the TIC7383_5 protein and
comprises amino acids 15 though 659 of TIC7383.
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 26
MQIIEPSSDS LLYSHNNYPY ATDPNTVLEQ RNYKEWLNKC TDNYTDLQG PEATAISKGA 60
VSAAISISTK VLGLLGVPPFA AQIGQLWTFI LNALWPSDNT QWEEFMRHVE ELINQRIADY 120
ARNKALAEIT GLGNNLDLYI EALDEDKWRNP TSQQAKDRVI DRFRIADGLF EGYMPSFRVS 180
GYEVPLLTGY AAAANLHLLL RDCSIYGIQ WGFQSTNVNE NYNRQIRHTA EYANHCTTWY 240
QTGLERLGRG NAFSWINYNR FRREMTLTVL DVCSLFSNYD YRTYPTVRA ELTREIYTD 300
IGFQNSPLPG VVPNWYDYAR SFAEIEINIAI RAPRTVTWLN STTIYTGRLN GYNNNSNYWA 360
GFRQNFSETN SGSSFNQPD LGLTPNYRIE TLDMVNRDIY SIYSRVVSQS WPIGNVKLFG 420
VSSSTLSFRD LNNSSGTLV YENPTNFSQ YLTTEPPGN SERPTFTDYS HRLTCLTRIG 480
AGNYGLVLCA GWTSSSVVER NLRQPKITQ YPAVKGFNLD GFTVVKGTGF TGGNWLRSSR 540
VTFSPRLNVY SPSVQTYRMR IRYASPLGNS TLGISSTDAG ISFTSPFLPS TIGSMPTSTV 600
YEAFRVLDIP ITVTVASORN YNFIEDILNP SVGAVYIDRI EFVVPVG 646

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SEQ ID NO: 27      moltype = DNA length = 2001
FEATURE           Location/Qualifiers
misc_feature       1..2001
                   note = A nucleic acid encoding a TIC7383_6 protein which
                   comprises an N-terminal and C-terminal truncation relative
                   to the TIC7383 protein.
source             1..2001
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 27
atgcaaatca tagaaccttc aagtgattct ttgctttata gtcacaacaa ttatccgtat 60
gcaactgata caaatacagt attagaaggt aggaattata aagagtgggt aaataagtgt 120
acagataaatt atacagagcg tttacagggg cccgaagcta ctgctatatc aaaaggagct 180
gtctctgctg cgatttctat cagcaccaaa gttcttggtt tattaggtgt tccatttgca 240
gctcaaatcg ggcaactttg gaccttcata ttgaatgcgt tatggccttc agacaatact 300
caatgggaag agttcatgag acatgtagaa gaactcataa accaacgaat agccgattat 360
gcaagaaaata aggcacttgc agaattaacg ggggttagta ataacttaga ttatatata 420
gaggctcttg aagattggaa acgaaatcct actagtcaac aagcgaaaga cgtgtaata 480
gatagattcc gtatagcaga tgggtttatt gaagggtata tgccttcatt tagagtatca 540
ggttatgaag taccattatt aacagtgtat gcagccgctg caaatctcca ttactttta 600
ttaagagatt gctctattta cggaaatcca tggggattta gtcaaacgaa tgttaacgag 660
aattacaatc gccaaataag acacaccgca gagtatgcaa atcattgtac aacttggtag 720
caaaactggt tagaaagatt gcgaggtacc aatgctttca gttggatcaa ttataatcga 780
ttccgtagag aaatgacgtt aaccgtattg gatgtttgtt cattattttc aaattatgat 840
tatcgtagct atccaacaga ggtaagagca gagcttacga gagaatttta tacggaccca 900
ataggttttc aaaatagctc tcttctctgc gttgttccta attggtacga ttacgcacga 960
tcgttcgcag agatagaaaa tatagccatt cgagcgccac gaactgttac ttggttaaat 1020
tctacaacta ttatcacagg tagattaaat ggctataaca atagtaatta ttattgggca 1080
ggtttcaggc aaaatttttc agaaaccaat tcaggcagtt catttaacgg tcctgactta 1140
ggggatttaa cacctaatta tcgtatagaa acattggata tgggtgaatcg ggatatttac 1200
tccattttat caagagtgtg ttcacaatct tggccaattg gcaacgttaa attgtttgga 1260
gtctcttcac ctactcttct attcagagat cttacaacata attcttcagg gacgctggta 1320
tatgaaaacc cgacaaaatt tagtagccaa tatctaacta cgaattccc tggggaaaac 1380
tcagaaagac caacttttac cgattatagt cacagactaa cttgtctcac acgtattggg 1440
gctggaaatt atggattggt tctatgcgcc ggctggacat ctagttagtg tgagcgtgac 1500
aacagactcc agccggacaa aataacacaa taccggctg ttaaaggatt caacctgat 1560
ggttttacag tagtaaaagg tactgggttt acagggggaa attggttgag atctagtcgt 1620
gttacaggtg gctttagact aaatgtttat tcaccgtctg tccaaactta tcgcatgcgt 1680
atacgttatg cttctcact gggaaattct actctaggta tatcttctac tgatgctggt 1740
attagtttca caagtttttc acttccctca acaataggat caatgccatc aactgtacca 1800
tacgaagctt tttagagtct agatatacct atcactgtta cagtagctag tcaagaaaat 1860
tataatttta ttctcgatat tctaatcca tcagtcggag cagtatacat tgacagaatt 1920
gaatcgcttc cagttgggtc tctgttattt gaatatgaaa ccaagcatga gctagaaaaa 1980
gctaagaaag cggtgaacta a 2001

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SEQ ID NO: 28      moltype = AA length = 666
FEATURE           Location/Qualifiers
REGION            1..666
                   note = The amino acid sequence of the TIC7383_6 protein,
                   comprising amino acids 15 through 679 of TIC7383.
source            1..666
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 28
MQIIEPSSDS LLYSHNNYPY ATPDNTVLEG RNYKEWLKNC TDNYTDALQG PEATAISKGA 60
VSAAISISTK VLGLLGVPFA AQIGQLWTFI LNALWPSDNT QWEEFMHRVE ELINQRIADY 120
ARNKALAELE GLGNLDLYI EALDWKRNK TSQQAQDRVI DRFRIADGLF EGYMPSFRVS 180
GYEVPLLTIV AAAANLHLLL LRDCSIYGIQ WGFSQTNVNE NYNRQIRHTA EYANHCTTWY 240
QTGLERLRGT NAFSWINYNR FRREMTLTVL DVCSLFSNYD YRTYPTVRA ELTREIYTD 300
IGFQNSPLPG VVPNWYDYAR SFAEINIENI RAPRTVTWLN STTIYTGRLN GYNNSNYYWA 360
GPRQNFSETN SGSSFNGPDL GDLPNRYRIE TLDMVNRDIY SIYSRVVSQS WPIGNVKLFG 420
VSSSTLSFRD LNNNSSGTLV YENPTNFSSQ YLTTEFPGEN SERPTFTDYS HRLTCLTRIG 480
AGNYGLVLCA GWTSSSVVERD NRLQPDKITQ YPAVKGFNLD GFTVVKGTFG TGNWLRSSR 540
VTGSFRLNVY SPSVQTYRMR IRYASPLGNS TLGISSTDAG ISFTSFPLPS TIGSMPSTVP 600
YEAFRVLDIP ITVTVASQRN YNFIFDILNP SVGAVYIDRI EFVPVGSSVF EYETKHELEK 660
AKKAVN 666

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SEQ ID NO: 29      moltype = DNA length = 3840
FEATURE           Location/Qualifiers
misc_feature       1..3840
                   note = A nucleic acid encoding a TIC7386 pesticidal protein
                   sequence obtained from Brevibacillus laterosporus species
                   DSC007651.
source             1..3840
                   mol_type = unassigned DNA
                   organism = Brevibacillus laterosporus

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SEQUENCE: 29
atgaatcaga atcaaaataa aaatgaactg caaatcatag aaccttcaag cgattctttt 60

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ctttatagtc acaacaatta tccgtatgct actgatccag atacaatatt aaaaggtagg 120
aattataaag agtgggctaaa tatgtgtaca gatacagacg attcacgaag tcccgagcgt 180
gcttctactg caagatcagc tatttctgtt gcgattacta taagcaccac aattccttggc 240
ttactaggty ttcggtttgc atctcaaatt ggagcatttt ataacttcgt attgaatacg 300
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SEQ ID NO: 30      moltype = AA length = 1279
FEATURE           Location/Qualifiers
REGION            1..1279
                  note = The amino acid sequence of the TIC7386 protein.
source            1..1279
                  mol_type = protein
                  organism = Brevibacillus laterosporus

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NERIADYARD KALAEALTGLG NNLNLYREAF EDWRRNPTSQ EAKTRVIDRF RILDGLFETY 180
MPSFAVRNFE VQLLTVYASA ANIHLFLLRD SSIYGLDWGL SQTNVNENYN RQIRHAETYA 240
NHCTTWYQTG LQRLQGTNAT SWVAYNRFR EMTLTVLDIC SLFSNYDYRS YPTEVRGELT 300
REIYTDPVGR NWQNVAPSPA EIENLTIRAP RTVTWLNSTR IFTGTLTWS GSNRYWAAHM 360
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NGTFRLTITS	FSSQSYRIRL	RYATSVGNTS	LVISSSDAGI	SSTTIPLTST	ITSLPQTPVY	600
QAFRVVDLPI	TFTTPTTQRN	YTFDFRLQNP	SNANVFIDRF	EFVPIGGSLS	EYETKHQLEK	660
ARKAVNDLFT	NESKNVLKDD	TTDYDIDQAA	NLVECVSDEC	ANAKMILLDE	VKYAKQLSEA	720
RNLLLNNGNFE	YQDRDGENPW	KTSPNVTIQE	NNPIFKGRYL	SMSGANNIEA	TNEIFPTYVY	780
QKIDESKLKP	YTRYKVRGEV	GNSKDLLELV	TRYDEEVDAI	LNVPNDIPHA	PPFPCGEFDR	840
CKPHSYPPIN	PECHHDVINN	IEISSPCQHN	KMVDNADISY	RHSRLSKKHG	ICHESHHEFF	900
HIDTGKIDLA	GNLGIWVIFK	ICSTDGYATL	DNLEVEIEGP	LGAESLERVK	RREKKWKHHM	960
EHKCKETKHA	YHAAKQAVVA	LFTNSKYDRL	KFETTISNIL	FADYLVQSIP	VYVYNKWLPGV	1020
PGMNYDIYTE	LKNLFTGAFN	LYDQRNIIGN	GDFNRGLMHV	HATPHARVEQ	IIDNRSVLVL	1080
PNYATNVSQE	VCLEHNRGYV	LRVTAKKEGP	GIGYVTFSDC	ANHIEKLFTT	SCDYGTNVVP	1140
YEQSNYPTDG	VPYQGHGCNI	DGVPEYQSGY	RTDGVPEYQS	GHRTDGVPEY	QSGYRTDGEP	1200
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SEQ ID NO: 31 moltype = DNA length = 3810
FEATURE Location/Qualifiers
misc_feature 1..3810
 note = A nucleic acid encoding a TIC7388 pesticidal protein
 sequence obtained from *Brevibacillus laterosporus* species
 DSC007962.
source 1..3810
 mol_type = unassigned DNA
 organism = *Brevibacillus laterosporus*

SEQUENCE: 31

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SEQ ID NO: 32 moltype = AA length = 1269
 FEATURE Location/Qualifiers
 REGION 1..1269
 note = The amino acid sequence of the TIC7388 protein.
 source 1..1269
 mol_type = protein
 organism = Brevibacillus laterosporus

SEQUENCE: 32

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LINQRIADYA	RNKALAEITG	LGNLDLYIE	ALDDWKRNP	SQEAKTVID	RFRIVDGLFE	180
AYIPSAVSG	YVQQLTVYA	AAANLHLLLL	RDSTIYGIDW	GLSQTNVNDN	YNRQIRLTAT	240
YANHCTTWYQ	TGLERLRGSN	ASSWVTYNRF	REMTLTVLVD	ICSLFSNYDY	RSYPAEVRGE	300
ITREIYTDVP	GVGVWDSAPS	FGEIENLAIR	APRTVTWLNS	TRIFTGRLQG	WSGTNNYWAA	360
HMQNFSETNS	GNIQFEGPLY	GSTVGTIHR	DDYDMGNRDI	YTITSQAVLG	LWATGQVRVG	420
VASARFTLRN	LFNNLTQVLV	YENPISSTFG	SSTLTHELGS	ENSDRPTSSD	YSHRLTSITG	480
FRAGANGTVP	VFGWTSATVD	RNNIERNKI	TQFPGVKSHT	LNNCQVVRGT	GFTGGDWLRP	540
NNNGTFRLLI	TSFSSQSYRI	RLRYATSVGN	TSLVISSSDA	GISSTTIPLT	STITSPLPQT	600
PYQAFRVVDL	PITFTPTTQ	RNYTFDFRLQ	NPSNANVFID	RFEFVPIGGS	LSEYETKHQL	660
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EARNLLLNNG	FEYQDRDGEN	PWKTSNPVTI	QENNPIFKGR	YLSMSGANNI	EATNEIFPTY	780
VYQKIDESKL	KPYTRYKVRG	FVGNSKDLLE	LVTRYDEEVD	AILNVPNDIP	HAPPPFCGEF	840
DRYKPHSYPP	INPECHHDI	NNIEISSPCQ	HNKMVDNADI	SYRHSRLSKK	HGICHESHHP	900
EPHIDTGKID	LVENLGIWVV	FKICSTDGYA	TLDNLEVIEE	GPLGAESLER	VKRREKKWKH	960
HMEHKCSETK	HAYHAAKQAV	VALFTNSKYD	RLKFETTISN	ILFADYLVQS	IPYVYNKWL	1020
GVPGMNYDIY	TELKNLFTGA	FNLYDQQRNI	KNGDFNRLGM	HWHATPHARV	EQIIDNRSVL	1080
VLPNYAANVS	QEVCELEHNRG	YVLRVTAKKE	GPYGIVYVTS	DCANHIEKLT	FTSCDYGTNV	1140
VPYEQSNYPT	DGVPGYQHG	NIDGVPEYEQS	GYRTDGVPEY	QSGHRTDGV	YEQSGYRTDG	1200
VPCEQHGCHT	DGLPHIQHGC	RTDRSRDELL	GVYTKTIDVF	PNTDKVRIDI	GETEGTFKVE	1260
SVELICMEE						1269

SEQ ID NO: 33 moltype = DNA length = 3960
 FEATURE Location/Qualifiers
 misc_feature 1..3960
 note = A nucleic acid encoding a TIC7389 pesticidal protein
 sequence obtained from Brevibacillus laterosporus species
 DSC006878.
 source 1..3960
 mol_type = unassigned DNA
 organism = Brevibacillus laterosporus

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SEQ ID NO: 34      moltype = AA length = 1319
FEATURE           Location/Qualifiers
REGION            1..1319
                  note = The amino acid sequence of the TIC7389 protein.
source            1..1319
                  mol_type = protein
                  organism = Brevibacillus laterosporus

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SEQUENCE: 34
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EELINQRIAD YARSKALAEI TGLGNLNDLY IEALEDWKRN PTSQQAQDRV KDRFRIADGL 180
FEAYMPSFRV SGYEVPLLTV YAAAANLHLL LLRDCSIYGI QWGFSTQNVN ENYNRQIRHT 240
AEYANHCTTW YQTGLERLRG TNASSWVPYN RFRREMTLTV LDICSLFSNY DYRSYPAEVR 300
AELTREIYTD PVVSTSLWVN NAPSFGIEIN LAIRAPRTVT WLNSTRISTG TLQWSGSNR 360
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IVLGVASARF TLRNLNNNLT QALVYENPIS SSFNRSTLTR ELPGENSDRP TSSDYSHRLT 480
SITAFRAGSN GTIPVFGWTS ISVNRDNILE RNKITQFPGV KSHTLNNCQV VRGTGFTGGD 540
WLRPNNGSFG RLITTSFSSQ SYRIRLRYAS AANTSRLISS SAAGISSTTV PLASTITSLP 600
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WLPDVPGMNY YITELKNLF TGAFLYDQR NIIKNGDFNR GLMHWHATPH ARVEQIDNRS 1080
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DGVPEYQHG CHTDGVPEYQ NYPTDGVPEY QHGCHTDGVP YEQHGCHTDG VPYKQHGCHT 1260
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SEQ ID NO: 35      moltype = DNA length = 3780
FEATURE           Location/Qualifiers
misc_feature       1..3780
                  note = A synthetic coding sequence,
                  CR-BREla.TIC7040.nno_Mc:1 encoding a TIC7040 pesticidal
                  protein used for expression in plant cells.
source            1..3780

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mol_type = other DNA
organism = synthetic construct

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SEQ ID NO: 36      moltype = DNA   length = 3888
FEATURE            Location/Qualifiers
misc_feature        1..3888
                    note = A synthetic coding sequence,
                    CR-BREla.TIC7040_10.nno_Mc:1 encoding a TIC7040HT
                    pesticidal protein used for expression in plant cells.
source              1..3888
                    mol_type = other DNA
                    organism = synthetic construct

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SEQ ID NO: 37      moltype = DNA length = 3888
FEATURE           Location/Qualifiers
misc_feature       1..3888
                   note = A synthetic coding sequence,
                   CR-BRE1a.TIC7040_10.nno_Mc:3 encoding a TIC7040HT
                   pesticidal protein used for expression in plant cells.
source            1..3888
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 37
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SEQ ID NO: 38      moltype = DNA length = 3888
FEATURE            Location/Qualifiers
misc_feature       1..3888
                    note = A synthetic coding sequence,
                    CR-BRE1a.TIC7040_10.nno_Mc:4 encoding a TIC7040HT
                    pesticidal protein used for expression in plant cells.
source             1..3888
                    mol_type = other DNA
                    organism = synthetic construct

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SEQ ID NO: 39      moltype = DNA length = 3888
FEATURE           Location/Qualifiers
misc_feature      1..3888
                  note = A synthetic coding sequence,
                  CR-BRE1a.TIC7040_10.nno_Mc:5 encoding a TIC7040HT
                  pesticidal protein used for expression in plant cells.
source            1..3888
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 39
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SEQ ID NO: 40      moltype = DNA length = 3888
FEATURE           Location/Qualifiers
misc_feature       1..3888
                   note = A synthetic coding sequence,
                   CR-BRE1a.TIC7040_10.nno.Mc:6 encoding a TIC7040HT
                   pesticidal protein used for expression in plant cells.
source             1..3888
                   mol_type = other DNA
                   organism = synthetic construct

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accggggcct tcaacctgta cgatcagagg aacattatca agaaccggca cttcaacagg 3180
ggcctcatgc actggcacgc gacgccccac gcccgagtgg agcagatcga caaccgttct 3240
gtgctggttc tacccaacta gcgggccaaac gtcagccaaag aggtgtgctt ggagcacaac 3300
cggggttacg tgctgcgagt gacggccaag aaggaaggtc cagggatcgg ctatgtgacg 3360
ttctcggaat cgcgccaaca catcgagaag ctacggttca cgtcctgcga ctacggcacc 3420
aacgaggtta cttacgagca aagcaattac catacggagc gcgttcctga tgaacagtca 3480
aactaccgca cggacggcgt gccctacgag cagcacgggt gccacacga cggcgtgccc 3540
tacagcgaga cgaactaccc caccgacggc gtgcccgtac agcagcacgg ctgccacacc 3600
gacggagtgc cgtacgagca gcacggctgc cacaccgacg gcgtgccgta cgagcagcac 3660
gggtgccata ccgacggcgt gccgtacaag cagcacggct gccgcacga cggggtgctg 3720
tacaagcagc acggttgccg cacagatcgg agccgtgacg agcagctcga ttacgtgacg 3780
aagaccatcg acgttttccc ggacaccgac aaggttcgta ttgacatcgg cgaacgggaa 3840
gggacattca agtgggagtc cgtcgagcta atcttcatgg agggatag 3888

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SEQ ID NO: 41      moltype = DNA length = 3888
FEATURE            Location/Qualifiers
misc_feature       1..3888
                    note = A synthetic coding sequence,
                    CR-BRE1a.TIC7040_10.nno_Mc:7 encoding a TIC7040HT
                    pesticidal protein used for expression in plant cells.
source             1..3888
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 41
atgaaccaga accagaacca gaacaagaac gagcttcaga tcatcgagcc gtccctccgac 60
agcctgtctc acagccacaa caattaccct tatgccaccg acccgaacac cgtgctggag 120
gggcgcgaact acaaggagtg gctgaacaag tgcaccgaca actacacgga cgcgctccag 180
ggccccgagg ccaccgcgat cagcaaaagg gccgtgtcgg cggcgatctc catcagcagc 240
aaagtgtcga gccgtgtcgg ggtgccattc gcggcccaga tcgggcagct ctggacgttc 300
atcctcaacg cctgtgtggc gtccgacaac acccagtggt aggagttcat gcggcacgtc 360
gaggaaactta ttaaccagag gatcgcggtc tacgcgcgct cgaaggccct cgcggagcta 420
accggggtcg gcaacaacct ggacctctac attgaggccc ttgaggactg gaagcgtaac 480
ccaacgtcgc aacaggcgaa ggaccgcgtg aaggaccggt tccgtatcgc cgacggactg 540
ttcgaggcgt acatgccgag ctcccgctgc tcaggctacg aagtgccttt gctcaccgtc 600
tatcgggcgg cggcgaacct ccacctcctt ctgctccggg actgctcgat ctacggtatc 660
caatggggct tctcccagag aaacgtgaat gagaactaca atcgtcagat ccggcacacc 720
gcagagtacg cgaaccactg cacaacctgg tatcagactg ggttggagcg gctccgaggg 780
acgaatgcct catcctgggt gccgtacaac cgtttccggc gtgagatgac attaacggtc 840
cttgacatct gttccctgtt cagcaattac gattaccgca gttaccgggc ggaggtccgg 900
cggaggttga cgcgcgaaat ctacacagac ccagtggta gcacttccct ctggatgaac 960
aacgtccga gcttcggcga gatcgagAAC ctggccatcc gggccccgcg cagggtgacg 1020

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tggctcaact	caaccccgat	ctccacagge	acgctacagg	ggtgggtctgg	tagtaatcgc	1080
tactgggccc	cgacacttgc	gaacttcagc	gagacgaaca	gcgggaacat	cggttctgac	1140
ggcccgcctc	acgggtccac	cgtgggcacc	ataatcaggg	atgataacta	cgagatggtc	1200
aaccgggaca	tctacaccat	caccagcgag	gccgtggccc	ccctgtggcc	gacagggcag	1260
atagtgctcg	gcgtggcgag	cgcccgtttt	accctgcgga	acctcaacaa	taacctgacc	1320
caggctctag	tgtacgagaa	cccaatcagc	tccagtttca	accgaagtac	cttgacgagg	1380
gaactgccgg	gcgagaactc	tgaccgtccc	acgtcgagcg	actacagcca	cagacttacc	1440
tctattacgg	ccttcocgtg	gggcagcaac	ggtacgatcc	cggtcttcgg	gtggaccagc	1500
ataagcgtga	atcgggacaa	catcctggaa	cgcaacaaga	tcacccagtt	tcttggcgtg	1560
aagtacaca	cactgaacaa	ctgccaaagt	gtgaggggca	ccggcttcac	gggcggggac	1620
tggctccgcc	cgaacaacaa	tggctccttt	cgctcaacaa	tcacctctt	ctcgtcgag	1680
agctaccgga	tcaggctccg	gtacgcttcc	gcgcgaata	cttcactacg	catcagctcc	1740
tcggccgccc	gaatctctcc	tactactgtg	ccattgactt	cgacgatcac	cagcctcccg	1800
cagacagcgg	tgccctacga	agcgttccgc	gtcatcgacc	tcctatcac	tttccagacc	1860
gcgacgcaga	gtaactacac	attcgacttc	gtactccaga	accgttccaa	cgcaacgtc	1920
ttcattgac	ggtttgagtt	cgtgccaaata	ggcgggtcac	tcagcgagta	cgagacgaag	1980
caccaattgg	aaaaggctcg	gaaagctgtg	aatgatctgt	ttacaaacga	aagcaagaat	2040
gtgctcaaga	agtacactac	tgattacgac	atcgaccaag	ccgccaacct	cgtggagtg	2100
gtctccgacg	agtgccgcaa	cgtcaaagat	atcctgctcg	atgaggtgaa	gtacgccaac	2160
caactaagcg	aggcacgcga	tctctctctc	aacggcaact	tcgagtacca	ggaccgggac	2220
ggggagaaac	cctggaagac	ctcgccgaac	gtgaccatcc	aggagaacaa	ccccatcttc	2280
aaggggagat	acttgagcat	gagcggggct	aactacatcg	aggccaccaa	cgacaccttc	2340
ccgacctacg	cttaccagaa	aatcgacgag	gccaaattga	agccctacac	gcgatacaag	2400
gtgcgcggct	tcgtcgggaa	cagcaaaagc	cttgagctgc	tcacaaagcg	gtacgacgag	2460
gaggtggatg	cgatcctcaa	gctcccgaac	gacatcccgc	acgcgcccac	cccgtttctg	2520
ggcggtctcg	accgttgcaa	gccccactcc	tacattccga	tgaacccgga	atgtcatcac	2580
gatgtcatta	acaacattga	aatcagcagc	ccgtgccagc	ataacaagat	gctggacaac	2640
gcggacatct	catcccgcga	cagtgaactt	ggcaagaaac	gcgggatctg	ccacgagtc	2700
caccacttcg	agttccacat	cgacacgggg	aagatcgact	tgaacgagaa	tctcgggac	2760
tgggtgatct	tcaagatttg	cagtacggac	ggctacgcca	cgctcgacaa	cctggagggt	2820
atcgaggagg	ggcgcgtcgg	agccgagtc	ctggagcgcg	tcaaacgccg	cgagaagaag	2880
tggagcacc	acatggagca	caagtgcagc	gagacgaagc	tcgctacca	cgagctaaa	2940
caagctctgg	tgggctgtgt	caccaacacg	aagtacgacc	ggctcaagtt	cgagaccacg	3000
atcagcaaca	tctgttctgc	ggactacctg	gtgcagtcga	taccgtatgt	gtacaacaaa	3060
tggctcccgg	acgtgccggg	catgaactac	gacatctaca	ccgaacttaa	gaacctattc	3120
acaggagcat	tcaacctgtg	cgaccagcga	aacattatca	agaacggcga	cttcaatagg	3180
ggcctgatgc	actggcagc	cactccccac	gcaagggtcg	agcaaataga	caaccggagc	3240
gtgctcgttc	taccgaatta	cgccgccaat	gtaagccagg	aggtctgcct	tgagcacaac	3300
cgtgtgtacg	tcttgccgtg	gacggccaag	aaggaagggc	cagggatcgg	ctacgtgacg	3360
ttctccgact	gcgccaacaa	catcgagaag	ctcactttta	cctcctgcga	ctacggcaca	3420
aacgaggtca	cttacgagca	gtcgaactac	cacaccgatg	gcgttccgta	cgaacaaagc	3480
aactacccca	ccgacggagt	gcctacgag	cagcacggct	gccacacaga	cggcgtgccg	3540
tacgagcagt	caaacatccc	gacggacggc	gtgccctacg	agcagcacgg	gtgccacacg	3600
gacggcgtgc	cttacgagca	gcacgggtgc	cacacggacg	gggtgcccta	cgagcagcac	3660
ggctgccaca	cggacggcgt	gocgtacaag	cagcacggct	gcgcacagga	tggcgtgctg	3720
tacaagcagc	acggatgtag	aacagatcgg	tcccgcgacg	agcaactgga	ctatgttacc	3780
aagacaatag	acgtctctcc	ggacaccgac	aagggtcgga	ttgacatcgg	ggagactgag	3840
ggcacgttca	aggtcgagtc	cgtggagctg	atcttcatgg	aggagtga		3888

SEQ ID NO: 42 moltype = DNA length = 1923
 FEATURE Location/Qualifiers
 misc_feature 1..1923
 note = A synthetic coding sequence used for expression in
 plant cells, CR-BRE1a.TIC7040_1.nno.Mc:1 which encodes a
 protein having an N-terminal and C-terminal truncation
 relative to the TIC7040HT protein.
 source 1..1923
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 42

atgcagatca	tcgagcctag	ctccgactcg	ctcctgtaca	gccacaacaa	ctaccggtac	60
gcgacggacc	cgaataactgt	gctggagggc	cgcaactaca	aggagtggct	caacaagtgc	120
acggacaact	acacggagcg	cctccaaggc	ccggaggcca	cggcgattag	caagggcgcg	180
gtctcggcgg	caatttcgat	ctcgactaag	gtgctctcgc	tgctcggcgt	gccgttttga	240
gcgcagatcg	gccagctctg	gaccttcata	ctcaacgcgc	tctggccgag	cgacaacacc	300
cagtgggagg	agttcatgcg	tcacgtcgag	gaactgataa	accagcgcat	cgcgactac	360
gccaggagca	aggccctcgc	ggagctgacc	gggctcgcca	ataacctgga	cttgtacatc	420
gaggcccttg	aggactggaa	acgaaccccg	accagccaac	aagccaagga	ccgggtgga	480
gaccgggttc	ggatcgccga	cggcctgttc	gaggcgta	tgcttctctt	ccgggtctcg	540
ggctatgaag	tgccgctcct	cacggtgtac	gccgcagcgg	ccaacctgca	tctcctgctg	600
ctaagggatt	gctcgatcta	cggcatccag	tggggtttct	cccagaccaa	cgtaaacgag	660
aactacaacc	gtcagatccg	gcacacggct	gagtagcga	accactgcac	gacgtgggtat	720
cagacgggac	tggagcggct	ccgtggcacc	aacgcatctt	cgtgggttcc	gtacaaccgt	780
ttccgcagag	agatgactct	taccgtcctg	gacatctgct	ccctcttctc	caactatgac	840
tacagaagct	accggcgcca	agtgcggggc	gagctaacc	gcgaaatcta	cacggaccgg	900
gtggtctcca	catcgctgtg	gatgaacaac	gcaccatcgt	tcggtgagat	cgagaacctg	960
gcaatccgcg	ctccgcgcac	ggtgacgtgg	ttgaactcaa	ccaggatctc	taccggcacc	1020
ctccagggct	ggtagggcag	taaccgatac	tgggcggcac	acatgcagaa	cttcagcgag	1080
accaactccg	ggaacattgg	attcgacggc	ccactgtacg	gctctacctg	gggcactatc	1140

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atcaggagc acaactacga gatggtcaac agggacatct acacgattac gtccgaagcg 1200
gtggccgctc tttggccac agggcagata gtgctggcg tggcctcgc acgtttcaca 1260
ctccgtaatc tcaacaacaa tctgacccag gccctgggtt acgagaaccc gatttctccc 1320
agcttcaacc gcagcacgct cacacgggaa ctacctggcg agaacagtga tcgcccacag 1380
tccagtgtac acagccatag gctgacatca atcacggcgt tccgggcgg gagtaacggc 1440
accatcccggt tcttcggctg gacctccatc tcggtcaacc gggataacat cctggagcgc 1500
aacaagataa cccagtctcc tggcgtaac agccacactc tcaacaactg ccaagtcgtc 1560
agagggaccg gcttcaactg cggcgactgg ctcagaccca acaacaacgg gtctttccgc 1620
ctgacgatca ccagctttctc ctgcagctcc taccggataa ggctccggta cgcctctgcg 1680
gccaacacat cctccggatg ttcttctccc gctgcgggca taagctctac aaccgtgccc 1740
ctgacctcca ctattactag ccttccgcaa accgctgtac cctacgaggc gttcccgctc 1800
atcgacctcc cgatcacctt cactaccgct acgcagagta actacacctt cgactttgtc 1860
ctccagaacc cgtccaatgc taacgtcttc atcgaccggt tcgagttcgt ccctattgga 1920
tga 1923

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SEQ ID NO: 43      moltype = AA length = 640
FEATURE            Location/Qualifiers
REGION             1..640
                   note = The amino acid sequence of the
                   CR-BRE1a.TIC7040_1.nno_Mc:1 protein, comprising amino
                   acids 15 through 651 of TIC7040HT.
source             1..640
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 43
MQIIEPSSDS LLYSHNNYPY ATPDNTVLEG RNYKEWLNKC TDNYTDALQG PEATAISKGA 60
VSAAISISTK VLSLLGVFFA AQIGQLWTFI LNALWPSDNT QWEEFMHRVE ELINQRIADY 120
ARSKALAELE GLGNLNDLYI EALDWRKRP TSQAKDRVK DRFRIADGLF EAYMPSFRVS 180
GYEVLPLTVY AAAANLHLLL LRDCSIYGIQ WGFSTQNVNE NYNRQIRHTA EYANHCTTWY 240
QTGLERLRGT NASSWVPYNR FRREMTLTVL DICSLFSNYD YRSYPAEVRA ELTREIYTD 300
VYSTSLWMNN APSFGEIENL AIRAPRTVTW LNSTRISTGT LQWSGSNRY WAAHMQNFSE 360
TNSGNIGFDG PLYGSTVGTI IRDDNYEMVN RDIYITITSEA VAALWPTGQI VLGVASARFT 420
LRNLNNNLQ ALVYENPISS SFNRSTLTRE LPGENSDRPT SSDYSHRLTS ITAFRAGSNG 480
TIPVFGWTSI SVNRDNLER NKITQFPGVK SHTLNNCQVQ RGTGFTGGDW LRPNNNGSFR 540
LTITSFSSQS YRIRLRYASA ANTSRLRISS AAGISSTTVP LTSTITSLPQ TAVPYEAFRV 600
IDLPTITFTA TQSNYTFDFV LQNPNSNANVF IDRFEFVPIG 640

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SEQ ID NO: 44      moltype = DNA length = 1983
FEATURE            Location/Qualifiers
misc_feature        1..1983
                   note = A synthetic coding sequence used for expression in
                   plant cells, CR-BRE1a.TIC7040_2.nno_Mc:1 encoding a
                   TIC7040_6 (SEQ ID NO:10) pesticidal protein sequence which
                   comprises an N-terminal and C-terminal truncation relative
                   to the TIC7040HT protein.
source             1..1983
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 44
atgcagatca tcgagcctag ctcgactcg ctcctgtaca gccacaacaa ctacccgtag 60
gcgacggacc cgaatactgt gctggagggc cgcaactaca aggagtggct caacaagtgc 120
acggacaact acacggacgc cctccaaggc cgggagggcca cggcgattag caaggcgcv 180
gtctcggcgg caatttcgat ctcgactaag gtgctctcgc tgcctggcgt gccgtttgca 240
gcgcagatcg gccagctcgt gacctccata ctcaacgcgc tctggccgag cgacaacacc 300
cagtgggagg agttcatgct tcacgtcgag gaactgataa accagcgcat cgcggactac 360
gccaggagca agtgcctcgc ggagctgacc gggctcggca ataacctgga cttgtacatc 420
gaggcccttc aggaactggaa acgaaacccg accagccaac aagccaagga ccgggtgaag 480
gaccgggttc ggatcgccga cggcctgttc gaggcgtaaa tgccttcctt ccgggtctcg 540
ggctatgaag tgcgctcct caccgtgtac gccgcagcgg ccaacctgca tctcctgctg 600
ctaagggatt gctcgatcta cggcatccag tggggtttct cccagaccaa cgtaaacgag 660
aactacaacc gtcagatccg gcacacggct gactacgcaa accactgcac gacgtgggat 720
cagaccggac tggagcggct ccgtggcacc aacgcactct cgtgggttcc gtacaaccgt 780
ttccgcagag agatgactct tacctgcctg gacatctgct cctctctctc caactatgac 840
tacagaagct acccggggga agtgcggggc gagtaacccc gcgaaatcta caccggaccg 900
gtggtctcca catcgctgtg gatgaacaac gcaccactgt tcggtgagat cgagaacctg 960
gcaatccgcg ctcgcgcac ggtgacgtgg ttgaactcaa ccaggatctc taccggcacc 1020
ctccagggct ggtcaggcag taaccgatac tgggcggcac acatgcagaa cttcagcgag 1080
accaactccg ggaacattgg attcgacggc ccaactgtac gctctaccgt gggcactatc 1140
atcaggagc acaactacga gatggtcaac agggacatct acacgattac gtccgaagcg 1200
gtggccgctc tttggccac agggcagata gtgctggcg tggcctcgc acgtttcaca 1260
ctccgtaatc tcaacaacaa tctgacccag gccctgggtt acgagaaccc gatttctccc 1320
agcttcaacc gcagcacgct cacacgggaa ctacctggcg agaacagtga tcgcccacag 1380
tccagtgtac acagccatag gctgacatca atcacggcgt tccgggcgg gagtaacggc 1440
accatcccggt tcttcggctg gacctccatc tcggtcaacc gggataacat cctggagcgc 1500
aacaagataa cccagtctcc tggcgtaac agccacactc tcaacaactg ccaagtcgtc 1560
agagggaccg gcttcaactg cggcgactgg ctcagaccca acaacaacgg gtctttccgc 1620
ctgacgatca ccagctttctc ctgcagctcc taccggataa ggctccggta cgcctctgcg 1680
gccaacacat cctccggatg ttcttctccc gctgcgggca taagctctac aaccgtgccc 1740
ctgacctcca ctattactag ccttccgcaa accgctgtac cctacgaggc gttcccgctc 1800

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atcgacctcc cgateacctt cactaccgct acgcagagta actacacctt cgactttgtc 1860
ctccagaacc cgtccaatgc taacgtcttc atcgaccggt tcgagttcgt cctatttga 1920
ggcagccttt cagagtacga gacgaagcac cagcttgaga aggcgaggaa agctgtcaac 1980
tga 1983

SEQ ID NO: 45      moltype = DNA length = 1752
FEATURE           Location/Qualifiers
misc_feature       1..1752
                   note = A synthetic coding sequence used for expression in
                   plant cells, CR-BRE1a.TIC7040_11.nno_Mc:1 which encodes a
                   protein having an N-terminal and C-terminal truncation
                   relative to the TIC7040HT protein.
source             1..1752
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 45
atgaactaca aggagtggct gaacaagtgt accgacaact acaccgacgc gctccagggc 60
cctgaggcca cgcgccatcg caagggcgcg gtgtcagccg cgatctccat cagcacaaaag 120
gtgctctcgt tactggggcgt gcccttcgcg gcgcagatcg gccagctctg gacctttatc 180
ctgaacgccc tatggcccag cgacaacacc cagtgggagg agtttatgcg ccacgtggag 240
gaactcatta accagcgaat tgctgactac gccaggagca aggccttggc cgagctgacg 300
ggtcttgga acaacctcga cctctacatc gaggcccttg aagactggaa gcggaacccg 360
accagccaac agccaaaaga ccgtgtgaag gaccgggttc ggatcgccga cgggctgttc 420
gaggcgta taacctcttt ccgcgtcagc ggctacgagg tgccactgtt gaccgtttac 480
gcggcagcgc ctaacctcca ctgtgtgctc ctgcgggatt gctccatcta cggtatccag 540
tggggcttca gccagacaaa cgtcaacgag aactacaacc gacagatccg gcacacggcc 600
gagtacgcaa accactgcac gacatggtat cagacagggc ttgaacgcct tcgtggcacc 660
aacgcctcgt catgggtgct gtacaaccgg ttccgcccag agatgacct caccgttctt 720
gacatctgct cactgttcag caactacgac taccggctcg accctgcgga ggttcgggcc 780
gaactgacgc gggagatcta cactgaccgc gtggtcagca cctcgttgtg gatgaacaac 840
gcgcccagct tcggtgagat cgagaacctc gccataagag caccacgcac cgtcacttgg 900
ctcaacagca ccggtatctc cacgggcacc ttacagggct ggagcggctc caaccgatac 960
tgggcagcgc acatgcagaa cttctccgag actaactccg gcaacatcgg cttcgacggc 1020
ccgctctacg gcagcaccgc cttggaccatc atccgggacg acaactatga gatggtaac 1080
aggacatct acacgattac ctgcggagcc gtggctgccc tgtggccaac cggtcaaata 1140
gtgttgggag tcgcctccgc ccgctttacg ctgcgtaacc tgaacaacaa ccttaccagg 1200
gcgctcgtgt acgagaaccc gatctccagc tccttcaata ggtccacctc cagcagggag 1260
ctgcggggtg agaacagcga caggccgacg tcctcggact acagtcaccg gctcaccctg 1320
atcacagcgt tccgagccgg gagcaacgga acgatcccat tcttcggatg gaccagcatc 1380
agcgtcaacc gcgacaacat cctggaaacgc aacaagatca ctcagttccc tggagttaag 1440
tctcacacgc tcaacaactg ccaagtgggt cgtggcacag gggtttaccg aggcgactgg 1500
cttcgccccc acaacaatgg gtcattccgg ctgaccatta ccagcttctc atcgcaatca 1560
taccgcatcc ggctacggta aacctcggcg gcgaacacta gccttcgcat ctctcttca 1620
gccgcgggga tcagctgcac aaccgtcccg ctgaccagca cgatcacgag cctgccacaa 1680
actgctgtgc cttacgaggc ttccagagtg atcgacctgc ccattacctt caccacggca 1740
acccaatcgt ga 1752

SEQ ID NO: 46      moltype = AA length = 583
FEATURE           Location/Qualifiers
REGION            1..583
                   note = The amino acid sequence of the
                   CR-BRE1a.TIC7040_11.nno_Mc:1 protein, comprising amino
                   acids 14 through 671 of TIC7040HT.
source             1..583
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 46
MNYKEWLKNC TDNYTDALQG PEATAISKGA VSAAISISTK VLSLLGVPPFA AQIGQLWTFI 60
LNALWPSDNT QWEEFMHRVE ELINQRIADY ARSKALAEIT GLGNLDLYI EALWDKRNPN 120
TSQQAKDRVK DRFRIADGLF EAYMPSFRVS GYEVPLLTIV AAAANLHLLL LRDCSIYGIQ 180
WGFSTQTNVNE NYNRQIRHTA EYANHCTTWY QTGLERLRGT NASSWVPYNR FRREMTLTVL 240
DICSLFSNYD YRSYPAEVRA ELTREIYTDI VVSTSLWMNN APSFGEIENL AIRAPRTVTW 300
LNSTRISTGT LQGWGSGSNRY WAAHMQNFSE TNSGNIGFDG PLYGSTVGTI IRDDNYEMVN 360
RDIYTITSEA VAALWPTGQI VLGVASARFT LRNLNNNLQ ALVYENPISS SFNRSTLTRE 420
LPGNSDRPT SSDYSHRLTS ITAFRAGSNG TIPVFGWTSI SVNRDNILR NKITQFPVGK 480
SHTLNNCQV RGTGFTGGDW LRPNNNGSFR LTITSFSSQS YRIRLRYASA ANTSLRISSS 540
AAGISSTTVP LTSTITSLPQ TAVPYEAFRV IDLPITFTTA TQS 583

SEQ ID NO: 47      moltype = DNA length = 1983
FEATURE           Location/Qualifiers
misc_feature       1..1983
                   note = A synthetic coding sequence used for expression in
                   plant cells, CR-BRE1a.TIC7040_12.nno_Mc:2 which encodes a
                   protein having a C-terminal truncation relative to the
                   TIC7040HT protein.
source             1..1983
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 47

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-continued

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atgaaccaga accagaatca gaacaagaac gagcttcaga taatcgagcc ctccctcgga 60
tccctcctct actcgcataa caactaccgg tacgctaccg acccgaaacac cgtgctggaa 120
ggccgcaact acaaggagtg gctgaacaag tgtaccgaca actacaccga cgcgctccag 180
ggccctgagg ccaccgccat cagcaagggc gcggtgtcag ccgcatctc catcagcaca 240
aagggtgctct cgttactggg cgtgcccttc gcggcgagca tcggccagct ctggacctt 300
atcctgaacg ccctatggcc cagcgacaac acccagtggg aggagtttat gcgccacgtg 360
gaggaactca ttaaccagcg aattgctgac tacgccagga gcaaggccct ggccgagctg 420
acgggtcttg gcaacaacct cgacctctac atcgaggccc ttgaagactg gaagcggaac 480
ccgaccagcc aacaggccaa agaccgtgtg aaggaccggt tccggatcgc cgacgggctg 540
ttcgaggcgt acatgccttc ttcccgctc agcggctacg aggtgccact gttgaccgtt 600
tacgcgccag ccgctaacct ccacttctg ctccctgccc attgctccat ctacggtatc 660
cagtgagggt tcagccagac aaacgtcaac gagaactaca accgacagat ccggcacacg 720
gccgagtacg caaaccactg cagcagatgg tatcagacag ggcttgaacg ccttcgtggc 780
accaacgcct cgtcatgggt gccgtacaac cggttccgcc gagagatgac cctcaccgtt 840
cttgacatct gctcactgtt cagcaactac gactaccggt cgtacctcgc ggaggttcgg 900
gccgaactga cgcgggagat ctacactgac ccggtgggtc gcacctcgtt gtggatgaac 960
aacgcgcccc gcttcgggtg gatcgagaac ctccgcaata gagcaccacg caccgtcact 1020
tggtctcaaca gcccccggat ctccacgggc acctacagg gctggagcgg ctccaaccga 1080
tactgggcag cgcacatgca gaacttctcc gagactaact ccggcaacat cggttcgac 1140
ggcccgctct ccgcaagcac cgtcgggacc atcatccggg acgacaacta tgagatggtc 1200
aacagggaca tctacacgat taactcggag gccgtgggtg ccctgtggcc aaccggtcaa 1260
atagtgttgg gactcgctc cgcgcgttt acgctgcgta acctgaacaa caaccttacc 1320
caggcgctcg tgtacgagaa cccgatctcc agctccttca atagggtccac cctcacgagg 1380
gagctgcccgg gtgagaacag cgcagggccg acgtcctcgg actacagtc cgggtccacc 1440
tcgatacagc cgttcggagc cgggagcaac ggaacgatcc cagtcttcgg atggaccagc 1500
atcagcgtca accgcgacaa catcctggaa cgcaacaaga tcaactcagt ccttgaggtt 1560
aagttctaca cgctcaacaa ctgccaaagt gtgcgtggca cagggtttac cggaggcgac 1620
tggttcgcc ccgaacaacaa tgggtcattc cggctgacca ttaccagctt ctcatcgcaa 1680
tcataccgca tccggctacg gtaagcctcg gcggcgaaac ctagccttcg catctcctct 1740
tcagccgccc ggatcagctc gacaaccgtc ccgctgacca gcacgatcac gaggcctgcc 1800
caaaactgctg tgccttacga ggttttcaga gtgatcgacc tgcccattac cttaccacg 1860
gcaacccaat gcaactacga ttctgacttc gtgttacaga atccgagcaa cgccaacgtc 1920
ttcatcgacc ggttcagatt cgtgcccatc ggccgggagcc tctccagta cgagaccaag 1980
tga 1983

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SEQ ID NO: 48      moltype = AA length = 660
FEATURE           Location/Qualifiers
REGION            1..660
                  note = The amino acid sequence of the
                  CR-BREla.TIC7040_12.nno Mc:2 protein, consisting of amino
                  acids 1 through 660 of TIC7040HT.
source            1..660
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 48
MNQNQNQNK ELQIEPSSD SLLYSHNNYP YATDPNTVLE GRNYKEWLNK CTDNYTDALQ 60
GPEATAISK AVSAAISIST KVLSELLGVFP AAQIGQLWTF ILNALWPSDN TQWEEFMRHV 120
EELINQRIAD YARSKALAE LGLGNLGLY IEALDWDKRN PTSQQAQDRV KDRFRIADGL 180
FEAYMPSFRV SGYEVPLLTV YAAAANLHL LRLDCSIYGI QWFGSQTNVN ENYNRQIRHT 240
AEYANHCTTW YQTGLERLRG TNASSWVPYN RFRREMTLTV LDICSLFSNY DYRSYPAEVR 300
AELTREIYTD PVVSTSLWMN NAPSFGIEEN LAIRAPRTVT WLNSTRISTG TLQWWSGSNR 360
YWAHMNQNS ETNSGNIGFD GPLYGSTVGT IIRDDNYEMV NRDIYTTITSE AVAALWPTGQ 420
IVLGVASARF TLRNLNNNLT QALVYENPIS SSFNRSTLTR ELPGENSDRP TSSDYSHRLT 480
SITAFRAGSN GTIPVFGWTS ISVNRDNILE RNKITQFPQV KSHTLNNQCV VRGTGFTGGD 540
WLRPNNGNSF RLTTTSFSSQ SYRIRLRYAS AANTSLRISS SAAGISSTTV PLTSTITSLP 600
QTAVPYEAPR VIDLPITFTT ATQSNYTFDF VLQNPNSNAV FIDRFEFVPI GGSLSLEYETK 660

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SEQ ID NO: 49      moltype = DNA length = 1875
FEATURE           Location/Qualifiers
misc_feature       1..1875
                  note = A synthetic coding sequence used for expression in
                  plant cells, CR-BREla.TIC7040_13.nno Mc:1 which encodes a
                  protein having a C-terminal truncation relative to the
                  TIC7040HT protein.
source            1..1875
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 49
atgaaccaga accagaatca gaacaagaac gagcttcaga taatcgagcc ctccctcgga 60
tccctcctct actcgcataa caactaccgg tacgctaccg acccgaaacac cgtgctggaa 120
ggccgcaact acaaggagtg gctgaacaag tgtaccgaca actacaccga cgcgctccag 180
ggccctgagg ccaccgccat cagcaagggc gcggtgtcag ccgcatctc catcagcaca 240
aagggtgctct cgttactggg cgtgcccttc gcggcgagca tcggccagct ctggacctt 300
atcctgaacg ccctatggcc cagcgacaac acccagtggg aggagtttat gcgccacgtg 360
gaggaactca ttaaccagcg aattgctgac tacgccagga gcaaggccct ggccgagctg 420
acgggtcttg gcaacaacct cgacctctac atcgaggccc ttgaagactg gaagcggaac 480
ccgaccagcc aacaggccaa agaccgtgtg aaggaccggt tccggatcgc cgacgggctg 540
ttcgaggcgt acatgccttc ttcccgctc agcggctacg aggtgccact gttgaccgtt 600
tacgcgccag ccgctaacct ccacttctg ctccctgccc attgctccat ctacggtatc 660

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cagtggggct tcagccagac aaacgtcaac gagaactaca accgacagat ccggcacacg 720
gccgagtagc caaaccactg cagcagatgg tatcagacag ggcttgaacg ccttcgtggc 780
accaacgcct cgtcatgggt gccgtacaac cggttccgcc gagagatgac cctcaccggt 840
cttgacatct gctcactggt cagcaactac gactaccggt cgtaccctgc ggagggtcgg 900
gccgaactga cgcggggagat ctacactgac ccggtgggtca gcacctcgtt gtggatgaac 960
aacgcgcccc gcttcggtga gatcgagaac ctgcgcataa gagcaccacg caccgtcact 1020
tggctcaaca gcacccggat ctccacgggc accttacagg gctggagcgg ctccaaccga 1080
tactggggcag cgcacatgca gaacttctcc gagactaact ccggcaacat cggtctcgac 1140
ggcccgctct acggcagcac cgtcgggacc atcatccggg acgacaacta tgagatgggtc 1200
aacagggaca tctacacgat tacctcggag gccgtggctg cctgtgggcc aaccgggtcaa 1260
atagtgttgg gagtcgcctc gccccgcttt acgctgcgta acctgaacaa caaccttacc 1320
caggcgctcg tgtacgagaa ccgatctctc agctccttca ataggtccac cctcaccagg 1380
gagctgcagg gtgagaacag cgacaggccg acgtcctcgg actacagtca ccggctcacc 1440
tcgatcacag cgttcggagc cgggagcaac cagttctcgg cagttctcgg atggaccagc 1500
atcagcgcta acccgacaaa catcctggaa cgcaacaaga tcaactcagt ccttgagggt 1560
aagttcaca cgctcaacaa ctgccaaagt gtgcgtggca cagggtttac cggaggcgac 1620
tggcttcgcc cgaacaacaa tgggtcattc cggctgacca ttaccagctt ctcactcgaa 1680
tcataccgca tccggctacg gtacgcctcg cggcgcaaca ctacgctcgg catctcctct 1740
tcagccggcg ggatcagctc gacaaccgtc ccgctgacca gcacgatcac gagcctgcca 1800
caaaactgctg tgcttacga ggcttcaga gtgatcgacc tgccattac cttcaccacg 1860
gcaacccaat cgtga                                     1875

SEQ ID NO: 50      moltype = AA length = 624
FEATURE           Location/Qualifiers
REGION            1..624
                  note = The amino acid sequence of the
                  CR-BRE1a.TIC7040_13.nno_Mc:1 protein, consisting of amino
                  acids 1 through 627 of TIC7040HT.
source            1..624
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 50
MNQNQNQNKN ELQIIEPSSD SLLYSHNNYP YATDPNTVLE GRNYKEWLNK CTDNYTDALQ 60
GPEATAISKG AVSAAISIST KVLSSLGVFP AAQIGQLWTF ILNALWPSDN TQWEEFMRHV 120
EELINQRIAD YARSKALAEI TGLGNLDLY IEALEDWKRN PTSQQAQDRV KDRFRIADGL 180
FEAYMPSFRV SGYEVPLLTV YAAAAANLHL LRRDCSIYGI QWGFSTQNVN ENYNRQIRHT 240
AEYANHCTTW YQTGLERLRG TNASSWVPYN RFRREMTLV LDICSLFSNY DYRSYPAEVR 300
AELTREIYTD PVVSTSLWMN NAPSFGIEIN LAIRAPRTVT WLNSTRISTG TLQWGSNSNR 360
YWAAHMQNFS ETNSGNIGFD GPLYGSTVGT IIRDDNYEMV NRDIYITSE AVAALWPTGQ 420
IVLGVASARF LRLNLNNLTL QALVYENPIS SSFNRSTLTR ELPGENSDRP TSSDYSRLT 480
SITAFRAGSN GTIPVFGWTS ISVNRDNILE RNKITQFPQV KSHTLNNCQV VRGTGFTGGD 540
WLRPNNGNSF RLTITSFSSQ SYRIRLRYAS AANTSLRISS SAAGISSTTV PLTSTITSLP 600
QTAVPYEAFR VIDLPITFTT ATQS                                     624

SEQ ID NO: 51      moltype = DNA length = 3801
FEATURE           Location/Qualifiers
misc_feature       1..3801
                  note = A synthetic coding sequence used for expression in
                  plant cells, CR-BRE1a.TIC7042.nno_Mc:1 which encodes a
                  TIC7042 protein (SEQ ID NO:12).
source            1..3801
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 51
atgaatcaga accagaacaa gaacgagatg cagatcatcg agccgtcgag cgactccttc 60
ctctattccc acaacaacta cccgtacgcc actgaccggg acacgacccg caaggggccg 120
aactacaagg agtgggtcaa catgtgcacc gatacggacg actcgcgctc gcccgaggcc 180
gcgagcacgg ctcgcgcacc gatctccgtc gccatcacca ttagcaccac catcctggga 240
ctccttggag ttcccttcgc gtcccagatc ggtgccttct acaacttcgt gctcaacacc 300
gtgtggcctc agggaaataa ccagtgaggg gagttcatga ggcacgtgga gaacctcatc 360
aacgagcgca ttgctgacta cgcgcgcgac aaagccttgg ccgagctgac cggcctgggc 420
aacaacctga atctctatcg ggaggcttcc gaggactggc ggcggaaccc gacctcccag 480
gaagccaaga ctcgggtgat cgaccgcttt cgtattctcg acggcctgtt cgagacttac 540
atgcctctct tcgcggtccg caactttgag gttcagctcc tgactgtgta cgctcggct 600
gcaaacatac acctgttctc gctgcgggac agctcgatct acggtttgga ctggggcctg 660
tcacagacca acgtgaacga gaactacaac cggcagatcc gccacgcgcg cactatgca 720
aaccactgca ccacttggtg tcagacaggg ctccagcgcc tccagggcac taacgcaacg 780
tcattgggtg catacaaccg gtttcggcgg gagatgacac tcacgctgtt ggacatctgc 840
tcgctgttct ccaactacga ctaccggtcg taccgactg aggtgaaggg cgagttagac 900
agggaaatct acacagaccc tgtcggcccg aactggcaga acgtcgcgcg ttcatctcgg 960
gagattgaga acctgacctc tcgcgctccg cgtaccgtca cttgggtcaa ctccactcgg 1020
atcttcacag ggacgtgac cggctgggtc ggggtccaac gctactgggc cgcccatg 1080
cagaacttca gcgagacaaa ctctggcaac atcggattcg atggccaca gtatgggtcc 1140
acggtcggga cgatcacacc gaccgacgac tacgacatgg tgaaccggga catctacac 1200
ataacgagcc aggcagtcgc gcgccctctg gccacggccc agacagtcct gggcgtggcg 1260
agcaccgcgt tcacactgag gaattcgaac aacaactcga cggaggcact cgtgtacgag 1320
aacgcatctc cgtcctcctt cgttagcagc actctcactc atgagctgac gggcgagaa 1380
agcgaccgcc caacgagcag cgattacagc caccgtcttt cctccatcac gggcttcgg 1440
gctggagcca acgggaccgt gcccggttcc ggttgagaca gcgcgacagt tgaccgtaac 1500

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aacatcatcg agcagaacaa gatcacccag tteccctggcg ttaagtcaca cactcttaac 1560
aactgtcaag tgggtgcgtgg caccgggttc accggcgcgcg actggcttcg cccaaacaac 1620
aatggcacct ttcggctgac cattaccagc ttctcgtccc agtcctaccg catcaggctt 1680
cgatacgcta ccagcgtcgg caaacctct ctgtcatct cgtcatccga cgcgggcate 1740
tccagtagca caatocctct gacatccacc atcactgcgc tcccgagac cgttcctgtac 1800
caggcggttc cgcgtcgtga tctcccgatc actttcacta ctccgaccac tcagcgcaat 1860
tacaccttcg atttcgcct ccagaacccg agcaacgcta acgtattcat cgaccgcttc 1920
gagttcgtac caatcggcgg ctactgtcc gagtacgaga cgaagcacca actggagaag 1980
gctcgcaagg cagtgaacga cctcttcaact aacgagagca agaactgct caagaaggac 2040
acgacggatt acgacattga ccaggcggcg aatctggtgg aatgcgtgtc tgacgagtgc 2100
gccaacgcca agatgatact cctcgatgag gttaagtacg ccaaacagct cagcgaggct 2160
cgaaacctcc ttcttaattgg caactttgaa taccaagacc gtgacggcga gaacctgtg 2220
aagacctcgc caaacgtcac catccaggag aacaatccga tcttcaaggg ccgctacctt 2280
agcatgtccg gacgaacaaa catcgaggcc accaacgaaa tcttcccaac ttatgtttat 2340
cagaagatcg acgagcaaaa gctaaaggcca tacacctcgt acaagggtcg cggctttgtg 2400
ggcaactcca aggatctcga actgctggtc actcgtttacg acgaggagggt ggacgcgata 2460
ctgaatgtcc ccaatgacat accgcacgcg ccaccgcat tctgcggcga gttcgaccgc 2520
tgcaagccac actcctaccc tctatttaac ccggagtgc accacagcgt catatacaac 2580
atcgaaatct ctagcccatg ccagcacaa acagatggtgg acagcgcgga catcagttac 2640
cggcatagcc ggatcagtaa gaagcacggc atctgtcatg aatcacacca cttcgagttc 2700
cacattgaca ccggcaagat agacctggtc gagaacctgg gcatctgggt gatcttcaag 2760
atttgacgca cggatggcta cgcgactctc gacaatctgg aggtcatcga ggagggccca 2820
ctcggagccg agtccctaga cgcgcttaag cggcgcgaga agaagtggaa gcaccacatg 2880
gagcacaaat gctcggagac gaagcacgca taccatgcag cgaagcaagc ggtggtggca 2940
ctcttcacca attccaagta cgataggctc aagttcgaaa ccacatctc gaacatctg 3000
ttcgcggact acctcgttca gtcaatcccg tacgtttaca ataagtggct gcccggtgtg 3060
ccgggcatga actacgacat ctacaccgaa ctaaaagaacc tggtaacggg cgcattcaac 3120
ctttacgata agcggaaatc catcaagaac ggcgatttca accgtggcct catgcaactg 3180
cacgctacgc cgacgcctcg ggtggagcag atcactgcga accggtcagt cctggctcctg 3240
cccaactatg cggccaatgt ttctcaggag gtgtgccttg agcacaatcg cgggtatgtt 3300
ctccgggtga cagccaagaa ggagggccct ggcatcgggt acgtcacctt ctccgactgc 3360
gccaaccaca tcgagaaact tacgtttacg agctgcgact atggcaccaa cgtggctccg 3420
tacgagcagt caaattaccc tacagacggc gtgcgctacg ggcagcagcg ctgcaacatc 3480
gacggcgctc cgtacgagca acgcggctac cgcaccgacg cgtcccgta cgagctgggc 3540
caccgcaccg accgcgctac gtacgagcag agcggatacc gcaccgacgg cgtgccctgc 3600
gagcagcagc gttgtcacac cgacgggctg ccgcacatcc agcaccgctg ccgcacggac 3660
cgctccagag acgagctggt gggctacgtc accaagacta ttgatgtgtt cccgaacacg 3720
gacaagggtg ggatcgacat tggcgagacc gagggaaacct tcaagggtga gtcggtggaa 3780
ctgatctgca tggaggagta g 3801

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SEQ ID NO: 52 moltype = DNA length = 1914
FEATURE Location/Qualifiers
misc_feature 1..1914
 note = A synthetic coding sequence used for expression in
 plant cells, CR-BRE1a.TIC7042_1.nno Mc:1 which encodes a
 protein having an N-terminal and C-terminal truncation
 relative to the TIC7042 protein.
source 1..1914
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 52

```

atgcagatca tcgagccgtc gagcgactcc ttctcttatt cccacaacaa ctaccgtagc 60
gccactgacc cggacacgat cctcaagggc cggaaactaca aggagtgggt caacatgtgc 120
accgatacgg acgactcgcg ctgcgccgag gcccgagaca cggctcgcag cgcgatctcc 180
gtcgccatca ccattagcac caccatcttg ggaactcctt gagttccctt cgcgtccag 240
atcgggtgct tctacaactt cgtgtcgaac accgtgtggc ctccaggaaa taaccagtgg 300
gaggagttca tgaggcagct ggagaacctc atcaacgagc gcattgtcta ctacgcgcg 360
gacaaagcct tggccgagct gaccggcctg ggcaacaacc tgaatctcta tcgggagggt 420
ttcgaggact gggggcgga cccgacctcc cagggaagcca agactcgggt gatcgaccgc 480
tttcgtattc tcgacggcct gttcgagact tacatgccct ccttcgcggt ccgcaacttt 540
gaggttcagc tctgactgt gtacgcctcg gctgcaaaaca tacacctgtt cctgctgcgg 600
gacagctcga tctacggttt ggactggggc ctgtcacaga ccaactgtga cgagaactac 660
aaccggcaga tccgccacgc cgccacctat gcaaacact gcaccacttg gtatcagaca 720
gggttcagc gccctcaggg cactaacgca acgtcatggg ttgcatacaa ccggtttcgg 780
cgggagatga cactcaccgt gttggacatc tgctcgtgt tctccaacta cgactaccgg 840
tcgtaccgca ctgaggtgaa gggcgagttg accagggaaa tctacacaga cctgtcggc 900
cggaactggc agaactcgc gcccttcatt cgggagattg agaactcgtc cattcgcgct 960
ccggtaccg tcacttgggt caactccact cggatcttca cagggacgtg gacggctgtg 1020
tccgggtcca accgtactg gcccgccac atgcagaact tcagcgagac aaactctggc 1080
aacatcggat tcgatggccc acagatggg tccacggctg ggacgataca ccggaccgac 1140
gactacgaca tggggaaccg gacatctac accataacga gccaggcagt gcgcgccctc 1200
tggcccaccg gccagacagt cctgggcgtg gcgagcacc ccttcacact gaggaatctg 1260
aacaacaact cgacggaggc actcgtgtac gagaacgcca tctcgtctc cttcgtttagc 1320
agcatcttca ctctagagct gccgggcgag aacagcgacc gcccaacgag cagcgattac 1380
agccaccgtc tttcctccat caccggcttc cgggctggag ccaacgggac cgtgccctgt 1440
ttcgggttga ccagcgcgac agttgaccgt aacaacatca tcgagcgaaa caagatcacc 1500
cagttccctg gcgttaagtc acacactctt aacaactgtc aagtggtggt tggcaccggg 1560
tccaccggcg cgcactgggt tcgcccaaac ccttccggt gaccattacc 1620
agcttctcgt cccagtccta ccgcatcagg cttcgatagc ctaccagcgt cggcaacacc 1680

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tctcttgtca tctcgtcatc cgaacgcggc atctccagta ccacaatccc tctgacatcc 1740
accatcaccgt cgctcccgca gaccgttccg taccaggcgt tccgcgtcgt cgatctcccg 1800
atcacttttca ctactccgac cactcagcgc aattacacct tcgatttccg cctccagaac 1860
ccgagcaacg ctaacgtatt catcgaccgc ttcgagttcg taccaatcgg ctag 1914

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SEQ ID NO: 53      moltype = AA length = 637
FEATURE           Location/Qualifiers
REGION            1..637
                  note = The amino acid sequence of the
                  CR-BREla.TIC7042_1.nno_Mc:1 protein, comprising amino
                  acids 11 through 646 of TIC70.
source            1..637
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 53
MQIEPSSDS FLYSHNNYPY ATDPDTILKG RNYKEWLNMC TDTDSRSPE AASTARSAIS 60
VAITISTTIL GLLGVPPASQ IGAFYNFVLN TVWPQGNQW EEFMRHVENL INERIADYAR 120
DKALAEITGL GNNNLNLYREA FEDWRRNPTS QEAKTRVIDR FRILDGLFET YMPSPFAVRNF 180
EVQLLTIVYAS AANIHLFLLR DSSIYGLDWG LSQTNVNNENY NRQIRHAATY ANHCTTWYQT 240
GLQRLQGTNA TSWVAYNRFR REMTLTVLDI CSLFSNYDYR SYPTEVKGEL TREIYTDVPG 300
RNWQNVAPSF AEIENLTIRA PRTVTWLNST RIFTGTLTGW SGSNRYWAAH MQNFSETNSG 360
NIGFDGPQYG STVGTIHRD DYDMVNRDIY TITSQAAVAL WPTGQTVLGV ASTRFTLRNL 420
NNNSTEALVY ENAISSSPVS STLTHELPGE NSDRPTSSDY SHRLSSITGF RAGANGTVPV 480
FGWTSATVDR NNIEQNKIT QFPGVKSHTL NNCQVVRGTG FTGGDWLRPN NNGTFRLLTIT 540
SFSSQSYRIR LRYATSVGNT SLVSSSDAG ISSTTIPLTS TITSLPQTVP YQAFRVVDLP 600
ITFTTPTTQR NYTFDFRLQN PSNANVFIDR FEFVPIG 637

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SEQ ID NO: 54      moltype = DNA length = 1974
FEATURE           Location/Qualifiers
misc_feature       1..1974
                  note = A synthetic coding sequence used for expression in
                  plant cells, CR-BREla.TIC7042_2.nno_Mc:1 which encodes a
                  protein having an N-terminal and C-terminal truncation
                  relative to the TIC7042 protein.
source            1..1974
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 54
atgcagatca tcgagccgtc gagcgactcc ttcctctatt cccacaacaa ctacccgtag 60
gccactgacc cggacacgat cctcaagggc cggaaactaca aggagtggct caacatgtgc 120
accgataccg acgactcgcg ctcgcccagag gccgcgagca cggtctgcag cgcgatctcc 180
gtcgccatca ccattagcac caccatcctg ggactccttg gagttccctt cgcgtcccag 240
atcgggtgct tctacaactt cgtgctcaac accgtgtggc ctcagggaata taaccagtgg 300
gaggagttca tgaggcacgt ggagaacctc atcaacgagc gcattgctga ctacgcgcgc 360
gacaaagcct tggccgagct gaccggcctg ggcaacaacc tgaatctcta tcgggagggt 420
ttcgaggact ggcggcggaa cccgacctcc cagggaagcca agactcgggt gatcgaccgc 480
tttctgtatc tcgacggcct gttcgagact tacatgccct ccttcgcggg ccgcaacttt 540
gaggttcacg tctcgactgt gtacgcctcg gctgcaaaac tacacctgtt cctgctgcgg 600
gacagctcga cactcaacct ggactggggc ctgtcacaga ccaacgtgaa cgagaactac 660
aaccggcaga tccgccacgc gcgccacctat gcaaacactt gcaccacttg gtatcagaca 720
gggtctccag gccctccagg cactaacgca acgtcatggg ttgcatacaa ccgggttccg 780
cgggagatga cactcaacct tggtgacatc tgctcgtgtt tctccaaacta cgactaccgg 840
tcgtaccgca ctgaggtgaa gggcgagttg accagggaata tctacacaga cctgtcggc 900
cggaactggc agaacgtcgc gccttcattc cgggagattg agaacctgac cattcgcgct 960
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tcgggttcca accgctactg ggcgcgccac atgcagaact tcagcgagac aaactctggc 1080
aacatcggat tcgatggccc acagtatggg tccacggctg ggacgataca ccggaccgac 1140
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tggcccaccg gccagacagt cctggggcgtg gcgagcaccg gcttcacact gaggaatctg 1260
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cagttccctg gcgttaagtc acacactctt aacaactgtc aagtggtgcg tggcacggcg 1560
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SEQ ID NO: 55      moltype = AA length = 657
FEATURE           Location/Qualifiers
REGION            1..657
                  note = The amino acid sequence of the
                  CR-BREla.TIC7042_2.nno_Mc:1 protein, comprising amino
                  acids 11 through 665 of TIC7042.
source            1..657

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mol_type = protein
organism = synthetic construct

SEQUENCE: 55
MQIIEPSSDS  FLYSHNNYPY  ATDPDTILKG  RNYKEWLNMC  TDTDDSRSP  AASTARSAIS  60
VAITISTTIL  GLLGVFPFASQ  IGAFYNFVLN  TVWPQGNQW  EEFMRHVENL  INERIADYAR  120
DKALAEITGL  GNNLNLYREA  FEDWRRNPTS  QEAKTRVIDR  FRILDGLFET  YMPSPAVRNF  180
EVQLLTIVYAS  AANIHLFLLR  DSSIYGLDWG  LSQTNVNNENY  NRQIRHAATY  ANHCTTWYQT  240
GLQRLQGTNA  TSWVAYNRFR  REMTLTVLDI  CSLFSNYDYR  SYPTEVKGEL  TREIYTDVPV  300
RNWQNVAPSF  AEIENLTIRA  PRTVTWLNST  RIFTGTLTGW  SGSNRYWAAH  MQNFSSTNSG  360
NIGFDGPQYG  STVGTIHRD  DYDMVNRDIY  TITSQAAAL  WPTGQTVLGV  ASTRFTLRNL  420
NNNSTEALVY  ENAISSSFVS  STLTHELPGE  NSDRPTSSDY  SHRLSSITGF  RAGANGTVPV  480
FGWTSATVDR  NNIEBQNKIT  QPPGVKSHTL  NNCQVVRGTG  FTGGDWLRPN  NNGTFRLLIT  540
SFSSQSYRIR  LRYATSVGNT  SLVISSSDAG  ISSTTIPLTS  TITSLPQTV  YQAFRVVDLP  600
ITFTTPTTQR  NYTFDFRLQN  PSNANVFIDR  FEFVPIGGSL  SEYETKHQLE  KARKAVN  657

SEQ ID NO: 56      moltype = DNA  length = 3813
FEATURE            Location/Qualifiers
misc_feature        1..3813
                    note = A synthetic coding sequence used for expression in
                    plant cells, CR-BREla.TIC7381_1.nno_Mc:1 which encodes a
                    TIC7381 protein wherein an additional alanine codon is
                    inserted immediately following the initiating methionine
                    codon.
source              1..3813
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 56
atggctaacc  agaaccagaa  caagaacgag  ctacagataa  tcgagccgtc  ctcggtatcc  60
tgtctgtact  agcacacaac  ctacccgtag  gccaccgtag  cgaacacggt  gctagagggc  120
cgcaactaca  aggagtggtc  gaataagtgc  acagacaact  acaccgatgc  actgcaaggc  180
ccagaggcaa  cggctatcag  caagggtgcc  gtgtcggccg  ctatctctat  cagcaccaaa  240
gtcttgagcc  ttctggaggt  gcccttcgca  gctcagattg  ggcaattgtg  gacgtttatt  300
ctgaacgctc  tttggccctc  ggacaacacc  cagtgaggag  agttcatgcg  ccatgtcgag  360
gagctcatca  accagcgtag  cgcgcgactac  ggcggagtag  aggcgctcgc  cgagctcacg  420
ggcctgggca  acaacctcga  tctgtacatc  gaagctcttg  aggactggaa  gcgcaaccgc  480
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gaagcgtaga  tgcctagctt  ccgctgagc  ggctatgaag  tgcattgct  caccgtatat  600
ggcgcgccg  ccaacctgca  cctcctcctc  ctcccgact  gctctatcta  cggcattcag  660
tggggtttct  cccagaccaa  tgtcaacgag  aactataatc  ggcagatccg  tcacaccgcc  720
gagtagcgga  accactcgac  gacatggtag  cagacgggcc  tggagcgact  gcgcgccacg  780
aacgcctcaa  gctgggtgcc  gtacaacgt  ttcaggagag  agatgactct  caccgtgctc  840
gacatctgca  gtctgttctc  caactatgac  tacaggagtt  acccgccaga  ggtgcccggc  900
gagctgacca  gggagatata  cagggaaccg  gtggtagcga  cctctctgtg  ggtgaacaac  960
gcgcgctcct  tcggcgaaat  cgagaatctg  gcaatccgag  cgccacgcac  cgtgacatgg  1020
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gtgctggcgg  tggcttcgcg  aagattcacc  cttcgaaacc  tcaacaacaa  cctgaccacg  1320
gcgctgggtc  acgagaaccc  gatataaagc  agtttcaacc  ggtcaactct  caccacgcag  1380
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tcccataccc  tgaacaattg  ccaggctcgt  ccggttagag  gctttaccgg  cggcgactgg  1620
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tacagaatca  ggctgcggta  cgcgtccgca  gcgaatacct  ccctgcgaat  ctgctcttcg  1740
gcggccggga  tctcgtcaac  aaccgtccca  ctccctcca  cgatcactag  cctcccgcag  1800
acagccgtcc  catacagggc  gttccgggtg  atcgacctgc  caatcacgtt  caccacgcgc  1860
acgcagtcta  actacacttt  gacttcgtc  ctccagaatc  cttccaacgc  caacgtgttc  1920
atcgaccgct  tcgagtttgt  ccgattgga  ggttcgctct  cagagtacga  gacgaagcac  1980
cagctggaga  aggcacgtaa  ggcggtgaac  gatctgttta  ctaacgagtc  aaagaacgtt  2040
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tcggacgagt  gcgctaaccg  aaagatgatc  ctccagagcg  aggtgaagta  tcggaagcaa  2160
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ggattcgacc  gttgcaagcc  gcaactctac  atcccgatga  atcccgaatg  ccaccacgac  2580
gtgatcaata  acatcgagat  cagctcgcca  tgtcagcata  acaagatgct  cgacaacgcg  2640
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cacttcgagt  tccacatcga  cactggcaag  atccacctca  acgagaacct  gggcatctgg  2760
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gaagagggac  cgcttggtgc  ggagtcctc  gagcgcgtca  agcgacgcga  gaagaagtg  2880
aagcaccaca  tggagcacia  atgttccgag  acgaaacttg  cgtaccatgc  cgccaagcag  2940
gcccttgttg  ggctgttcac  gaacaccaag  tacgataggg  tcaagttcga  gaccaccatc  3000
tcgaacatcc  tctttgcgca  ctacctgtg  cagtcfaatc  catacgttta  caacaatgg  3060
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ggcgcccttca acctgtacga ccagcgtaac atcataaaga acggcgactt caaccgtggc 3180
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ggctacgtgc tgcgggtaac tgccaagaag gagggccctg gcattggata cgttaccttc 3360
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ggcgcttcctt acaagcagca cggctgccgc accgacggag tcccgtaaca gcagcatggg 3660
tgtcgaccgc accggtcccg cgacgagcag ctggattacg taacgaagac gatcgacgtc 3720
ttcccggaac cggacaaggt caggatcgac attggcgaaa cggagggaac cttcaaagtc 3780
gagtcagtgag agcttatctt catggaggag tga 3813

SEQ ID NO: 57      moltype = AA length = 1270
FEATURE           Location/Qualifiers
REGION            1..1270
                  note = The amino acid sequence of
                  CR-BRE1a.TIC7381_1.nno_Mc:1 wherein an additional alanine
                  amino acid is inserted immediately following the
                  initiating methionine relative to the TIC7381 protein
                  sequence.
source            1..1270
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 57
MANQNQNKNE LQIIEPSSDS LLYSHNNYPY ATDPNTVLEG RNYKEWLNKC TDNYTDALQG 60
PEATAISKGA VSAAISISTK VLSLLGVPPA AQIGQLWTFI LNALWPSDNT QWEEFMRHVE 120
ELINQRIADY ARSKALAEIT GLGNLDDLYI EALDWKRNK TSQQAQKRVK DRFRIADGLF 180
EAYMPSFRVS GYEVPLLTIV AAAANLHLLL LRDCSIYGIQ WGFQSTNVNE NYNRQIRHTA 240
EYANHCTTWY QTGLERLRGT NASSWVPYNR FRREMTLTVL DICSLFSNYD YRSYPAEVRA 300
ELTREIYTDP VVSTSLWVNN APSFGEIENL AIRAPRTVTW LNSTRISTGT LQWWSGSNRY 360
WAAHMQNFSE TNSGNIGFDG PLYGSTVGTI IRDDNYEMVN RDIYITITSEA VAALWPTGQI 420
VLGVASARFT LRNLNNLTQ ALVYENPISS SFNRSTLTRE LPGENSDRPT SSDYSHRLTS 480
ITAFRAGSNG TIPVFGWTSI SVNRDNILER NKITQFPGVK SHTLNNCQV RGTGFTGGDW 540
LRPNNGSFR LTITSFSSQS YRIRLRYASA ANTSLRISSE AAGISSTTVP LASTITSLPQ 600
TAVPYEAFRV IDLPITFTTA TQSNYTFDFV LQNPNSANVF IDRFEPVPIG GSLSEYETKH 660
QLEKARKAVN DLFTNESKNV LKKYTTDYDI DQANLVECV SDECANAKMI LLDEVKYAKQ 720
LSEARNLLL GNFEYQDRDG ENPWKTSNPV TIQENNPPIK GRYLSMSGAN YIEATNDTFP 780
TYAYQKIDEA KLPKYTRYKV RGVVGNKSDL ELLIKRYDEE VDAILNVPND IPHAPTPFCG 840
GFDRCCKPHSY IYTMPECHND VINNIEISS CQHNKMLDNA DISSRHSSELG KKHGICHESH 900
HPEFHIDTGG IHLNENLGIW VIFKICSTDG YATLDNLEVI EEGPLGAESL ERVKREKKW 960
KHMEHKCSE TKLAYHAAKQ ALVGLFTNTK YDRLKFETI SNILFADYLV QSIPIVYNKW 1020
LPDVPGMNYD IYTELKNLFT GAFNLVDQRN IIKNGDFNRG LMHWHPHFA RVEQIDNRV 1080
LVLPNYAANV SQEVCLEHNR GYVLRVTAKK EGPVIGYVTF SDCANNIEKL TFTSCDYGTN 1140
EVTYEQSNYP TDGVPYQHQH CHTDGVPEYQ HGCHTDGVPEY BQHGCHTDGV PYQHGCHTD 1200
GVVPYQHGCR TDGVPYQHQH CRTDRSRDEQ LDYVTKTIDV PPDTDKVRID IGTEGTFKV 1260
ESVELIFMEE 1270

SEQ ID NO: 58      moltype = DNA length = 3744
FEATURE           Location/Qualifiers
misc_feature       1..3744
                  note = A synthetic coding sequence used for expression in
                  plant cells, CR-BRE1a.TIC7382_1.nno_Mc:1 which encodes a
                  TIC7382 protein wherein an additional alanine codon is
                  inserted immediately following the initiating methionine
                  codon.
source            1..3744
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 58
atggctaacc agaaccagaa ccagaacaag aacgagatgc agatcatcga gccctcctcc 60
gactcattcc tctactcgca caacaactac cctacgcgca cggaccccaa cactgttctc 120
gaagggcgga actacaagga gtgggctaaac aagtgcacta acaattacac cgacgcgctc 180
cagtcctcgg aagccacagc catttcgaag gggcggtctc cggcgcgcat cagcatcagc 240
accaaggtgc tcggcctcct cggcggtgcc ttcggcgctc agataggaca gctctggaca 300
ttcattctga acgcccgtgt gccgtccgac aacacccagt gggaggagtt catgcggcac 360
gtcagaggagc tcactcaaca cgggatcgca gactatgcc gtaacaaggc cctggccgag 420
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aaccgccacca gccaggaggc caagactcgc gtcactgacc gcttcgccat cgtcgacggg 540
ctgttcgagg cctacattcc gagcttcgcc gtgagcggtc atcaagtcca gctgctcact 600
gtctacggcg ctcagcaaaa cctccacctc ctgctgctgc gcgactccac gatctacgga 660
atcgactggg gcccttccca gactaacgtc aacgataact acaaccggca gattcgtctc 720
accgcagcgt acgctaacca ctgtacaacg tggtagcaga caggactgga gcggctccgc 780
gggagcaacg cgtcttcctg ggtgacctac aaccgtttcc ggagagagat gactctcaca 840
gtcttggaac gtctgttcct gttcagcaac tacgactatc gcagctaccc ggccgaggtt 900
cgccgagaga ttacgaggga aatctacaca gacctgtcgc gcgtcggttg ggttgacagc 960
gcgccttctt tcggcgagat cgagaacctg gcgacccggc cgccacggac tgtgacctgg 1020
ctgaactcca cccggatctc caccgcacc ttaagcggct ggtcgggctc taaccggtac 1080
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ccgctgtacg ggagcacagt gggcaccatc caccgcaactg atgactacga catgggcaac 1200
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tcgcacacgc taaacaactg ccagggtggg aggggaaccg gtttcacggg aggtgactgg 1620
ttaagaccaa ataacaacgg cactttccgc ctcactatta cgagcttctc ctcccaaagt 1680
taccgcatcc gccttcgcta cgccacgtca gtggggaaca cctcgctggg gatattcctc 1740
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cagacgggtc cgtatcaggc cttcagagtg gtggaccttc ctattacggt cagcagcccc 1860
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atcagcaaca tccattatgc ggactactta gtccagagca tcccttacgt ttacaacaaa 3060
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tacaagcagc agggctgcgc gaccgacggc gtgccctacg agcagagcgg ctgcccacag 3600
gaccggagca gggatgagca gctggactac gtgaccaaga cgatcgacgt cttcccggac 3660
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gagctattct gcattggagga gtag 3744

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SEQ ID NO: 59      moltype = AA length = 1247
FEATURE           Location/Qualifiers
REGION            1..1247
                  note = The amino acid sequence of
                  CR-BREla.TIC7382_1.nno_Mc:1 wherein an additional alanine
                  amino acid is inserted immediately following the
                  initiating methionine relative to the TIC7381 protein
                  sequence.
source            1..1247
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 59
MANQNQNQNK NEMQIIEPSS DSFLYSHNNY PYATDPNTVL EGRNYKEWLN KCTNNYTDAL 60
QSPEATAIK  GAVSAAISIS TKVLGLLGVP FAAQIGQLWT FILNALWPSD NTQWEEFMRH 120
VEELINQRIA DYARNKALAE LTGLGNLNDL YIEALDDWKR NPTSQEAETR VIDRFRIVDG 180
LFEAYIPSPA VSGYQVQLLT VYAAAANLHL LLLRDSTIYG IDWGLSQTNV NDNYNRQIRL 240
TATYANHCTT WYQTLGLERLR GSNASSWVTY NFRFREMTLT VLDICSLFSN YDYSYPAEV 300
RGEITREIYT DPVGVGWVDS APSFGEIENL AIRAPRTVTW LNSTRISTGT LSGWSGSNRY 360
WAAHMQNFSE TNSGNIGFDG PLYGSTVGTI HRTDDYDMGN RDIYTITSQA VLGLWPTGQR 420
VLGVASARPT LRNLFNNTLQ VLVYENPISS NFGSSTLTHE LSGENS DRPT SSDYSHRLTS 480
ITGFRAGANG TVPVFGWTS A TVDRNNIIER NKITQFPVK SHTLNNCQV RGTGFTGGDW 540
LRPNNGTFR LTTISFSSQS YRIRLRYATS VGNTSLVISS SDAGISSTI PLTSTITSLP 600
QTVPYQAFRV VDLPIITPT TQQRNYTFDF RLQNPSNANV FIDRFEFVPI GGLSEYETK 660
HQLKARKAV NDLFTNESKN VLKKTETD YD IDQAANLVEC ISDECANAKM ILLDEVKYAK 720
QLSEARNLLL NGNFDNLDNR GENPWTSPN VTIQENNPFI KGRYLSMSG NNI EATNEIF 780
PTYVYQKIDE AKLKPYTRYK VRGFVGS SKD LELLVTRYDE EVDAILNLVN DIPHAPPPFC 840
GGFDRCKPHS YPPMNPECHH DVINNIEISS PCQHNKMLDN ADIFSRHSEL GKKGHCICHES 900
HHFEPHIDTG KIDLNLNLGI WVFVKICSTD GYATLDNLEV IE EGPLGAES LERVKREKK 960
WKHHMEHKCS ETKLAYHAAK QALVGLFTNT KYDR LKFETT ISNLFADYL VQSIPVYVYNK 1020
WLPDVPGMNY DIYTELKNLF TGAFNLYDQR NIIKNGDFNR GLMHWHATPH ARVEQIDNRS 1080
VLVLPNYAAN VSQVCLLEHN RGVVLRVTAK KEGPGIGYVT FSDCANNIEK LFTSCDYGT 1140
NEVPYEQSGY GTNEVPYEQS GYGTVNEVPY EQSGYRTDGPV YKQHGCR TDG VPYEQSGCRT 1200
DRSRDEQLDY VTKTIDVFPD TDKVRIDIGE TEGTFKVESV ELFCMEE 1247

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-continued

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SEQ ID NO: 60      moltype = DNA length = 1983
FEATURE           Location/Qualifiers
misc_feature       1..1983
                  note = A synthetic coding sequence used for expression in
                  plant cells, CR-BRE1a.TIC7382_2.nno_Mc:1 which encodes a
                  protein comprising a C-terminal truncation relative to the
                  TIC7382 protein and wherein an alanine codon is inserted
                  after the initiating methionine.
source            1..1983
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 60
atggctaacc agaaccagaa ccagaacaag aacgagatgc agatcatcga gccctcctcc 60
gactcattcc tctactcgca caacaactac ccctacgcga cggaccccaa cactgttctc 120
gaagggcgga actacaagga gtgggtaaac aagtgcacta acaattacac cgacgcgctc 180
cagtctccgg aagccacagc catttcgaag ggggcggtct cggcgcgcat cagcatcagc 240
accaaggtgc tcggcctcct cggcggtgctc agataggaca gctctggaca 300
ttcattctga acgcccgtgt gccgtccgac aacacccagt gggaggagtt catgcggcac 360
gtcagaggagc tcatcaacca gcggatcgca gactatgccc gtaacaaggc cctggccgag 420
ctgacgggac tcggcaacaa cctcgacctc tacatcgagg ccctggatga ctggaagagg 480
aaccaccacca gccaggaggg caagactcgc gtcacgcacc gcttcgcgat cgtcgacggg 540
ctgttcgagg cctacattcc gagcttcgcc gtgagcggct atcaagtta gctgctcact 600
gtctacgccc ctcgagcaaa cctccacctc ctgctgctgc gcgactccac gatctacgga 660
atcgactggg gccttttccc gactaacgtc aacgataact acaaccggca gattcgtctc 720
accgcgacgt acgctaacca ctgtacaacg tggtaaccga caggactgga gcggctccgc 780
gggagcaacg cgtcttctct ggtgacctac aaccgtttcc ggagagagat gactctcaca 840
gtcttggaac tctgttctct gttcagcaac tacgactatc gcagctaccc ggccgaggtt 900
cgccgcgaga ttacgaggga aattacaca gaccctgtcg gcgtcggttg ggttgacagc 960
gcgccttctt tcggcgagat cgagaacctg gcgacccggg cgccacggag tgtgacctgg 1020
ctgaactcca cccggtatct caccggcacc ttaagcggct ggtcgggctc taaccggtag 1080
tggcgggccc acatgcagaa tttctccgag acaaactcgg gtaacatcgg cttegatggg 1140
cgcgtgtacg agtgcaacgt gggcaccatc caccgcactg atgactacga catgggcaac 1200
cgggacatct acacgataac gtcgcaggcc gttctcggtc tgtggccac ggccagcgg 1260
tcctctggag tcgcgcgagg agcaaatggt accgttcccg tctttgggtg gaccagcgc 1320
gtgctggtat acgagaatcc tattagctcc aacttcggct cttcaacgct aaccacgag 1380
ttatccggcg agaacagcga taggcccacc agcagcgatt actcacatcg actcacaagc 1440
atcacgggat tccgcgcagg atcgagagcg aacaagatca cgcagttccc aggcgtcaaa 1500
acagtcgacc ggaacaacat cctcgagcgc aacaagatca cgcagttccc aggcgtcaaa 1560
tcgcacacgc taaacaactg ccaggtgggt aggggaaccc gtttcaccgg aggtgactgg 1620
ttaagaccaa ataacaacgg cactttccgc ctactatta cgagcttctc ctccaaagt 1680
taccgcatcc gccttcgcta cggccagctc gtggggaaca cctcgctggt gatatactcc 1740
tcagacgccc ggatcagcag caccacgata ccaactaac ccacaataac ctcatgccc 1800
cagacggtcc cgtatcaggc ttccagagtg gtggaccttc ctattacgtt cacgacgccc 1860
acgacccagc gcaactacac attcgatttc cgggtgcaga acccgagtaa cgccaatgtc 1920
ttcatcgatc gtttcagatt cgtcccagata ggccgctccc ttcggagta cgaacgaag 1980
tga                                                1983

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SEQ ID NO: 61      moltype = AA length = 660
FEATURE           Location/Qualifiers
REGION            1..660
                  note = The amino acid sequence of
                  CR-BRE1a.TIC7382_2.nno_Mc:1 comprising a C-terminal
                  deletion, an additional alanine amino acid, and comprises
                  amino acids 1 through 659 of TIC7382.
source            1..660
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 61
MANQNQNQN NEMQIIEPSS DFLYSHNNY PYATDPNTVL EGRNYKEWLN KCTNNYTDAL 60
QSPEATAIK GAVSAAISIS TKVLGLLGVF FAAQIGQLWT FILNALWPSD NTQWEEFMRH 120
VEELINQRIA DYARNKALAE LTGLGNLDDL YIEALDDWKR NPTSQEAQTR VIDRFRIVDG 180
LFEAYIPSPA VSGYQVQLLT VYAAAANLHL LLLRDSTIYG IDWGLSQTNV NDYNNRQIRL 240
TATYANHCTT WYQTGLERLR GSNASSWVTY NFRFRREMTLT VLDICSLFSN YDYRSYPAEV 300
RGEITREIYT DPVGVGWVDS APSFGEIENL AIRAPRTVTW LNSTRISTGT LSGWGSNNRY 360
WAAHMQNFSE TNSGNIGFDG PLYGSTVGTI HRTDDYDMGN RDIYTITSQA VLGLWPTGQR 420
VLGVASARPT LRNLFNLTQ VLVYENPISS NFGSSTLTHE LSGSNSDRPT SSDYSHRLTS 480
ITGFRAGANG TVPVFGWTSFA TVDRNNIIEK NKITQFPGVK SHTLNQCQV RGTGFTGGDW 540
LRPNNGTFR LTITSFSSQS YRIRLRYATS VGNTSLVISS SDAGISSTTI PLTSTITSLP 600
QTPVYQAFRV VDLPTITFTT TQQRNYTFDF RLQNPSSNANV FIDRFEFVPI GGSLSLEYETK 660

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SEQ ID NO: 62      moltype = DNA length = 3774
FEATURE           Location/Qualifiers
misc_feature       1..3774
                  note = A synthetic coding sequence used for expression in
                  plant cells, CR-BRE1a.TIC7383_1.nno_Mc:1 which encodes a
                  TIC7383 protein wherein an additional alanine codon is
                  inserted immediately following the initiating methionine
                  codon.
source            1..3774

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-continued

```

mol_type = other DNA
organism = synthetic construct

SEQUENCE: 62
atggctaacc agaaccagaa ccaaaaccag aacaagaacg agctgcagat cattgaacc 60
tcctcggaca gcctcctgta cagccataac aactaccgtg acgccaactga cccaaacacc 120
gtgctcgagg ggcgcaacta caaggagtggt ctaacaacat gcaccgacaa ctacaccgac 180
gcactccagg gcccgagggc gacggccatc agcaagggtg ccgtctccgc ggccatctcc 240
atctcgacga aggtcctagg gctcctaggg gtcccgttcg ctgctcgagat tggccagctg 300
tggacgttta tccttaacgc actctggccc tcagacaaca cccagtgagg gaggttcatg 360
cgccacgtcg aggaactgat caaccagcgc atagcggact acgcgaggaa caaggccctg 420
gcccagctga cagggtctgg caacaacctg gacctotaca tcgaggcgct cgaggactgg 480
aagcgggaacc ctacctccca acaagctaag gaccgcgtta tcgaccgctt ccgaatcgcg 540
gacggcctct tcgaggggta catgcccagc ttccgggtct cgggctacga ggtgcccctc 600
tcacgggtgt acgcccgggc ggctaatttg catctcctgc tgcctcaggga ttgctcgatc 660
tacggaatcc agtggggctt ctcccagact aacgttaacg agaactacaa tcggcagatt 720
cggcacaccg ccaggtacgc taacctactgc acaacatggt atcagaccgg cctcgagagg 780
ctgcgcggta ccaatgcttt ctcatggatc aactacaacc gcttcgcggc cgagatgaca 840
ctgactgtct tggacgtctg ctccctcttc tccaactatg actaccgcac ttaccccact 900
gagggtccggg ccgagcttac gcgggagatc tacacagacc ccacgggtt ccagaactcc 960
ccgtctcccg cgtgtggtcc caattggtag gactacgaga ggtcgttcgc ggaaatcgaa 1020
aacatcgcca tacgagcacc tcggacagtc acgtggctga actcaaccac catctacacc 1080
gggagactga acggctacaa caatagcaac tactactggg cggggttcgc ccagaacttc 1140
tcagagacca actctggcag cagcttcaac ggcccagacc taggcgacct cactccgaac 1200
tacaggatag agaccctcga catggtcaac agggacatct acagcatcta ctcccgggtc 1260
gtgtcgcaat cctggcccat cggcaacgtc aagctcttcg gcgtaagctc ctctacctta 1320
agtttccggg atctcaacaa caacagctcc ggcacctcgg tctacgagaa cccgactaac 1380
ttcagctccc agtacctaac cacagagttt ccaggggaga actccgagcg gccacacctc 1440
acggactaca gtcaccgctt aacctgcctt acgcccagtc gggccgggaa ctacggcctc 1500
gtgtcttggt cgggttggtg ttcttcgagt gtcgaaaagg acaaccgctt acaaccggat 1560
aagattaccc agtaccgccg cgtcaagggt ttcaacttgg acggtctcac agtggtgaaa 1620
gggactggtt tcaccggggg caactggctg cgggtccagca gggtcaccgg gtcattccgc 1680
ctgaacgtat actcacctag tgtgcagact tatcggatga ggatcaggta cgctccccg 1740
cttggaactc cgaagcttgg gataagttcg acagacgccc gcctctcgtt cagcagtttc 1800
cctctgcctc ccactatcgg gagcatgccc tccactgtgc cctacgaggg gttcgtgtc 1860
ctcgacatcc ccacacgggt caccgtagcc tcccaaagga actacaactt catcttcgac 1920
atccttaacc cctcgggttg ggcctgtgac atcgatcgta ttgagttcgt gccctggggc 1980
tcttcagtgt tcgagtacga aaccaagcac gagcttgaaa aggcgaaaaa agcagttaac 2040
gacctcttca cgaacgagtc caaaaacatg ttaaagaaag acacgaccga ctacgacatc 2100
gaccaggctg cggacctggt ggagtgctga agcagcaggt gcgcccacgc taagatgata 2160
ctgctggacg aagtgaagta cgcgaagcag ctgagcagag ctcggaaact gctccagaac 2220
gggaacttca aggtcttaga cgttgacaac aacaaccctg ggacgacctc accgaacgta 2280
acgatccagg agaacaaccc gatcttcaaa ggtcactacc tctccatgtc cgggtgcgaac 2340
gcaatcgagg ccaccaacga ggtgttccca acgtacgtgt accaaaagat tgaggagagt 2400
aaactcaagc cctacacagc gtacaagggt cgggggttca tcggccagag caaggacgtc 2460
gagctgctcg tcacgcgtta cgacgaggag gtggacgcta ttctgaacgt gcctaacgac 2520
ctaaagtacg caagtccgag gcacctgtcg ggagagttca accgctgcaa gcctcacact 2580
taccggcca cagaccctcg atgccacgac gacgtgatcg acaagatcga catctcctcg 2640
ccgtgtcaga acaacattat gctgtcggac gcggacatct ctagtcttca tagcggactt 2700
ggcaagaagc acgggatctg ccacgagagc catcacttcg agttccacat cgacacgggc 2760
aaaatcgacc tcgtggagaa ccttggcatc tgggtgatct tcaagatctg ctctaccgac 2820
ggatacgcca cgctcgacaa tctggaagta atagaggaa gcccgttggt cgcagagtcc 2880
ctagagcggg tgaagcgtag ggagaagaag tggaaagcata acatggagca caagtgtctg 2940
gaaacgaaac acgcatacca cgcggcgaag cagggtgtcg aagcactctt cactgaactc 3000
aaggacgaga gggtcaagtt cgagacgacg attagtaaca tcctgtcagc tgagtacct 3060
gtccagtcca tcccatagct ctacaacaag tggctgtcgg acgtacctgg gatgaactac 3120
gacatctaca ccgagctcaa gaaccgcac tcggcaagcgt tcaacttgta cgaccagagg 3180
aacatcatca agaacggcca cttcaatcat ggtttaatgc attggcacgc gacaccgcac 3240
gcgaacgtgc agcaaataga cggatatttc gtgctggtcc tccaaaactg gggggccaac 3300
gtcagccagg aggtgtgctt caagcacaac cggggctacg tgctgcgagt caccggccaag 3360
gaggagggtc acggcaaggg ctacgtgacg atttcggact gcgctaacca ggtcgagaag 3420
ctgtcgttta ctagcctacc gacctctacc gacggcgctc catacgaaca gtccaactac 3480
ccgaccgacg gagtgtctta cgggcagcac ggggtgcaaca tcgaccaggt gccctacgag 3540
caatctggct accccaccga cgggggttcct tacgaacagt cgggctaccg tacggacggg 3600
gtcccgtaca agcagcatgg atgccactcg gacggaagcc gtgaggagca gcacgggtac 3660
gttacgaaga ctatcgacgt cttcccgcac acggacaagg tccggatcga catcggcgag 3720
accgagggga cgttcaaggt tgagagcgtg gagctcatct gcatggagga gtag 3774

SEQ ID NO: 63      moltype = AA  length = 1257
FEATURE            Location/Qualifiers
REGION             1..1257
note = The amino acid sequence of
CR-BRE1a.TIC7383_1.nno_Mc:1 wherein an additional alanine
amino acid is inserted immediately following the
initiating methionine relative to the TIC7383 protein
sequence.
source             1..1257
mol_type = protein
organism = synthetic construct

SEQUENCE: 63

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MANQNQNQ	NKNEQLIIEP	SSDSLLYSHN	NYPYATDPNT	VLEGRNYKEW	LNKCTDNYTD	60	
ALQGPEATAI	SKGAVSAAIS	ISTKVLGLLG	VPFAAQIGQL	WTFILNALWP	SDNTQWEEFM	120	
RHVEELINQR	IADYARNKAL	AELTGLGNNL	DLYIEALEDW	KRNPTSQQAQ	DRVIDRFRIA	180	
DGLFEGYMP	PRVSGYEVPL	LTVYAAAANL	HLLLLRDCSI	YGIQWGSFQT	NVNNYNNRQI	240	
RHTAERYANHC	TTWYQTGLER	LRGTNAFSWI	NYNRRFRREMT	LTVLDVCSLF	SNYDYRTYPT	300	
EVRAELTREI	YTDPIGFQNS	PLPGVVPNWY	DYARSAEIE	NIARAPRTV	TWLNSTTIYT	360	
GRLNGYNNNS	YYWAGFRQNF	SETNSGSSFN	GPDLGDLTPN	YRIETLDMVN	RDIYSIYSRV	420	
VSQSWPIGNV	KLFGVSSSTL	SFRDLNNNSS	GTLVYENPTN	FSSQYLTFEF	PGENSERPTF	480	
TDYSHRLTCL	TRIGAGNYGL	VLCAGWTSSS	VERDNRLQPD	KITQYPVAVG	FNLDGFTVVK	540	
GTGFTGGNWL	RSSRVTSFSR	LNVSPTSQVT	YRMRIRYASP	LGNSTLGISS	TDAGISFTSF	600	
PLPSTIGSMP	STVPYEAFRV	LDIPITVTVA	SQRNYNFIFD	ILNPSVGAVY	IDRIEFVPGV	660	
SSVFYEYETH	ELEKAKKAVN	DLFTNESKNM	LKKDSTDYDI	DQAADLVECV	SDECAHAKMI	720	
LLDEVKYAKQ	LSEARNLLQN	GNFKVLDVDN	NNPWTSPNV	TIQENNPPIFK	GHYLSMSGAN	780	
AIEATNEVFP	TYVYQKIEES	KLKPYTRYKV	RGFIGQSKDV	ELLVTRYDEE	VDAILNVPND	840	
LKYAVPTHL	GEFNRCFKPT	YPATDPRCHD	DVIDKIDISS	PCQNNIMLSD	ADISLSHSL	900	
GKHKGI	CHES	HHFEFHIDTG	KIDLVENLGI	WVIFKICSTD	GYATLDNLEV	IEEGPLGAES	960
LERVKRREKK	WKHNMEHKCS	ETKHAYHAAK	QAVEALFTNF	KDERLKFETT	ISNLSAEYL	1020	
VQSIPIVYVYNK	WLSVDPGMNY	DIYTELKNRI	WQAFNLYDQR	NIKNGHFNH	GLMHWATHPH	1080	
ANVQIDG	IS	VLVLPNWGAN	VSQEVCLKHN	RGYVLRVTAK	EEGHGKGYVT	ISDCANQVEK	1140
LSFTSRDYST	DGVPEQSNY	PTDGVSYGQH	GCNIDRVPE	QSGYPTDGV	YEQSGYRTDG	1200	
VYPKQHGCHS	DGSREEQHG	VTKTIDVFPD	TDKVRIDIGE	TEGTFKVESV	ELICMEE	1257	

SEQ ID NO: 64 moltype = DNA length = 1854
 FEATURE Location/Qualifiers
 misc_feature 1..1854
 note = A synthetic coding sequence used for expression in
 plant cells, CR-BRE1a.TIC7383_7.nno_Mc:1 which encodes a
 protein comprising an N-terminal and C-terminal truncation
 relative to the TIC7383 protein with an alanine codon
 after initiating methionine.
 source 1..1854
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 64

atggctaccg	acaactacac	cgacgcactc	caggggcccg	aggcgacggc	catcagcaag	60
ggtgccgtct	ccgcggccat	ctccatctcg	acgaaggtcc	tagggctcct	aggggtcccg	120
ttcgtctgcg	agattggcca	gctgtggacg	ttcatcctta	acgcactctg	gccctcagac	180
aacacccagt	gggagagatt	catcgccac	gtcgaggaa	tgatcaacca	gcgcatacgc	240
gactacgcga	ggaacaaggc	cctggccgag	ctgacagggc	tgggcaacaa	cctagacctc	300
tacatcgagg	cgctcgagga	ctggaagcgg	aacctacct	ccaacaacgc	taaggaccgc	360
gttatcgacc	gcttcggaat	cgcggacggc	ctcttcgagg	ggatcatgcc	gagcttcggg	420
gtctcgggct	acgaggtgcc	ctctctcacc	gtgtacgcgg	cgggcggtta	tttgcattct	480
ctgctgctca	gggattgctc	gattctacga	atccagtggg	gcttctccca	gactaacgtt	540
aacgagaact	acaatcgga	gattcggcac	accgcccagt	acgctaacca	ctgcacaaca	600
tggatcacga	ccggcctcga	gagggtgcgc	ggcaccaatg	ctttctcatg	gatcaactac	660
aaaccgttcc	gccgcgagat	gacactgact	gtcttgacg	tctgtctcct	cttctccaac	720
tatgactacc	gcacttacc	cactgaggtc	cgggccgagc	ttacgcggga	gatctacaca	780
gaccccatcg	ggttcagaa	ctcccgcgtc	cctggcgtgg	tcccaaatg	gtacgactac	840
gcgaggtcgt	tcgcggaaa	cgaaaacatc	gccatacgag	cacctcggac	agtcacgtgg	900
ctgaactcaa	ccaccatcta	caccggggaga	ctgaacggct	acaacaatag	caactactac	960
tgggcccggg	tccgccagaa	ctctcagag	accaactctg	gcagcagctt	caacggccca	1020
gacctagggc	acctcaactc	gaactacagg	atagagaccc	tcgacatggt	caacagggac	1080
atctacagca	tctactcccg	ggctgtgtcg	caatcctggc	ccatcgga	cgtaagctc	1140
tcggcgctaa	gctcctctac	cttaagtctt	cgggatctca	acaacaacag	ctccggcacc	1200
ctggtctacg	agaacccgac	taacttcagc	tccagtagc	taaccacaga	gtttccaggg	1260
gagaactccg	agcggccac	cttcacggac	tacagtcacc	gcctaacctg	ccttacggcg	1320
atccggggccg	ggaaactacg	cctcgtgctt	tgtgcgggtt	ggacttcttc	gagtgctcaa	1380
agggacaacc	gcctacaacc	ccgataagatt	accagtagc	ccgcgctcaa	ggggttcaac	1440
ttggacggct	tcacagtggt	gaaagggact	ggtttcaccg	ggggcaactg	gctgcgggtc	1500
agcaggggtca	ccgggtcatt	ccgcctgaac	gtatactcac	ctagtgtgca	gacttatcgg	1560
atgaggatca	ggtacgcctc	ccgcgttggc	aactcgacgc	ttgggataag	ttcgacagac	1620
gccggcatct	cgttcacgag	tttccctctg	ccctccacta	tcgggagcat	gccgtccact	1680
gtgccctacg	agcggttccg	tgtcctcgac	atccccatca	cggtcaccgt	agcctcccaa	1740
aggaactaca	acttcattct	cgacatcctt	aacctcctcg	ttggggccgt	gtacatcgat	1800
cgtattgagt	tcgtgccctg	gggctcttca	gtgttcagat	acgaaaccaa	gtga	1854

SEQ ID NO: 65 moltype = AA length = 617
 FEATURE Location/Qualifiers
 REGION 1..617
 note = The amino acid sequence of
 CR-BRE1a.TIC7383_7.nno_Mc:1 comprising an N-terminal and
 C-terminal deletion, an additional alanine amino acid, and
 comprises amino acids 54 through 668 of TIC7383.
 source 1..617
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 65

MATDNYTDAL	QGPEATAISK	GAVSAAISIS	TKVLGLLGV	FAAQIGQLWT	FILNALWPSD	60
NTQWEEPMRH	VEELINQRIA	DYARNKALAE	LTGLGNLDDL	YIEALEDWKR	NPTSQQAADR	120

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VIDRFRIADG LFEYGMPSEF VSGYEVPLLT VYAAAAANLHL LLLRDCSIYQ IQWGFSTQNTV 180
NENYNRQIRH TAEYANHCTT WYQTGLERLR GTNAFWSWNY NRFRREMTLT VLDVCSLFSN 240
YDRTYPTTEV RAELETREIYT DFIGFQNSPL PGVVPNWYDY ARSFAEIE NI AIRAPRTVTW 300
LNSTTIYTGR LNGYNNSNYW WAGFRQNFSE TNSGSSFNPG DLGDLTPNYR IETLDMVNRD 360
IYSIYSRVVS QSWPIGNVKL FGVSSSTLSF RDLNNNSSTG LVYENPTNFS SQYLTTTEFP 420
ENSERPTFTD YSHRLTCLTR IGAGNYGLVL CAGWTSSSVE RDNRLQPDKI TQYPAVKGFN 480
LDGFTVVKGK GFTGGNWLRS SRVTGSFRLN VYSPSVQTYR MRIRYASPLG NSTLGISSTD 540
AGISFTSFPL PSTIGSMPST VPYEAFRVLD IPITVTVASQ RNYNFIFDIL NPSVGAVYID 600
RIEFVPVGSS VFEYETK 617

SEQ ID NO: 66      moltype = DNA length = 1989
FEATURE           Location/Qualifiers
misc_feature       1..1989
                  note = A synthetic coding sequence used for expression in
                  plant cells, CR-BRE1a.TIC7383_8.nno_Mc:1 which encodes a
                  protein comprising a C-terminal truncation relative to the
                  TIC7383 protein with an additional alanine codon is insert
                  after initiating methionine.
source             1..1989
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 66
atggctaacc agaaccagaa ccaaaaccag aacaagaacg agctgcagat cattgaaccc 60
tcctcggaca gcctcctgta cagccataac aactaccctg acgccactga cccaaacacc 120
gtgctcgagg ggcgcgaact caaggagtg gacggccatc agcaaggggt cgcgtctcgc ggccatctcc 240
gcactccagg gcccgagggc gacggccatc agcaaggggt cgcgtctcgc ggccatctcc 240
atctcgacga aggtcctagg gctcctagg gtcctcgttc ctgcgcagat tggccagctg 300
tggacgttca tccttaacgc actctggccc tcagacaaca ccagtgagg gaggttcatg 360
cgccacgtcg aggaactgat caaccagcgc atagcggact acgcgaggaa caaggccctg 420
gccgagctga cagggtctgg caacaaccta gacctctaca tcgaggcgct cgaggactgg 480
aagcggaaac ctacctccca caaagctaag gaccgcgtta tcgaccgctt ccgaatcgcg 540
gacggcctct tcgaggggta catgcccagc ttccgggtct cgggctacga ggtgccccct 600
ctcagcgtgt acgcggcggc ggctaatttg catctcctgc tgctcaggga ttgctcgatc 660
tacggaatcc agtggggctt ctcccagact aacgttaacg agaactacaa tcggcagatt 720
cggcacaccg ccgagtacgc taaccactgc acaacatggt atcagaccgg cctcgagagg 780
ctgcgcggca ccaatgcttt ctcatggatc aactacaacc gcttcgcggc cgagatgaca 840
ctgactgtct tggacgtctc ctccctcttc tccaactatg actaccgcac ttacccact 900
gaggtccggg ccgagcttac gcgggagatc tacacagacc ccatcggggt ccagaactcc 960
ccgctccctg gcgtggtccc caattggtac gactacgcga ggtcggtcgc ggaatcgaa 1020
aacatcgcca tacgagcacc tcggacagtc acgtggctga actcaaccac catctacacc 1080
gggagactga acggctacaa caatagcaac tactactggg cggggttcgc ccagaacttc 1140
tcagagacca actctggcag cagcttcaac ggcccagacc taggcgacct cactccgaac 1200
tacaggatag agaccctcga catggtcaac agggacatct acagcatcta ctcccgggtc 1260
gtgtcgcaat cctggcccat cggcaacgtc aagctcttcg gcgtaagctc ctctacctta 1320
agtttcgggg atctcaacaa caacagctcc ggcaccctgg tctacgagaa cccgactaac 1380
ttcagctccc agtacctaac cacagagttt ccaggggaga actccgagcg gccacacct 1440
acggactaca gtcaccgect aacctgcctt acgcggatcg gggccgggaa ctacggcctc 1500
gtgctttgtg cgggttgagc ttcttcgagt gtcgaagggg acaaccgcct acaaccgat 1560
aagattaccg agtaccgccg cgtcaagggg ttcaacttgg acggttcac agtggtgaaa 1620
gggactggtt tcaccggggg caactggcgt cgggtccagca gggtcaccgg gtcattccgc 1680
ctgaacgtat actcacctga tgtgcagact tatcgatga ggatcaggta cgctccccg 1740
cttgccaact cgacgtctgg gataagttcg acagacggcg gcatctcgtt cacgagtttc 1800
cctctgcctt ccactatcgg gagcatgcgg tccactgtgc cctacgaggg gttccgtgtc 1860
ctcgacatcc ccacacgggt caccgtagcc tcccaaagga actacaact catcttcgac 1920
atccttaacc cctcggttgg ggcgtgtgac atcgatcgta ttgagttcgt gcccggtggc 1980
tcttcatga 1989

SEQ ID NO: 67      moltype = AA length = 662
FEATURE           Location/Qualifiers
REGION            1..662
                  note = The amino acid sequence of
                  CR-BRE1a.TIC7383_8.nno_Mc:1 comprising a C-terminal
                  deletion, an additional alanine amino acid, and comprises
                  amino acids 1 through 661 of TIC7383.
source             1..662
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 67
MANQNQNQNQ NKNELQIIEP SSDSLLYSHN NYPYATDPNT VLEGRNYKEW LNKCTDNYTD 60
ALQGPEATAI SKGAVSAAIS ISTKVLGLLG VPFAAQIGQL WTFILNALWP SDNTQWEEFM 120
RHVEELINQR IADYARNKAL AELTGLGNL DLYIEALEDW KRNPTSQQAQ DRVIDRFRIA 180
DGLFEGYMPS PRVSGYEVPL LTVYAAAAANL HLLLLRDCSI YGIQWGFSTQ NVNENYNRQI 240
RHTAEYANHC TTWYQTGLER LRGTNAFWSI NYNRRFRREMT LTVLDVCSLF SNYDRTYPT 300
EVRAELETREI YTDPIGFQNS PLPGVVPNWY DYARSFAEIE NIAIRAPRTV TWLNSTTIYT 360
GRLNGYNNSN YYWAGFRQNF SETNSGSSFN GPDLDLTPN YRIETLDMVN RDIYSIYSRV 420
VSQSWPIGNV KLFGVSSSTL SFRDLNNNS GTLVYENPTN FSSQYLTTTEF PGENSERPTF 480
TDYSHRLTCL TRIGAGNYGL VLCAGWTSSS VERDNRLQPD KITQYPAVKG FNLDGFTVVK 540
GTGFTGGNWL RSSRVTSFGR LNVYSPSVQT YMRIRYASP LGNSTLGISS TDAGISFTSF 600
PLPSTIGSMP STVPYEAFRV LDIPITVTVS QRNYNFIFD ILNPSVGAVY IDRIEFVPVG 600

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-continued

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SS                                     662

SEQ ID NO: 68                       moltype = DNA length = 2010
FEATURE                             Location/Qualifiers
misc_feature                         1..2010
                                     note = A synthetic coding sequence used for expression in
                                     plant cells, CR-BRE1a.TIC7383_9.nno_Mc:1 which encodes a
                                     protein comprising a C-terminal truncation relative to the
                                     TIC7383 protein with an additional alanine codon insert
                                     after initiating methionine.
source                              1..2010
                                     mol_type = other DNA
                                     organism = synthetic construct

SEQUENCE: 68
atggctaacc agaaccagaa ccaaaaccag aacaagaacg agctgcagat cattgaaccc 60
tcctcggaca gctcctctgta cagccataac aactaccctg acgccaactga cccaaacacc 120
gtgctcgagg ggcgcaacta caaggagtag ctcaacaaat gcaccgacaa ctacaccgac 180
gcactccagg gcccgagggc gacggccatc agcaagggtg ccgtctccgc ggccatctcc 240
atctcgacga aggtcttagg gctcctaggg gtcccggtcg ctgacgcagat tggccagctg 300
tggacgttca tctttaacgc actctggccc tcagacaaca cccagtggga ggagttcatg 360
cgccacgtcg aggaactgat caaccagcgc atagcggact acgcgaggaa caaggccctg 420
gccgagctga cagggttggg caacaacctg gacctctaca tcgaggcgct cgaggactgg 480
aagcggaaacc ctacctccca acaagctaag gaccgcgtta tcgaccgctt ccgaatcgcg 540
cacggcctct tcgaggggta catgcccagc ttccgggtct cgggctacga ggtgcccctc 600
ctcacggtgt acgcggcggc ggctaatttg catctctctg tgctcaggga ttgctcgatc 660
tacggaatcc agtggggctt ctcccagact aacgtttaacg agaactacaa tcggcgagatt 720
cggcacaccg ccgagtagcg taaccactgc acaacatggt atcagaccgg cctcgagagg 780
ctgcgcggca ccaatgcttt ctcatggatc aactacaacc gcttcgcgcg cgagatgaca 840
ctgactgtct tggacgtctg ctccctcttc tccaaactatg actaccgcac ttacccact 900
gaggtccggg ccgagcttac cggggagatc tacacagacc ccacggtt ccagaactcc 960
ccgctccctg gcgtggtcac caattggtac gactacgcga ggtcgttcgc ggaaatcgaa 1020
aacatcgcca tacgagcacc tcggacagtc acgtggctga actcaaccac catctacacc 1080
gggagactga acggtacaaa caatagcaac tactactggg ccgggttcgc ccagaacttc 1140
tcagagacca actctggcag cagcttcaac ggcccagacc taggcgacct cactccgaac 1200
tacaggatag agaccctcga catggtcaac agggacatct acagcatcta ctcccgggtc 1260
gtgtcgcaat cctggcccat cggcaacgtc aagctcttcg gcgtaagctc ctctacctta 1320
agtttcggg atctcaacaa caacagctcc ggcaccctgg tctacgagaa cccgactaac 1380
ttcagctccc agtacctaac cacagagttt ccaggggaga actccgagcg gccaccttc 1440
acggactaca gtcaccgcct aacctgcctt acgcggatcg gggccgggaa ctacggcctc 1500
gtgctttgtg cgggttgtag ttcttcgagt gtcgaaaggg acaaccgcct acaaccgat 1560
aagattaccc agtacccgcg cgtcaagggt ttcaacttgg acggcttcac agtggtgaaa 1620
gggactggtt tcaccggggg caactggctg cgttcacgca gggtaaccgg gtcattccgc 1680
ctgaacgtat actcacctg tgtgcagact tatcggtatg ggtacaggtt cgctccccc 1740
cttggaactc cgacgcttgg gataagttcg acagacgcgc gcatctcgtt cagcagtttc 1800
cctctgcctc ccactatcgg gagcatgccg tccactgtgc cctacgagcg gttccgtgtc 1860
ctcgacatcc ccacacggt caocgtagcc tcccaaagga actacaactt catcttcgac 1920
atccttaacc cctcggttgg ggcggtgtac atcgatcgta ttgagttcgt gcccggtggg 1980
tcttcagtg tcgagtagca aaccaagtga 2010

SEQ ID NO: 69                       moltype = AA length = 669
FEATURE                             Location/Qualifiers
REGION                             1..669
                                     note = The amino acid sequence of
                                     CR-BRE1a.TIC7383_9.nno_Mc:1 comprising a C-terminal
                                     deletion, an additional alanine amino acid, and comprises
                                     amino acids 1 though 668 of TIC7383.
source                              1..669
                                     mol_type = protein
                                     organism = synthetic construct

SEQUENCE: 69
MANQNQNQ NKNELQIIEP SSDSLLYSHN NYPYATDPNT VLEGRNYKEW LNKCTDNYTD 60
ALQGPETA I SKGAVSAAIS ISTKVLGLLG VPFAAQIGQL WTFILNALWP SDNTQWEEFM 120
RHVEELINQ IADYARNKAL AELTGLGNL DLYIEALEDW KRNPSSQAK DRVIDRFRIA 180
DGLFEGYMP FRVSGYEVPL LTVYAAANL HLLLRDCSI YGIQWFSQT NVNENYNRQI 240
RHTAEYANH CTTWYQTGLER LRGTNAFSWI NYNRFRREMT LTVLDVCSLF SNYDYRTYPT 300
EVRaelTREI YTDPIGFQNS PLPGVVPNWY DYARSAEIE NIAIRAPRTV TWLNSTTIYT 360
GRLNGVNN N YYWAGFRQNF SETNSGSSFN GPDLDLTPN YRIETLDMVN RDIYSYISRV 420
VSQSWPIGN V KLFGVSSSTL SFRDLNNNSS GTLVYENPTN FSSQYLTFE PGENSERPTF 480
TDYSHRLTCL TRIGAGNYGL VLCAGWTSSS VERDNRLQPD KITQYPAVKG FNLDGFTVVK 540
GTGFTGGNWL RSSRVTSFRL LNVYSPSVQT YMRIRYASP LGNSTLGISS TDAGISFTSF 600
PLPSTIGSM P STVPYEAFRV LDIPITVTV SQRNYNFIFD ILNPSVGAVY IDRIEFVVPV 660
SSVFYEYETK

SEQ ID NO: 70                       moltype = DNA length = 1746
FEATURE                             Location/Qualifiers
misc_feature                         1..1746
                                     note = A coding sequence encoding the Tryptic core of the
                                     TIC7040HT protein as determined by mass spectrometry.
source                              1..1746

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mol_type = other DNA
organism = Brevibacillus laterosporus

SEQUENCE: 70
aattataaag agtgggctaaa taagtgcaca gataattata cagacgcttt acaggggtccc 60
gaagctactg ctatatcaaa aggagctgtc tctgctgcga tttctatcag caccaaagt 120
cttagtttat taggtgttcc gtttgcagct caaatcggac aactttggac cttcatattg 180
aatgcgttat ggccttcaga caatactcaa tgggaagagt tcatgagaca tgtagaagaa 240
ctcataaacc aacgaatagc cgattatgca agaagtaagg cacttgacga attaacgggt 300
ttaggtaata acttagattt atatatagaa gctcttgaag attggaacg aaatcctact 360
agtcaacaag cgaagaccg tgtaaaagat agattccgta tagcagatgg tttatttgaa 420
gcgtatatgc cttcatttag agtatcaggt tatgaagtac cattattaac agtgtatgca 480
gocgctgcaa atctccattt acttttatta agagattgct ctatttacgg aatccaatgg 540
ggatttagtc aaacgaatgt taacgagaat tacaatcgcc aaataagaca caccgcagag 600
tatgcaaatc attgtacaaa ttggtaacca actggttttag aaagattgcg aggtaccaat 660
gcttccagtt ggggccctta taatcgtttc cgaagagaaa tgacgttaac tgtattggat 720
atttgttcct tattttcaaa ttatgattat cgtagtattc cagcagaggt aagggcagag 780
cttacaagag aaattttatac ggaccagtt gtaagcacta gcttggtgat gaataatgca 840
ccatcattcg gagaataga aaatctagca attagggcgc caagaacgt tacttggtta 900
aattctacaa gaatttctac agggaccttg cagggctgga gtggttctaa cagatattgg 960
gcagctcaca tgcaaaactt ttcagaaaac aattcaggaa atataggatt tgacggtcct 1020
ctctatgggt cgacggtagg tactattatt cgtgatgata attacgaaat ggtgaaccga 1080
gatatttaca ccattacttc aagggctggt gccgccttt ggccaactgg tcaaatgtg 1140
ttgggagtcg cttcggttag atttacttta agaaatctta acaataatct tacacaggcg 1200
ctggtgatg agaaccaaat aagttcaaagt tttaaatagg caactttaac tcgtgaatta 1260
cctggggaaa actcagatag gccaaacttca agcgactata gtcatagact aacgtctatt 1320
acagcttttc gagctggaag taatgggacg attccggttt ttggatggac atctataagt 1380
gttaatcgtg acaatatact tgagcgaaac aaaataaac aattcccagg cgttaagtca 1440
cacactctca acaattgtca agtattaga ggtactggat ttacaggagg agactggtt 1500
agaccaaata ataattgttc atttagatta actattactt cattctcgag ccaactttac 1560
cgaatccgct tacgttatgc ttccgcagca aatacttctt tgcgtatata ttcttctgca 1620
gccggtattt cttccacaac cgttccgctt acctcaacaa taacatcact gccacaaact 1680
gctgtaccat atgaagcttt tagagttata gatttaccta ttacttttac aacagctacc 1740
caaaagt 1746

SEQ ID NO: 71      moltype = AA length = 582
FEATURE           Location/Qualifiers
REGION            1..582
                  note = The amino acid sequence of the Tryptic core of the
                  TIC7040HT protein as determined by mass spectrometry, and
                  comprises amino acids 43 through 624 of TIC7040HT.
source            1..582
                  mol_type = protein
                  organism = Brevibacillus laterosporus

SEQUENCE: 71
NYKEWLKCT DNYTDALQGP EATAISKGA SAAISISTKV LSLLGVPFAA QIGQLWTFIL 60
NALWPSDNTQ WEEFMRHVEE LINQRIADYA RSKALAEITG LGNNLDLYIE ALEDWKRNP 120
SQQAKDRVKD RFRIADGLFE AYMPSPRVSG YEVPLLTVYA AAANLHLLLL RDCSIYGIQW 180
GFSQTNVVEN YNRQIRHTAE YANHCTTWYQ TGLERLRGTN ASSWVPYNRF RREMTLTVD 240
ICSLFSNYDY RSYPAEVRAE LTREIYTDPV VSTSLWMNNA PSFGEIENLA IRAPRTVTWL 300
NSTRISTGTL QGWSGSNRYW AAHMQNFSET NSGNIGFDGP LYGSTVGTII RDDNYEMVNR 360
DIYTTITSEV AALWPTGQIV LGVASARFTL RNLNNNLQQA LVYENPISS FNRSTLTREL 420
PGENSDRPTS SDYSHRLTSI TAFRAGSNGT IPVFGWTSIS VNRDNLERN KITQFPGVKS 480
HTLNNCQVVR GTGFTGGDWL RPNNGSFRL TITSPSSQSY RIRLRYASAA NTSRLISSA 540
AGISSTTVPL TSTITSLPQT AVPYEAFRVI DLPITFTTAT QS 582

SEQ ID NO: 72      moltype = DNA length = 1791
FEATURE           Location/Qualifiers
misc_feature      1..1791
                  note = A coding sequence encoding the Chymotryptic core of
                  the TIC7040HT protein as determined by mass spectrometry.
source            1..1791
                  mol_type = other DNA
                  organism = Brevibacillus laterosporus

SEQUENCE: 72
aaagagtggc taaataagtg cacagataat tatacacagc ctttacaggg tcccgaagct 60
actgctatat caaaggagc tgtctctgct gcgatttcta tcagcaccaa agttcttagt 120
ttattagggt ttccggttgc agctcaaatc ggacaacttt ggaccttcatt attgaatgcg 180
ttatggcctt cagacaatac tcaatgggaa gagttcatga gacatgtaga agaactcata 240
aaccaacgaa tagccgatta tgcaagaagt aaggcacttg cagaattaac gggtttaggt 300
aataacttag atttatatat agaagctctt gaagattgga aacgaaatcc tactagtcaa 360
caagcgaaaag accgtgtaaa agatagattc cgtatagcag atggtttatt tgaagcgtat 420
atgccttcatt ttagagtatc aggttatgaa gtaccattat taacagtgtg tgcagccgct 480
gcaaatctcc atttactttt attaaagagt tgctctattt acggaatcca atggggattt 540
agtcaaacga atgttaacga gaattacaat cgccaaataa gacacaccgc agagtatgca 600
aatcattgtg caacttggtg ccaaaactgg ttgaaagat tgcgaggtac caatgcttcc 660
agttgggtcc cttataatcg ttccgaaga gaaatgacgt taactgtatt ggatatttgt 720
tccttatttt caaattatga ttatcgtagt tatccagcag aggtaagggc agagcttaca 780
agagaaattt atacggaccc agttgtaagc actagcttgt ggatgaataa tgcaccatca 840
ttcggagaaa tagaaaatct agcaattagg gcgccaagaa ccgttactgt gttaaattct 900

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acaagaattt ctacagggac cttgcagggc tggagtgggt ctaacagata ttgggcagct 960
cacatgcaaa acttttcaga aaccaattca ggaaatatag gatttgacgg tcctctctat 1020
gggtcgacgg taggtactat tattcgtgat gataattacg aaatggtgaa ccgagatatt 1080
tacaccatta cttcagaggg tgttgccggc ctttggccaa ctggtcaaat tgtgttgagg 1140
gtcgcttcgg ctagatttac ttttaagaaat ctttaacaata atcttacaca ggcgctgggtg 1200
tatgagaacc caataagttc aagttttaat aggtcaactt taactcgtga attacctggg 1260
gaaaactcag ataggccaac ttcaagcgac tatagtcata gactaacgtc tattacagct 1320
tttcgagctg gaagtaatgg gacgattccg gtttttggat ggacatctat aagtgttaat 1380
cgtgacaata tacttgagcg aaacaaaata acacaattcc caggcgttaa gtcacacact 1440
ctcaacaatt gtcaagtagt tagaggtact ggatttacag gaggagactg gttgagacca 1500
aataataatg gttcatttag attaactatt acttcattct cgagccaatc ttaccgaatc 1560
cgcttacgtt atgcttcgcg agcaataact tctttgcgta tatcttcttc tgcagccggt 1620
atcttctcca caaccgttcc gcttacctca acaataacat cactgccaca aactgctgta 1680
ccatagaaag cttttagagt tatagattta cctattactt ttacaacagc taccctaaag 1740
aattatactt ttgattttgt tctccaaaat ccatcaaacg caaatgtatt c 1791

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SEQ ID NO: 73 moltype = AA length = 597
FEATURE Location/Qualifiers
REGION 1..597
 note = The amino acid sequence of the Chymotryptic core of
 the TIC7040HT protein as determined by mass spectrometry,
 and comprises amino acids 45 through 641 of TIC7040HT.
source 1..597
 mol_type = protein
 organism = Brevibacillus laterosporus

SEQUENCE: 73

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KEWLNKCTDN YTDALQGPEA TAISKGAWSA AISISTKVLV LLGVPFAAQI GQLWTFILNA 60
LWPSDNTQWE EFMRHVERLI NQRIADYARS KALAEITGLG NNLDLYIEAL EDWKRNPSTQ 120
QAKDRVKDRF RIADGLFEAY MPSFRVSGYE VPLLTVYAAA ANLHLLLLRD CSYGIQWGF 180
SQTNNVNNYN RQIRHTAAYA NHCTTWYQTG LERLRGTNAS SWVPYNRFRF EMTLTVLDDC 240
SLFSNYDYRS YPAEVRALRT REIYTDVVS TSLWMNNAPS FGEIENLAIR APRTVTWLNS 300
TRISTGTLQG WSGSNRYWAA HMQNFSETNS GNIGFDGPLY GSTVGTIIRD DNYEMVNRDI 360
YTITSEAVAA LWPTGQIVLG VASARFTLRN LNNNLTAQAL YENPISSSFN RSTLTRELPG 420
ENSDRPTSSD YSHRLTSITA FRAGSNGTIP VFGWTSISVN RDNILERNKI TQFPGVKSHT 480
LNNCQVVRGT GFTGGDWLRP NNNGSFRLTI TSFSSQSYRI RLRYSAAANT SLRISSSAAG 540
ISSTTVPLTS TITSLPQTAV PYEAFRVIDL PITFTTATQS NYTFDFVLQN PSNANVF 597

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SEQ ID NO: 74 moltype = DNA length = 1842
FEATURE Location/Qualifiers
misc_feature 1..1842
 note = A coding sequence encoding the Tryptic core of the
 TIC7383 protein as determined by mass spectrometry.
source 1..1842
 mol_type = other DNA
 organism = Brevibacillus laterosporus

SEQUENCE: 74

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gataattata cagacgcttt acagggtccc gaagctactg ctatatcaaa aggagctgtc 60
tctgctgcga tttctatcag caccaaaagt cttggtttat tagtgtgtcc atttgcagct 120
caaatcgggc aactttggac ctctcatatt aatgcgttat ggccttcaga caataactca 180
tggaagagat tcatgagaca tgtagaagaa ctcataaacc aacgaatagc cgattatgca 240
agaaataagg cacttgacga attaacgggg ttaggtaata acttagattt atatatagag 300
gctcttgaag attggaacgg aatccctact agtcaacaag cgaaagaccg tgtaatatag 360
agattccgta tagcagatgg tttatttgaa gggatatatg cttcatttag agtatcaggt 420
tatgaagtac cattattaac agtgatgca gccgctgcaa atctccattt acttttatta 480
agagattgct ctatttcagg aatccaatgg ggatttagtc aaacgaatgt taacgagaat 540
tacaatcgcc aaataagaca caccgcagag tatgcaaatc attgtacaac ttggtaccaa 600
actggtttag aaagattgcg aggtaccaat gcttccagtt ggatcaatta taatcgattc 660
cgtagagaaa tgacgttaac cgtattggat gtttgttcat tattttcaaa ttatgattat 720
cgtacgtatc caacagaggt aagagcagag cttacgagag aaattttatac ggaccaata 780
ggttttcaaa atagtccctc tcttgccgtt gttcctaatt ggtacgatta cgcacgatcg 840
ttcgcagaga tagaaaatat agccattcga gcgccacgaa ctgttacttg gttaaattct 900
acaactattt atacaggtag attaaatggc tataacaata gtaattatta ttgggcaggt 960
ttcaggcaaa atttttcaga aaccaattca ggcagttcat ttaacggtcc tgacttaggy 1020
gatttaacac ctaattatcg tatagaaca ttggatatgg tgaatcggga tatttactcc 1080
atattattca gagttgtttc acaatcttgg ccaattggca acgttaaatt gtttggagtc 1140
tcttcatcta ctctttcatt cagagatctt aacaataatt cttcagggac gctggtatat 1200
gaaaaccgca caaattttag tagccaatat ctaactaccg aatccctgg ggaaaactca 1260
gaaagacca cttttaccga ttatagtcac agactaaact gtctcacacg tattggggct 1320
ggaaattatg gattggttct atgcgccggc tggacatcta gtagtgttga gcgtgacaac 1380
agactccagc cggacaaaaa aacacaatac ccggctgtta aaggattcaa cctcgatggt 1440
tttacagtag taaaaggtac tgggtttaca gggggaattt ggttgagatc tagtcgtgtt 1500
acaggtagct ttagactaaa tgtttattca ccgtctgtcc aaacttatcg catgcgtata 1560
cgttatgctt ctccactggg aaattctact ctaggatat cttctactga tgcgtgtatt 1620
agtttcacaa gttttccact tccctcaaca ataggatcaa tgccatcaac tgtaccatac 1680
gaagctttta gagttctaga tatacctatc actgttacag tagctagtca aagaaattat 1740
aattttattt tcgatattct aaatccatca gtccggagcag tatacatgta cagaattgaa 1800
ttcgttccag ttgggtcttc tgtatttgaa tatgaacca ag 1842

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SEQ ID NO: 75 moltype = AA length = 614

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FEATURE Location/Qualifiers
 REGION 1..614
 note = The amino acid sequence of the Tryptic core of the TIC7383 protein as determined by mass spectrometry, and comprises amino acids 55 through 668 of TIC7383.
 source 1..614
 mol_type = protein
 organism = Brevibacillus laterosporus

SEQUENCE: 75

DNYTDALQGP	EATAISKAV	SAAISISTKV	LGLLGVPFAA	QIGQLWTFIL	NALWPSDNTQ	60
WEEFMRHVEE	LINQRIADYA	RNKALAEITG	LGNNLDLYIE	ALEDWKRNP	SQQAQKDRVID	120
RFRIADGLFE	GYMPSFRVSG	YEVPLLTVYA	AAANLHLLLL	RDCSIYGIQW	GFSQTNVNEN	180
YNRQIRHTAE	YANHCTTWYQ	TGLERLRGTN	AFSWINYNRF	RREMTLTVLD	VCSLFSNYDY	240
RTYPTVEVRAE	LTRREIYTDPI	GFQNSPLPGV	VPNWYDYARS	FAEIENIAIR	APRTVTWLN	300
TTIYTGRNLG	YNNSNYYWAG	FRQNFSETNS	GSSFNGPDLG	DLTPNYRIET	LDMVNRDIYS	360
IYSRVVSQSW	PIGNVKLFGV	SSSTLSFRDL	NNSSGTLVY	ENPTNFSSQY	LTTEFPGENS	420
ERPTFTDYSH	RLTCLTRIGA	GNVGLVLCAG	WTSSSVERDN	RLQPDKITQY	PAVKGFNLDG	480
FTVVKGTTGT	GGNWLRRSRV	TGSFRLNVYS	PSVQTYRMRI	RYASPLGNST	LGISSTDAGI	540
SPTSFPPLPST	IGSMPSTVPY	EAPRVLDIPI	TVTVASQRNY	NFIPDILNPS	VGAVYIDRIE	600
FVPVGSVFPE	YETK					614

SEQ ID NO: 76 moltype = DNA length = 1836
 FEATURE Location/Qualifiers
 misc_feature 1..1836
 note = A synthetic coding sequence used for expression in a plant cell, encoding a CR-BREla.TIC7040_14.nno_Mc:1 protein having an N-terminal and C-terminal truncation relative to the TIC7040HT protein, with an additional alanine codon.
 source 1..1836
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 76

atggctaccg	acaactacac	cgacgcgctc	cagggccctg	aggccaccgc	catcagcaag	60
ggcgcggtgt	cagccgcgat	ctccatcagc	acaaagggtg	tctcggtact	ggcgctgccc	120
tccgcggcgc	agatcggcca	gctctggacc	ttatccctga	acgcacctatg	gccacgcgac	180
aacacccagt	gggaggagtt	tatgcgccac	gtggagggaac	tcattaacca	gcgaattgct	240
gactacgcca	ggagcaaggg	cctggccgag	ctgacgggtc	ttggcaacaa	cctcgacctc	300
tacatcgagg	cccttgaaga	ctggaagcgg	aaccgcacca	gccaacaggc	caaagaccgt	360
gtgaaggacc	ggttcggagt	cgcgcacggg	ctggtcgagg	cgtacatgcc	ctctttccgc	420
gtcagcgggt	acgaggtgcc	actgttgacc	gtttacgcgg	cagccgctaa	cctccacttg	480
ctgctcctgc	gggattgctc	catctacggt	atccagtggt	gcttcagcca	gacaaacgtc	540
aacgagaact	acaaccgaca	gatccggcac	acggccgagt	acgcaaacca	ctgcacgaca	600
tggatcaga	cagggtctga	agcccttcgt	ggcaccaacg	cctcgctcatg	ggtgccgtac	660
aaccgggttc	gccgagagat	gacccctacc	gttcttgaca	tctgctcact	gttcagcaac	720
tacgactacc	ggtcgtaccc	tgccggaggt	cgggcccgaac	tgacgcggga	gatctacact	780
gaccgcgttg	tcagcacctc	gttgtggatg	aacaacgcgc	ccagcttcgg	tgagatcgag	840
aacctcgcca	taagagcacc	acgcaccgtc	acttggtcca	acagcacccg	gatctccacg	900
ggcaccttac	agggtcggag	cggtctccaa	cgatactggg	cagcgcacat	gcagaacttc	960
tccgagacta	actccggcaa	catcggttc	gacggcccgc	tctacggcag	caccgtcggg	1020
accatcatcc	gggacgacaa	ctatgagatg	gtcaacaggg	acatctacac	gattacctcg	1080
gaggccgttg	ctgcctgtgt	gccaaaccgt	caaatagtgt	tgggagtcgc	ctccgcccgc	1140
tttacgctgc	gtaacctgaa	caacaacctt	acccaggcgc	tcgtgtacga	gaacccgatc	1200
tcagactcct	tcaataggct	caccctcacc	agggagctgc	cgggtgagaa	cagcgacagg	1260
ccgacgtcc	cggactacag	tcaccggctc	acctcgatca	cagcgttcgc	agccgggagc	1320
aacggaacga	tcccagtcct	cggatggacc	agcatcagcg	tcaaccgcga	caacatcctg	1380
gaacgcaaca	agatcaactca	gttccctgga	gttaagtctc	acacgctcaa	caactgccaa	1440
gtggtgcgtg	gcacaggggt	taccggaggc	gactgggttc	gcccgaacaa	caatgggtca	1500
ttccgggtga	ccattaccag	cttctcatcg	caateatacc	gcateccggt	acgggtacgcc	1560
tcggcgcgga	acactagcct	tcgcatctcc	tcttcagccg	ccgggatcag	ctcgacaacc	1620
gtcccgctga	ccagcacgat	cacgagcctg	ccacaaaactg	ctgtgcctta	cgaggctttc	1680
agagtgatcg	acctgcccac	taccttcacc	acggcaaccc	aatcgaaacta	cactttcgac	1740
ttcgtgttac	agaatccgag	caacgccaac	gtcttcatcg	accggttcga	gttcgtgccc	1800
atcggcgggga	gcctctccga	gtacgagacc	aagtga			1836

SEQ ID NO: 77 moltype = AA length = 611
 FEATURE Location/Qualifiers
 REGION 1..611
 note = The amino acid sequence of the CR-BREla.TIC7040_14.nno_Mc:1 protein, wherein an additional alanine amino acid is inserted immediately following the initiating methionine, and comprises amino acids 52 through 660 of TIC7040HT.
 source 1..611
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 77

MATDNYTDAL	QGPEATAISK	GAVSAAISIS	TKVLSLLGVP	FAAQIGQLWT	FILNALWPSD	60
NTQWEEFMRH	VEELINQRIA	DYARSKALAE	LTGLGNNLDL	YIEALEDWKR	NPTSQAQKDR	120

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VKDRFRIADG LFEAYMPSFR VSGYEVPLLT VYAAAAANLHL LLLRDCSIYG IQWGFSTQTNV 180
NENYNRQIRH TAEYANHCTT WYQTGLERLR GTNASSWVPY NRFRREMTLT VLDICSLFSN 240
YDYSYPAEV RAEITREIYT DPVVSTSLWM NNAPSFGEIE NLAIRAPRTV TWLNSTRIST 300
GTLQGWSGSN RYWAAHMQNF SETNSGNIGF DGPLYGSTVG TIIRDNYEM VNRDIYTITS 360
EAVAALWPTG QIVLGVASAR FTLRNLNNL TQALVYENPI SSSFNRSTLT RELPGENSDR 420
PTSSDYSHRL TSITAFRAGS NGTIPVFGWT SISVNRDNL ERNKITQFPG VKSHTLNNCQ 480
VVRGTGTGG DWLRPNNGS FRLTITSFSS QSYRIRLRYA SAANTSLRIS SSAAGISST 540
VPLTSTITSL PQTAVPYEAF RVIDLPITFT TATQSNYTFD FVLQNPSNAN VFIDRFEFVP 600
IGGSLSEYET K 611

SEQ ID NO: 78      moltype = DNA length = 1980
FEATURE            Location/Qualifiers
misc_feature        1..1980
                    note = A synthetic coding sequence used for expression in a
                        plant cell, encoding a CR-BRE1a.TIC7381_2.nno_Mc:1 protein
                        with a C-terminal truncation, and an additional alanine
                        codon after the initiating methionine codon.
source              1..1980
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 78
atggctaacc agaaccagaa caagaacgag ctacagataa tcgagccgtc ctcggtattcc 60
ttgctgtact cgcacaacaa ctacccgtag gccaccgtag cgaacacggg gctagagggc 120
cgcaactaca aggagtggtc gaataagtgc acagacaact acaccgatgc actgcaaggc 180
ccagaggcaa cggctatcag caagggtgcc gtgtcggccg ctatctctat cagcaccaaa 240
gtcttgagcc ttctgggagt gcccttcgca gctcagattg ggcaattgtg gacgtttatt 300
ctgaacgctc tttggccctc ggacaacacc cagtgaggag agttcatgcg ccatgtcgag 360
gagctcatca accagcgtat cgccgactac gcgcggagta aggcgctcgc cagcctcacg 420
ggcctgggca acaacctega cctctcctc gaagctcttg aggactggaa gcgcaaccgc 480
acctctcaac aagccaagga cgggtgaag gaccgcttcc gcctcgccga cggcctcttc 540
gaagcgtaca tgcctagctt ccgcgtgagc ggctatgaag tgccattgct caccgtatat 600
gcggcgccgg ccaacctgca cctcctcctc ctcccgagct gctctatcta cggcattcag 660
tggggtttct cccagaccaa tgtcaacgag aactataatc ggcagatccg tcacaccgcc 720
gagtagcgga accactgcac gacatggtag cagacgggcc tggagcgact gcgcggcacg 780
aacgcctcaa gctgggtgcc gtacaacctt ttcaggagag agatgactct caccgtgctc 840
gacatctgca gtctgttctc caactatgac tacaggagtt acccgccaga ggtgcgggct 900
gagctgacca gggagatata cacggaccgc gtggtcagca cctctctgtg ggtgaacaac 960
gcgcgctcct tcggcgaaat cgagaatctg gcaatccgag cgccacgcac cgtgacatgg 1020
ctcaattcga cgcgtattag cactggcaca ctccaaggtt ggtctgggtc caacaggtag 1080
tgggcccgcc acatcgagaa cttctcggag actaactccg gcaacatagg ctccgacgce 1140
ccgctgtacg ggtccactgt ggggtacaatc attcgcgatg acaattacga gatgggtcaac 1200
cgggacattt acaccatcac ttccggaggcg gttgcggctt tgtggccacg gggccagatt 1260
gtgctgggag tggcttcgca aagattcacc cttcgaaacc tcaacaacaa cctgaccag 1320
gcgctggtct acgagaaccc gatataaagc agtttcaacc ggtcaactct cacacgcgag 1380
ttgcggggcg agaacagcga caggcccacc tcttcggact actcccacgc tctcaactagc 1440
atcaactgctg tccgcgcagg cagtaacggg acgattcctg tgttcggctg gaccagtagc 1500
tccgtcaaca gggacaacat actggagcgg aacaagatca cccagttccc tggcggttaa 1560
tcccataccc tgaacaattg ccaggctcgt cgcggtacag gctttaccgg cggcgactgg 1620
cttcggccta acaacaacgg gtccgttcga ttgactatca cctcgttcag ctcaacagag 1680
tacagaatca ggctgcggta cgcgtccgca gcgaatacct ccctgcgaat ctctgtcttc 1740
cgggccggga tctcgtcaac aaccgtccca ctccgctcca cgatcaactag cctcccgcag 1800
acagccgtcc catcagagcg gttccgggtg atcgacctgc caatcacggt caccaccgcg 1860
acgcagtcta actacacttt gacttcgtc ctccagaatc cttccaacgc caacgtgttc 1920
atcgaccgct tcgagtttgt cccgattgga gggtcgctct cagagtacga gacgaagtga 1980

SEQ ID NO: 79      moltype = AA length = 659
FEATURE            Location/Qualifiers
REGION              1..659
                    note = The amino acid sequence of the
                        CR-BRE1a.TIC7381_2.nno_Mc:1 protein with a C-terminal
                        truncation, an additional alanine residue, and comprises
                        amino acids 1 through 658 of TIC7381.
source              1..659
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 79
MANQNQNKNE LQIIEPSSDS LLYSHNNYPY ATDPNTVLEG RNYKEWLNKC TDNYTDALQG 60
PEATAISKGA VSAAISISTK VLSLLGVPPA AQIGQLWTFI LNALWPSDNT QWEEFMRHVE 120
ELINQRIADY ARSKALAEIT GLGNLNDLYI EALDQWKRNP TSQQAQDRVK DRFRIADGLF 180
EAYMPSFRVS GYEVPLLTIVY AAAANLHLLL LRDCSIYGIQ WGFSTQTNVNE NYNRQIRHTA 240
EYANHCTTWY QTGLERLRGT NASSWVPYNR FRREMTLTVL DICSLFSNYD YRSYPAEVRA 300
ELTREIYTDP VVSTSLWVNN APSFGEIENL AIRAPRTVTW LNSTRISTGT LQGWSGSNRY 360
WAHMQNFSE TNSGNIGFDG PLYGSTVGTI IRDDNYEMVN RDIYITISEA VAALWPTGQI 420
VLGVASARTF LRNLNNLTQ ALVYENPISS SFNRSTLTRE LPGENSDRPT SSDYSHRLTS 480
ITAFRAGSNG TIPVFGWTSI SVNDRNILER NKITQFPGVK SHTLNNCQVV RGTGFTGGDW 540
LRPNNGSFR LTITSFSSQS YRIRLRYASA ANTSLRISS AAGISSTVLP LASTITSLPQ 600
TAVPYEAFRV IDLPITFTTA TQSNYTFDFV LQNPSNANVF IDRFEFVPIG GSLSEYETK 659

SEQ ID NO: 80      moltype = DNA length = 1836

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FEATURE Location/Qualifiers
misc_feature 1..1836
note = A synthetic coding sequence used for expression in plant cells encoding a CR-BREla.TIC7381_3.nno_Mc:1 with an N-terminal and C-terminal truncation and additional alanine codon after the initiating methionine relative to TIC7381.
source 1..1836
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 80

atgggctacag	acaactacac	cgatgcactg	caaggcccag	aggcaacggc	tatcagcaag	60
ggtagccgtgt	cgggccgctat	ctctatcagc	accaaagtct	tgagccttct	gggagtggcc	120
ttcgagcgtc	agattgggca	attgtgggac	tttattctga	acgctctttg	gccctcggac	180
aacacccag	gggaggagtt	catgcgccat	gtcagaggagc	tcatcaacca	gcgtatcgcc	240
gactacggcg	ggagtaaggc	gctcgccgag	ctcacggggc	tgggcaacaa	cctcgatctg	300
tacatcgaa	ctcttgagga	ctggaagcgc	aaccgcacct	ctcaacaagc	caaggaccgg	360
gtgaaggacc	gcttcgcgat	cgcgcagcgc	ctcttcgaag	cgtacatgcc	tagcttcgcg	420
gtgagcggtc	atgaagtggc	attgtcacc	gtatatgcgg	cggcgggcaa	cctgcacctc	480
ctctctctcc	gcgactgctc	tatctacggc	attcagtggg	gtttctccca	gaccaatgtc	540
aacgagaact	ataatcgcca	gatccgtcac	accgcccagt	acgcgaacca	ctgcacgaca	600
tggtagcaga	cgggcttgga	gcgactgcgc	ggcacgaacg	cctcaagctg	ggtagccgtac	660
aaccgtttca	ggagagagat	gactctcacc	gtgctcgaca	tctgcagtct	gttctccaac	720
tatgactaca	ggagttaccc	ggcagagggt	cgggctgagc	tgaccaggga	gatatacacg	780
gacccggtgg	tcagcacctc	tctgtgggtg	aacaacgcgc	cgtccttcgg	cgaatcgag	840
aatctggcaa	tccgagcgcc	acgcacgcgt	acatggctca	attcgacggc	tattagcact	900
ggcacactcc	aaggttggtc	tgggtccaac	aggtagtggt	cgcgccacat	gcagaacttc	960
tcggagacta	actccggcaa	cataggcttc	gacggccgcg	tgtacgggtc	cactgtgggt	1020
acaatcattc	gcgatgacaa	ttacgagatg	gtcaaccggg	acatttacac	catcacttcg	1080
gaggcggttg	cggtttgtg	gccacggggc	cagattgtgc	tgggcgtggc	ttccgcaaga	1140
ttcacccctc	gaaacctcaa	caacaacctg	accagggcgc	tggtctacga	gaacccgata	1200
tcaagcagtt	tcaaccggtc	aactctcaca	cgcgagttgc	cgggcgagaa	cagcgacagg	1260
cccacctctt	cggactactc	ccaccgtctc	actagcatca	ctgcgttcgc	cgcaggcagt	1320
aacggtagca	ttcctgtggt	cggtcggacc	agtatctccg	tcaacaggga	caacatactg	1380
gagcggaaca	agatcaccca	gttccttggt	gttaagtccc	ataccctgaa	caattgccag	1440
gtcgtcccg	gtacaggctt	taccggcggc	gactggcttc	ggcctaacaa	caacgggtcg	1500
ttccgattga	ctatcacctc	gttcagctca	cagagttaca	gaatcaggct	gcggtacgcg	1560
ttccgagcga	atacctccct	gcgaatctcg	tcttcggcgg	cggggatctc	gtcaacaacc	1620
gtccactcgc	cctccacgat	cactagcctc	cgcgagacag	ccgtcccata	cgaggcggtc	1680
cgggtgatcg	acctgccaat	acggttcacc	accgcgacgc	agtctaacta	cactttcgac	1740
ttcgtctctc	agaatccttc	caacgccaac	gtgttcacgc	accgcttcga	gtttgtcccg	1800
attggaggtt	cgctctcaga	gtacgagacg	aagtga			1836

SEQ ID NO: 81 moltype = AA length = 611
FEATURE Location/Qualifiers
REGION 1..611
note = The amino acid sequence of the CR-BREla.TIC7381_3.nno_Mc:1 protein, comprising an N-terminal and C-terminal truncation, an additional alanine residue, and comprises amino acids 50 through 658 of TIC7381.
source 1..611
mol_type = protein
organism = synthetic construct

SEQUENCE: 81

MATDNYTDAL	QGPEATAISK	GAVSAAISIS	TKVLSLLGVP	FAAQIGQLWT	FILNALWPSD	60
NTQWEEFMRH	VEELINQRIA	DYARSKALAE	LTGLGNLNDL	YIEALEDWKR	NPTSQQAKDR	120
VKDRFRIADG	LFEAYMPSFR	VSGYEVPLLT	VYAAANLHL	LLLRDCSIYG	IQWGFSTQTNV	180
NENYNRQIRH	TAEYANHCTT	WYQTGLERLR	GTNASSWVPY	NRFRREMTLT	VLDICSLFSN	240
YDYSYPAEV	RAELTREIYT	DPVVSTSLWV	NNAPSFGEIE	NLAIRAPRTV	TWLNSTRIST	300
GTLQGWSGSN	RYWAAHMQNF	SETNSGNIGF	DGPLYGSTVG	TIIRDNYEM	VNRDIYTITS	360
EAVAALWPTG	QIVLGVASAR	FTLRNLNNNL	TQALVYENPI	SSSFNRSTLT	RELPGENSDR	420
PTSSDYSHRL	TSITAFRAGS	NGTIPVFGWT	SISVNRDNIL	ERNKITQFPG	VKSHTLNNCQ	480
VVRGTGFTGG	DWLRPNNNGS	FRLTITSFSS	QSYRIRLRYA	SAANTSLRIS	SSAAGISSTT	540
VPLASTITSL	PQTAVPYEAF	RVIDLPITFT	TATQSNYTFD	FVLQNPNSAN	VFIDRFEFVP	600
IGGSLSEYET	K					611

SEQ ID NO: 82 moltype = DNA length = 1833
FEATURE Location/Qualifiers
misc_feature 1..1833
note = A synthetic coding sequence used for expression in plants encoding a CR-BREla.TIC7382_3.nno_Mc:1 protein with an N-terminal and C-terminal truncation, and an additional alanine codon after the initiating methionine relative to TIC7382.
source 1..1833
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 82

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atggccacta acaattacac cgaacgcgtc cagtctccgg aagccacagc catttcgaag 60
ggggcggtct cggcgcgcat cagcatcagc accaagggtc tcggcctcct cggcggtgccc 120
ttcgcggtctc agataggaca gctctggaca ttcattctga acgcccctgtg gccgtccgac 180
aacacccagt gggaggagtt catgcggcac gtccgaggagc tcatcaacca gcggatcgca 240
gactatgccc gtaacaaggg cctggccgag ctgacgggac tcggcaacaa cctcgacctc 300
tacatcgagg cccctggatga ctggaagagg aacccccacca gccaggaggc caagactcgc 360
gtcatcgacc gcttcgcgat cgtcgacggg ctgttcgagg cctacattcc gagcttcgcc 420
gtgagcggtc atcaagtcca gctgctcact gtctacgccc ctgcagcaaa cctccacctc 480
ctgctgctgc gcgactccac gatctacgga atcgactggg gccctttccca gactaacgtc 540
aacgataact acaacccggca gattcgtctc accgcgacgt acgctaacca ctgtacaacg 600
tggtaaccaga caggactgga gcggctccgc gggagcaacg cgtcttcctg ggtgacctac 660
aaccgtttcc ggagagagat gactctcaca gtcttggaca tctgttctct gttcagcaac 720
tacgactatc gcagctaccc ggccgaggtt cgcggcgaga ttacgaggga aatctacaca 780
gacctgtgc gcgtcggttg ggttgacagc gcgcttctct tcggcgagat cgagaacctg 840
gcgatccggg cgccacggac tgtgacctgg ctgaaactca cccggatctc caccggcacc 900
ttaagcggtc ggtcggtctc taaccggtag tggcgggccc acatgcagaa tttctccgag 960
acaaactcgg gtaacatcgg ctctgatggg ccgctgtacg ggagcacagt gggcaccatc 1020
caccgcactg atgactacga catgggcaac cgggacatct acacgataac gtccgaggcc 1080
gttctcggtc tgtggcccac gggccagcgg gtctctggag tcgcatctgc gcgtttccac 1140
ctcagggaacc tattcaataa cctgacccag gtgctgggat acgagaatcc tattagctcc 1200
aacttcggct cttcaacgct aacccacgag ttatccggcg agaacagcga taggcccacc 1260
agcagcgatt actcacatcg actcacaagc atcacgggat tccgcgcagg agcaaatggt 1320
accgttcccg tctttgggtg gaccagcgcc acagtcgacc ggaacaacat catcgagcgc 1380
aacaagatca cgcagttccc aggcgtcaaa tcgcacacgc taacaacact ccagggtggtg 1440
atggggaaccg gtttcacggc aggtgactgg ttaagaccaa ataacaacgg cactttccgc 1500
ctcactatta cgagcttctc ctcccaaagt taccgcaccc gccttcgcta cgccacgtca 1560
gtgggggaaca cctcgctggt gatctctccc tcagacgccc ggatcagcag caccacgata 1620
ccactaacat ccactaatcc ctcatcgcc cagacgggtc cgtatcaggc cttcagagtg 1680
gtggaccttc ctattacgtt cagcagcgcc acgacccagc gcaactacac attcgatttc 1740
cggttgcaga acccgagtaa cgcgaatgtc ttcacgatgc gtttcgagtt cgtcccagata 1800
ggcggtctccc tttcggagta cgaaacgaag tga 1833

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SEQ ID NO: 83      moltype = AA length = 610
FEATURE           Location/Qualifiers
REGION            1..610
note = The amino acid sequence of the
CR-BREla.TIC7382_3.nno_Mc:1 protein, with an N-terminal
and C-terminal truncation, an additional alanine residue,
and comprises amino acids 52 through 659 of TIC7382.
source            1..610
mol_type = protein
organism = synthetic construct

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SEQUENCE: 83
MATNNYTDAL QSPEATAIK  GAVSAAISIS  TKVLGLLGVP  FAAQIGQLWT  FILNALWPSD  60
NTQWEEFMRH VEEELINQRI  DYARNKALAE  LTGLGNLDDL  YIEALDDWKR  NPTSQEAQTR  120
VIDRFRIVDG LFEAYIPSA  VSGYQVQLLT  VYAAANLHL  LLLRDSTIYG  IDWGLSQTNV  180
NDNYNRQIRL TATYANHCTT  WYQTGLERLR  GSNASSWVTY  NRFRREMTLT  VLDICSLFSN  240
YDYSYPAEV  RGEITREIYT  DPVGVGWVDS  APSFGEIENL  AIRAPRTVTW  LNSTRISTGT  300
LSGWSGSNRY WAAHMQNFSE  TNSGNIGFDG  PLYGSTVGTI  HRTDDYDMGN  RDIYTTISQA  360
VLGLWPTGQR VLGVASARFT  LRNLFNNTLQ  VLVYENPISS  NFGSSTLTHE  LSGENSDRPT  420
SSDYSHRLTS ITGFRAGANG  TVPVFGWTS  TVDRNNIER  NKITQFPVK  SHTLNNCQVV  480
RGTGFTGDW  LRPNNNGTFR  LTTTSFSSQS  YRIRLRYATS  VGNTSLVISS  SDAGISSTTI  540
PLTSTITSLP QTPYQAFRV  VDLPIFTTTP  TTQRYNTFDF  RLQNPSNANV  FIDRFEFVPI  600
GSLSEYETK  610

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SEQ ID NO: 84      moltype = DNA length = 1968
FEATURE           Location/Qualifiers
misc_feature       1..1968
note = A synthetic coding sequence used for expression in a
plant cell encoding a CR-BREla.TIC7383_19.nno_Mc:1
protein, with an N-terminal and C-terminal truncation
relative to the TIC7383 protein.
source            1..1968
mol_type = other DNA
organism = synthetic construct

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SEQUENCE: 84
atgcagatca ttgaaccctc ctccgacagc ctctgtgata gccataacaa ctaccctgtac 60
gccactgacc caaacaccgt gctcgagggg cgcaactaca aggagtggtc caacaatgc 120
accgacaact acacccagcg actccagggg ccggaggcga cggccatcag caagggtgcc 180
gtctccgcgg ccatctccat ctccagcaag gtccctagggc tcctaggggt cccgttcgct 240
gcgcagattg gccagctgtg gacgttcac cttaacgcac tctggccctc agacaacacc 300
cagtgaggag agttcatcgc ccacgtcgag gaactgatca accagcgcat agcggactac 360
gcgaggaaca aggcctggc cgagctgaca gggctgggca acaacctaga cctctacatc 420
gaggcgctcc aggtcggaa gcggaacctc acctcccaac aagctaagga ccgcgttatc 480
gaccgcttcc gaatcgcgga cggcctcttc gaggggtaca tgccgagctt ccgggtctcg 540
ggctacgagg tgccctctct caccgtgtac gcggcgggcg ctaatttgca tctcctgctg 600
ctcagggtat gctcgatcta cggaatccag tggggcttct cccagactaa cgttaacgag 660
aactacaatc ggcagattcg gcacaccgcc gactacgcta accactgcac aacatggtat 720
cagaccggcc tcgagaggct gcgcggcacc aatgctttct catggatcaa ctacaaccgc 780

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ttccgcccgc agatgacact gactgtcttg gacgtctgct cccctctctc caactatgac 840
taccgcactt accccactga ggtccggggc gagcttacgc gggagatcta cacagacccc 900
atcgggttcc agaactcccc gctccctggc gtggtcccca attggtacga ctacgcgagg 960
tcgttcgcgg aaatcgaaaa catcgccata cgagcacctc ggacagtcac gtggctgaac 1020
tcaaccacca tctacaccgg gagactgaac ggctacaaca atagcaacta ctactgggcc 1080
gggttccgcc agaacttctc agagaccaac tctggcagca gcttcaacgg cccagaccta 1140
ggcgacctca ctccgaacta caggatagag accctcgaca tggccaacag ggacatctac 1200
agcatctact cccgggtcgt gtcgcaatcc tggcccatcg gcaacgtcaa gctcttcggc 1260
gtaagctcct ctactttaag ttccggggat ctcaacaaca acagctccgg caccctggtc 1320
tacgagaacc cgactaactt cagctcccag tacctaacca cagagtttcc aggggagAAC 1380
tccgagcggc ccacottcac ggactacagt caccgcctaa cctgccttac gcggatcggg 1440
gccgggaact acggcctcgt gctttgtgcg ggttggaact cttcgagtgt cgaaggggac 1500
aaccgcctac aaccggataa gattacccag taccggcgcc tcaagggggt caacttgga 1560
ggcttcacag tggtgaaagg gactggtttc accgggggca actggctgcg gtccagcagg 1620
gtcaccgggt cattccgcct gaacgtatac tcacctagtg tgcagactta tcggatgagg 1680
atcaggtacg cctccccgtt tggcaactcg acgcttggga taagtctgac agacgcccgc 1740
atctcgttac tctgccctcc actatcgga gcatgcccgt cactgtgccc 1800
tacgagcggt tccgtgtcct cgactcccc atcacgttca ccgtagcctc ccaaaggAAC 1860
tacaacttca ccttcgacat ccttaacccc tcggttgggg ccgtgtacat cgatcgatt 1920
gagttcgtgc ccgtgggctc ttccagtgtt gagtacgaaa ccaagtga 1968

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SEQ ID NO: 85      moltype = AA length = 655
FEATURE           Location/Qualifiers
REGION            1..655
                  note = The amino acid sequence of the
                  CR-BREla.TIC7383_19.nno_Mc:1 protein, and comprises amino
                  acids 15 through 668 of TIC7383.
source            1..655
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 85
MQIIEPSSDS LLYSHNNYPY ATPDNTVLEG RNYKEWLKNC TDNYTDALQG PEATAISKGA 60
VSAAISISTK VLGLLGVFFA AQIGQLWTFI LNALWPSDNT QWEEFMHRVE ELINQRIADY 120
ARNKALAEIT GLGNLNDLYI EALEDWKRNQ TSQQAQDRVI DRFRIADGLF EGYMPSFRVS 180
GYEVPLLTIV AAAANLHLLL LRDCSIYGIQ WGFSTQNVNE NYNRQIRHTA EYANHCTTWY 240
QTGLERLRGT NAFSWINYNR FRREMTLTVL DVCSLFSNYD YRTYPTEVRA ELTREIYTD 300
IGFQNSPLPG VVPNWYDYAR SFAEIEINIAI RAPRTVTWLN STTIYTGRLN GYNNSNYWA 360
GPRQNFSETN SGSSFNGPDL GDLPNRYRIE TLDMVNRDIY SIYSRVVSQS WPIGNVKLFG 420
VSSSTLSFRD LNNSSGTLV YENPTNFSSQ YLTTEFPGEN SERPTFTDYS HRLTCLTRIG 480
AGNYGLVLCA GRLSSSVVERD NRLQPDKITQ YPAVKGFNLD GFTVVKGTGF TGNWLRRSSR 540
VTGSFRLNVY SPSVQTYRMR IRYASPLGNS TLGISSTDAG ISFTSFPLPS TIGSMPSTVP 600
YEAFRVLDIP ITVTVASQRN YNFIIDILNP SVGAVYIDRI EFVPGVSSVF EYETK 655

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SEQ ID NO: 86      moltype = DNA length = 1947
FEATURE           Location/Qualifiers
misc_feature       1..1947
                  note = A synthetic coding sequence used for expression in
                  plant cells encoding a CR-BREla.TIC7383_20.nno_Mc:1
                  protein, with an N-terminal and C-terminal truncation
                  relative to the TIC7383 protein.
source            1..1947
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 86
atgcagatca ttgaaccctc ctccggacagc ctccgtgtaca gccataacaa ctaccgtac 60
gccactgacc caaacaccgt gctcagaggg cgcaactaca aggagtggct caacaaatgc 120
accgacaact acaccgacgc actccagggc ccggaggcga cggccatcag caagggtgcc 180
gtctccgcgg ccattcccat ctccgacgaag gtccataggc tctcagggtt cccgttcgct 240
gctcagattg gccagctgtg gacgttcac cttaacgcac tctggccctc agacaacacc 300
cagtgaggag agttcatgcg ccacgtcgag gaactgatca accagcgcat agcggactac 360
gcgaggaaca aggccctggc cgagctgaca gggctgggca acaacctaga cctctacatc 420
gaggcgctcg aggactggaa gcggaaccct acctcccaac aagctaagga ccgcgttatc 480
gaccgcttcc gaatcgcgga cggcctcttc gaggggtaca tgccgagctt ccgggtctcg 540
ggctacgagg tgccctcctc caccggtgtac gcggcgcgcg ctaatttgca tctcctgctg 600
ctcagggatt gctcgatcta cgggaatccag tggggtctct cccagactaa cgtaaacgag 660
aactacaatc ggcagattcg gcacaccgcc gactacgcta accactgcac aacatgggat 720
cagaccggcc tcgagaggct gcgcggcacc aatgctttct catggatcaa ctacaaccgc 780
ttccgcccgc agatgacact gactgtcttg gacgtctgct cctctctctc caactatgac 840
taccgcactt accccactga ggtccggggc gagcttacgc gggagatcta cacagacccc 900
atcgggttcc agaactcccc gctccctggc gtggtcccca attggtacga ctacgcgagg 960
tcgttcgcgg aaatcgaaaa catcgccata cgagcacctc ggacagtcac gtggctgaac 1020
tcaaccacca tctacaccgg gagactgaac ggctacaaca atagcaacta ctactgggcc 1080
gggttccgcc agaacttctc agagaccaac tctggcagca gcttcaacgg cccagaccta 1140
ggcgacctca ctccgaacta caggatagag accctcgaca tggccaacag ggacatctac 1200
agcatctact cccgggtcgt gtcgcaatcc tggcccatcg gcaacgtcaa gctcttcggc 1260
gtaagctcct ctactttaag ttccggggat ctcaacaaca acagctccgg caccctggtc 1320
tacgagaacc cgactaactt cagctcccag tacctaacca cagagtttcc aggggagAAC 1380
tccgagcggc ccacottcac ggactacagt caccgcctaa cctgccttac gcggatcggg 1440
gccgggaact acggcctcgt gctttgtgcg ggttggaact cttcgagtgt cgaaggggac 1500

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-continued

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aaccgcctac aacccgataa gattaccacg taccgcgcgc tcaagggggt caacttggac 1560
ggcttcacag tggtgaaagg gactggtttc accgggggca actggctgcg gtccagcagg 1620
gtcacccgggt cattccgcct gaacgtatac tcacactagt tgcagactta tcggatgagg 1680
atcagggtacg cctccccgct tggcaactcg acgcttggga taagtctgac agacgcccgc 1740
atctcgttca cgagtttccc tctgccctcc actatcgga gcatgccgtc cactgtgccc 1800
tacgaggcgt tccgtgtcct cgacatcccc atcacggcca ccgtagcctc ccaaaggaa 1860
tacaacttca tcttcgacat ccttaacccc tcgggtgggg ccgtgtacat cgatcgatt 1920
gagttcgtgc ccgtgggctc ttcattga 1947

```

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SEQ ID NO: 87      moltype = AA length = 648
FEATURE           Location/Qualifiers
REGION           1..648
                  note = The amino acid sequence of the
                  CR-BREla.TIC7383_20.nno_Mc:1 protein, and comprises amino
                  acids 15 through 661 of TIC7383.
source           1..648
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 87
MQIIEPSSDS LLYSHNNYPY ATPDNTVLEG RNYKEWLKNC TDNYTDALQG PEATAISKGA 60
VSAAISISTK VLGLLGVPFA AQIGQLWTFI LNALWPSDNT QWEEFMRHVE ELINQRIADY 120
ARNKALAEIT GLGNLMDLYI EALWDKRNK TSQQAQKDRVI DRFRIADGLF EGYMPSFRVS 180
GYEVPLLTIV AAAANLHLLL LRDCSIYGIQ WGFSSQTNVNE NYNRQIRHTA EYANHCTTWY 240
QTGLERLRGT NAFSWINYNR FRREMTLTVL DVCSLFSNYD YRTYPTEVRA ELTREIYTDY 300
IGFQNSPLPG VVPNWYDYAR SFAEINIAT RAPRTVTWLN STTIYTGRLN GYNNSNYWA 360
GPRQNFSETN SGSSFNPGDL GDLTNPYRIE TLDMVNRDIY SIYSRVVSQS WPIGNVKLFG 420
VSSSTLSFRD LNNSSSGTLV YENPTNFSSQ YLTTEFPGEN SERPTFTDYS HRLTCLTRIG 480
AGNYGLVLCA GWTSSSVVERD NRLQPKITQ YPAVKGFNLD GFTVVKGTGF TGNWLRSSR 540
VTGSPRLNVY SPSVQTYRMR IRYASPLGNS TLGISSTDAG ISFTSPPLPS TIGSMPSTVP 600
YEAFRVLDIP ITVTVASQRN YNIFIDILNP SVGAVYIDRI EFVVPVGS 648

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SEQ ID NO: 88      moltype = DNA length = 1833
FEATURE           Location/Qualifiers
misc_feature      1..1833
                  note = A synthetic coding sequence used for expression in a
                  plant cell, encoding a CR-BREla.TIC7383_21.nno_Mc:1
                  protein with an N-terminal and C-terminal truncation, and
                  an additional alanine codon after the initiating methionine
                  relative to TIC7383.
source           1..1833
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 88
atggctaccg acaactacac cgacgcactc caggggcccg aggcgacggc catcagcaag 60
gggtccgtct ccgcggccat ctccatctcg acgaaggtcc tagggctcct aggggtccc 120
tcgctgcgcg agattggcca gctgtggacg ttcctcctta acgcactctg gccctcagac 180
aacacccagt gggaggaggt catcgcgccac gtcgaggaac tgatcaacca gcgcatacgc 240
gactacgcga ggaacaaggc cctggccgag ctgacagggc tgggcaacaa cctagacctc 300
tacatcgagg cgctcgagga ctggaagcgg aacctacct cccaacaagc taaggaccgc 360
gttatcgacc gcttccgaat cgcggacggc ctcttcgagg ggtacatgcc gagcttccg 420
gtctcgggct acgaggtgcc cctcctcacc gtgtacgcgg cggcggttaa ttgcatctc 480
ctgctgtctc gggatctgct gatctacgga atccagtggg gcttctccca gactaacgtt 540
aacgagaact acaatcgcca gattcggcac accgccagt acgctaacca ctgcacaaca 600
tggtatcaga ccggcctcga gaggtgcgc gccaccaatg ctttctcatg gatcaactac 660
aacgccttcc gccgcgagat gacactgact gtcttgacg tctgtccctc cttctccaac 720
tatgactacc gcacttcccc cactgaggtc cggggccgagc ttacgcggga gatctacaca 780
gaccccatcg gggtccagaa ctcccgcgtc cctggcgtgg tcccaattg gtacgactac 840
gcgaggtcgt tcgcggaagt cgaaaacatc gccatacgag cacctcggac agtcacgtgg 900
ctgaactcaa ccaccatcta caccgggaga ctgaacggct acaacaatag caactactac 960
tgggccgggt tccgccagaa cttctcagag accaactctg gcagcagctt caacggccca 1020
gacctaggcg acctcaactc gaactacagg atagagacct tcgacatggt caacagggac 1080
atctacagca tctactcccg ggtcgtgtcg caatcctggc ccatcggaac cgtcaagctc 1140
ttcggcgtaa gctcctctac cttaagtttc cgggatctca acaacaacag ctccggcacc 1200
ctgggtctacg agaacccgac taacttcagc tccagtagc taaccacaga gtttccagg 1260
gagaactccg agcggccccc cttcacggac tacagtcacc gcctaacctg ctttacgcgg 1320
atcggggccg ggaactacgg cctcgtgctt tgtcggggtt ggacttcttc gagtgtcgaa 1380
agggacaacc gcctacaacc cgataagatt acccagtagc ccgcgcgtca ggggttcaac 1440
ttggacggct tcacagtggt gaaggagact ggtttcaccc ggggcaactg gctgcgggtc 1500
agcagggtca ccgggtcatt ccgcctgaac gtatactcac ctagtgtgca gacttatcgg 1560
atgaggatca ggtacgcctc ccgccttggc aactcgacgc ttgggataag ttcgacagac 1620
gccggcatct cggtcaacgag ttccctctg cctccacta tcgggagcat gccgtccact 1680
gtgccctacg aggcgttccg tgtcctcgac atccccatca cggtcacctg agcctcccaa 1740
aggaactaca acttcattct cgacatcctt aacctcctcg ttggggccgt gtacatcgat 1800
cgtattgagt tcgtgccctg gggctcttca tga 1833

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SEQ ID NO: 89      moltype = AA length = 610
FEATURE           Location/Qualifiers
REGION           1..610
                  note = The amino acid sequence of the

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CR-BREla.TIC7383_21.nno_Mc:1 protein, with an N-terminal and C-terminal truncation, an additional alanine residue, and comprises amino acids 54 through 661 of TIC7383.

source 1..610
mol_type = protein
organism = synthetic construct

SEQUENCE: 89

MATDNYTDAL	QGPEATAISK	GAVSAAISIS	TKVLGLLGVP	FAAQIGQLWT	FILNALWPSD	60
NTQWEEFMRH	VEELINQRIA	DYARNKALAE	LTGLGNLDDL	YIEALEDWKR	NPTSQQAKDR	120
VIDRFRIADG	LFEGYMPSEF	VSGYEVPLLT	VYAAANLHL	LLLRDCSIYG	IQWGFSTQNV	180
NENYNRQIRH	TAEYANHCTT	WYQTGLERLR	GTNAFSWINY	NRFRREMTLT	VLDVCSLFSN	240
YDRTYPTTEV	RAELTREIYT	DPIGFQNSPL	PGVVPNWYDY	ARSAEAIENI	AIRAPRTVTW	300
LNSTTIYTGR	LNGYNNSNY	WAGFRQNFSE	TNSGSSFNGP	DLGDLTPNYR	IETLDMVNRD	360
IYSIYSRVVS	QSWPIGNVKL	FGVSSSTLSF	RDNNNSSGT	LVYENPTNFS	SQYLTTTEFPG	420
ENSERPTFTD	YSHRLTCLTR	IGAGNYGLVL	CAGWTSSSVE	RDNRLQPDKI	TQYPAVKGFN	480
LDGFTVVKGT	GFTGGNWLR	SRVTGSFRLN	VYSPSVQTYR	MRIRYASPLG	NSTLGISSTD	540
AGISFTSFPL	PSTIGSMPST	VPYEAFRVLD	IPITVTVASQ	RNYNFIFDIL	NPSVGAVYID	600
RIEFVPVGSS						610

SEQ ID NO: 90 moltype = DNA length = 1851
FEATURE Location/Qualifiers
misc_feature 1..1851
note = A synthetic coding sequence used for expression in plant cells, encoding a CR-BREla.TIC7383_22.nno_Mc:1 protein, with an N-terminal and C-terminal truncation relative to the TIC7383 protein.

source 1..1851
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 90

atgaccgaca	actacaccga	cgcactccag	ggccccggagg	cgacggccat	cagcaaggggt	60
gccgtctccg	cgcccatctc	catctcgacg	aaggctctag	ggctcctagg	ggtcccgttc	120
gctcgcgaga	ttggccagct	gtggacgttc	atccttaacg	cactctggcc	ctcagacaac	180
acccagtggt	aggagttcat	gcgccacgtc	gaggaaactga	tcaaccagcg	catagcggac	240
tacgcgagga	acaagggcct	ggccgagctg	acagggctgg	gcaacaacct	agacctctac	300
atcgaggcgc	tccgaggact	gaagcggaac	cctacctccc	aacaagctaa	ggaccgctgt	360
atcgaccgct	tccgaatcgc	ggacggcctc	ttcgaggggt	acatgccgag	cttcgggttc	420
tccggctacg	agggtccctc	cctcacgggt	tacgcggcgg	cggtctaat	gcctctctcg	480
ctgctcaggg	attgctcgat	ctacggaatc	cagtggggct	tctcccagac	taacgttaac	540
gagaactaca	atcggcagat	tcggcacacc	gccgagtacg	ctaaccactg	cacaacatgg	600
tatcagaccg	gcctcgagag	gctgcgcggc	accaaagctt	tctcatggat	caactacaac	660
cgcttccgcc	gcgagatgac	actgactgtc	ttggacgtct	gctccctctt	ctccaactat	720
gactaccgca	cttaccaccac	tagggtccgg	gccgagctta	cgccggagat	ctacacagac	780
cccatcgggt	tccagaactc	cccgtccctc	ggcggtgtcc	ccaattggta	cgactacgcg	840
agggtcggtc	cggaatccga	aaacatcgcc	atacagcac	ctcgagacgt	cacgtggctg	900
aactcaacca	ccatctacac	cgggagactg	aacggctaca	acaatagcaa	ctactactgg	960
gccgggttcc	gccagaactt	ctcagagacc	aactctggca	gcagcttcaa	cggcccagac	1020
ctaggcgacc	tcactccgaa	ctacaggata	gagaccctcg	acatgggtcaa	caggggacatc	1080
tacagcatct	ctcccggtg	ctgttcgcaa	tcttggccca	tcggcaacgt	caagctcttc	1140
ggcgtaagct	cctctacctt	aagtttccgg	gatctcaaca	acaacagctc	cggcaccctg	1200
gtctacgaga	accggaacta	cttcagctcc	cagtaacctaa	ccacagagtt	tccaggggag	1260
aaactccgag	ggccaccact	cacggactac	agtcaccgcc	taacctgtct	tacgcggatc	1320
ggggccggga	actacggcct	ctgtgcttgt	gcgggttgga	cttctctcag	tgtcgaaagg	1380
gacaaccgcg	tacaaccgga	taagattacc	cagtaaccgc	ccgtcaaggg	gttcaacttg	1440
gacggcttca	cagtgtgtga	agggaactgt	ttcacggggg	gcaactggct	gcggtccagc	1500
agggtcaccg	gggtcatctc	cctgaacgta	tactcaccta	gtgtgcagac	ttatcggatg	1560
aggatcagg	acgcctcccc	gcttggaac	tcgacgcttg	ggataagttc	gacagacgcc	1620
ggcatctcgt	tcacgagttt	ccctctgccc	tccactatcg	ggagcatgcc	gtccactgtg	1680
ccctacgagg	cgttccgtgt	cctcgacatc	cccatcacgg	tcaccgtagc	ctcccaagg	1740
aactacaact	tcactctcga	catccttaac	ccctcggttg	gggcccgtga	catcgatcgt	1800
attgagttcg	tgccccgtgg	ctcttcagtg	ttcgagtacg	aaaccaagtg	a	1851

SEQ ID NO: 91 moltype = AA length = 616
FEATURE Location/Qualifiers
REGION 1..616
note = The amino acid sequence of the CR-BREla.TIC7383_22.nno_Mc:1 protein, and comprises amino acids 54 through 668 of TIC7383.

source 1..616
mol_type = protein
organism = synthetic construct

SEQUENCE: 91

MTDNYTDALQ	GPEATAISKG	AVSAAISIST	KVLGLLGVPF	AAQIGQLWTF	ILNALWPSDN	60
TQWEEFMRHV	EELINQRIAD	YARNKALAE	TGLGNLDDL	IEALEDWKR	PTSQQAKDRV	120
IDRFRIADGL	FEGYMPSEF	VSGYEVPLTV	YAAANLHL	LLLRDCSIYG	QWGFSTQNVN	180
ENYNRQIRHT	AEYANHCTTW	YQTGLERLRG	TNAFSWINYN	RFRREMTLTV	LDVCSLFSNY	240
DYRTYPTTEVR	AELTREIYTD	PIGFQNSPLP	GVVPNWYDYA	RSFAEAIENIA	IRAPRTVTWL	300
NSTTIYTGR	NGYNNSNYW	AGFRQNFSET	NSGSSFNGPD	LGDLTPNYRI	ETLDMVNRDI	360
YSIYSRVVSQ	SWPIGNVKLF	GVSSSTLSFR	DLNNNSGTL	VYENPTNFSS	QYLTTTEFPG	420

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NSERPTFTDY SHRLTCLTRI GAGNYGLVLC AGWTSSSVR DNRLQPDKIT QYPAVKGFNL 480
DGFTTVKGTG FTGGNWLRS RVTSFRLNV YSPSVQTYRM RIRYASPLGN STLGISSTDA 540
GISFTSFPLP STIGSMPSTV PYEAFRVLDI PITVTVASQR NYNFIFDILN PSVGAVIDR 600
IEFVPGSSV FEYETK 616

SEQ ID NO: 92      moltype = DNA length = 1830
FEATURE           Location/Qualifiers
misc_feature       1..1830
                   note = A synthetic coding sequence used for expression in a
                   plant cell encoding a CR-BREla.TIC7383_23.nno_Mc:1
                   protein, with an N-terminal and C-terminal truncation
                   relative to the TIC7383 protein.
source             1..1830
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 92
atgaccgaca actacaccga cgcactccag ggcccgagg cgacggccat cagcaagggt 60
gccgtctccg cggccatctc catctcgacg aaggtcctag ggctcctagg ggtcccgttc 120
gctgcgcaga ttggccagct gtggacgttc atccttaacg cactctggcc ctacagacaac 180
accagtgagg aggagttcat gcgccacgtc gaggaactga tcaaccagcg catagcggac 240
tacgcgagga acaaggccct gcccgagctg acagggtctg gcaacaacct agacctctac 300
ctcgaggcgc tcgaggactg gaagcggaac cctacctccc aacaagctaa ggaccgcgtt 360
atcgaccgct tccgaatcgc ggaacggcctc ttcgaggggt acatgccgag cttccgggtc 420
tcgggctacg aggtgccctc cctcacgggtg tacgcggcgg cggttaattt gcatctcctg 480
ctgctcaggg attgctcgat ctacggaaatc cagtggggct tctcccagac taacgttaac 540
gagaactaca atcggcagat tcggcacacc gccgagtaac ctaaccactg cacaacatgg 600
tatcagaccg gccctcgagag gctgcgcggc accaatgctt tctcatggat caactacaac 660
cgcttcgccg gcgagatgac actgactgtc ttggacgtct gctccctctt ctccaactat 720
gactaccgca cttaccccac tgagggtccgg gccgagctta cgcgggagat ctacacagac 780
cccctcgggt tccagaactc cccgctccct gccgtgggtc ccaattggta cgactacgcg 840
aggtcggtcg cggaaatcga aaacatcgcc atacgagcac ctccggacgt cacgtggctg 900
aactcaacca ccattctacac cgggagactg aacggctaca acaatagcaa ctactactg 960
gccgggttcc gccagaactt ctacagagac aactctggca gcagcttcaa cggcccagac 1020
ctaggcgacc tcaactcggaa ctacaggata gagaccctcg acatgggtcaa caggggacatc 1080
tacagcatct actcccgggt cgtgtcgcaa tcttggccca tcggcaacgt caagctcttc 1140
ggcgtaagct cctctacctt aagtttcctg gatctcaaca acaacagctc cggcacccctg 1200
gctctacgag acccgactaa cttcagctcc cagtacctaa ccacagagtt tccaggggag 1260
aactccgagc ggcccacctt cacggactac agtcaccgcc taacctgctt tacgcggatc 1320
ggggccggga actacggcct cgtgctttgt gcgggttgga cttcttcgag tgtcgaaagg 1380
gacaaccgcc tacaaccoga taagattacc cagtaccctg ccgtcaaggg gttcaacttg 1440
gacggcttca cagtgggtga agggactggt ttcaccgggg gcaactggct gcggtccagc 1500
agggtcaccg ggtcattccg cctgaacgta tactcaccta gtgtgcagac ttatcggatg 1560
aggatcaggt acgcctcccc ccttgccaac tcgacgcttg ggataagttc gacagacgcc 1620
ggcatctcgt tcacgagttt cctctcgccc tccactatcg ggagcatgcc gtccactgtg 1680
ccctacgagg cgttcctgtg cctcgacatc cccatcacgg tcaccgtagc ctcccaaagg 1740
aactacaact tcattcttga catccttaac cctcgggttg gggccgtgta catcgatcgt 1800
attgagttcg tgcccggtgg cttctcatga 1830

SEQ ID NO: 93      moltype = AA length = 609
FEATURE           Location/Qualifiers
REGION            1..609
                   note = The amino acid sequence of the
                   CR-BREla.TIC7383_23.nno_Mc:1 protein, and comprises amino
                   acids 54 through 661 of TIC7383.
source             1..609
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 93
MTDNYTDALQ GPEATAISK AVSAAISIST KVLGLLGVPF AAQIGQLWTF ILNALWPSDN 60
TQWEEFMRHV EELINQRIAD YARNKALAE TGLGNLDLY IEALEDWKRN PTSQQAQDRV 120
IDRFRIADGL FEGYMPFRV SGYEVPLLTV YAAAANLHLL LLRDCSIYGI QWGFSTQNVN 180
ENYNRQIRHT AEYANHCTTW YQTGLERLRG TNAPSWINYN RFRREMTLTV LDVCSLFSNY 240
DYRTYPTFVR AELTREIYTD PIGFQNSPLP GVVPNWYDYA RSPAEIENIA IRAPRTVTWL 300
NSTTYTGRL NGYNSNYW AGFRQNFSET NSGSSFNGPD LGDLTPNYRI ETLDMVNRDI 360
YSIYSRVVSQ SWPIGNVKLF GVSSSTLSFR DLNNSSGTL VYENPTNFSS QYLTEFPGE 420
NSERPTFTDY SHRLTCLTRI GAGNYGLVLC AGWTSSSVR DNRLQPDKIT QYPAVKGFNL 480
DGFTTVKGTG FTGGNWLRS RVTSFRLNV YSPSVQTYRM RIRYASPLGN STLGISSTDA 540
GISFTSFPLP STIGSMPSTV PYEAFRVLDI PITVTVASQR NYNFIFDILN PSVGAVIDR 600
IEFVPGSS 609

SEQ ID NO: 94      moltype = DNA length = 1773
FEATURE           Location/Qualifiers
misc_feature       1..1773
                   note = A synthetic coding sequence used for expression in a
                   plant cell encoding a CR-BREla.TIC7383_24.nno_Mc:2
                   protein, with an N-terminal and C-terminal truncation
                   relative to the TIC7383 protein.
source             1..1773
                   mol_type = other DNA

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organism = synthetic construct

SEQUENCE: 94
atggccgtct cccggcccat ctccatctcg acgaaggctc tagggctcct aggggtcccg 60
ttcgtctgctc agattggcca gctgtggacg ttcctcctta acgcactctg gccctcagac 120
aacacccagt gggaggaggt catgcgccac gtccagggaac tgatcaacca gcgcatagcg 180
gactacgcga ggaacaaggc cctggccgag ctgacagggc tgggcaacca cctagacctc 240
tacatcgagg cgctcgagga ctggaagcgg aacctacct cccaacaagc taaggaccgc 300
gttatcgacc gcttcggaat cgcggacggc ctcttcgagg ggtacatgcc gagcttcggy 360
gtctcgggct acgaggtgcc cctcctcagc gtgtacgcgg cggcggttaa ttgcatctc 420
ctgctgtctca gggattgctc gatctacgga atccagtggg gcttctccca gactaacgtt 480
aacgagaact acaatcgga gattcggcac accgcgaggt acgctaacca ctgcacaaca 540
tggtatcaga ccggcctcga gaggtgcgc ggcaccaatg ctttctcatg gatcaactac 600
aacgccttcc gcccgagat gacactgact gtcttgagc tctgctccct ctttcccaac 660
tatgactacc gcacttacc cactgaggtc cgggcgagc ttacgcggga gatctacaca 720
gaccccatcg ggttcagaa ctcccgcgtc cctggcggtg tccccaattg gtacgactac 780
gcgaggtcgt tcgcggaatc gaaaacatc gccatacgag cactcggac agtcacgtgg 840
ctgaactcaa ccacatcta caccgggaga ctgaacggct acaacaatag caactactac 900
tgggcccggg tccgccagaa cttctcagag accaactctg gcagcagctt caacggccca 960
gacctaggcg acctcactcg gaactacagg atagagacct tcgacatggt caacagggac 1020
atctacagca tctactcccg ggtcgtgtcg caatcctggc ccacggcaa cgtcaagctc 1080
ttcggcgtaa gctcctctac ctttaagttt cgggatctca acaacaacag ctccggcacc 1140
ctggtctacg agaaccgcac taacttcagc tccagtagc taaccacaga gtttccaggg 1200
gagaactccg agcggccacc cttcaccgac tacagtcacc gcctaacctg ctttacgagg 1260
atccggggcc ggaactacgg cctcgtgctt tgtgcggggt ggacttcttc gagtgtcgaa 1320
agggacaacc gccacaacc cgaataagatt acccagtagc ccgccgtcaa ggggttcaac 1380
ttggacgggt tcacagtggt gaaagggact ggtttcacgg ggggcaactg gctgcgggtc 1440
agcagggtca ccgggtcatt ccgcctgaac gtatactcac ctagtgtgca gacttatcgg 1500
atgaggatca ggtacgcctc ccgccttgcc aactcgacgc ttgggataag ttcgacagac 1560
gccggcatct cgttcacgag tttcctctcg cctccacta tcggggagcat gccgtccact 1620
gtgcctctac aggcgttcct gtctcctgac atccccatca cggtcaccgt agcctcccaa 1680
aggaactaca acttcatctt gacatcctt aaccctctcg ttggggccgt gtacatcgat 1740
cgtattgagt tcgtgcctcg gggctcttca tga 1773

SEQ ID NO: 95      moltype = AA  length = 590
FEATURE           Location/Qualifiers
REGION            1..590
                  note = The amino acid sequence of the
                  CR-BREla.TIC7383_24.nno_Mc:2 protein, and comprises amino
                  acids 73 through 661 of TIC7383.
source            1..590
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 95
MAVSAAISIS TKVLGLLGVF FAAIGQLWT FILNALWPSD NTQWEEFMRH VEELINQRIA 60
DYARKALAE LTGLGNLIDL YIEALEDWKR NPTSQQAQDR VIDRFRIADG LFEGYMPSPR 120
VSGYEVPLLT VYAAANLHL LLLRDCSIYG IQWFSQTNV NENYNRQIRH TAEYANHCTT 180
WYQTGLERLR GTNAFSWINY NRFRREMTLT VLDVCSLFSN YDYRTYPTTEV RAELETREIYT 240
DPIGFQNSPL PGVVPNWYDY ARSFAEINI AIRAPRTVTW LNSTTIYTRG LNGYNNNSNY 300
WAGFRQNFSE TNSGSSFNPG DLGDLTPNYR IETLDMVNDR IYSIYSRVVS QSWPIGNVKL 360
FGVSSSTLSF RDLNNNSSGT LVYENPTNFS SQYLTFEPPG ENSERPTFTD YSHRLTCLTR 420
IGAGNYGLVL CAGWTSSSVE RDNRLQPDKI TQYPAVKGFN LDGFTVVKGT GFTGGNWLRS 480
SRVTGSFRLN VYSPSVQTYR MRIRYASPLG NSTLGISSTD AGISFTSFPL PSTIGSMPST 540
VPYEAFRVLD IPITVTVASQ RNYNFIFDIL NPSVGAVYID RIEFVVPVGS 590

SEQ ID NO: 96      moltype = DNA  length = 1710
FEATURE           Location/Qualifiers
misc_feature       1..1710
                  note = A synthetic coding sequence used for expression in a
                  plant cell encoding a CR-BREla.TIC7383_25.nno_Mc:3
                  protein, with N-terminal and C-terminal truncation
                  relative to the TIC7383 protein.
source            1..1710
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 96
atggcgacga ttggccagct gtggacgttc atccttaacg cactctggcc ctacagacaac 60
acccagtggg aggagtctat gcgccacgtc gaggaactga tcaaccagcg catagcggac 120
tacgcgagga acaaggccct ggccgagctg acagggtcgg gcaacaacct agacctctac 180
atcgaggcgc tcgaggactg gaagcggaac cctacctccc aacaagctaa ggaccgcgtt 240
atcgaccgct tccgaatcgc ggcggcctc ttcgaggggt acatgcgcag cttccgggtc 300
tcgggctacg aggtgccccct cctcacgggt tacgcggcgg cggctaattt gcatctcctg 360
ctgctcaggg attgctcgat ctacggaatc cagtgggggt tctcccagac taacgttaac 420
gagaactaca atcggcagat tcggcacacc gccagtagac ctaaccactg cacaacatgg 480
tatcagacgg gccctgagag gctgcggcgc accaatgctt tctcatggat caactacaac 540
cgcttcgcgc gcgagatgac actgactgtc ttggacgtct gctccctctt ctccaactat 600
gactaccgca cttacccacc tgaggctccg gccagcttta cgcgggagat ctacacagac 660
cccatcgggt tccagaacte cccgctccct ggcgtgtgcc ccaattggta cgactacgcg 720
aggtcggttc cggaatcga aacatcgcc atacgagcac ctcgagacgt cagtggtctg 780
aactcaacca ccatctacac cgggagactg aacggtcata acaatagcaa ctactactgg 840

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gccgggttcc gccagaactt ctcagagacc aactctggca gcagcttcaa cggcccagac 900
ctaggcgacc tcaactccga ctacaggata gagacctcg acatgggtcaa cagggacatc 960
tacagcatct actcccgggt cgtgtcgcaa tcctggccca tcggcaacgt caagctcttc 1020
ggcgtaagct cctctacett aagtttcctg gatctcaaca acaacagctc cggcaccctg 1080
gtctacgaga acccgactaa cttcagctcc cagtacctaa ccacagagtt tccaggggag 1140
aactccgagc ggcccacett caccgactac agtcaccgcc taacctgcct tacgcggtac 1200
ggggccggga actacggcct cgtgctttgt gcgggttgga cttcttcgag tgtcgaaagg 1260
gacaaccgcc tacaaccgga taagattacc cagtaccgcc ccgtcaagggt gttcaacttg 1320
gacggcttca cagtgggtgaa agggactggt ttcaccgggg gcaactggct gcggtccagc 1380
agggtcaccg ggtcattccg cctgaacgta tactcaccta gtgtgcagac ttatcggtatg 1440
aggatcaggt acgcctcccc gcttggaac tcgacgcttg ggataagttc gacagacgcc 1500
ggcatctcgt tcacgagttt cctctgccc tccactatcg ggagcatgcc gtccactgtg 1560
ccctacgagg cgttcctgtg cctcgacatc cccatcacgg tcaccgtagc ctcccaaagg 1620
aactacaact tcattcttga catccttaac cctcgggttg gggccgtgta catcgatcgt 1680
attgagttcg tgcccgtggg cttctcatga 1710

SEQ ID NO: 97      moltype = AA  length = 569
FEATURE           Location/Qualifiers
REGION            1..569
                  note = The amino acid sequence of the
                  CR-BRE1a.TIC7383_25.nno_Mc:3 protein, and comprises amino
                  acids 94 through 661 of TIC7383.
source            1..569
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 97
MAIQIGQLWTF ILNALWPSDN TQWEEFMRHV EELINQRIAD YARNKALAEI TGLGNNLDLY 60
IEALEDWKRN PTSQQAQDRV IDRFRIADGL FEGYMPSPRV SGYEVPLTLV YAAAANLHLL 120
LLRDCSIYGI QWGFSSQTNVN ENYNRQIRHT AEYANHCTTW YQTGLERLRG TNAFSSWINYN 180
RFRREMTLTV LDVCSLFSNY DYRTYPTFVR AELTREIYTD PIGFQNSPLP GVVPNWYDYA 240
RSFAELENIA IRAPRTVFWL NSTTIYTRGL NGYNNSNYVW AGFRQNFSET NSGSSFNPGD 300
LGDLPNRYRI ETLDMVNRDI YSIYSRVVSQ SWPIGNVKLF GVSSSTLSFR DLNNNSSGTL 360
VYENPTNFSS QYLTFEPFGE NSERPTFTDY SHRLTCLTRI GAGNYGLVLC AGWTSSSVER 420
DNRLQPDKIT QYPAVKGFNL DGFTTVVKGTG FTGGNWLRSR RVTGSFRLNV YSPSVQTYRM 480
RIRYASPLGN STLGISSTDA GISFTSFPLP STIGSMPSTV PYEAFRVLDI PITVTVASQR 540
NYNFIFDLIN PSVGAVYIDR IEFVPVGSS 569

SEQ ID NO: 98      moltype = DNA  length = 1650
FEATURE           Location/Qualifiers
misc_feature       1..1650
                  note = A synthetic coding sequence used for expression in a
                  plant cell encoding a CR-BRE1a.TIC7383_26.nno_Mc:1
                  protein, with an N-terminal and C-terminal truncation
                  relative to the TIC7383 protein.
source            1..1650
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 98
atgcagtggtg aggagttcat gcgccacgtc gaggaactga tcaaccagcg catagcggac 60
tacgcgagga acaaggccct ggccgagctg acaggggtgg gcaacaacct agacctctac 120
atcgaggcgc tcgaggactg gaagcggaac cctacctccc aacaagctaa ggaccgcgtt 180
atcgaccgct tccgaatcgc ggacggcctc ttcgaggggt acatgcgagc cttccgggtc 240
tcgggctacg aggtgccctc cctcacggtg tacgcggcgg cggtctaatg gcatctctctg 300
ctgctcaggg attgctcgat ctacggaatc cagtgggggt tctcccagac taacgttaac 360
gagaactaca atcggcagat tcggcacacc gccgagtagc ctaaccactg cacaacatgg 420
tatcagaccg gccctcgagag gctgcgcggc accaatgctt tctcatggat caactacaac 480
cgcttcgccg gcgagatgac actgactgtc ttggacgtct gctccctctt ctccaactat 540
gactaccgca cttaccccac tgaagtcctg gccgagctta cgcgggagat ctacacagac 600
cccatcgggt tccagaactc cccgctccct ggcgtgggtc ccaattggta cgactacgcg 660
aggctcgttc cggaatcgca aaacatcgcc atacgagcac ctccggacgt cactgtggctg 720
aactcaacca ccatctacac cgggagactg aacggctaca acaatagcaa ctactactgg 780
gccgggttcc gccagaactt ctcagagacc aactctggca gcagcttcaa cggcccagac 840
ctaggcgacc tcaactccga ctacaggata gagacctcg acatgggtcaa cagggacatc 900
tacagcatct actcccgggt cgtgtcgcaa tcctggccca tcggcaacgt caagctcttc 960
ggcgtaagct cctctacett aagtttcctg gatctcaaca acaacagctc cggcaccctg 1020
gtctacgaga acccgactaa cttcagctcc cagtacctaa ccacagagtt tccaggggag 1080
aactccgagc ggcccacett caccgactac agtcaccgcc taacctgcct tacgcggtac 1140
ggggccggga actacggcct cgtgctttgt gcgggttgga cttcttcgag tgtcgaaagg 1200
gacaaccgcc tacaaccgga taagattacc cagtaccgcc ccgtcaagggt gttcaacttg 1260
gacggcttca cagtgggtgaa agggactggt ttcaccgggg gcaactggct gcggtccagc 1320
agggtcaccg ggtcattccg cctgaacgta tactcaccta gtgtgcagac ttatcggtatg 1380
aggatcaggt acgcctcccc gcttggaac tcgacgcttg ggataagttc gacagacgcc 1440
ggcatctcgt tcacgagttt cctctgccc tccactatcg ggagcatgcc gtccactgtg 1500
ccctacgagg cgttcctgtg cctcgacatc cccatcacgg tcaccgtagc ctcccaaagg 1560
aactacaact tcattcttga catccttaac cctcgggttg gggccgtgta catcgatcgt 1620
attgagttcg tgcccgtggg cttctcatga 1650

SEQ ID NO: 99      moltype = AA  length = 549
FEATURE           Location/Qualifiers

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REGION 1..549
 note = The amino acid sequence of the
 CR-BRE1a.TIC7383_26.nno_Mc:1 protein, and comprises amino
 acids 114 through 661 of TIC7383.

source 1..549
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 99

MQWEEFMRHV	EELINQRIAD	YARNKALAE	TGLGNLNDLY	IEALEDWKRN	PTSQQAKDRV	60
IDRFRIADGL	PEGYMPSFRV	SGYEVPLLT	VAAAAHLHL	LLRDCSIYGI	QWGFSTQNVN	120
ENYNRQIRHT	AEYANHCTTW	YQTGLERLRG	TNAFWSWYN	RFRREMTLTV	LDVCSLFSNY	180
DYRTPTEVR	AELTREIYTD	PIGFQNSPLP	GVVFNWYDYA	RSFAEINIA	IRAPRTVTWL	240
NSTTIYTGR	NGYNSNYW	AGFRQNFSET	NSGSSFNGPD	LGDLTPNYRI	ETLDMVNRDI	300
YSIYSRVVSQ	SWPIGNVKLF	GVSSSTLSFR	DLNNSSGTL	VYENPTNFSS	QYLTFEPGE	360
NSERPTFTDY	SHRLTCLTRI	GAGNYGLVLC	AGWTSSSVR	DNRLQDPKIT	QYPAVKGFNL	420
DGFTVVKGTG	FTGGNWLRS	RVTGSFRLNV	YSPSVQTYRM	RIRYASPLGN	STLGISSTDA	480
GISFTSFPLP	STIGSMPSTV	PYEAFRVLDI	PITVTVASQR	NYNFIFDILN	PSVGAVYIDR	540
IEFVVGSS						549

SEQ ID NO: 100 moltype = DNA length = 1824
 FEATURE Location/Qualifiers
 misc_feature 1..1824
 note = A synthetic coding sequence used for expression in a
 plant cell encoding a CR-BRE1a.TIC7383_27.nno_Mc:1
 protein, with an N-terminal and C-terminal truncation, and
 an additional alanine codon after the initiating
 methionine, relative to TIC7383.

source 1..1824
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 100

atggctaccg	acaactacac	cgacgcactc	cagggcccgg	aggcgacggc	catcagcaag	60
ggtgccgtct	ccgcggccat	ctccatctcg	acgaaggtcc	tagggctcct	aggggtcccg	120
ttcgtctgcg	agatttgcca	gctgtggacg	ttcatcctta	acgcactctg	gccctcagac	180
aacacccagt	gggagagatt	catcgccac	gtcgaggaa	tgatcaacca	gcgcatacgc	240
gactacgcga	ggaacaaggc	cctggccgag	ctgacagggc	tgggcaacaa	cctagacctc	300
tacatcgagg	cgctcgagga	ctggaagcgg	aacctacct	ccaacaagc	taaggaccgc	360
gttatcgacc	gcttccgaat	cgcggacggc	ctcttcgagg	ggtacatgcc	gagcttccgg	420
gtctcggtcg	acgaggtgcc	ctctcctcag	gtgtacgcgg	cggcggctaa	tttgcattct	480
ctgctgctca	gggattgctc	gattctacga	atccagtggg	gcttctccca	gactaacgtt	540
aacgagaact	acaatcgcca	gattcggcac	accgcccagt	acgctaacca	ctgcacaaca	600
tggtatcaga	ccggcctcga	gagggtgcgc	ggcaccaatg	ctttctcatg	gatcaactac	660
aaccgcttcc	gccgcgagat	gacactgact	gtcttgagac	tctgctcctt	cttctccaac	720
tatgactacc	gcacttaccc	cactgaggtc	cgggcccagc	ttacgcggga	gatctacaca	780
gaccccatcg	gggtccagaa	ctcccgcgtc	cctggcgtgg	tccccaattg	gtacgactac	840
gcgaggtcgt	tcgcggaaat	gcataacac	gccatcgag	cacctcggac	agtcacgtgg	900
ctgaactcaa	ccaccatcta	caccgggaga	ctgaacggct	acaacaatag	caactactac	960
tgggccgggt	tccgccagaa	cttctcagag	accaactctg	gcagcagctt	caacggccca	1020
gacctaggcg	acctcaactc	gaactacagg	atagagaccc	tcgacatggt	caacagggac	1080
atctacagca	tctactcccg	ggctgtgtcg	caatctggc	ccatcgga	cgtaacgtc	1140
ttcgccgtaa	gctcctctac	cttaagtttc	cgggatctca	acaacaacag	ctccggcacc	1200
ctggtctacg	agaaccagac	taacttcagc	tccagtagc	taaccacaga	gtttccagg	1260
gagaactccg	agcggccccc	cttcacggac	tacagtcacc	gcctaacctg	ccttacgcgg	1320
atcggggccg	ggaactacgg	cctcgtgctt	tgtgcgggtt	ggactcttct	gagtgtcgaa	1380
agggacaacc	gcatacaacc	cgataagatt	accagtagc	ccgcgctcaa	gggtgtcaac	1440
ttggacggct	tcacagtggt	gaaagggact	ggtttcacgc	ggggcaactg	gctgcgggtc	1500
agcaggggtc	ccgggtcatt	ccgcctgaac	gtatactcac	ctagtgtgca	gacttatcgg	1560
atgaggatca	ggtagccctc	cccgcctggc	aactcgacgc	ttgggataag	ttcgacagac	1620
gccggcatct	ggttcacag	tttccctctg	ccctccacta	tcgggagcat	gccgtccact	1680
gtgccctacg	aggcgttccg	tgtcctcgac	atccccatca	cggtcaccgt	agcctcccaa	1740
aggaactaca	acttcattct	cgacatcctt	aaccctcgg	ttggggccgt	gtacatcgat	1800
cgtattgagt	tcgtgccctg	gtga				1824

SEQ ID NO: 101 moltype = AA length = 607
 FEATURE Location/Qualifiers
 REGION 1..607
 note = The amino acid sequence of the
 CR-BRE1a.TIC7383_27.nno_Mc:1 protein, with an N-terminal
 and C-terminal truncation, an additional alanine residue,
 and comprises amino acids 54 through 658 of TIC7383.

source 1..607
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 101

MATDNYTDAL	QGPEATAISK	GAVSAAISIS	TKVLGLLGVP	FAAQIGQLWT	FILNALWPSD	60
NTQWEEFMRH	VEELINQRIA	DYARNKALAE	LTGLGNLNDL	YIEALEDWKR	NPTSQQAKDR	120
VIDRFRIADG	LFEGYMPSFR	VSGYEVPLLT	VAAAAHLHL	LLLRDCSIYG	IQWGFSTQNV	180
NENYNRQIRH	TAEYANHCTT	WYQTGLERLR	GTNAFWSWYN	NRFRREMTLT	VLDVCSLFSN	240
YDYRTPTEV	RAELTREIYT	DPIGFQNSPL	PGVVFNWYDY	ARSAEINIA	AIRAPRTVTW	300

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LNSTTIYTR  LNGYNNNSNY  WAGFRQNFSE  TNSGSSFNPG  DLGLTPNYR  IETLDMVNRD  360
IYSIYSRVVS  QSWPIGNVKL  FGVSSSTLSF  RDLNNSSTG  LVYENPTNFS  SQYLTTTEPPG  420
ENSERPTFTD  YSHRLTCLTR  IGAGNYGLVL  CAGWTSSSV  RDNRLQDPKI  TQYPAVKGFN  480
LDGFTVVKGT  GFTGWNWLR  SRVTGSFRLN  VYSPSVQTYR  MRIRYASPLG  NSTLGISSTD  540
AGISFTSPFL  PSTIGSMPST  VPYEAFRVLD  IPITVTVASQ  RNYNFIFDIL  NPSVGAVYID  600
RIEFVPV  607

SEQ ID NO: 102      moltype = DNA  length = 1938
FEATURE            Location/Qualifiers
misc_feature        1..1938
                    note = A synthetic coding sequence used for expression in a
                        plant cell encoding a CR-BREla.TIC7383_28.nno_Mc:1 protein
                        with an N-terminal and C-terminal truncation relative to
                        the TIC7383 protein.
source              1..1938
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 102
atgcagatca  ctgaaccctc  ctcggacagc  ctccgtgtaca  gccataacaa  ctacccgtag  60
gccactgacc  caaacaccgt  gctcgagggg  cgcaactaca  aggagtggtc  caacaaatgc  120
accgacaact  acaccgacgc  actccagggc  cgggagggcg  cgcccatcag  caagggtgcc  180
gtctccgcgg  ccatctccat  ctcgacgaag  gtctaggggc  tcctaggggt  cccgttcgct  240
gcgcagattg  cccagctgtg  gacgttcac  cttaacgcac  tctggccctc  agacaacacc  300
cagtgaggag  agttcatgcg  ccacgtcgag  gaactgatca  accagcgcat  agcggactac  360
gcgaggaaca  aggccttgcc  cgagctgaca  gggctgggca  acaacctaga  cctctacatc  420
gaggcgctcg  aggaactgaa  gcggaaacct  acctcccaac  aagctaagga  ccgcgttatc  480
gaccgcttcc  gaatcgcgga  cggcctcttc  gaggggtaca  tgccgagctt  ccgggtctcg  540
ggctacgagg  tgccctctct  cacggtgtac  gcggcgggcg  ctaatttgca  tctcctgctg  600
ctcagggatt  gctcgatcta  cgggaatccg  tggggcttct  cccagactaa  cgttaacgag  660
aactacaatc  ggcagattcg  gcacaccgcc  gactacgcta  accactgcac  aacatgggat  720
cagaccggcc  tcgagaggct  gcgcggcacc  aatgctttct  catggatcaa  ctacaaccgc  780
ttccgcccgc  agatgacact  gactgtcttg  gacgtctgct  cctctctctc  caactatgac  840
taccgcaact  accccactga  ggtccggggc  gagcttacgc  gggagatcta  cacagacccc  900
atcgggttcc  agaactcccc  gctccctggc  gtggtcacca  attggtacga  ctacgcgagg  960
tcgttcgcgg  aaatcgaaaa  catcgccata  cgagcacctc  ggacagtcac  gtggtgaaac  1020
tcaaccacca  tctacaccgg  gagactgaac  ggctacaaca  atagcaacta  ctactgggcc  1080
gggttcgcgc  agaacttctc  ctctggcagc  gcttcaacgg  cccagacctc  1140
ggcgacctca  ctccgaacta  caggatagag  accctcgaca  tggtaacacg  ggacatctac  1200
agcatctact  cccgggtcgt  gtcgcaatcc  tggcccatcg  gcaacgtcaa  gctcttcggc  1260
gtaagctcct  ctaccttaag  ttctcgggat  ctcaacaaca  acagctccgg  caccctggtc  1320
tcagagaacc  cgactaactc  cagctcccg  tacctaacca  cagagtttcc  agggggagaa  1380
tccgagcgcg  ccaccttcac  ggaactacgt  caccgcctaa  cctgccttac  gcggatcggg  1440
acggggaact  acggcctcgt  gctttgtgcg  ggttggactt  cttcgagtgt  cgaaagggac  1500
aaccgacctc  aaccogataa  gattaccgag  taccocggcg  tcaagggggt  caacttggac  1560
ggcttcacag  tggtgaaagg  gactggttct  accgggggca  actggctgcg  gtccagcagg  1620
gtcaccgggt  cattccgcct  gaacgtatac  tcacctagtg  tgcagactta  tcggatgagg  1680
atcaggtacg  cctccccgct  tggcaactcg  acgcttggga  taagttcgac  agacgcggcg  1740
atctcgttca  caggtttccc  tctgccctcc  actatcgggg  gcatgcgcgt  cactgtgccc  1800
tacgagcggt  ccgtgtcctc  cgacatcccc  atcaggttca  ccgtagcctc  ccaaagggaac  1860
tacaacttca  tcttcgacat  ccttaacccc  tcgggtgggg  ccgtgtacat  cgatcgattt  1920
gagttcgtgc  ccgtgtga  1938

SEQ ID NO: 103      moltype = AA  length = 645
FEATURE            Location/Qualifiers
REGION             1..645
                    note = The amino acid sequence of the
                        CR-BREla.TIC7383_28.nno_Mc:1 protein, and comprises amino
                        acids 15 through 658 of TIC7383.
source              1..645
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 103
MQIIEPSSDS  LLYSHNNYPY  ATDPNTVLEG  RNYKEWLKNC  TDNYTDALQG  PEATAISKGA  60
VSAAISISTK  VLGLLGVPFA  AQIGQLWTFI  LNALWPSDNT  QWEEFMHRVE  ELINQRIADY  120
ARNKALAEIT  GLGNLDLYI  EALDWKRN  TSQQAQDRVI  DRFRIADGLF  EGYMPSFRVS  180
GYEVPLLTIV  AAAANLHLL  LRDCSIYGIQ  WGFSTQNVNE  NYNRQIRHTA  EYANHCTTWY  240
QTGLERLRGT  NAFSWINYNR  FRREMTLTVL  DVCSLFSNYD  YRTYPTEVRA  ELTREIYTDP  300
IGFQNSPLPG  VFPNWYDYAR  SFAEINIAI  RAPRTVTWLN  STTIYTGRLN  GYNNNSYYWA  360
GFRQNFSETN  SGSSFNPGDL  GDLTPNYRIE  TLDMVNRDIY  SIYSRVVSQS  WPIGNVKLFG  420
VSSSTLSFRD  LNNSSSTLTV  YENPTNFSSQ  YLTTEFFPEN  SERPTFTDYS  HRLTCLTRIG  480
AGNYGLVLCA  GWTSSSVVER  NRLQDPKITQ  YPAVKGFNLD  GFTVVKGTF  TGGNWLRSR  540
VTGSFRLNVY  SPSVQTYRMR  IRYASPLGNS  TLGISSTDAG  ISFTSFPLPS  TIGSMPSTVP  600
YEAFRVLDIP  ITVTVASQRN  YNFIFDILNP  SVGAVYIDRI  EFVPV  645

SEQ ID NO: 104      moltype = DNA  length = 2895
FEATURE            Location/Qualifiers
misc_feature        1..2895
                    note = A synthetic coding sequence used for expression in a
                        plant cell encoding a CR-BREla.TIC7383_29.nno_Mc:1 protein

```

-continued

with, and an additional alanine codon, and a C-terminal truncation relative to the TIC7383 protein.

source 1..2895
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 104

atggctaacc	agaaccagaa	ccaaaaccag	aacaagaacg	agctgcagat	cattgaaccc	60
tcctcggaca	gcctcctgta	cagccataac	aactacccgt	acgccactga	cccaaacacc	120
gtgctcgagg	ggcgcaacta	caaggagtg	ctcaacaaat	gcaccgacaa	ctacaccgac	180
gcactccagg	gcccggaggc	gacggccatc	agcaagggtg	ccgtctccgc	ggccatctcc	240
atctcgacga	aggtcctagg	gtcctctagg	gtcccgttcg	ctgcgcagat	tggccagctg	300
tggacgttca	tccttaacgc	actctggccc	tcagacaaca	cccagtgga	ggagtccatg	360
cgccacgtcg	aggaactgat	caaccagcgc	atagcggact	acgcgaggaa	caaggccctg	420
gccgagctga	cagggtctgg	caacaacctt	gacctctaca	tcgaggcgct	cgaggactgg	480
aagcgggaacc	ctacctccca	acaagctaag	gaccgcgtta	tcgaccgctt	ccgaatcgcg	540
gacggcctct	tcgaggggta	catgccgagc	ttccgggtct	cggtctacga	ggtgcccctc	600
ctcacgggtg	acgcggcgcc	ggctaatttg	catctcctgc	tgtccaggga	ttgctcgatc	660
tacggaatcc	agtggggctt	ctcccagact	aacgttaacg	agaactacaa	tcggcagatt	720
cggcacaccg	cagagtacgc	taaccactgc	acaacatggt	atcagaccgg	cctcgagagg	780
ctgcgcggca	ccaatgcttt	ctcatggatc	aactacaacc	gcttcgcgcc	cgagatgaca	840
ctgactgtct	tggacgtctg	ctccctcttc	tccaactatg	actaccgcac	ttaccccact	900
gaggtccggg	cgcagctctg	gcgggagatc	tacacagacc	ccatcggttt	ccagaactcc	960
ccgctccctg	ccgtgggtcc	caattgggtg	gactacgcga	ggctcggtcg	ggaaatcgaa	1020
aacatcgcca	tacgagcacc	tcggcagatc	acgtgggtga	actcaaccac	catctacacc	1080
gggagactga	acggctacaa	caatagcaac	tactactggg	ccgggttcgc	ccagaacttc	1140
tcagagacca	actctggcag	caagcttcaac	ggcccagacc	taggcgacct	cactccgaac	1200
tacaggatag	acgacctcga	catggtcaac	agggacatct	acagcatcta	ctcccgggtc	1260
gtgtcgcaat	cctggccctg	cggaacgctc	aagctcttcg	gcgtaagctc	ctctacctta	1320
agtttccggg	atctcaacaa	caacagctcc	ggcaccctgg	tctacgagaa	cccgaactaa	1380
ttcagctccc	agtacctaac	cacagagttt	ccaggggaga	actccgagcg	gcccaccttc	1440
acggactaca	gtcacccgct	aacctgcctt	acgcggatcg	gggcccggaa	ctacggcctc	1500
gtgttttgtg	cggtttggac	ttcttcgagt	gtcgaaaagg	acaaccgcct	acaacccgat	1560
aagattaccg	agtaccccg	cgtcaagggtg	ttcaacttgg	acggcttcac	agtggtgaaa	1620
gggactgggt	tgcacggggg	caactggctg	cggtccagca	gggtccaccg	gtcattccgc	1680
ctgaacgtat	actcacctag	tgtgcagact	tatcggatga	ggatcaggta	cgctccccg	1740
cttggaact	cgacgcttgg	gataagttcg	acagacgccg	gcattctggt	cacgagtttc	1800
cctctgcctc	ccactatcgc	gagcatgccg	tccactgtgc	cctacgaggg	gttccgtgtc	1860
ctcgacatcc	ccatcacggg	caaccgtagc	tcccaaagga	actacaactt	catcttcgac	1920
atccttaacc	cctcggttgg	ggccgtgtac	atcgatcgta	ttgagttcgt	gcccgtgggc	1980
ttctcagttg	tcaggtacga	aaccaagcac	gagcttgaaa	aggcgaaaaa	agcagttaac	2040
gacctcttca	cgaacgagtc	caaaaacatg	ttaaagaaag	acacgaccga	ctacgacatc	2100
gaccaggctg	cggacctggt	ggagtgctga	agcgacgagt	gcgcccacgc	taagatgata	2160
ctgctggacg	aagtgaaagc	cgcgaagcag	ctgagcggag	ctcggaacct	gctccagaac	2220
gggaacttca	aggtcttaga	cgttgacaac	aacaacccgt	ggacgacctc	accgaacgta	2280
acgatccagg	agaacaaccc	gatcttcaaa	ggctcactacc	tctccatgtc	cggtgcgaac	2340
gcaatcgagg	ccaccaacga	ggtgttccca	acgtacgtgt	accaaaagat	tgaggagagt	2400
aaactcaagc	cctacacaag	gtacaagggt	cgggggttca	tcggccagag	caaggacgtc	2460
gagctgtctg	tcacgcgtta	cgacgaggag	gtggacgcta	ttctgaacct	gcctaacgac	2520
ctaaagtacg	cagtgccgac	gcacctgtcg	ggagagttca	accgctgcaa	gcctcacact	2580
taccggccca	cagacccccg	atgccacgac	gacgtgatcg	acaagatcga	catctcctcg	2640
ccgtgtcaga	acaacattat	gctgtcggac	cgggacatct	ctagtcttca	tagcggactt	2700
ggcaagaagc	acgggactct	ccacgagagc	catcacttcg	agttccacat	cgacacgggc	2760
aaaaatcgacc	tcgtggagaa	ccttggcatc	tgggtgatct	tcaagatctg	ctctaccgac	2820
ggatacgcca	cgctcgacaa	tctggaagta	atagaggaa	gcccgttggtg	cgcagagttc	2880
ctagagcggg	tgtag					2895

SEQ ID NO: 105 moltype = AA length = 964
FEATURE Location/Qualifiers
REGION 1..964
note = The amino acid sequence of the
CR-BRE1a.TIC7383_29.nno_Mc:1 protein, with an additional
alanine residue, and comprises amino acids 1 through 963
of TIC7383.
source 1..964
mol_type = protein
organism = synthetic construct

SEQUENCE: 105

MANQNQNQ	NKNEIIEP	SSDSLLYSHN	NYPYATDPNT	VLEGRNYKEW	LNKCTDNYTD	60
ALQGPEATAI	SKGAVSAAIS	ISTKVLGLLG	VPFAAQIGQL	WTFILNALWP	SDNTQWEEFM	120
RHVEELINQR	IADYARNKAL	AELTGLGNL	DLYIEALEDW	KRNPTSQQAK	DRVIDRFRIA	180
DGLFEGYMP	FRVSGYEVPL	LTVYAAAANL	HLLLLRDCSI	YGIQWGSQPT	NVNENYNRQI	240
RHTAEYANHC	TTWYQTGLER	LRGTNAFSWI	NYNRFRREMT	LTVLDVCSLF	SNYDYRTYPT	300
EVRAELTREI	YTDPIGQNS	PLPGVVPNWY	DYARSFAEIE	NIAIRAPRTV	TWLNSTTIYT	360
GRLNGYNN	SYWAGFRQNF	SETNSGSSFN	GPDLGDLTPN	YRIETLDMVN	RDIYSIYSRV	420
VSQSWPIGNV	KLFGVSSSTL	SFRDLNNNSS	GTLVYENPTN	FSSQYLTFEF	PGENSERPTF	480
TDYSHRLTCL	TRIGAGNYGL	VLCAGWTSSS	VERDNRLQPD	KITQYPAVKG	FNLDGFTVVK	540
GTGFTGGNWL	RSSRVTSFGR	LNVSFSPVQT	YRMRIYASFP	LGNSTLGISS	TDAGISFTSF	600
PLPSTIGSMP	STVPYEFARV	LDIPITVTVA	SQRNYNFIFD	ILNPSVGAVY	IDRIEFVPGV	660
SSVFYEYETH	BLEKAKKAVN	DLFTNESKNM	LKKDITDYDI	DQAADLVECV	SDECAHAKMI	720

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LLEDEVKYAKQ LSEARNLLQN GNFKVLVDVN NNPWTSPNV TIQENNPFIK GHYLSMSGAN 780
 AIEATNEVP TYVYQKIEES KLPYTRYKV RGFIGQSKDV ELLVTRYDEE VDAILNVPND 840
 LKYAVPTHLG GEFNRCKPHT YPATDPRCHD DVIDKIDISS PCQNNIMLSD ADISSLHSGL 900
 GKKGICHES HHFEFHIDTG KIDLVENLGI WVIFKICSTD GYATLDNLEV IEEGPLGAES 960
 LERV 964

SEQ ID NO: 106 moltype = DNA length = 3774
 FEATURE Location/Qualifiers
 misc_feature 1..3774
 note = A synthetic coding sequence used for expression in a
 plant cell encoding a CR-BREla.TIC7383_30.nno_Mc:1
 protein, comprising an additional alanine codon, and
 mutations to the codons corresponding amino acid positions
 964, 966, and 968 relative to
 source 1..3774
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 106
 atggctaacc agaaccagaa ccaaaaccag aacaagaacg agctgcagat cattgaacc 60
 tcctcggaca gcctcctgta cagccataac aactaccgt acgccactga cccaaacacc 120
 gtgctcgagg ggcgcaacta caaggagtggt ctcaacaaat gcaccgacaa ctacaccgac 180
 gcactccagg gcccgaggc gacggccatc agcaagggtg cgtctccgc ggccatctcc 240
 atctcgacga aggtcctagg gctcctagg gtcccgttcg ctgacgagat tggccagctg 300
 tggacgttca tccttaacgc actctggccc tcagacaaca cccagtgagg ggagttcatg 360
 cgccacgtcg aggaactgat caaccagcgc atagcggact acgcgaggaa caaggccctg 420
 gccagactga cagggtctgg caacaacctg gacctctaca tcgaggcgct cgaggactgg 480
 aagcggaaac ctacctccca acaagctaag gaccgcgtta tcgaccgctt ccgaatccgc 540
 gacggcctct tcgaggggta catgccgagc ttccgggtct cgggctacga ggtgccctc 600
 ctacgggtgt acgcggcgcc ggctaatttg catctcctgc tgcacaggga ttgctcgatc 660
 tacggaatcc agtggggctt ctcccagact aacgttaacg agaactaca tcggcagatt 720
 cggcacaccc ccaagtacgc taaccactgc acaacatggt atcagaccgg cctcgagagg 780
 ctgcgcggca ccaatgcttt ctcatggatc aactacaacc gcttcgcgcg cgagatgaca 840
 ctgactgtct tggacgtctg ctccctcttc tccaactatg actaccgcac ttacccact 900
 gaggtccggg cgaagcttac gcgggagatc tacacagacc ccacgggtt ccagaactcc 960
 ccgctccctg gcgtgggtcc caattgggtg gactacgcga ggtcggttcg ggaaatcgaa 1020
 aacatcgcca tacgagcacc tcggacagtc acgtggctga actcaaccac catctacacc 1080
 gggagactga acggtacga caatagcaac tactactggg ccgggttcg ccagaacttc 1140
 tcagagacca actctggcag cagcttcaac ggcccagacc taggcgacct cactccgaac 1200
 tacaggatag agaccctcga catggtcaac agggacatct acagcatcta ctcccggtc 1260
 gtgtcgcaat cctggcccat cgccaacgtc aagctcttcg cgtaagctc ctctacctta 1320
 agttcccggt atctcaacaa caacagctcc ggcaccctgg tctacgagaa cccgactaac 1380
 ttcagctccc agtacctaac cacagagttt ccaggggaga actccgagcg gccacccttc 1440
 accgactaca gtcaccgctt aacctgcctt acgcggatcg gggccgggaa ctacggctc 1500
 gtgctttgtg cgggttggtg ttcttcgagt gtcgaaaagg acaaccgctt acaaccgat 1560
 aagattaccg agtaccgcgc cgtcaagggt ttcaacttgg acggtctcac agtggtgaaa 1620
 gggactgggt tcaccggggg caactggctg cgggtccagca ggtccaccgg gtcattccgc 1680
 ctgaacgtat actcacctag tgtgcagact tatcggtatg ggtacaggta cgctcccg 1740
 ctgtgcaact cgacgcttgg gataagttcg acagacgccc gcatctcgtt cagcagtttc 1800
 cctctgcccc ccactatcgg gacgatgccg tccactgtgc cctacgaggg gttcgtgtc 1860
 ctgcacatcc ccatcagcgt caccgtagcc tccaaaagga actacaactt catcttcgac 1920
 atccttaacc cctcggttgg ggcggtgtac atcgatcgta ttgagttcgt gccctggggc 1980
 tcttcagttg tcagtagcga aaccaagcac gagcttgaaa aggcgaaaaa agcagttaac 2040
 gacctcttca cgaacgagtc caaaaacatg ttaaagaaag acacgaccga ctacgacatc 2100
 gaccaggctg cggacctggt ggagtgctga agcagcaggt gcgcccacgc taagatgata 2160
 ctgctggacg aagtgaaatg cgttgacaac aacaaccgtt ggacgacctc accgaacgta 2220
 gggaaactta aggtcttaga cgttgacaac aacaaccgtt ggacgacctc accgaacgta 2280
 acgatccagg agaacaaccc gatcttcaaa ggtcactacc tctccatgtc cgggtcgaaac 2340
 gcaatcgagg ccaccaacga ggtgttccca acgtacgtgt accaaaagat tgaggagagt 2400
 aaactcaagc cctacacaag gtacaagggt cgggggttca tcggccagag caaggacgtc 2460
 gagctgctcg tcacgcgtta cgacgaggag gtggacgcta ttctgaacgt gcctaacgac 2520
 ctaaagtacg cagtgcggac gcacctgtcg ggagagttca accgctgcaa gcctcacact 2580
 taccgggcca cagacccccg atgccacgac gacgtgatcg acaagatcga catctcctcg 2640
 ccgtgtcaga acaacattat gctgtcggac gcggacatct ctagtcttca tagcggactt 2700
 ggcaagaagc acgggactcg ccacgagagc catcacttcg agttccactc cgacacgggc 2760
 aaaatcgacc tcgtggagaa ccttggcatc tgggtgatct tcaagatctg ctctaccgac 2820
 ggatacgcga cgttcgacaa tctggaagta atagaggaa gcccgttggg cgacaggtcc 2880
 ctagagcggg tggcacgtgc ggaggctaag tggaagcata acatggagca caagtgtctg 2940
 gaaacgaaac acgcatacca cgcggcgaag caggctgtcg aagcaactct cagcaacttc 3000
 aaggacgaga ggctcaagtt cgagacgacg attagtaaca tctgtcagc tgagtacct 3060
 gtccagttca tcccatcgt ctacaacaag ttggtgtcgg acgtacctgg gatgaactac 3120
 gacatctaca ccagagctcaa gaaccgcatc tggcaagcgt tcaacttgta cgaccagagg 3180
 aacatcatca agaacggcca ctcaatcat ggtttaatgc attggcacgc gacaccgcac 3240
 gcgaacgtgc agcaaataga cgttatcttc gtgctgggtc tcccaaaactg gggggccaac 3300
 gtcagccagg aggtgtgctt caagcacaac cggggctacg tgcgtcgagt caggccaag 3360
 gagggaggtc acggcaagggt ctacgtgacg atttcggact gcgctaacca ggtcgagaag 3420
 ctgtcgttca ctaccccgca ctactctacc gacggcgttc catacgaaca gtccaactac 3480
 cgacccgacg gagtgtctta cgggcagcac ggggtcaaca tcgaccaggt gccctacgag 3540
 caatctggtc accccaacga cgggggttct tacgaacagt cgggctaccg tacggacggg 3600
 gtcccgtaac agcagcatgg atgccactcg gacggaagcc gtgaggagca gcacgggtac 3660

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gttacgaaga ctatcgacgt cttccccgac acggacaaggt tccggatcga catcgcgagag 3720
accgagggga cgttcaaggt tgagagcgtg gagctcatct gcatggaggga gtag 3774

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SEQ ID NO: 107      moltype = AA  length = 1257
FEATURE            Location/Qualifiers
REGION             1..1257
                   note = The amino acid sequence of the
                   CR-BREla.TIC7383_30.nno_Mc:1 protein, with an additional
                   alanine residue, and comprises the mutations, K964A;
                   R966A; K968A, relative to TIC7383.
source             1..1257
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 107
MANQNQNQNNQ NKNELQIIEP SSDSLLYSHN NYPYATDPNT VLEGRNYKEW LNKCTDNYTD 60
ALQGPEATAI SKGAVSAAIS ISTKVLGLLG VPFAAQIGQL WTFILNALWP SDNTQWEEFM 120
RHVEELINQR IADYARNKAL AELTGLGNL DLYIEALEDW KRNPTSQQAQ DRVIDRFRIA 180
DGLFEGYMPS FRVSGYEVPL LTVYAAAANL HLLLLRDCSI YGIQWGSQT NVNENYNRQI 240
RHTAAYANHC TTWYQTGLER LRGTNAFSWI NYNRPREMT LTVLDVCSLF SNYDYRTYPT 300
EVRAELTREI YTDPIGFQNS PLPGVVPNWY DYARSFAEIE NIAIRAPRTV TWLNSTTIYT 360
GRLNGYNNNS YYWAGFRQNF SETNSGSSFN GPDLDGLTPN YRIETLDMVN RDIYSIYSRV 420
VSQSWPIGNV KLFVSVSSSTL SFRDLNNNSS GTLVYENPTN FSSQYLTFEF PGENSERPTF 480
TDYSHRLTCL TRIGAGNYGL VLCAGWTSSS VERDNRLQPD KITQYPAVKG FNLDGFTVVK 540
GTGFTGGNWL RSSRVTSFNR LNVYSPSVQT YRMIRYASP LGNSTLGISS TDAGISFTSF 600
PLPSTIGMSP STVPYEAFFR LDIPITVTV SQRNYNFIED ILNPSVGAVY IDRIEFVPG 660
SSVFYETKH ELEKAKKAVN DLFTNESKNM LKKDTTDYDI DQAADLVECV SDECAHAKMI 720
LLDEVKYAQK LSEARNLLQN GNFKVLVDVN NNPWTSTPNV TIQENNPFIK GHYLSMSGAN 780
AIEATNEVPF TVVYQKIEES KLKPYTRYKV RGFIGQSKDV ELLVTRYDEE VDAILNVPND 840
LKYAVPHTLS GEFNRCKPHT YPATDPRCHD DVIDKIDISS PCQNNIMLSD ADISSLHSG 900
GKKHGI CHES HHFEFHIDTG KIDLVENLGI WVIKICSTD GYATLDNLEV IEEGPLGAES 960
LERVARAEAK WKHNMEHKCS ETKHAYHAAK QAVEALFTNF KDERLKFETT ISNLSAEYL 1020
VQSIPYVYNK WLSVDPGMNY DIYTELKNRI WQAFNLYDQR NIIKNGHFNH GLMHWATPH 1080
ANVQQIDGIS VLVLPNWGAN VSGEVCLKHN RGYVLRVTAQ EEGHGKGYVT ISDCANQVEK 1140
LSFTSRDYST DGVPEQSNY PTDGVSYGQH GCNIDRVPEY QSGYPTDGVV YEQSGYRTDG 1200
VPYKQHGCHS DGSREEQHG VTKTIDVFPD TDKVRIDIGE TEGTFKVESV ELICMEE 1257

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SEQ ID NO: 108      moltype = DNA  length = 3201
FEATURE            Location/Qualifiers
misc_feature        1..3201
                   note = A synthetic coding sequence used for expression in a
                   plant cell encoding a CR-BREla.TIC7383_31.nno_Mc:1 protein
                   with an alanine codon, C-terminal truncation, and
                   mutations to the codons corresponding amino acid positions
                   964, 966, and 968 relative to
source             1..3201
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 108
atggctaacc agaaccagaa ccaaaaccag aacaagaacg agctgcagat cattgaaccc 60
tctcgggaca gctcctctgta cagccataac aactaccgct acgccaactga cccaaacacc 120
gtgctcgaggg ggcgcaacta caaggagtggt ctcaacaagt gcaccgacaa ctacaccgac 180
gcactccagg gccgggagcc gacggccatc agcaagggtg cgtctccgc ggccatctcc 240
atctcgacga aggtcctagg gctcctagggt gtcccggtcg ctgctgcagat tggccagctg 300
tggacgttca tctctaaccg actctggccc tcagacaaca cccagtgagg gaagtctcat 360
cgccacgtcg aggaactgat caaccagcgc atagcggact acgagaggaa caaggccctg 420
gccgagctga caggggctggg caacaaccta gacctctaca tcgaggcgct cgaggactgg 480
aagcgggaacc ctacctccca acaagctaag gaccgcgtta tcgaccgctt ccgaatcgcg 540
gacggcctct tcgaggggta catgccgagc ttccgggtct cgggctacga ggtgcccttc 600
ctcacgggtg acgcgggcgc ggctaatttg catctcctgc tgctcaggga ttgctcgatc 660
tacggaatcc agtggggctt ctcccagact aacgttaacg agaactaca tcggcagatt 720
cggcacaccg ccgagtagcg taaccactgc acaacatggt atcagaccgg cctcgagagg 780
ctgctcgcca ccaatgcttt ctcatggatc aactacaacc gcttcgcgcg cgagatgaca 840
ctgactgtct tggacgtctg ctccctcttc tccaactatg actaccgcac ttacccact 900
gaggtccggg ccgagcttac gcgggagatc tacacagacc ccatcggtt ccagaactcc 960
ccgctccctg gcgtgggtccc caattgggtac gactacgcga ggtcggttcg ggaaatcgaa 1020
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gggagactga acggctacaa caatagcaac tactactggg ccgggttcgc ccagaacttc 1140
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ttcagctccc agtacctaac cacagagttt ccaggggaga actccgagcg gccacacctc 1440
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gtgctttgtg cgggttgtag ttcttcgagt gtcgaaaggg acaaccgcct acaaccgat 1560
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cttggaactc cgcgcttg gataagttcg acagacgcgg gcattctggt cagcagtttc 1800
cctctgcctc ccactatcgg gagcatgccg tccactgtgc cctacgagcg gttccgtgtc 1860

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-continued

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ctcgacatcc ccatcacggt caccgtagcc tcccaaagga actacaactt catcttcgac 1920
atccttaacc cctcggttgg ggccgtgtac atcgatcgta ttgagttcgt gcccggtgggc 1980
tcttcagtggt tcgagtagca aaccaagcac gagcttgaaa aggcgaaaaa agcagttaac 2040
gacctcttca cgaacgagtc caaaaacatg ttaaagaaa acacgaccga ctacgacatc 2100
gaccaggctg cggacctggt ggagtgcgta agcgacgagt gcgcccacgc taagatgata 2160
ctgctggagc aagtgaagta cgcgaagcag ctgagcgagg ctccgaacct gctccagaac 2220
gggaacttca aggtcttaga cgttgacaac aacaacccgt ggacgacctc accgaacgta 2280
acgatccagg agaacaaccc gatcttcaaa ggtcactacc tctccatgtc cgggtgcgaac 2340
gcaatcgagg ccaccaacga ggtgttccca acgtacgtgt accaaaagat tgaggagagt 2400
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gagctgctcg tcacgcgtta cgacgaggag gtggacgcta ttctgaacgt gcctaacgac 2520
ctaaagtacg cagtgcgacg gcacctgtcg ggagagtcca accgctgcaa gcctcacact 2580
taccgggcca cagacccccg atgccacgac gacgtgatcg acaagatcga catctcctcg 2640
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ggcaagaagc acgggatctg ccacgagagc catcacttgc agttccacat cgacacgggc 2760
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ctagagcggg tggcacgtgc ggaggctaag tggaaagcata acatggagca caagtgtctg 2940
gaaacgaaac cgcataacca cgcggcgaag caggctgtcg aagcactctt cacgaacttc 3000
aaggacgaga ggctcaagtt cgagacgacg attagtaaca tctgtcagc tgagtacct 3060
gtccagtcca tccatacagt ctacaacaag tggtgtcggc acgtacctgg gatgaactac 3120
gacatctaca ccgagctcaa gaaccgcata tggcaagcgt tcaacttgta cgaccagagg 3180
aacatcatca agaacgggta g

```

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SEQ ID NO: 109      moltype = AA length = 1066
FEATURE            Location/Qualifiers
REGION             1..1066
                    note = The amino acid sequence of the
                    CR-BREla.TIC7383_31.nno_Mc:1 protein, with an additional
                    alanine residue, and comprising amino acids 1 through
                    1065, and also comprises the mutations, K964A; R966A;
                    K968A, relative to TIC7383.
source             1..1066
                    mol_type = protein
                    organism = synthetic construct

```

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SEQUENCE: 109
MANQNQNQNK NKNELQIIEP SSDSLLYSHN NYPYATDPNT VLEGRNYKEW LNKCTDNYTD 60
ALQGPPEATAI SKGAVSAAIS ISTKVLGLLG VPFAAQIGQL WTFILNALWP SDNTQWEEFM 120
RHVEELINQR IADYARNKAL AELTGLGNNL DLYIEALEDW KRNPISQOAK DRVIDRFRFA 180
DGLFEGYMPFS FRVSGYEVPL LTVYAAAANL HLLLLRDCSI YGIQWGSQT NVNENYNRQI 240
RHTAEYANHC TTWYQTGLER LRGTNAFSWI NYNRFREMT LTVLDVCSLF SNYDYRTYPT 300
EVRAELTREI YTDPIGFQNS PLPGVVPNWY DYARSAEIE NIAIRAPRTV TWLNSTTIYT 360
GRLNGYNNNSN YYWAGFRQNF SETNSGSSFN GPDLDLTPN YRIETLDMVN RDIYSIYSRV 420
VSQSWPIGNV KLPVGSSTSL SFRDLNNSST GTLVYENPTN FSSQYLTFEF PGENSERPTF 480
TDYSHRLTCL TRIGAGNYGL VLCAGWTSSS VERDNRLQPD KITQYPVAVG FNLDDGFTVVK 540
GTGFTGWNWL RSSRVTGSFR LNVYSPSVQT YRMRIYASP LGNSTLGISS TDAGISFTSF 600
PLPSTIGSMP STVPYEAFRV LDIPITVTVS SQRYNFIFD ILNPSVGAVY IDRIEFVVPVG 660
SSVFYEYETH ELEKAKKAVN DLPFTNESKNM LKKDITDYDI DQADLVECV SDECAHAKMI 720
LLDEVKYAKQ LSEARNLLQN GNFKVLVDVN NNPWTSPNV TIQENNPFIK GHYLSMSGAN 780
AIEATNEVFP TYVYQKIEES KLKPYTRYKV RGFIGQSKDV ELLVTRYDEE VDAILNVPND 840
LKYAVPHTLS GFHNRCKPHT YPATDPRCHD DVIDKIDISS PCQNNIMLSD ADISSLSHSL 900
GKKHGI CHES GHFHFHIDTG KIDLVENLGI WVIFKICSTD GYATLDNLEV IEEGPLGAES 960
LERVARAEAK WKHNMHAKCS ETKHAYHAAK QAVEALFTNF KDERLKFETT ISNILSAEYL 1020
VQSIPYVYNK WLSVDPGMNY DIYTELKNRI WQAFNLYDQR NIIKNG 1066

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SEQ ID NO: 110      moltype = DNA length = 3756
FEATURE            Location/Qualifiers
misc_feature        1..3756
                    note = A synthetic coding sequence used for expression in a
                    plant cell encoding a CR-BREla.TIC7383_32.nno_Mc:1 protein
                    comprising an additional alanine codon, and a deletion of
                    the codons corresponding amino acid positions 964 through
                    969 relative to
source             1..3756
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 110
atggctaacc agaaccagaa ccaaaaccag aacaagaacg agctgcagat cattgaaccc 60
tcctcggaca gcctcctgta cagccataac aactaccctg acgccaatga cccaaacacc 120
gtgctcgagg ggcgcaacta caaggagtggt ctcaacaaat gcaccgacaa ctacaccgac 180
gcactccagg gcccgagagg cagcgccatc agcaagggtg ccgtctccgc ggccatctcc 240
atctcgacga aggtcctagg gctcctaggg gtcccgttcg ctgcgcagat tggccagctg 300
tggacgttca tccttaacgc actctggccc tcagacaaca cccagtgagg ggagttcatg 360
cgccacgtcg aggaagcgc caaccagcgc atagcggact acgagaggaa caaggccctg 420
gccgagctga cagggtctggg caacaacctt gacctctaca tcgaggcgct caggagctgg 480
aagcggaaac ctacctccca acaagctaag gaccgcgtta tcgaccgctt ccgaatcgcg 540
cagcgccctct tcgaggggta catgccgagc ttccgggtct cgggctacga ggtgcccctc 600
ctcacgggtg acgcgcgagg ggctaatttg catctcctgc tgctcaggga ttgctcgatc 660
tacggaatcc agtggggcct ctcccagact aacgttaacg agaactacaa tcggcagatt 720

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cggcacaccg cccagtagcgc taaccactgc acaacatggc atcagaccgg cctcagagagg 780
ctgcgcggca ccaatgcttt ctcatggatc aactacaacc gcttcgcgcg cgagatgaca 840
ctgactgtct tggacgtctg ctccctcttc tccaactatg actaccgcac ttacccact 900
gaggtcgggg cccagcttac gccgggagatc tacacagacc ccacggggtt ccagaactcc 960
ccgctccctg gcgtgggtccc caattggtag gactacgcga ggctggtcgc ggaaatcgaa 1020
aacatcgcca tacgagcacc tcggacagtc acgtggctga actcaaccac catctacacc 1080
gggagactga accggtacaa caatagcaac tactactggg ccgggttcgc ccagaacttc 1140
tcagagacca actctggcag cagcttcaac ggcccagacc taggcgacct cactccgaac 1200
tacaggatag agaccctcga catgggtcaac agggacatct acagcatcta ctcccggtc 1260
gtgtcgcaat cctggcccat cggcaacgtc aagctcttcg gcgtaagctc ctctacctta 1320
agtttcgggg atctcaacaa caacagctcc ggcacccctg tctacgagaa cccgactaac 1380
ttcagctccc agtacctaac cacagagttt ccaggggaga actccgagcg gccacacttc 1440
acggactaca gtcaccgctt aacctgcctt acgctggatc gggccgggaa ctacggcttc 1500
gtgcttttgg gcgtttggac ttcttcgagt gtcgaaaagg acaaccgcct acaaccgcgt 1560
aagattaccc agtaccgccg cgtcaagggg ttcaacttgg acggtctcac agtggtgaaa 1620
gggactgggt tcaccggggg caactggctg cgggtccagca gggtcaccgg gtcattccgc 1680
ctgaacgtat actctggcag tgtgcagact tatcggtaga ggatcaggta cgctccctcg 1740
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atccttaacc cctcggttgg ggcgctgtac atcgatcgta ttgagttcgt gccctcgggc 1980
tcttcagtgt tcgagtacga aaccaagcac gagcttgaag aggcgaaaaa agcagttaac 2040
gacctcttca cgaacgagtc caaaaacatg ttaaagaaag acacgaccca ctacgacatc 2100
gaccaggctg cggacctggt ggagtgctga agcgacgagt gcgcccacgc taagatgata 2160
ctgctggacg aagtgaagta cgcgaagcag ctgagcggag ctcggaaact gctccagaac 2220
gggaacttca aggtcttaga cgttgacaac aacaaccctg ggacgacctc accgaacgta 2280
acgatccagg agaacaaccc gatcttcaaa ggctactacc tctccatgtc cgggtgcgaac 2340
gcaatcgagg ccaccaacga ggtgttccca acgtacgtgt accaaaagat tgaggagagt 2400
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gagctgctcg tcacgcgtta cgacgaggag gtggacgcta ttctgaacgt gcctaacgac 2520
ctaaagtacg cagtgcgcag gcacctgtcg ggagagtcca accgctgcaa gcctcacact 2580
taccgggcca cagacccccg atgccacgac gacgtgatcg acaagatcga catctcctcg 2640
ccgtgtcaga acaacattat gctgtcggac gcggacatct ctagtcttca tagcggactt 2700
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aaaaatcgac tcgtggagaa ccttggcatc tgggtgatct tcaagatctg ctctaccgac 2820
ggatacgcca cgctcgacaa tctggaagta atagaggaa gcccgttggg cgagagttcc 2880
ctagagcggg tgtggaagca taactggag cacaagtgtc cggaaacgaa acacgcatac 2940
cacgcggcga agcaggctgt cgaagcactc ttcacgaact tcaaggacga gaggtccaag 3000
ttcgagacga cgatttagta catcctgtca gctgagtacc tagtccagtc catccatac 3060
gtctacaaca tctggctgtc ggaagtacct gggatgaact acgacatcta caccgagctc 3120
aagaaccgca tctggcaagc gttcaacttg tacgaccaga ggaacatcat caagaacggc 3180
cacttcaatc atggtttaat gcattggcac gcgacaccgc acgcaacgt gcagcaata 3240
gacgtattt cgtgtctggt cctcccaaac tggggggcca acgtcagcca ggaggtgtgc 3300
ctcaagcaca accggggcta cgtgtcgcga gtcacggcca agggaggagg tcacggcaag 3360
ggctacgtga cgatttcgga ctgcgctaac caggtcgaga agctgtcgtt cactagccgc 3420
gactactcta ccgacggcgt ccatcacgaa cagttccact acccgaccga cggagtgtct 3480
tacggggcagc accgggtgcaa catcgaccga gtgcccctag agcaatctgg ctacccacc 3540
cacgggggtt cttacgaaca gtcgggctac cgtacggagc gggtcccgta caagcagcat 3600
ggatgccact cggacgggag cagcaggggt acgttacgaa gactatcgac 3660
gtcttccccc acacgggcaa ggtccggatc gacatcggcg agaccgaggg gacgttcaag 3720
gttgagagcg tggagctcat ctgcatggag gagttag 3756

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SEQ ID NO: 111      moltype = AA length = 1251
FEATURE             Location/Qualifiers
REGION              1..1251
                    note = The the amino acid sequence of the
                        CR-BRE1a.TIC7383_32.nno_Mc:1 protein, with an additional
                        alanine residue, and comprising a deletion of amino acids
                        964 through 969 relative to TIC7383.
source              1..1251
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 111
MANQNQNQNK NKNELQIIEP SSDSLLYSHN NYPYATDPNT VLEGRNYKEW LNKCTDNYTD 60
ALQGPEATAI SKGAVSAAIS ISTKVLGLLG VPFAAQIGQL WTFILNALWP SDNTQWEEFM 120
RHVEELINQR IADYARNKAL AELTGLGNL DLYIEALEDW KRNPISQAK DRVIDRFRIA 180
DGLFEGYMPS FRVSGYEVPL LTVYAAAANL HLLLLRDCSI YGIQWGFSTQ NVNENYNRQI 240
RHTAEYANHC TRWYQTLER LRGTNAFSWI NYNRFREMT LTVLDVCSLF SNYDYRTYPT 300
EVRABELTREI YTDPIGFQNS PLPGVVPNWY DYARSAEIE NIAIRAPRTV TWLNSTTIYT 360
GRLNGYNNSN YYWAGFRQNF SETNSGSSFN GPDLDGLTPN YRIETLDMVN RDIYSIYSRV 420
VSQSWPIGNV KLFVSSSTL SFRDLNNNSN GTLVYENPTN FSSQYLTFEF PGENSERPTF 480
TDYSHRLTCL TRIGAGNYGL VLCAGWTSSS VERDNRLQPD KITQYPAVKG FNLDGFTVVK 540
GTGFTGGNWL RSSRVTSFRL LNVYSPSVQT YMRIRYASP LGNSTLGISS TDAGISFTSF 600
PLPSTIGMPS STVPYEAERV LDIPITVTV SQRNYNFIFD ILNPSVGAVY IDRIEFVPVG 660
SSVFYEYETH ELEKAKKAVN DLFNTESKNM LKKDITTDYDI DQAADLVECV SDECAHAKMI 720
LLDEVKYAKQ LSEARNLLQN GNFKVLDVDN NNPWTSPNV TIQENNPFIK GHYLSMSGAN 780
AIEATNEVFP TYVYQKIEES KLKPYTRYKV RGFIFQSKDV ELLVTRYDEE VDAILNVPND 840
LKYAVPHTLS GEFNRCKPHT YPATDPRCHD DVIDKIDISS PCQNNIMLSD ADISSLHSLG 900
GKKHGICHES HHFEFHIDTG KIDLVENLGI WVFIKICSTD GYATLDNLEV IEEGPLGAES 960

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LERVWKHNME HKCSETHKAY HAAKQAVEAL FTNFKDERLK FETTISNILS AEYLVSQIPY 1020
VYNKWLSDVP GMNYDIYTEL KNRIWQAFNL YDQRNIKNG HFNHGLMHWH ATPHANVQQI 1080
DGISVLVLPN WGANVSQEVLC LKHNRGYVLR VTAKEEGHKG GYVTISDCAN QVEKLSFTSR 1140
DYSTDGVPEY QSNYPDGVVS YGQHGCNIDR VPYEQSGYPT DGVPEQSGY RTDGVVPYKQH 1200
GCHSDGSREE QHGYVTKTID VFPDTPDKVRI DIGETEGTFK VESVELICME E 1251

SEQ ID NO: 112      moltype = DNA length = 1854
FEATURE            Location/Qualifiers
misc_feature       1..1854
                    note = A synthetic coding sequence used for expression in a
                    plant cell encoding a GOI-TIC10743.nno_Mc:1 chimeric
                    protein comprised of domains one and two of TIC7383 and
                    domain three of TIC7042.
source             1..1854
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 112
atggctaccg acaactacac cgagcgactc cagggcccgaggcgacggc catcagcaag 60
ggtgccgtct ccgcggccat ctccatctcg acgaagggtcc taggggtcctc aggggtcccg 120
ttcgctcgcg agattggcca gctgtggacg ttcactccta acgcactctg gccctcagac 180
aacacccagtg gggaggagtt catgcgccac gtcgaggaac tgatcaacca gcgcatacg 240
gactacgcga ggaacaaggc cctggccgag ctgacagggc tgggcaacaa cctagacctc 300
tacatcgagg cgctcgagga ctggaagcgg aacctacctc ccaacaagc taaggaccgc 360
gttatcgacc gcttcggaat cgcggacggc ctcttcgagg ggtacatgcc gagcttcagg 420
gtctcggggt acgaggtgct cctctcacg gtgtacgcgg cggcggtctaa ttgcatctc 480
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aacgagaact acaatcgcca gattcggcac accgcccagt acgtaacca ctgcacaaca 600
tggtatcaga ccggcctcga gaggtgcgc ggcaccaatg cttctcatg gatcaactac 660
aaccgcttcc gccgcgagat gacactgact gtcttgagc tctgctccct cttctccaac 720
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gaacccatcg ggttcagaa cctcccgtc cctggcgtgg tccccaattg gtacgactac 840
gcgaggtcgt tcgcggaaat cgaaaacatc gccatacgag cacctcggac agtcacgtgg 900
ctgaactcaa ccaccatcta caccgggaga ctgaacggct acaacaatag caactactac 960
tgggcccggg ttctctcaga accaactctg gcagcagctt caacggccca 1020
gaacctaggc acctcaactc gaactacagg atagagacc tcgacatggt caacagggac 1080
atctacagca tctactcccg ggtcgtgtcg caatcctggc ccatcggaac cgtaaggtc 1140
ttcggcgtaa gtcctctac cttaagtctt cgggatctca acaacaacag ctccggcacc 1200
ctgggtctac agaaccggac taacttcagc tccagtagc taaccacaga gtttccagg 1260
gagaactccg agcggccacc cttcacggac tacagtcacc gcctaacctg ccttcacgg 1320
atcggggcgg ggaactacgg cctcgtgctt tgtgcgggtt ggacttcttc gagtgtcgaa 1380
agggacaaca tcatcgagca gaacaagatc acccagttcc ctggcggtta gtcacacact 1440
cttaacaact gtcaagtgtt gcgtggcacc ggggtcaccg cggcgagact gcttcgcccc 1500
aacaacaatg gcaactttcg gctgaccatt accagcttct cgtcccagtc ctaccgcatc 1560
aggttcgat acgctaccag cctcggcaac acctctcttg tcatctctgc atccgacgcg 1620
ggcatctcca gtaccacaat cctcttgaca tccaccatca cgtcgtctcc gcagaccgtt 1680
ccgtaccagg cgttcccggt cctcgatctc ccgatcactt tcaactactc gaccactcag 1740
cgcaattaca ccttcgatct cgcctccag aacccgagca acgtaacctt attcatcgac 1800
cgcttcgagt tcgtaccaat cggcggctca ctgtccgagt acgagacgaa gtga 1854

SEQ ID NO: 113      moltype = AA length = 617
FEATURE            Location/Qualifiers
REGION            1..617
                    note = The amino acid sequence of the GOI-TIC10743.nno_Mc:1
                    chimeric protein.
source             1..617
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 113
MATDNYTDAL QGPEATAISK GAVSAAISIS TKVLGLLGVP FAAQIGQLWT FILNALWPSD 60
NTQWEEFMRH VEELINQRIA DYARNKALAE LTGLGNLNDL YIEALEDWKR NPTSQQAKDR 120
VIDRFRIADG LFEGYMPFSR VSGYEVPLLT VYAAANLHL LLLRDCSIYG IQWGFSTQTNV 180
NENYNRQIRH TAEYANHCTT WYQTGLERLR GTNAPSWINY NRFRREMTLT VLDVCSLFSN 240
YDYRTYPTVE RAELETREIYT DFIGFQNSPL PGVVPNWDYD ARSPAIEINI AIRAPRTVTW 300
LNSTTIYTRG LNGYNNSNYW WAGFRQNFSE TNSGSSFNGP DLGDLTPNYR IETLDMVNRD 360
IYSIYSRVVS QSWPIGNVKL FGVSSSTLSF RDLNNSSSGT LVYENPTNFS SQYLTFEFP 420
ENSERPTFTD YSHRLTCLTR IGAGNYGLVL CAGWTSSSVE RDNIIEQNKI TQFPVKSHS 480
LNNCQVVRGT GFTGGDWLRP NNGMTFRLTI TSFSSQSYRI RLRVATSVGN TSLVISSSDA 540
GISSTIPLT STITSLPQTV PYQAFRVVDL PITFTPTTQ RNYTFDFRLQ NPSNANVFID 600
RFEFVPIGGS LSEYETK 617

SEQ ID NO: 114      moltype = DNA length = 1851
FEATURE            Location/Qualifiers
misc_feature       1..1851
                    note = A synthetic coding sequence used for expression in a
                    plant cell encoding a GOI-TIC10744.nno_Mc:1 chimeric
                    protein comprised of domains one and two of TIC7383 and
                    domain three of TIC7381.
source             1..1851
                    mol_type = other DNA

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organism = synthetic construct

SEQUENCE: 114
atggctaccg acaactacac cgacgcactc cagggccccg aggcgcacggc catcagcaag 60
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ttcgctcgcg agattggcca gctgtggacg ttcataccta acgcactctg gccctcagac 180
aacacccagt gggaggagtt catgcccacc gtcgaggaac tgatcaacca gcgcatacgc 240
gactacgcga ggaacaaggc cctggccgag ctgacagggc tgggcaacaa cctagacctc 300
tacatcgagg cgctcgagga ctggaagcgg aacctacct cccaacaagc taaggaccgc 360
gttatcgacc gcttcgcaat cgcggacggc ctcttcgagg ggtacatgcc gagcttcocg 420
gtctcgggct acgaggtgcc cctcctcacg gtgtacgcgg cggcgggctaa tttgcatctc 480
ctgctgctca gggattgtct gatctacgga atccagtggg gcttctccca gactaacgtt 540
aacgagaact acaatcgcca gattcggcac accgccagat acgtaacca ctgcacaaca 600
tggtatcaga ccggcctcga gaggtgcgc ggcaccaatg ctttctcatg gatcaactac 660
aaccgcttcc gccgcgaagt gacactgact gtcttgagcg tctgtccctt cttctccaac 720
tatgactacc gcacttacc cactgaggtc cgggccgagc ttacgcggga gatctacaca 780
gaccccatcg ggttcagaa ctcccgcgtc cctggcgtgg tcccatttg gtaagactac 840
gcgaggtcgt tcgcggaagt cgaaaacatc gccatagcag cacctcggac agtcacgttg 900
ctgaactcaa ccaccatcta caccgggaga ctgaacggct acaacaatag caactactac 960
tgggccgggt tccgccagaa cctctcagag accaactctg gcagcagctt caacggccca 1020
gaactagcgc gaactacagg gaactacagg atagagacc tcgacatggg caacagggac 1080
atctacagca tctactcccg ggtcgtgtcg caatcctggc ccatcggcaa cgtaagctc 1140
ttcggcgtaa gtcctctac cttaaagttt cgggatctca acaacaacag ctccggcacc 1200
ctgggtctac agaaccggac taacttcagc tcccagtag taaccacaga gtttccagg 1260
gagaactccg agcggccacc cctcacggac tacagtacac gcctaacctg ccttacgcgg 1320
atcggggccg ggaactacgg cctcgtgctt tgtcggggtt ggacttcttc gactgtcgaa 1380
agggacaaca tactggagcg gaacaagatc acccagttcc ctggcgtaa gtccatacc 1440
ctgaacaatt gccaggtcgt ccgcggtaca ggctttaccg cggcgagctg gcttcggcct 1500
aacaacaacg ggtcgttcgc attgactatc acctcgttca gctcacagag ttacagaatc 1560
aggtgcgggt acgcgtccgc agcgaatacc tccctgcgaa tctcgtcttc ggcggccggg 1620
atctcgtcaa caaccgtccc actcgcctcc acgatcacta gcctcccga gacagccgtc 1680
ccatcagagg cgttcggggt ctgcgacctg ccaatcacgt tcaccaccgc gacgcagtct 1740
aactacactt tgcacttcgt cctccagaat ccttccaacg ccaacgtgtt catcgaccgc 1800
ttcgagtttg tcccgatttg aggttcgctc tcagagtacg agacgaagtg a 1851

SEQ ID NO: 115      moltype = AA  length = 616
FEATURE            Location/Qualifiers
REGION              1..616
                    note = The amino acid sequence of the GOI-TIC10744.nno_Mc:1
                    chimeric protein.
source              1..616
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 115
MATDNYTDAL QGPEATAISK GAVSAAISIS TKVLGLLGVP FAAQIGQLWT FILNALWPSD 60
NTQWEEFMRH VEELINQRIA DYARNKALAE LTGLGNLNDL YIEALEDWKR NPTSQQAKDR 120
VIDRFRIADG LFEGYMPSFR VSGYEVPLLT VYAAANLHL LLLRDCSIYG IQWGFQSTNV 180
NENYNRQIRH TAEYANHCTT WYQTGLERLR GTNAPSWINY NRFRREMTLT VLDVCSLFSN 240
YDYRTYPTFV RAEITREIYT DPIGFQNSPL PGVVPNWYDY ARSFAEINI AIRAPRTVTW 300
LNSTTIYTRG LNGYNNSNYI WAGFRQNFSE TNSGSSFNGP DLGLDTPNYR IETLDMVNRD 360
IYSIYSRVVS QSWPIGNVKL FGVSSSTLSF RDLNNNSSGT LVYENPTNFS SQYLTFEPPG 420
ENSERPTFTD YSHRLTCLTR IGAGNYGLVL CAGWTSSSVE RDNILERNKI TQFPVKSHS 480
LNNCQVVRGT GFTGDLWRP NNNGSFRLTI TSFSSQSYRI RLRVASAANT SLRISSAAG 540
ISSTTVPLAS TITSLPQTAV PYEAFRVIDL PITFTTATQS NYTFDFVLQN PSNANVFIDR 600
FEFVPIGSSL SEYETK 616

SEQ ID NO: 116      moltype = DNA  length = 1854
FEATURE            Location/Qualifiers
misc_feature        1..1854
                    note = A synthetic coding sequence used for expression in a
                    plant cell encoding a GOI-TIC10745.nno_Mc:1 chimeric
                    protein comprised of domains one and two of TIC7383 and
                    domain three of TIC7382.
source              1..1854
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 116
atggctaccg acaactacac cgacgcactc cagggccccg aggcgcacggc catcagcaag 60
ggtgccgtct ccgcggccat ctccatctcg acgaaggtec tagggctcct aggggtcccg 120
ttcgctcgcg agattggcca gctgtggacg ttcataccta acgcactctg gccctcagac 180
aacacccagt gggaggagtt catgcccacc gtcgaggaac tgatcaacca gcgcatacgc 240
gactacgcga ggaacaaggc cctggccgag ctgacagggc tgggcaacaa cctagacctc 300
tacatcgagg cgctcgagga ctggaagcgg aacctacct cccaacaagc taaggaccgc 360
gttatcgacc gcttcgcaat cgcggacggc ctcttcgagg ggtacatgcc gagcttcocg 420
gtctcgggct acgaggtgcc cctcctcacg gtgtacgcgg cggcgggctaa tttgcatctc 480
ctgctgctca gggattgtct gatctacgga atccagtggg gcttctccca gactaacgtt 540
aacgagaact acaatcgcca gattcggcac accgccagat acgtaacca ctgcacaaca 600
tggtatcaga ccggcctcga gaggtgcgc ggcaccaatg ctttctcatg gatcaactac 660
aaccgcttcc gccgcgaagt gacactgact gtcttgagcg tctgtccctt cttctccaac 720
tatgactacc gcacttacc cactgaggtc cgggccgagc ttacgcggga gatctacaca 780

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gaccccatcg ggttcagaa ctcctccgctc cctggcgctgg tccccaattg gtacgactac 840
gcgagggtcgt tcgcgggaaat cgaaaacatc gccatcacgag cacctcggac agtcacgtgg 900
ctgaactcaa ccaccatcta caccgggaga ctgaacggctt acaacaatag caactactac 960
tgggcccgggt tccgccagaa cttctcagag accaactctg gcagcagctt caacggccca 1020
gacctagggc acctcactcc gaactacagg atagagaccc tcgacatgggt caacaggggac 1080
atctacagca tctactcccg ggtcgtgtcg caatcctggc ccacgcggcaa cgtcaagctc 1140
ttcggcgtaa gtcctctctac cttaagtctt cgggatctca acaacaacag ctccggcacc 1200
ctggttctacg agaacccgac taacttcagc tcccagttac taaccacaga gtttcagggg 1260
gagaactccg agcggccac ctctcaggac tacagctcac gcctaacctg ccttacgcgg 1320
atcggggccg ggaactacgg cctcgtgctt tgtgcgggtt ggactctctc gagtgtcgaa 1380
agggacaaca tcatcgagcg caacaagatc acgcagttcc caggcgctca atcgcacacg 1440
ctaaacaact cgacagttgg ttgagggaacc gggtttcacg gagggtgaact ggtgaagca 1500
aataacaacg gcactttccg cctcactatt acgagctctt cctcccaaa gttaccgcac 1560
cgctctcgct acgccacgtc agtggggaac acctcgcgtg tgatattctc ctgcagcgcc 1620
gggatccagca gcaccaagat accctaataa tccacaataa cctcattgcc gcagacggtc 1680
cgctatcagg ccttcagagt ggtgtgacct cctattacgt tcacgacgct caccgaccag 1740
cgcaactaca catctcagtt cgggttgca g aaccggagc acgccaatgt cttctcgat 1800
cgtttcgagt tcgtcccgat aggcggctcc ctttcggagt acgaaacgaa gtga 1854

SEQ ID NO: 117          moltype = AA length = 617
FEATURE                Location/Qualifiers
REGION                1..617
                        note = The amino acid sequence of the GOI-TIC10745.nno_Mc:1
                        chimeric protein.
source                1..617
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 117
MATDNYTDAL QGPEATAISK GAVSAAISIS TKVLGLLGPV FAAQIGQLWT FILNALWPSD 60
NTQWEEMMRH VEBELINQRIA DYARNKALAE LTGLGNLNDL YIEALDEWKR NPTSQQAKDR 120
VIDRFRIADG LFEGYMPSFR WSQYEVPLLT VYAAAAANLHW LLLRDCSIY IQWGFSSQTNV 180
YDRYTYPTFV TAHEYANHCTT VGTGTGLERLR GTNAFSLWNY NRFRREMTLT IOWDLSLFSN 240
LNNYQVRQIRH RAEALTREIYT DFIGQNSPL PGVVPNWYDY ARFPAEINI AIRAPRTVTW 300
LNSTTIYTRG LNGYNNSNYY WAGFRQNFSE TNSGSSFNPG DLGDLTPNYR IETLDMVNRD 360
IYSIYSRVVS QSWPIGNVKL FGVSSSTLSF RDLNNNSSGT LVYENPTNFS SQYLTTTEFPG 420
ENSRPPTFTD YSHRLTCLTR IAGNYYGLVL CAGWTSSSVE RDNIIERNKI TQFPVGKSHS 480
LNNYQVRQIRH GFTGGDWLRP NNNNGTFRLLT TSFSSQSRYI RLRYATSVGN TSLVISSSDA 540
GISSTTIPLT STITSLPQTV PYQAFRVVDL PITFTPTPTQ RNYTFDFRLQ NPSNANVFID 600
RFEFVPIGGS LSEYETFK                                     617

SEQ ID NO: 118          moltype = DNA length = 1833
FEATURE                Location/Qualifiers
misc_feature          1..1833
                        note = A synthetic coding sequence used for expression in a
                        plant cell encoding a GOI-TIC10746.nno_Mc:1 chimeric
                        protein comprised of domains one and two of TIC7382 and
                        domain three of TIC7383.
source                1..1833
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 118
atggccacta acaattacac cgacgcgctc cagtcctccg aagccacagc catttcgaag 60
ggggcggtct cggcgccgat cagcatcacg accaaggtgc tcggcctcct cggcgctgccg 120
ttcgcggctc agataggaca gctctgggac ttcattctga acgcctctgt gccgtccgac 180
aacaccagct ggaggagatt catcgggcac gtcgaggagc tcataacca cgcgatcgca 240
gactatgccc gtaacaaggc cctggccgag ctgacgggac tcggcaacaa cctcgacctc 300
tactctgagg ccttgatga ctggaagagg aacccccaga gccaggaggg caagactcgc 360
gtcatcgacc gcttcgcgat cgtcgacggg ctggttcgagg cctacattcc gagcttcgcc 420
gtgagcggtt atcaagttca gctgctcact gtctacgcgc ctcgacacca cctccacctc 480
ctgctgctgc gcgactccac gatctacgga atcgactggg cgtcttccca gttcacagct 540
aacgataact acaaccggca gattcgtctc accgcgacgt acgctaacca ctgtacaacg 600
tggtaccaga caggactgga gcggctccgc gggagcaacg cgtcttccgt ggtgacctac 660
aacggtttcc gcgagagagt gactctcaca gtcttgaca tctgttccct gttcacgca 720
tacgactatc gcagctaccc ggccgaggtt cgcggcgaga ttacgagga aatctacaca 780
gacctctgct gcctcgggtt ggttgacagc gcgccttctt tcggcgagat cgagaaacct 840
gcgacccggt cgcacgggac ttgtgacctg ctgaactcca cccggatctc caccggcacc 900
ttaagcggct ggtcgggctc taaccgggtac tgggcgggcc acatgcagaa tttctccgag 960
acaaactcgg gtaaatcctg ctctgatggg ccgctgtacg ggagcagagt gggcaccatc 1020
caccgcactg atgatacga catggggcaac cgggacatct acagcataac gtcgcaggcc 1080
gttctcggtc tgtggcccac gggccagcgg gtctctggag tcgcatctgc gcggttcacc 1140
ctcagggaac tattcaataa cctgacccag gtgctggtat acgagaatcc tattagctcc 1200
aacctcggct ctccaacgct aaccacaagc ttatccggcg agaacagcga tagggccacc 1260
agcagcgatt atccacatcg atccacaagc atcacgggat tcccgcgagg agcaaatggt 1320
accgttcccc tctttgggtg gaccagcgcc acagctgacc ggaacaaccc cctacaaccc 1380
gataagatta cccagctaccc cggcctcaag ggggttcaat tggacggctt cacagtgtgtg 1440
aaagggactg gtttcacgg ggccgaactgg ctcggtgtcca cagggttcac cgggtcattc 1500
cgctggaacg tatactcacc tagtgtgcag acttatcgga tgaggatcag gtacgcctcc 1560
ccgctgtggca actcgacgct tgggaatagt tcgacagcag ccggcatctc gttcacgagt 1620
ttccctctgc cctccactat cgggaacatg ccgtccactg tqccctacqa qcgcctccat 1680

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gtctctcgaca tccccatcac ggtccaccgta gcctcccaaa ggaactacaa ctctcatcttc 1740
gacatcctta acccctcggt tggggccggt tacatcgatc gtattgagtt cgtgcccggt 1800
ggctcttcag tgttcagagta cgaaccaag tga 1833

SEQ ID NO: 119      moltype = AA length = 610
FEATURE            Location/Qualifiers
REGION            1..610
                  note = The amino acid sequence of the GOI-TIC10746.nno_Mc:1
                  chimeric protein.
source            1..610
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 119
MATNNYTDAL QSPEATAIK GAVSAAISIS TKVLGLLGVP FAAQIGQLWT FILNALWPSD 60
NTQWEEFMRH VEELINQRIA DYARNKALAE LTGLGNNLDL YIEALDDWKR NPTSQEAQTR 120
VIDRFRIVDG LFEAYIPSEA VSGYQVQLLT VYAAANLHL LLLRDSITYG IDWGLSQTNV 180
NDNYNRQIRL TATYANHCTT WYQTGLERLR GSNASSWVTY NRRFRREMTLT VLDICSLFSN 240
YDYSYPAEV RGEITREIYT DPVGVGWVDS APSFGEIENL AIRAPRTVTW LNSTRISTGT 300
LSGWSGSNRY WAAHMQNFSE TNSGNIGFDG PLYGSTVGTI HRTDDYDMGN RDIYTITSQA 360
VLGLWPTGQR VLGVASARFT LRNLFNNTLQ VLVYENPISS NFGSSTLTHE LSGENSDRPT 420
SSDYSHRLTS ITGFRAGANG TVPVFGWTSV TVDRNNRLQP DKITQYPAVK GFNLDGFTTV 480
KGTGPTGGMW LRSSRVTSF RLNVYSPSVQ TYRMRIRYAS PLGNSTLGIS STDAGISFTS 540
FPLPSTIGSM PSTVPYEAFR VLDIPITVTV ASQRNYNFIF DILNPSVGAV YIDRIEFPVP 600
GSSVFEYETK 610

SEQ ID NO: 120      moltype = DNA length = 1839
FEATURE            Location/Qualifiers
misc_feature       1..1839
                  note = A synthetic coding sequence used for expression in a
                  plant cell encoding a GOI-TIC10747.nno_Mc:1 chimeric
                  protein comprised of domains one and two of TIC7381 and
                  domain three of TIC7383.
source            1..1839
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 120
atggctacag acaactacac cgaatgcactg caaggcccag agggcaacggc tatcagcaag 60
ggcgccggtg cgcccgctat ctctatcagc accaaagtct tgagccttct gggagtgccc 120
tcgcagctc agattgggca attgtggacg tttattctga acgctctttg gccctcggac 180
aacacccagt gggaggagtt catcgcccat gtcgaggagc tcatcaacca gcgtatcgcc 240
gactacgcgc gggagtaaggc gctcgccgag ctcaacgggc tgggcaacaa cctcgatctg 300
tacatcgaag ctcttgagga ctgggaagcg aaccgcacct ctcaacaagc caaggaccgg 360
gtgaaggacc gcttcgcgat cgccgacggc ctcttcgaag cgtacatgcc tagcttcgcg 420
gtgagcgggt atgaagtgc attgtcacc gtatatgcgg cggcgcccaa cctgcacctc 480
ctcctcctcc gcgactgcct tatctacggc attcagtggt gtttctccca gaccaatgtc 540
aacgagaact ataactcgga gatccgtcac accgcgagc acgcgaacca ctgcacgaca 600
tggtaccaga cgggcctgga gcgactgcgc ggcacgaacg cctcaagctg ggtgccgtac 660
aaccgtttca ggagagagat gactctcacc gtgctcgaca tctgcagctt gttctccaac 720
tatgactaca ggagttaccc ggcagaggtg cgggctgagc tgaccaggga gatatacacg 780
gaccgcgttg tcagcacctc tctgtgggtg aacaacgcgc cgtccttcgg cgaatcgag 840
aatctggcaa tccgagcgcc acgcaccggt acatggctca attcgacgct tattagcact 900
ggcacactcc aaggttggtt tgggtccaac aggtactggg cgcgccacat gcagaacttc 960
tcggagacta actcggcga cataggcttc gacggccgcg tgtaggggtc cactgtgggt 1020
acaatcattc gcgatgacaa ttacgagatg gtcaacggcg acatttacac catcacttcg 1080
gaggcggttg gggctttgtg gccacgggac cagattgtgc tggcggtggc ttccgcaaga 1140
ttcacccctc caaacctcaa caacaacctg acccaggcgc tgggtctacga gaaccgata 1200
tcaacgagtt tcaacgggtc aactctcaca cgcgagttgc cgggcgagaa cagcgacagg 1260
cccacctctt cggactactc ccaccgtctc actagcatca ctgcgttcgc cgcaggcagt 1320
aacggtacga ttctgtgtgt cggctggacc agtatctccg tcaacaggga caaccgccta 1380
caaccgata agattaccca gtaccccgcc gtcaagggtt tcaacttgga cggcttcaca 1440
gtggtgaaa ggactggttt caccgggggc aactggctgc ggtccagcag ggtcaccggg 1500
tcattccgcc tgaacgtata ctacactagt gtgcagactt atcggatgag gatcaggtac 1560
gcctccccgc ttggcaactc gacgcttggg ataagttcga cagacgcggc catctcggtc 1620
acgagtttcc ctctgccttc cactatcggg agcatgcccgt cactgtgccc ctacgaggcg 1680
ttccgtgtcc tcgacatccc catcacgggt accgtagcct cccaaaggaa ctacaacttc 1740
atcttcgaca tccttaaccc ctcggttggg gccgtgtaca tcgatcgat tgagttcggt 1800
cccggtgggt ctctcagttt cgagtacgaa accaagtga 1839

SEQ ID NO: 121      moltype = AA length = 612
FEATURE            Location/Qualifiers
REGION            1..612
                  note = The amino acid sequence of the GOI-TIC10747.nno_Mc:1
                  chimeric protein.
source            1..612
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 121
MATDNYTDAL QGPEATAIK GAVSAAISIS TKVLSLLGVP FAAQIGQLWT FILNALWPSD 60
NTQWEEFMRH VEELINQRIA DYARKALAE LTGLGNNLDL YIEALDDWKR NPTSQAQADR 120

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VKDRFRIADG LFEAYMPSFR VSGYEVPLLT VYAAAAANLHL LLLRDCSIYG IQWGFSTQNTV 180
NENYNRQIRH TAEYANHCTT WYQTGLERLR GTNASSWVPY NRRFRREMTLT VLDICSLFSN 240
YDYSYPAEV RAEALTREIYT DPVVSTSLWV NNAPSFGEIE NLAIRAPRTV TWLNSTRIST 300
GTLQGWSGSN RYWAAHMQNF SETNSGNIGF DGPLYGSTVG TIIRDNDYEM VNRDIYTITS 360
EAVAALWPTG QIVLGVASAR FTLRNLNNL TQALVYENPI SSSFNSTLT RELPGENSDR 420
PTSSDYSHRL TSITAFRAGS NGTIPVFGWT SISVNRDNRL QPDKITQYPA VKGFNLDGFT 480
VVKGTGTGG NWLRSRVTVG SFRLNVYSPS VQTYRMIRY ASPLGNSTLG ISSTDAGISF 540
TSFPLPSTIG SMPSTVPYEA FRVLDIPITV TVASQRNYNF IFDILNPSVG AVYIDRIEFV 600
PVGSSVFYEY TK 612

SEQ ID NO: 122      moltype = DNA length = 1830
FEATURE            Location/Qualifiers
misc_feature        1..1830
                    note = A synthetic coding sequence used for expression in a
                    plant cell encoding a GOI-TIC10748.nno_Mc:1 chimeric
                    protein comprised of domains one and two of TIC7042 and
                    domain three of TIC7383.
source              1..1830
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 122
atggcttgca cgcatacgga cgactcgcgc tcgcccaggg ccgcgagcac ggctcgcage 60
gcgatctccg tcgccaatcac cattagcacc accatcctgg gactccttgg agttcccttc 120
gcgtcccaga tcggtgcctt ctacaacttc gtgctcaaca ccgtgtggcc tcagggaagt 180
aaccagtggg aggagttcat gaggcacgtg gagaacctca tcaacgagcg cattgctgac 240
tacgcgcgcg acaaaagcctt gcccgagctg accggcctgg gcaacaacct gaatctctat 300
cgggaggctt tcgaggactg gcggcggaac ccgacctccc aggaagccaa gactcgggtg 360
atcgaccgtc tcgtattctt cgacggcctg ttcgagactt acatgccctc cttcgcggtc 420
cgcaactttg aggttcagct cctgactgtg tacgcctcgg ctgcaaacat acacctgttc 480
ctgctgcggg acagctcgat ctacggtttg gactggggcc tgtcacagac caacgtgaac 540
gagaactaca accgcagat cccccacgcc gccacctatg caaacactg caccacttgg 600
tatcagacag ggtccagcg actccagggc actaacgcaa cgtcatgggt tgcatacaac 660
cgggttcggc gggagatgac actcaccgtg ttggacatct gctcgctgtt ctccaactac 720
gactaccggt cgtaccgcac tgaggtgaag ggcgagttga ccagggaagt ctacacagac 780
cctgtcggcc ggaactggca gaacgtcgcg ccttcattcg cggagattga gaacctgacc 840
attcgcgctc cgcgtaccgt cacttggttc aactccactc ggatcttcac agggacgctg 900
accggtggtt ccgggtccaa ccgctactgg gccgccaca tgcagaactt cagcgagaca 960
aactctggca acatcggatt cgatggccca cagtatgggt ccacggtcgg gacgatacac 1020
cggaccgacg actacgacat ggtgaaccgg gacatctaca ccataacgag ccaggcagtc 1080
gcgcgccctc ggcgcccgcc ccagacagtc ctgggcgttg cgaaccccg cttcacactg 1140
aggaatctga acaacaactc gacggaggca ctcggtgacg agaacgccat ctcgtcctcc 1200
tcggttagca gcactctcac tcgatgctg ccgggcgaga acagcgacgg cccaacgagc 1260
agcgattaca gccaccgtct ttctccatc acgggcttcc gggctggagc caacgggacc 1320
gtgcccggtg tcggttggac cagcgcgaca gttgaccgta acaaccgctt acaaccggat 1380
aagattaccg agtaccgcgc cgtcaagggg ttcaacttgg acggcttcac agtggtgaaa 1440
gggactgggt tcaccggggg caactggctg cgggtccagca gggtcaccgg gtcattccgc 1500
ctgaacgtat actcacctag tgtgcagact tatcggatga ggatcaggta cgctccccg 1560
cttggaactc cgacgcttgg gataagttcg acagacgccg gcatctcgtt cagcagtttc 1620
cctctgccct ccactatcgg gagcatgccg tccactgtgc cctacgaggg gttccgtgtc 1680
ctcgacatcc ccatcacggt caccgtagcc tcccaaggga actacaactt catcttcgac 1740
atccttaacc cctcggttgg gcccggtgtac atcgatcgta ttgagttcgt gcccggtggc 1800
tcttcagtgt tcgagtaaga aaccaagtga 1830

SEQ ID NO: 123      moltype = AA length = 609
FEATURE            Location/Qualifiers
REGION              1..609
                    note = The amino acid sequence of the GOI-TIC10748.nno_Mc:1
                    chimeric protein.
source              1..609
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 123
MACTDTDDSR SPEAASTARS AISVAITIST TILGLLVGPF ASQIGAFYNF VLNTVVPQGN 60
NQWEEFMRHV ENLINERIAD YARDKALAEI TGLGNLNLNY REAFEDWRRN PTSQEAQTRV 120
IDRFRLDGL FETYMPSFAV RNFEVQLLTV YASAANIHLF LLDSSSIYGL DWGLSQTNVN 180
ENYNRQIRHA ATYANHCTTW YQTGLQLRQG TNATSWVAYN RFRREMTLTV LDICSLFSNY 240
DYRSYPTVKV GELTREIYTD PVGRNWQNVV PSFAEINLT IRAPRTVIWL NSTRIFTGTL 300
TGWSGSNRYW AAHMQNFSET NSGNIGFDGP QYGSTVGTIH RTDDYDMVNR DIYTITSQAV 360
AALWPTGQTV LGVASTRETL RNLNNNSTEA LVYENAISSS FVSSTLTHEL PGENSDRPTS 420
SDYSHRLSSI TGFRAGANGT VPVFGWTSAT VDRNNRLQPD KITQYPAVKG FNLDGFTVVK 480
GTGFTGGNWL RSSRVTSFSR LNVYSPSVQT YRMIRYASPL LGNSTLGISS TDAGISFTSF 540
PLPSTIGSMP STVPYEAFRV LDIPITVTVA SQRYNYFIFD ILNPSVGAVY IDRIEFVVPV 600
SSVFYETK 609

SEQ ID NO: 124      moltype = DNA length = 1980
FEATURE            Location/Qualifiers
misc_feature        1..1980
                    note = A synthetic coding sequence used for expression in a
                    plant cell encoding a TIC10746NTermExt1chimeric protein

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-continued

comprised of domains one and two of TIC7382, domain three of TIC7383, and the N-terminal extension of TIC7382.

source 1..1980
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 124

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gggcggaact acaaggagtg gctaaacaag tgcactaaca attacaccga cgcgctccag 180
tctccggaag ccacagccat ttcgaagggg gcggtctcgg cggcgatcag catcagcacc 240
aagggtgctcg gcctcctcgg cgtgccgttc gcggctcaga taggacagct ctggacattc 300
attctgaacg cctctgtggc gtcgcacaac acccagtggg aggagttcat gcggcacgtc 360
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acgggactcg gcaacaacct cgacctctac atcgaggccc tggatgactg gaagaggaaac 480
cccaccagcc aggaggccaa gactcgcgtc atcgaccgct tccgcatcgt cgacgggctg 540
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gactggggcc tttcccagac taacgtcaac gataactaca accggcagat tctgtctacc 720
gcgacgtacg ctaaccactg tacaacgtgg taccagacag gactggagcg gctccgcggg 780
agcaacgcgt cttcctgggt gacctacaac cgtttccgga gagagatgac tctcacagt 840
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SEQ ID NO: 125 moltype = AA length = 659
FEATURE Location/Qualifiers
1..659
note = The amino acid sequence of the TIC10746NTermExt1 chimeric protein.
source 1..659
mol_type = protein
organism = synthetic construct

SEQUENCE: 125

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MNQNQNQKN EMQIEPSSD SFLYSHNNYP YATDPNTVLE GRNYKEWLNK CTNNYTDALQ 60
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EELINQRIAD YARKNALAEL TGLGNLDDL Y IEALDDWKRN PTSQEAQTRV IDRFRIVDGL 180
FEAYIPSPAV SGYQVQLTV YAAAAANLHLL LLRDSTIYGI DWGLSQTNVN DNYNRQIRLT 240
ATYANHCTTW YQTGLERLRG SNASSWVTYN RFRREMTLTV LDICSLFSNY DYRSYPAEVR 300
GEITREIYTD PVGVGWVDSA PSFGEIENLA IRAPRTVTWL NSTRISTGTL SGWSGSNRYW 360
AAHMQNFSET NSGNIGFDGP LYGSTVGTIH RTDDYDMGMR DIYTITSQAV LGLWPTGQRV 420
LGVASARFTL RNLFPNNLTQV LVYENPISSN FGSSTLTHEL SGENS DRPTS SDYSHRLTSI 480
TGFRAGANGT VPVFGWTSAT VDRNNRLQPD KITQYPAVKG FNLDGFTVVK GTGFTGNNWL 540
RSSRVTSFPR LNVYSPSVQT YRMRIYASP LGNSTLGISS TDAGISFTSF PLPSTIGSMP 600
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SEQ ID NO: 126 moltype = DNA length = 1986
FEATURE Location/Qualifiers
misc_feature 1..1986
note = A synthetic coding sequence used for expression in a plant cell encoding a TIC10746NTermExt2 chimeric protein comprised of domains one and two of TIC7382, domain three of TIC7383, and the N-terminal extension of TIC7383.
source 1..1986
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 126

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ctccagtctc cggaagccac agccatttcg aaggggggcg tctcggcgcg gatcagcatc 240
agcaccaagg tgctcggcct cctcggcggt cggttcgcgg ctcagatagg acagctctgg 300
acattcatc tgaacgcctt gtggccgtcc gacaacaccc agtgggagga gttcatgccc 360
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aagtga 1986

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SEQ ID NO: 127      moltype = AA length = 661
FEATURE             Location/Qualifiers
REGION              1..661
                    note = The amino acid sequence of the TIC10746TermExt2
                    chimeric protein.
source              1..661
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 127
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HVEELINQRI ADYARNKALA ELTGLGNLND LYIEALDDWK RNPTSQEAKE RVIDRFRIVD 180
GLFEAYIPSF AVSGYQVQLL TVYAAAANLH LLLLRDSTIY GIDWGLSQTN VNDNYNRQIR 240
LTATYANHCT TWYQTGLERL RGSNASSWVT YNRFREMTL TVLDICSLFS NYDYRSYPAE 300
VRGEITREIY TDPVGVGVVD SAPSFGIEIN LAIRAPRTVT WLNSTRISTG TSGWGSNSR 360
YWAHMQNPS ETNSGNIGFD GPLYGSTVGT IHRDDYDMG NRDIYTITSQ AVLGLWPTGQ 420
RVLGVASARF TLRNLFNNLT QVLVYENPIS SNFGSSTLTH ELSGENSDRP TSSDYSHRLT 480
STGTFRAGAN GTVPVFGWTS ATVDRNNRLQ PDKITQYPAV KGFNLDGFTV VKGTGFTGGN 540
WLRSSRVTS PRLNVYSPSV QTYRMIRIYA SPLGNSTLGI SSTDAGISFT SFPLPSTIGS 600
MPSTVPYEAF RVLDIPITVT VASQRNINFI FDILNPSVGA VYIDRIEFVP VGSSVFEYET 660
K

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What is claimed is:

1. A recombinant nucleic acid molecule comprising a heterologous promoter operably linked to a polynucleotide segment encoding a pesticidal protein or pesticidal fragment thereof, wherein:

- a. said pesticidal protein comprises the amino acid sequence of SEQ ID NO:57, SEQ ID NO: 79, or SEQ ID NO:81;
- b. said pesticidal protein comprises an amino acid sequence having at least 95% identity to the amino acid sequence of SEQ ID NO:57, SEQ ID NO:79, or SEQ ID NO: 81; or
- c. said polynucleotide segment comprises the polynucleotide sequence of SEQ ID NO:56, SEQ ID NO:78, or SEQ ID NO:80.

2. The recombinant nucleic acid molecule of claim 1, wherein:

- a. the recombinant nucleic acid molecule comprises a sequence that functions to express the pesticidal protein in a plant; or
- b. the recombinant nucleic acid molecule is expressed in a plant cell to produce a pesticidally effective amount of the pesticidal protein; or

c. the recombinant nucleic acid molecule is in operable linkage with a vector, and said vector is selected from the group consisting of a plasmid, phagemid, bacmid, cosmid, and a bacterial or yeast artificial chromosome.

3. The recombinant nucleic acid molecule of claim 1, comprised within a host cell, wherein said host cell is selected from the group consisting of a bacterial and a plant cell.

4. The recombinant nucleic acid molecule of claim 3, wherein the bacterial host cell is from a genus of bacteria selected from the group consisting of: *Agrobacterium*, *Rhizobium*, *Bacillus*, *Brevibacillus*, *Escherichia*, *Pseudomonas*, *Klebsiella*, *Pantoea*, and *Erwinia*.

5. The recombinant nucleic acid molecule of claim 4, wherein the *Bacillus* species is *Bacillus cereus* or *Bacillus thuringiensis*, said *Brevibacillus* is *Brevibacillus laterosporus*, or said *Escherichia* is *Escherichia coli*.

6. The recombinant nucleic acid molecule of claim 3, wherein said plant cell is a dicotyledonous or a monocotyledonous plant cell.

7. The recombinant nucleic acid molecule of claim 6, wherein said plant host cell is selected from the group consisting of an alfalfa, banana, barley, bean, broccoli,

cabbage, *Brassica*, carrot, cassava, castor, cauliflower, celery, chickpea, Chinese cabbage, citrus, coconut, coffee, corn, clover, cotton, a cucurbit, cucumber, Douglas fir, eggplant, *eucalyptus*, flax, garlic, grape, hops, leek, lettuce, Loblolly pine, millets, melons, nut, oat, olive, onion, ornamental, palm, pasture grass, pea, peanut, pepper, pigeonpea, pine, potato, poplar, pumpkin, *Radiata* pine, radish, rapeseed, rice, rootstocks, rye, safflower, shrub, sorghum, Southern pine, soybean, spinach, squash, strawberry, sugar beet, sugarcane, sunflower, sweet corn, sweet gum, sweet potato, switchgrass, tea, tobacco, tomato, triticale, turf grass, watermelon, and wheat plant cell.

8. The recombinant nucleic acid molecule of claim 1, wherein said pesticidal protein exhibits activity against a Coleopteran insect.

9. The recombinant nucleic acid molecule of claim 8, wherein said insect is Western Corn Rootworm, Southern Corn Rootworm, Northern Corn Rootworm, Mexican Corn Rootworm, Brazilian Corn Rootworm, Colorado Potato Beetle, Brazilian Corn Rootworm complex consisting of *Diabrotica viridula* and *Diabrotica speciosa*, Crucifer Flea Beetle, Striped Flea Beetle, or Western Black Flea Beetle.

10. The recombinant nucleic acid molecule of claim 1, wherein said pesticidal protein exhibits activity against an insect species of the order of Lepidoptera.

11. The recombinant nucleic acid molecule of claim 10, wherein said insect is Black Cutworm, Corn Earworm, Diamondback Moth, European Corn Borer, Fall Armyworm, Southern Armyworm, Soybean Looper, Southwestern Corn Borer, Tobacco Budworm, Velvetbean Caterpillar, Sugarcane Borer, Lesser Cornstalk Borer, Black Armyworm, Beet Armyworm, Old World Bollworm, Oriental leaf Worm, or Pink Bollworm.

12. A plant or part thereof comprising the recombinant nucleic acid molecule of claim 1.

13. The plant or part thereof of claim 12, wherein said plant is a monocot plant.

14. The plant or part thereof of claim 12, wherein the plant is selected from the group consisting of an alfalfa, banana, barley, bean, broccoli, cabbage, *Brassica*, carrot, cassava, castor, cauliflower, celery, chickpea, Chinese cabbage, citrus, coconut, coffee, corn, clover, cotton, a cucurbit, cucumber, Douglas fir, eggplant, *eucalyptus*, flax, garlic, grape, hops, leek, lettuce, Loblolly pine, millets, melons, nut, oat, olive, onion, ornamental, palm, pasture grass, pea, peanut, pepper, pigeon pea, pine, potato, poplar, pumpkin, *Radiata* pine, radish, rapeseed, rice, rootstocks, rye, safflower, shrub, sorghum, Southern pine, soybean, spinach, squash, strawberry, sugar beet, sugarcane, sunflower, sweet corn, sweet gum, sweet potato, switchgrass, tea, tobacco, tomato, triticale, turf grass, watermelon, and wheat.

15. A seed of the plant of claim 12, wherein said seed comprises said recombinant nucleic acid molecule.

16. An insect inhibitory composition comprising the recombinant nucleic acid molecule of claim 1.

17. The insect inhibitory composition of claim 16, further comprising a nucleotide sequence encoding at least one other pesticidal agent that is different from said pesticidal protein.

18. The insect inhibitory composition of claim 17, wherein said at least one other pesticidal agent is selected from the group consisting of an insect inhibitory protein, an insect inhibitory dsRNA molecule, and an ancillary protein.

19. The insect inhibitory composition of claim 17, wherein said at least one other pesticidal agent exhibits activity against one or more pest species of the orders Lepidoptera, Coleoptera, or Hemiptera.

20. The insect inhibitory composition of claim 19, wherein said at least one other pesticidal protein is selected from the group consisting of Cry1A, Cry1Ab, Cry1Ac, Cry1A.105, Cry1Ae, Cry1B, Cry1C, Cry1D, Cry1E, Cry1F, Cry1G, Cry1H, Cry1I, Cry1J, Cry1K, Cry1L, Cry2A, Cry2Ab, Cry2Ae, Cry3, Cry3B, Cry4B, Cry6, Cry7, Cry8, Cry9, Cry15, Cry34, Cry35, Cry43A, Cry43B, Cry51Aa1, ET29, ET33, ET34, ET35, ET66, ET70, TIC400, TIC407, TIC417, TIC431, TIC800, TIC807, TIC834, TIC853, TIC900, TIC901, TIC1201, TIC1415, TIC2160, TIC3131, TIC836, TIC860, TIC867, TIC869, TIC1100, VIP3A, VIP3B, VIP3Ab, AXMI-88, AXMI-97, AXMI-102, AXMI-112, AXMI-117, AXMI-100, AXMI-115, AXMI-113, AXMI-005, AXMI134, AXMI-150, AXMI-171, AXMI-184, AXMI-196, AXMI-204, AXMI-207, AXMI-209, AXMI-205, AXMI-218, AXMI-220, AXMI-221z, AXMI-222z, AXMI-223z, AXMI-224z, AXMI-225z, AXMI-238, AXMI-270, AXMI-279, AXMI-345, AXMI-335, AXMI-R1, IP3, DIG-3, DIG-5, DIG-10, DIG-657, DIG-11, Cry71Aa1, Cry72Aa1, PHI-4, PIP-72, PIP-45, PIP-64, PIP-74, PIP-75, PIP-77, Axmi422, Dig-305, Axmi440, PIP-47, Axmi281, BT-009, BT-0012, BT-0013, BT-0023, BT0067, BT-0044, BT-0051, BT-0068, BT-0128, DIG-17, DIG-90, DIG-79, Cry1JP578V, Cry1JPS1, and Cry1 JPS1P578V.

21. The insect inhibitory composition of claim 16, comprising a plant cell that expresses said recombinant nucleic acid molecule.

22. A commodity product produced from the plant or part thereof of claim 12, wherein the commodity product comprises a detectable amount of said recombinant nucleic acid molecule or a pesticidal protein encoded thereby.

23. The commodity product of claim 22, selected from the group consisting of commodity corn bagged by a grain handler, corn flakes, corn cakes, corn flour, corn meal, corn syrup, corn oil, corn silage, corn starch, corn cereal, whole or processed cotton seed, cotton oil, lint, seeds and plant parts processed for feed or food, fiber, paper, biomasses, and fuel products such as fuel derived from cotton oil or pellets derived from cotton gin waste, whole or processed soybean seed, soybean oil, soybean protein, soybean meal, soybean flour, soybean flakes, soybean bran, soybean milk, soybean cheese, soybean wine, animal feed comprising soybean, paper comprising soybean, cream comprising soybean, soybean biomass, and fuel products produced using soybean plants and soybean plant parts.

24. A method of producing seed comprising:

- planting at least the first seed according to claim 15;
- growing a plant from the seed; and
- harvesting seed from the plant, wherein said harvested seed comprises said recombinant nucleic acid molecule.

25. A plant resistant to insect infestation, wherein the cells of said plant comprise the recombinant nucleic acid molecule of claim 1.

26. A method for controlling a Coleopteran or Lepidopteran species pest or pest infestation, said method comprising:

- contacting the pest with an insecticidally effective amount of a pesticidal protein comprising the sequence of SEQ ID NO:57, SEQ ID NO:79, or SEQ ID NO:81; SEQ ID NO:63 or SEQ ID NO:65; or
- contacting the pest with an insecticidally effective amount of one or more pesticidal proteins comprising an amino acid sequence having at least 95% identity to the sequence of SEQ ID NO:57, SEQ ID NO:79, or SEQ ID NO:81.

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27. The plant or part thereof of claim 12, wherein said plant is a dicot plant.

28. The recombinant nucleic acid molecule of claim 1, wherein the pesticidal protein comprises the amino acid sequence of SEQ ID NO:57. 5

29. The recombinant nucleic acid molecule of claim 1, wherein the pesticidal protein comprises the amino acid sequence of SEQ ID NO:79.

30. The recombinant nucleic acid molecule of claim 1, wherein the pesticidal protein comprises the amino acid sequence of SEQ ID NO:81. 10

31. The recombinant nucleic acid molecule of claim 1, wherein the polynucleotide segment comprises the polynucleotide sequence of SEQ ID NO:56, SEQ ID NO:78, or SEQ ID NO: 80. 15

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